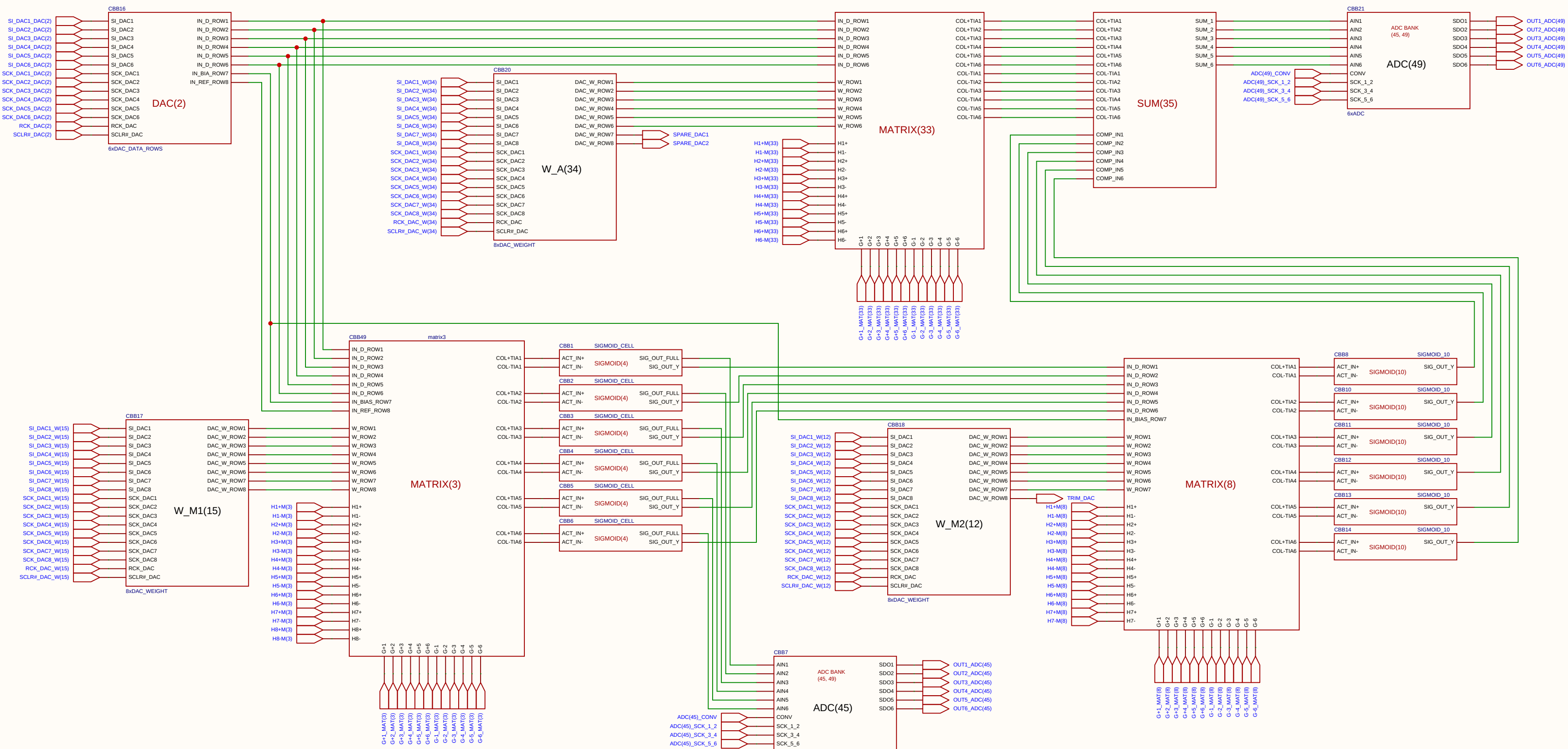
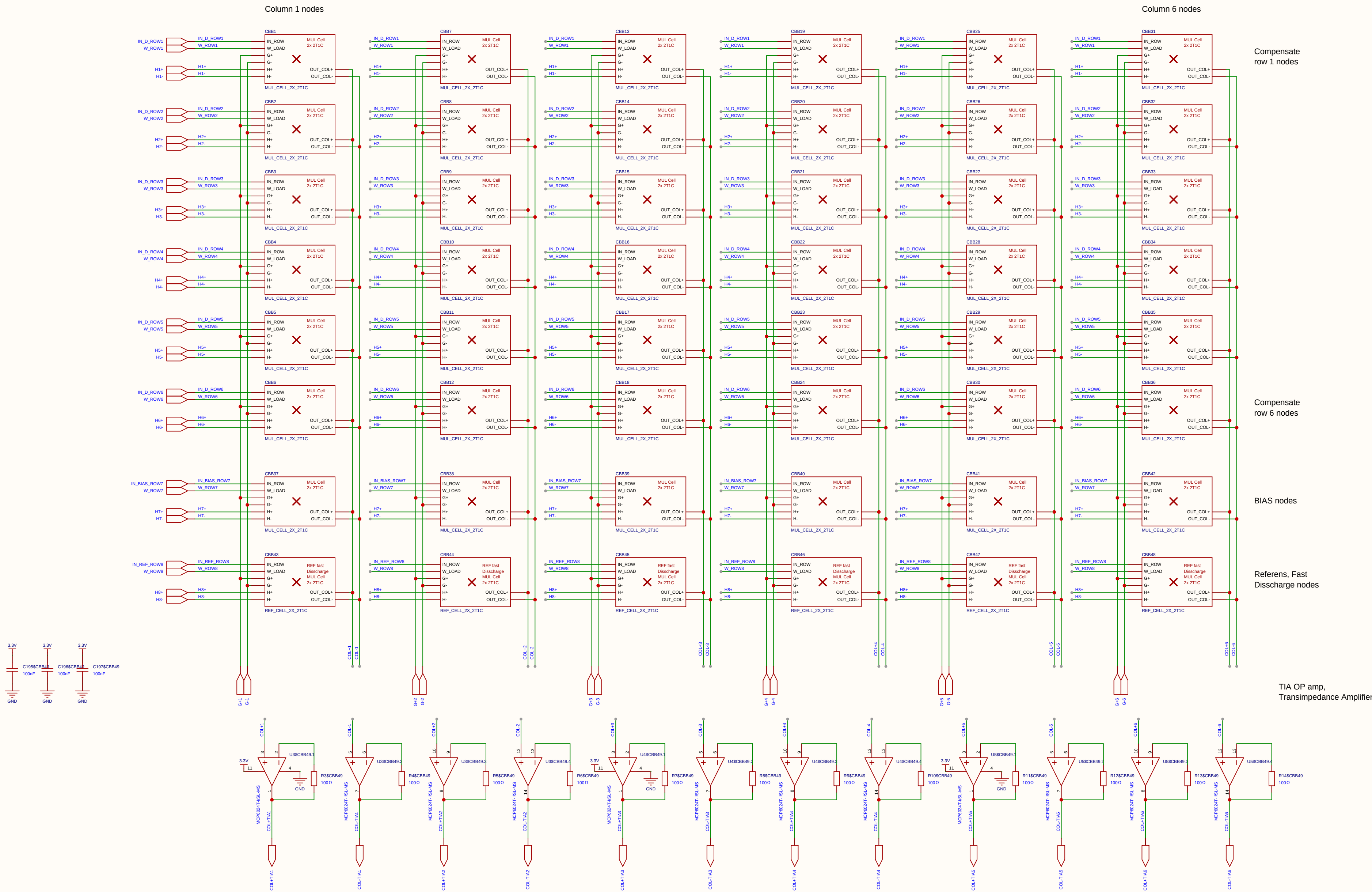




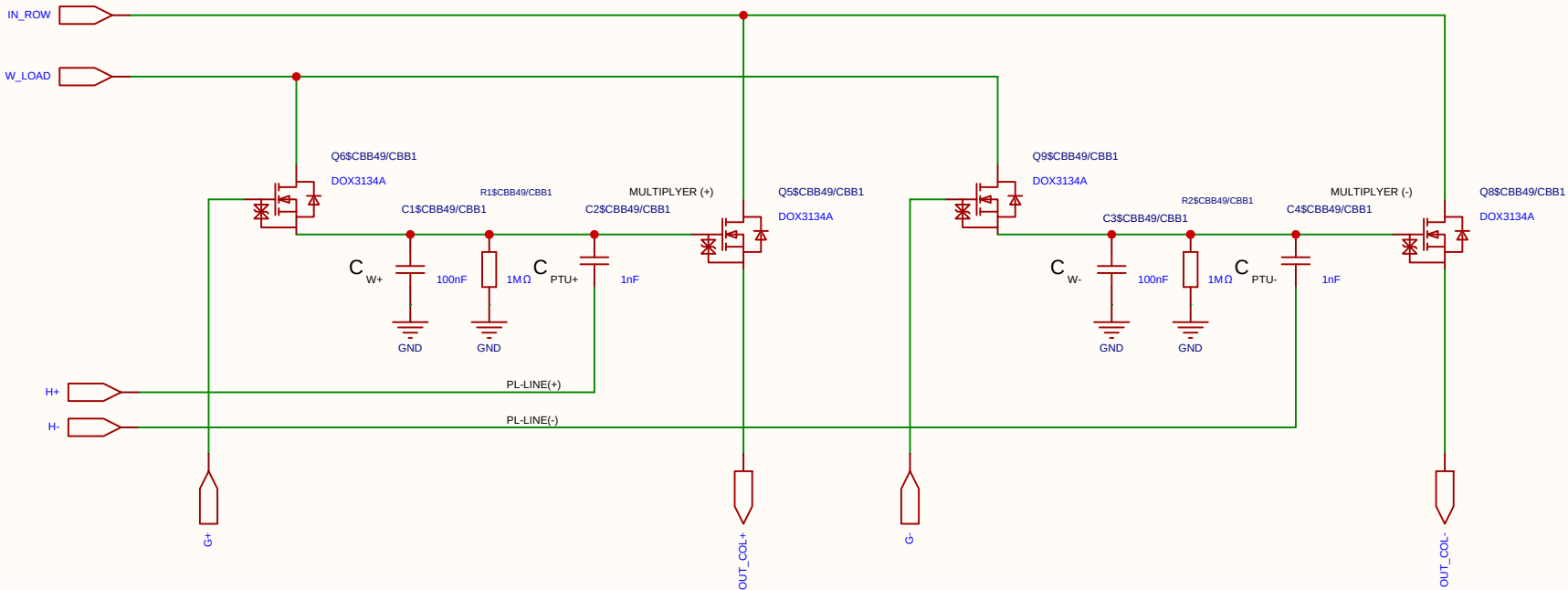
Schematic	Inverted_MAML			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	main
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 2 Total 2	
		V1.0	A4	EasyEDA.com	

$$V_{ut_TIA} = 1.65\text{ V} - (5\text{ mA} \times 100\ \Omega) = 1.65\text{ V} - 0.5\text{ V} = 1.15\text{ V}$$



Schematic	matrix3			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matx3
Drawn		Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

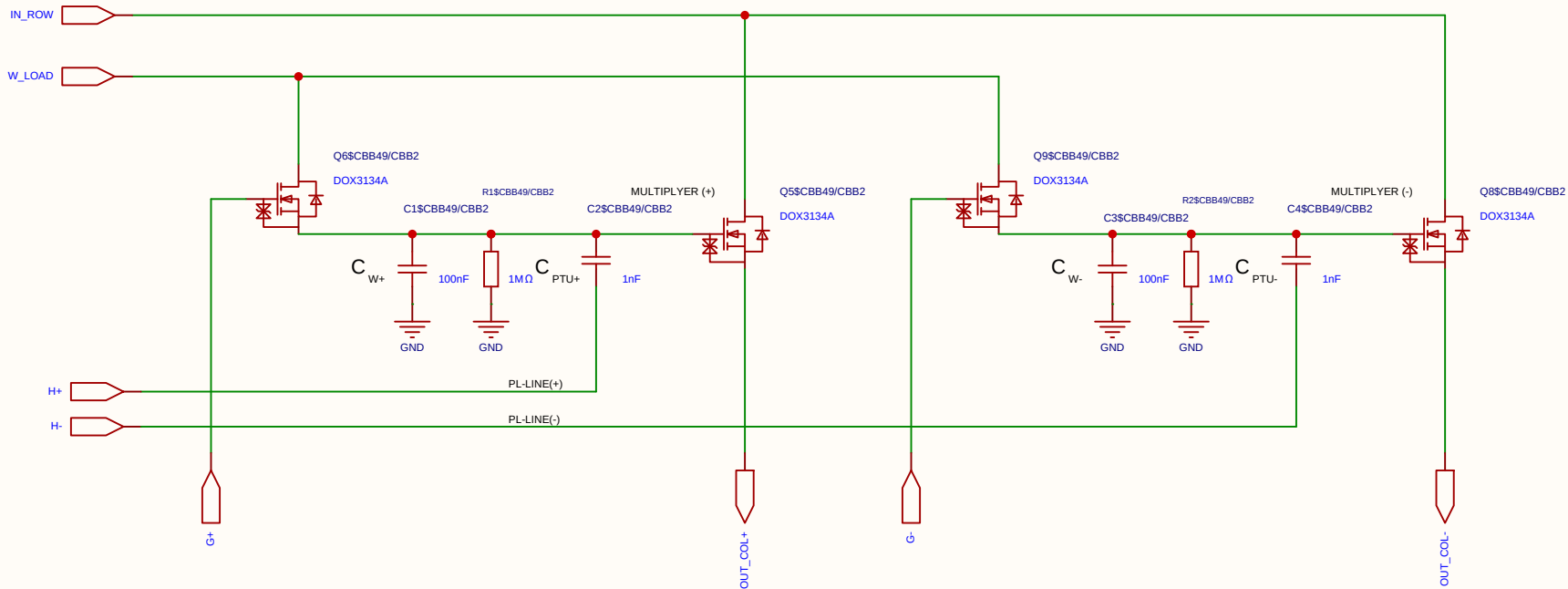
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

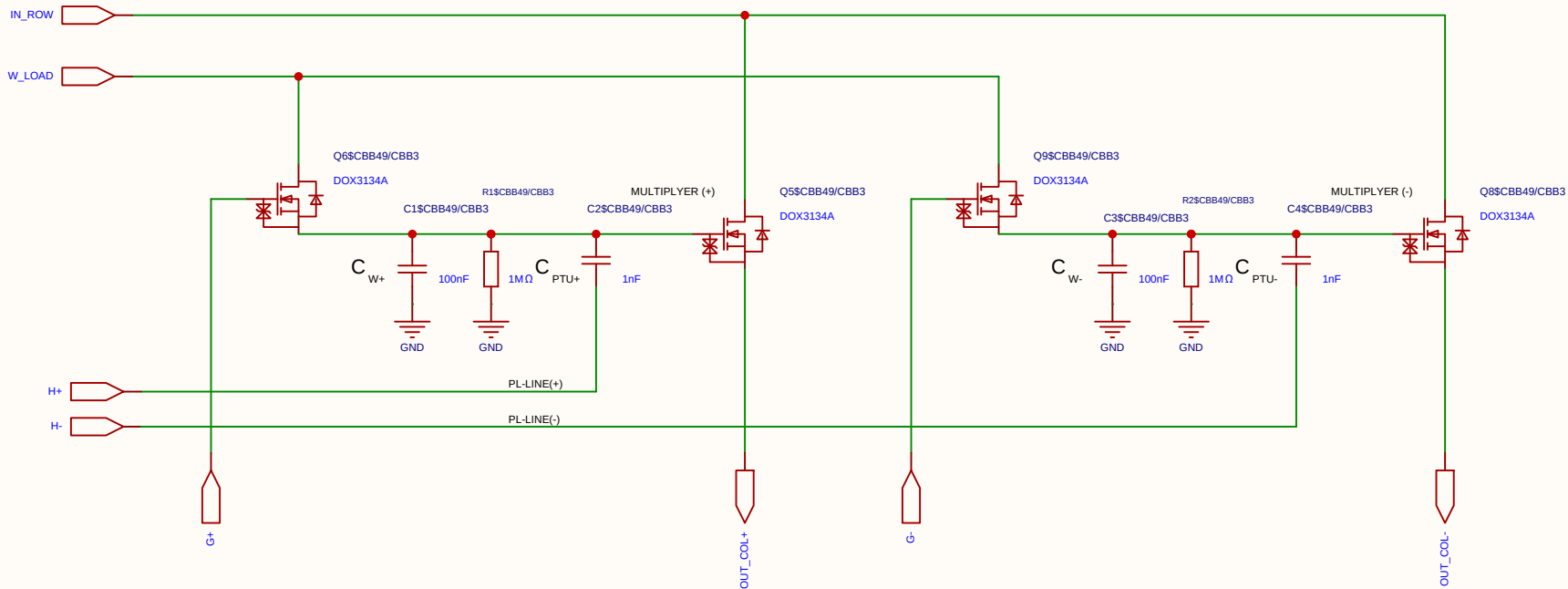
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

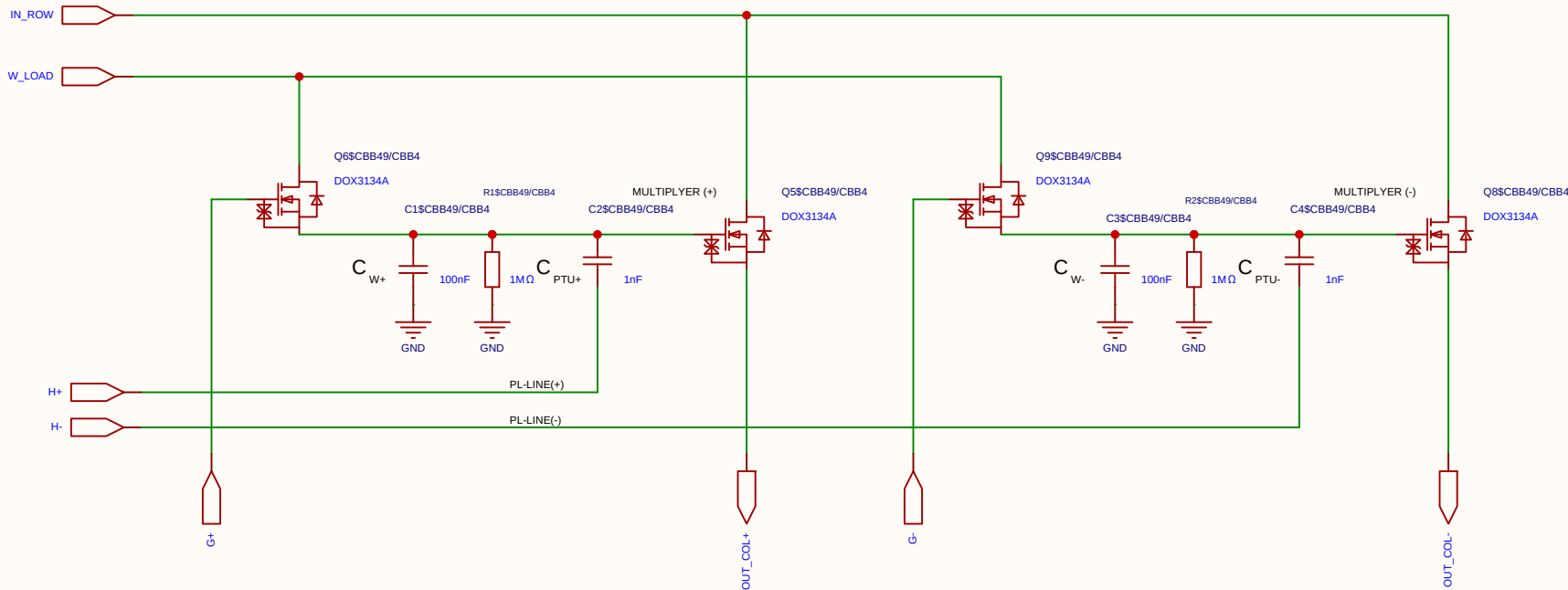
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

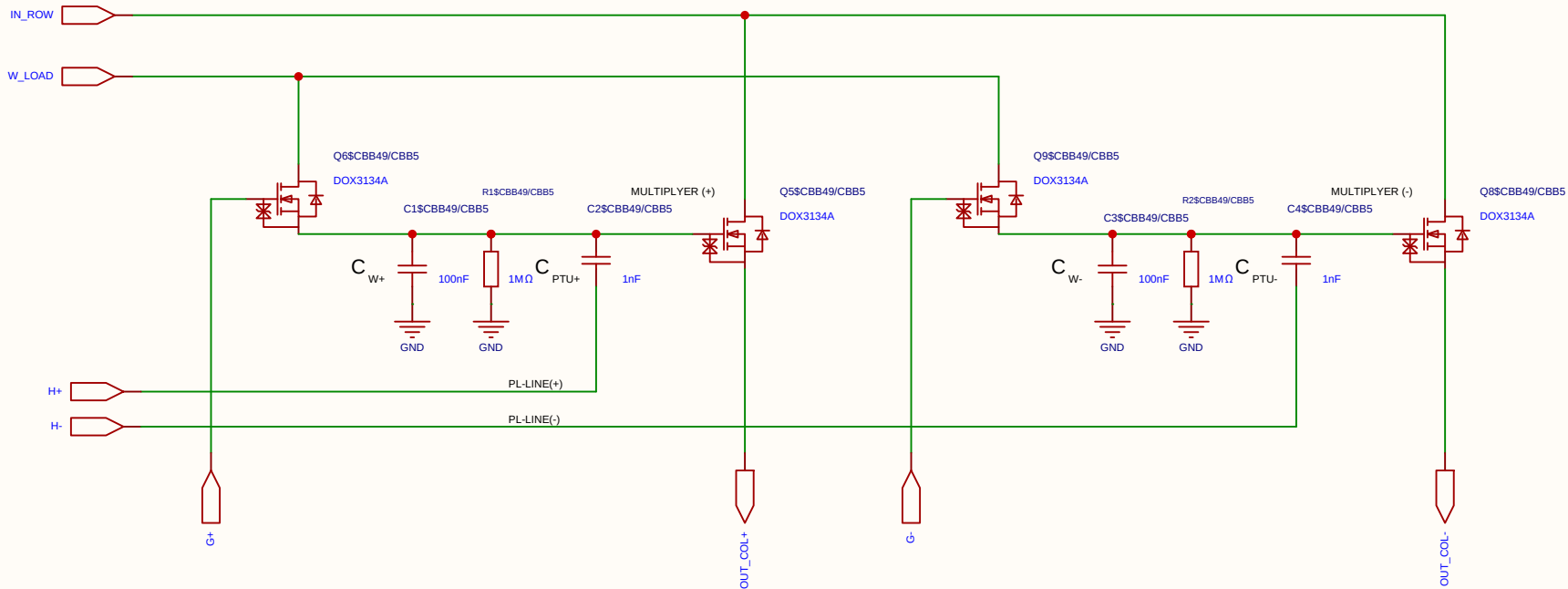
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

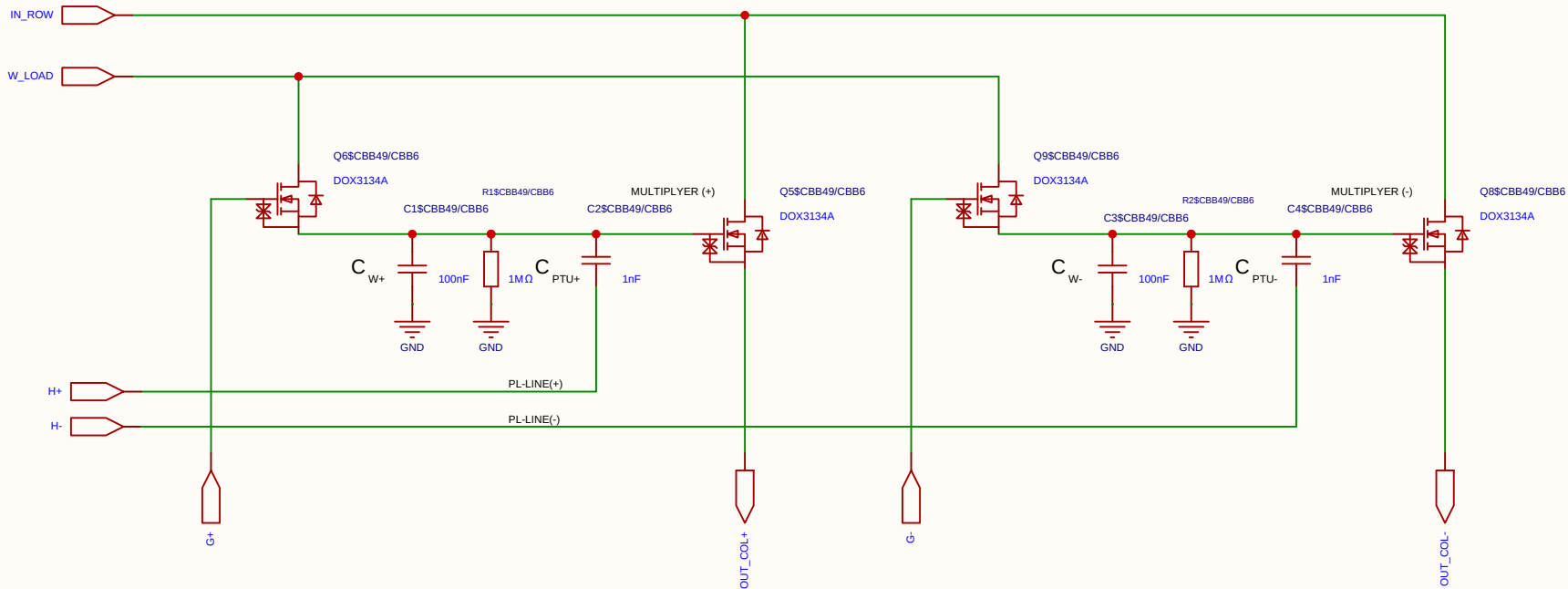
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

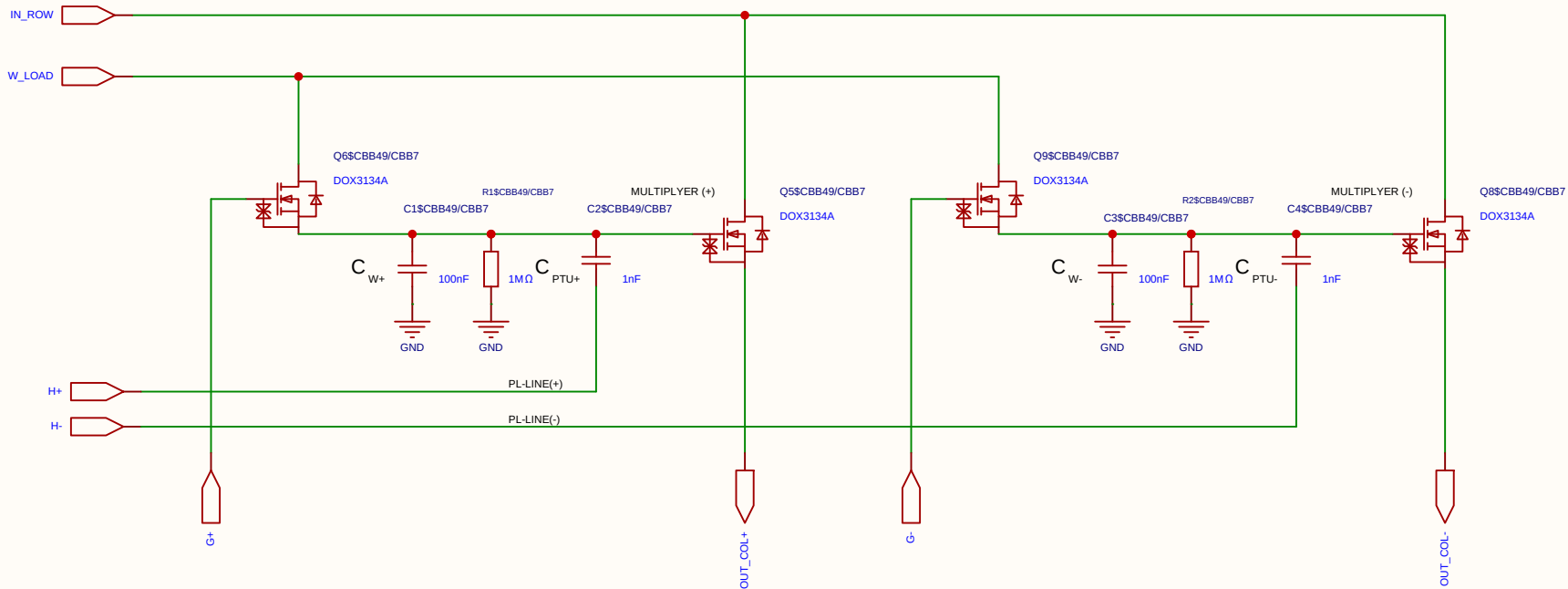
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

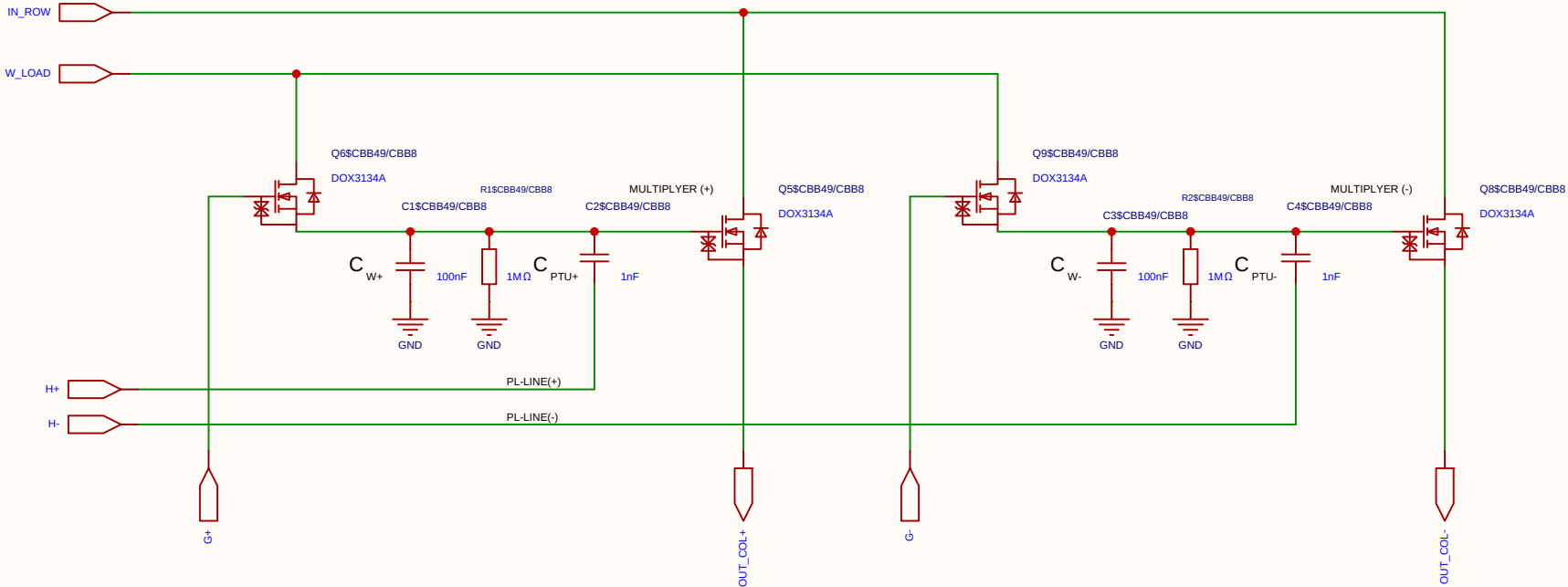
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

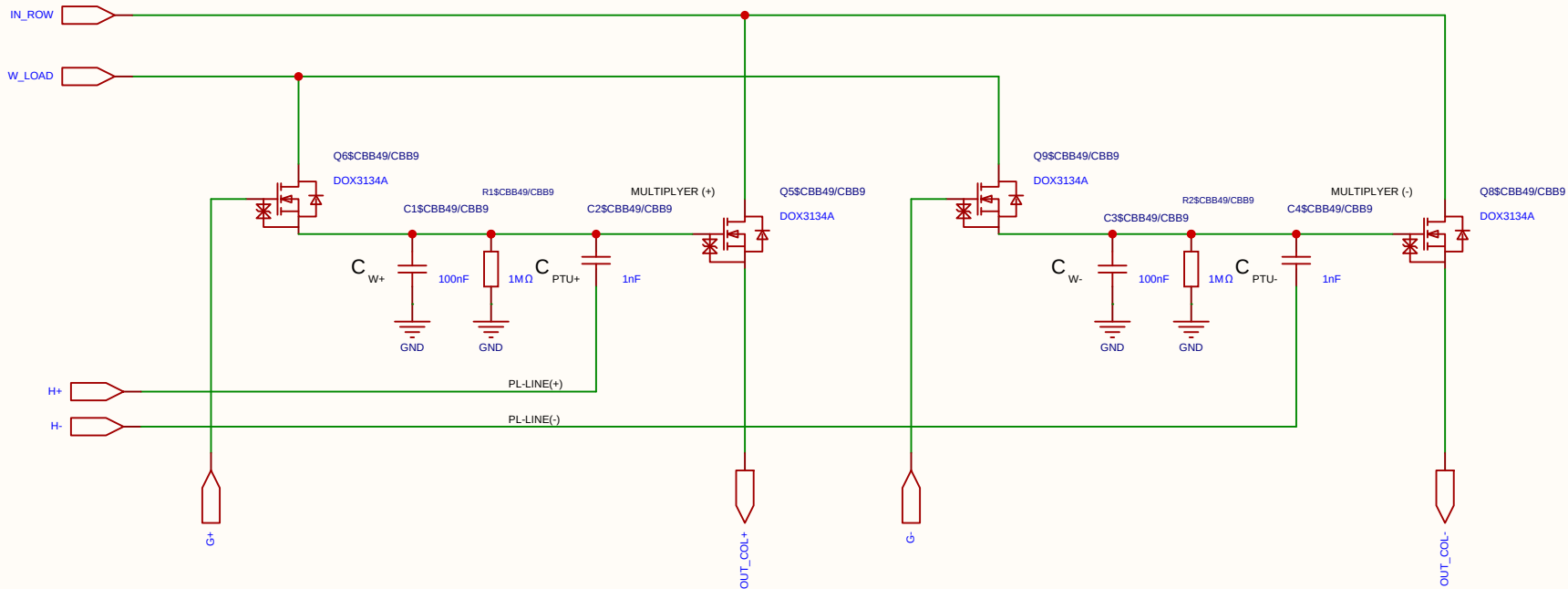
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

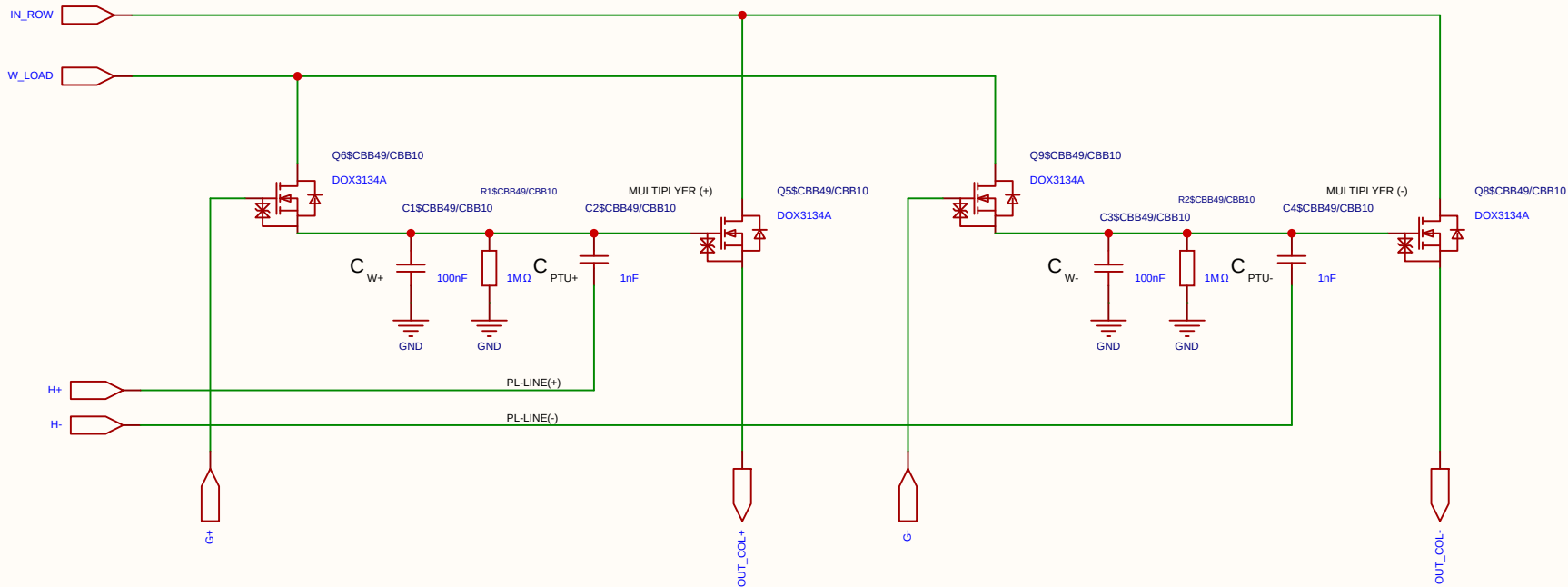
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
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		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

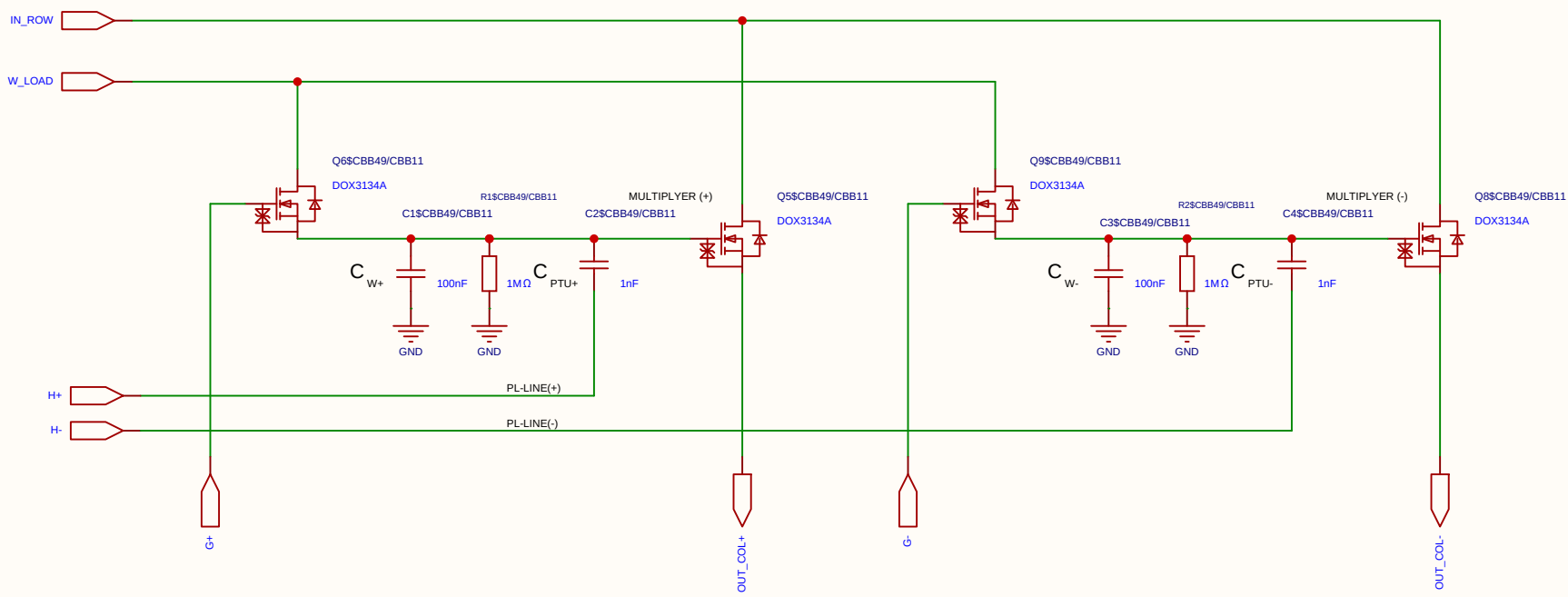
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
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Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

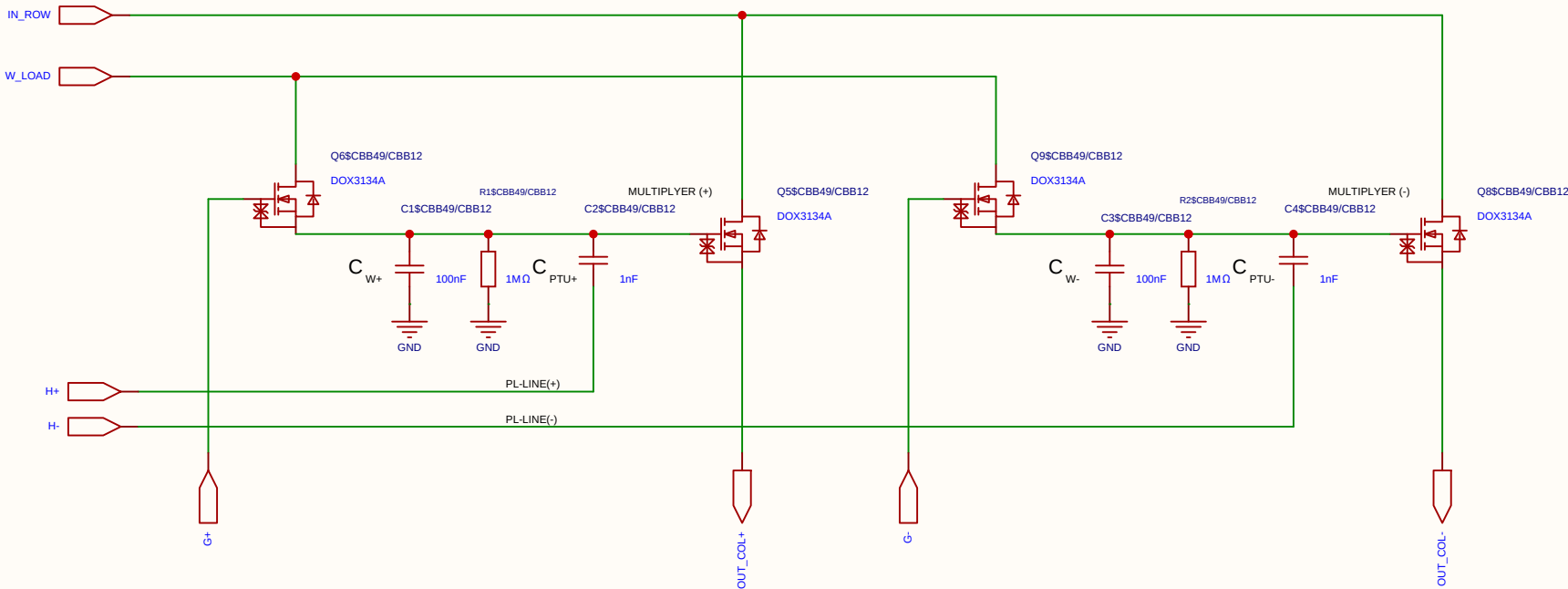
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

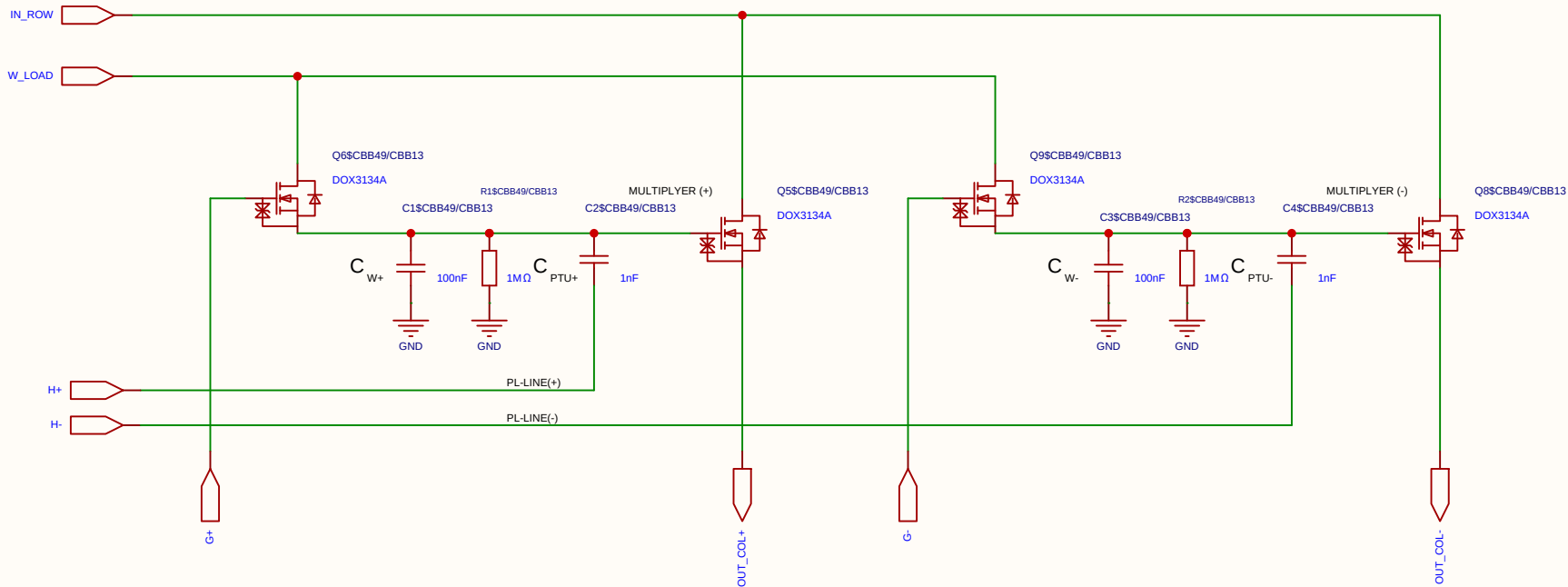
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

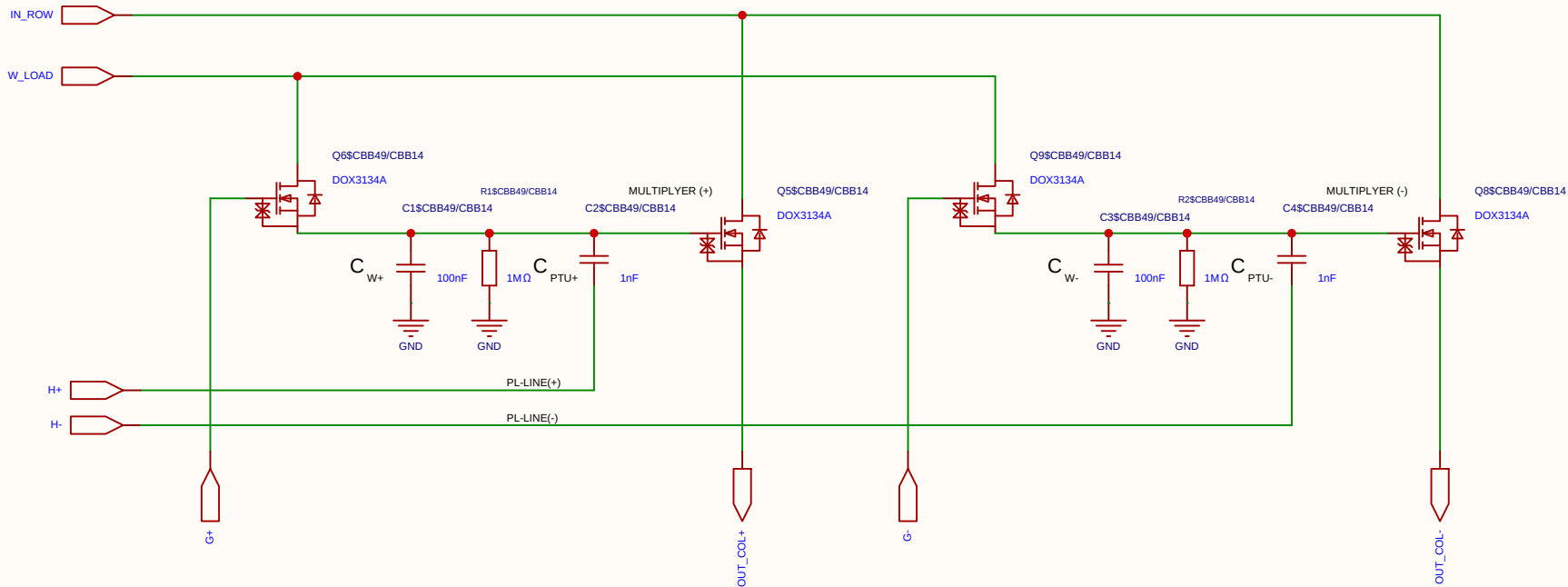
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

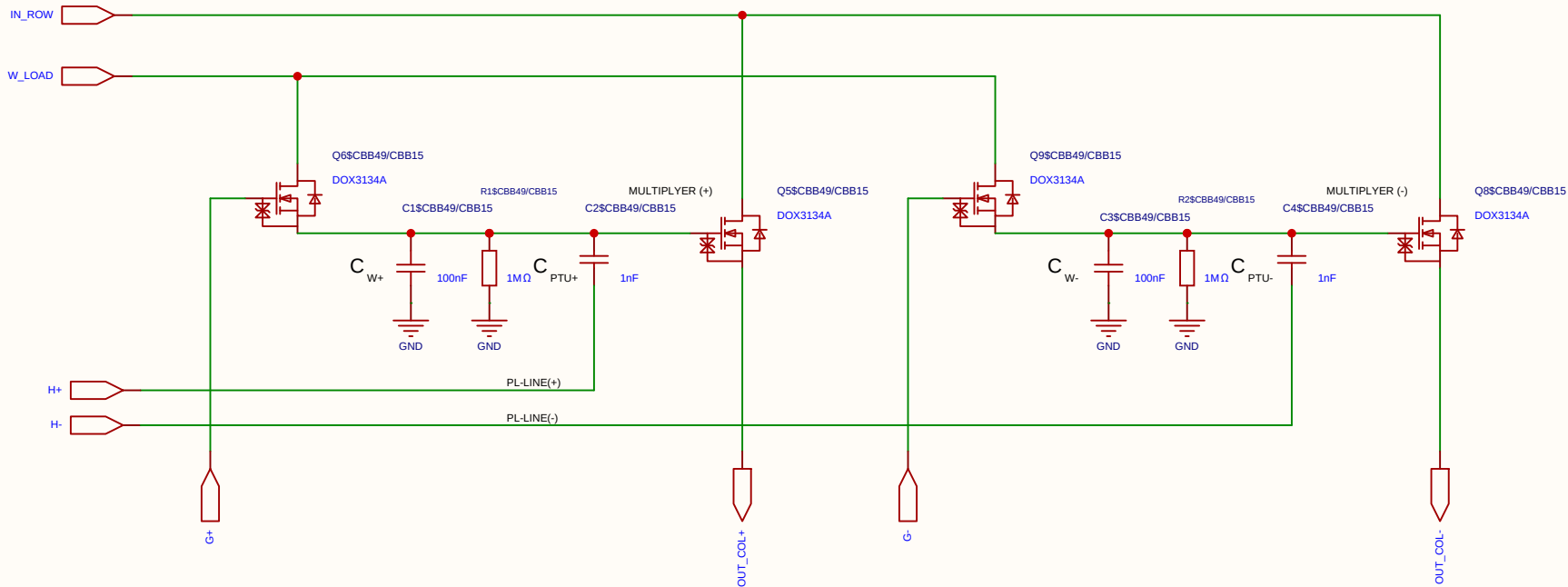
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

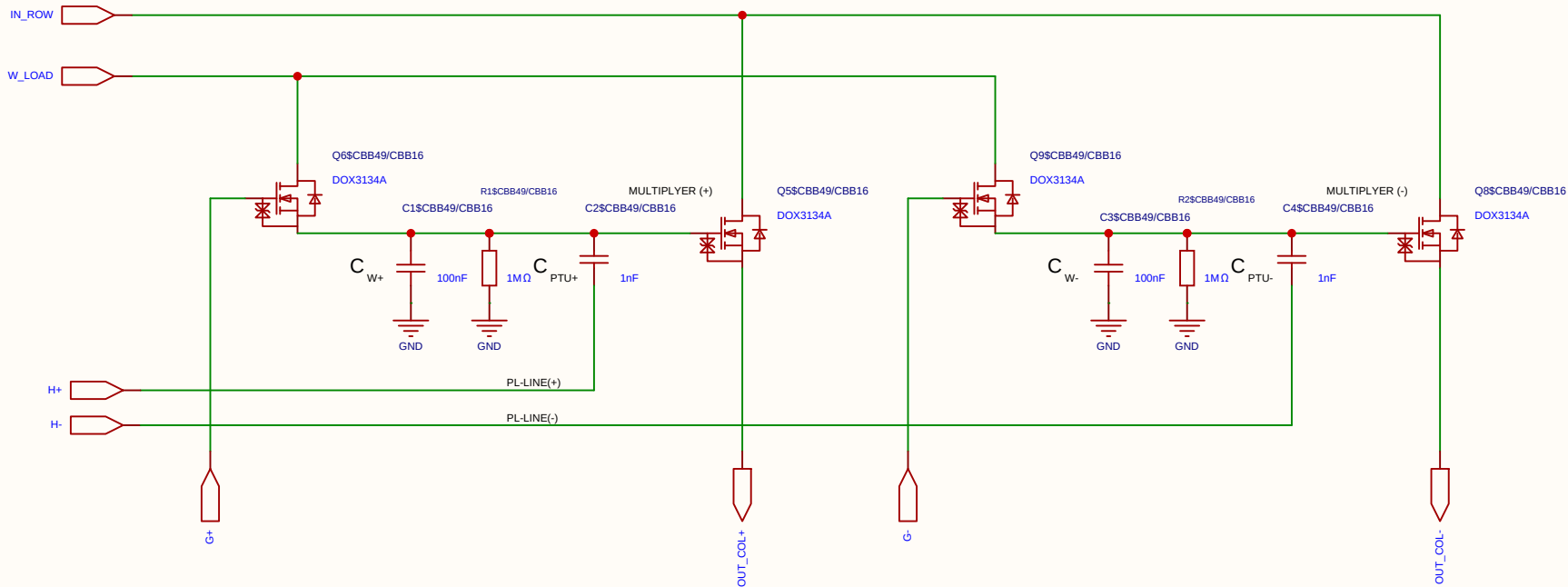
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

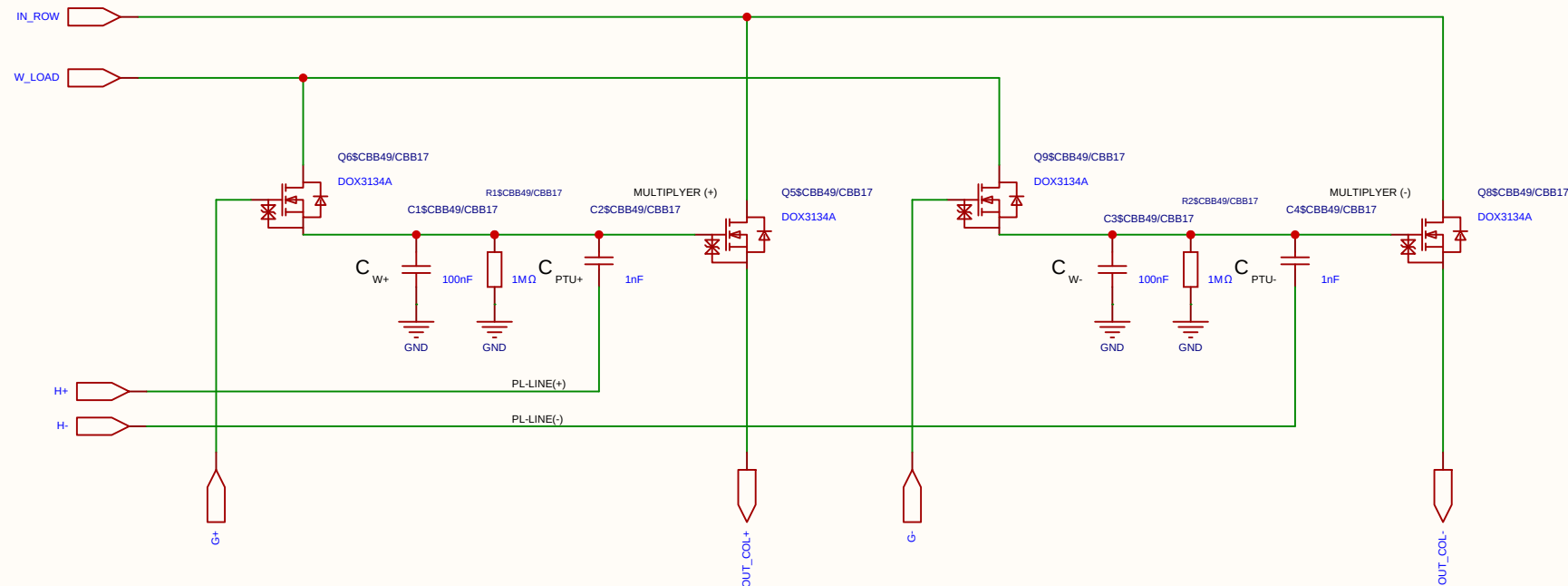
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Board				Page	matrix_3_8_MUL_CELL
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Analog Multiplication CELL Kernel



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
Resistors	Discharge resistor (R)	1 M Ω	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

Component	Value	Description / Function
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a $\sim 1/101$ voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

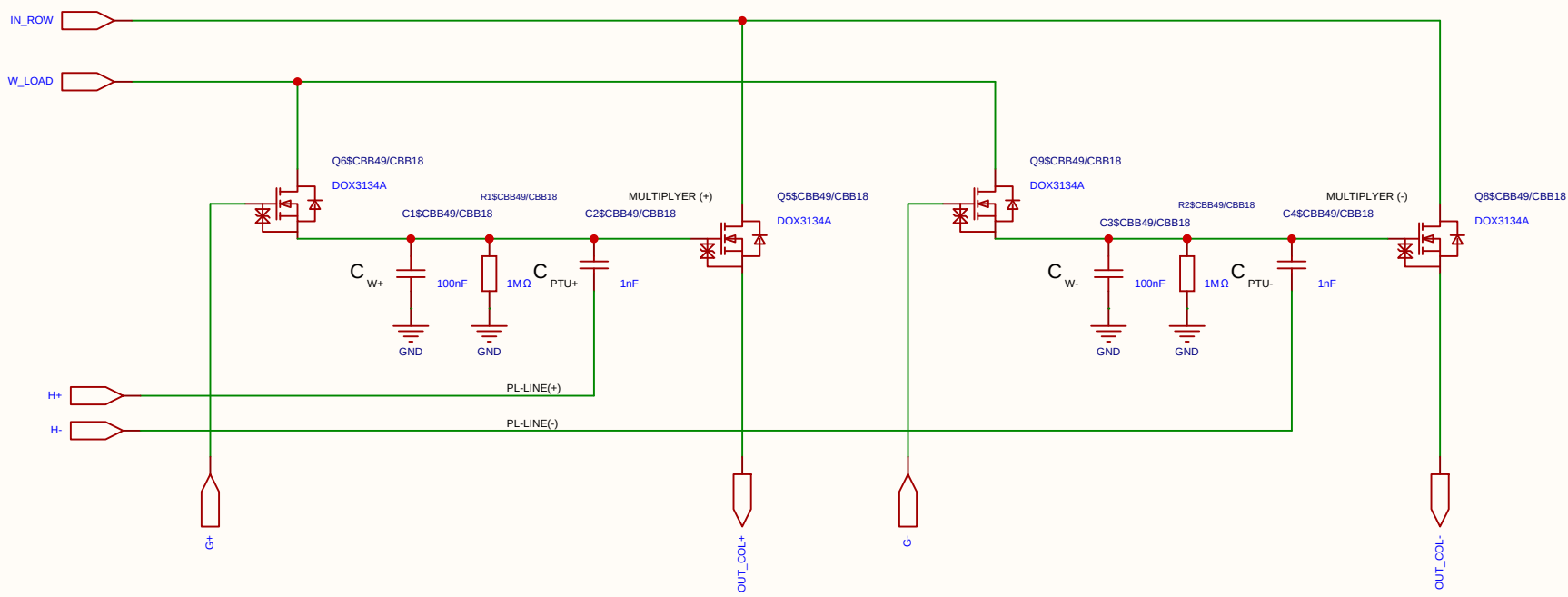
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_I row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

Component	Value	Description / Function
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

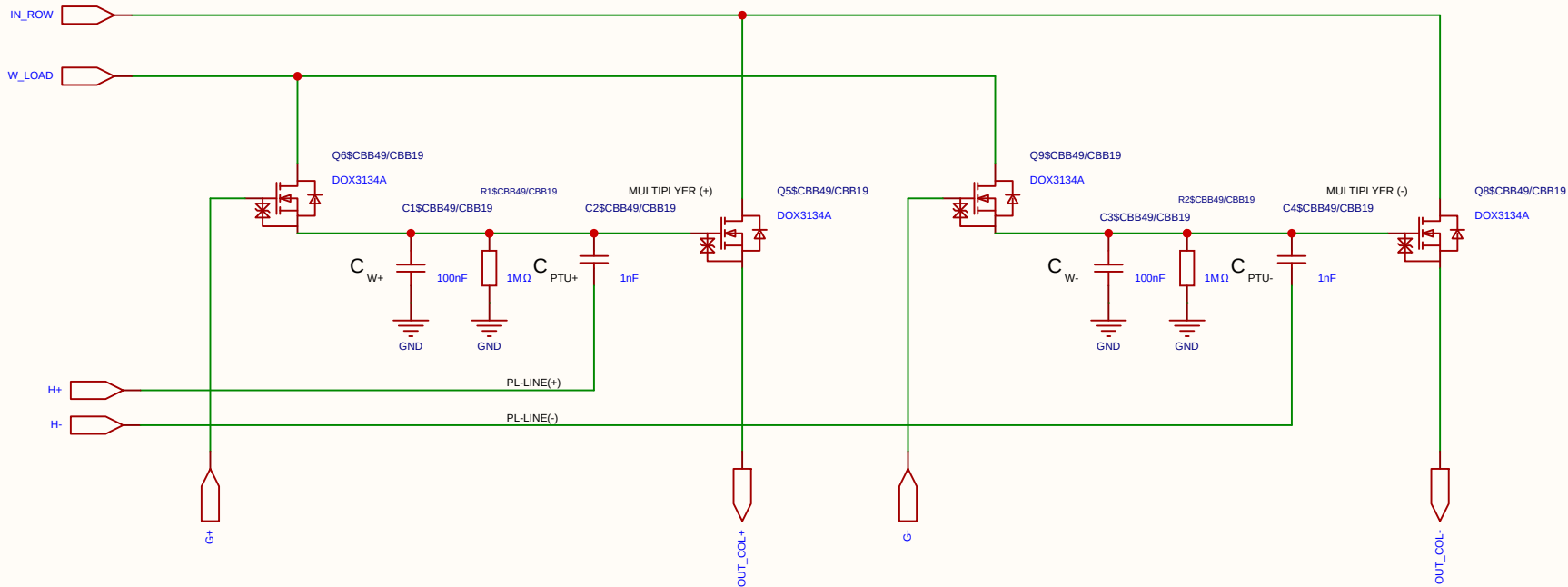
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

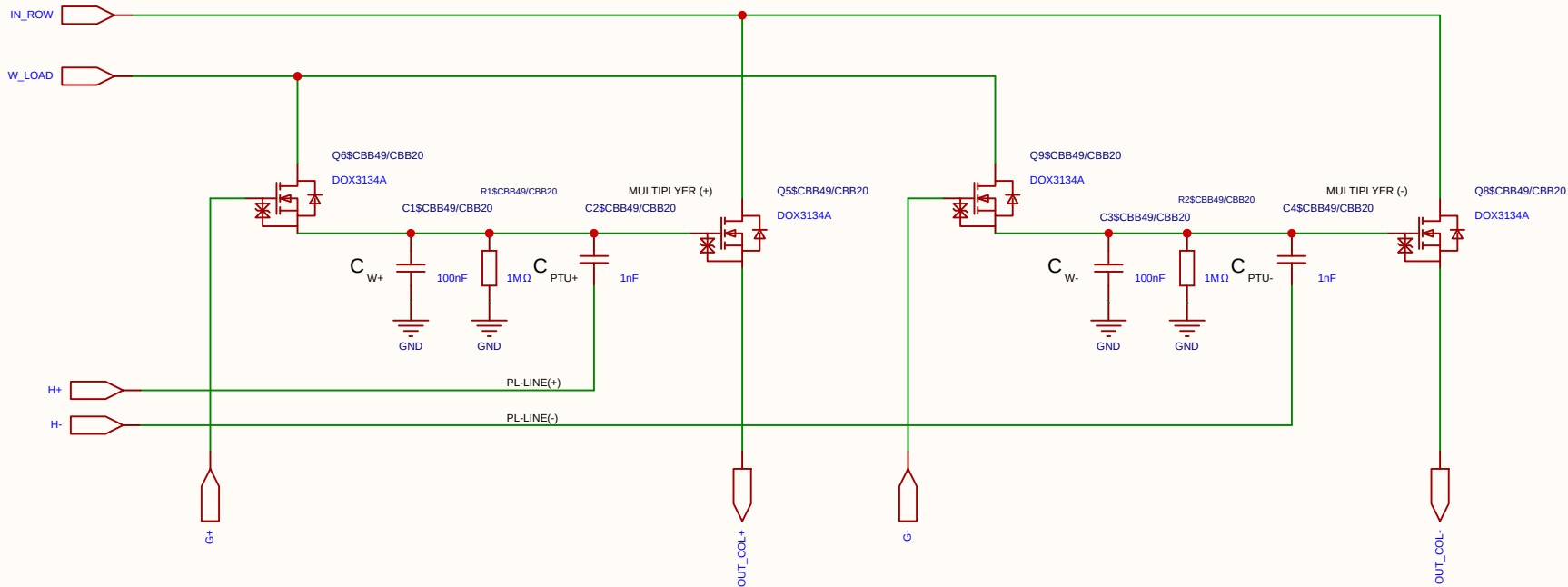
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

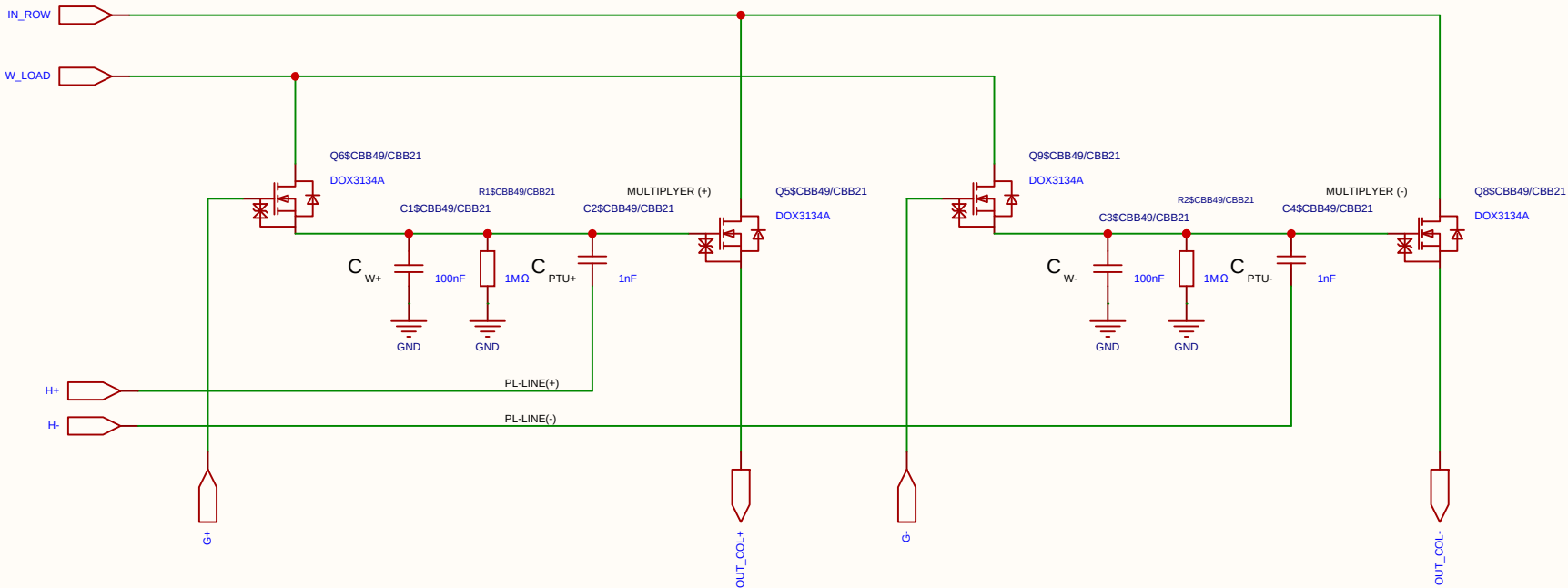
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

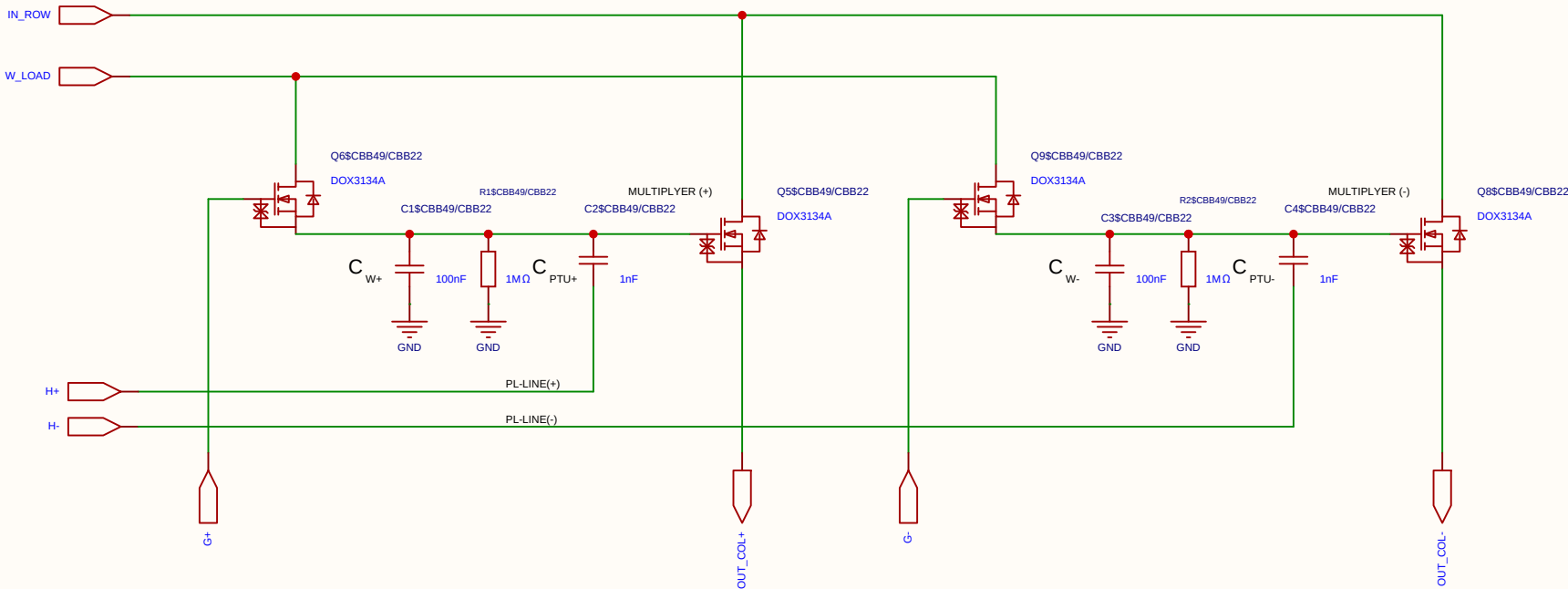
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

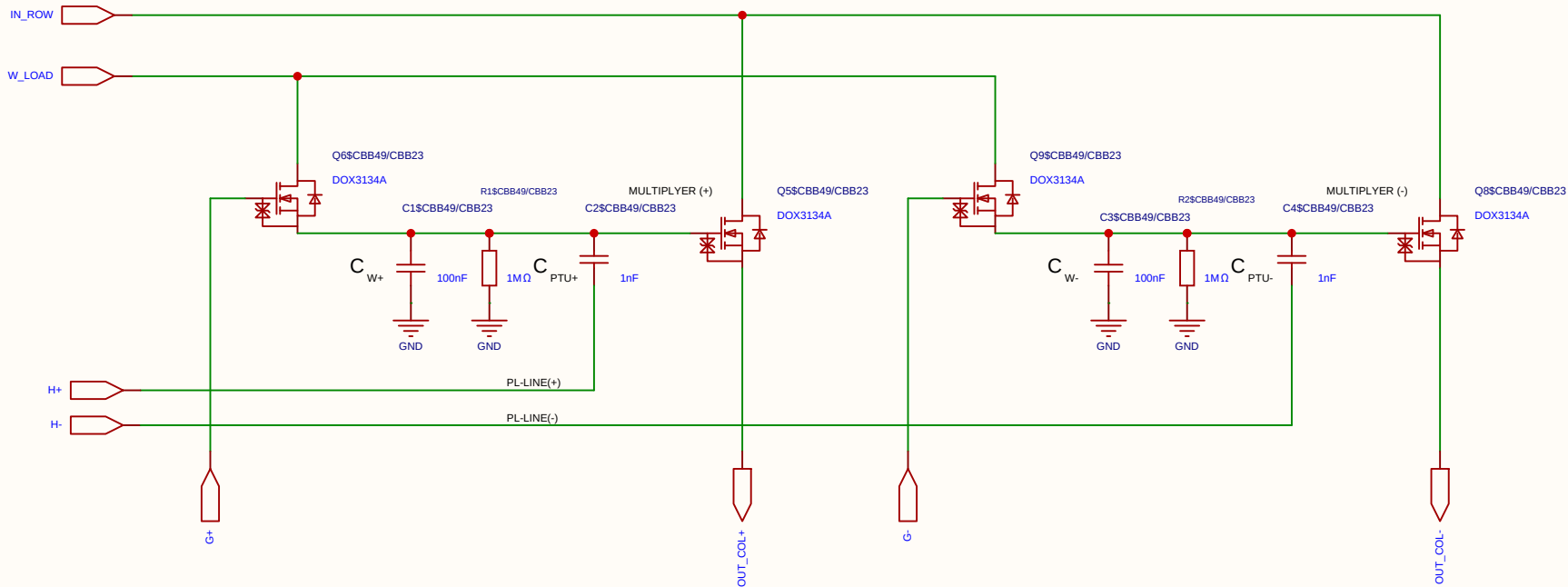
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

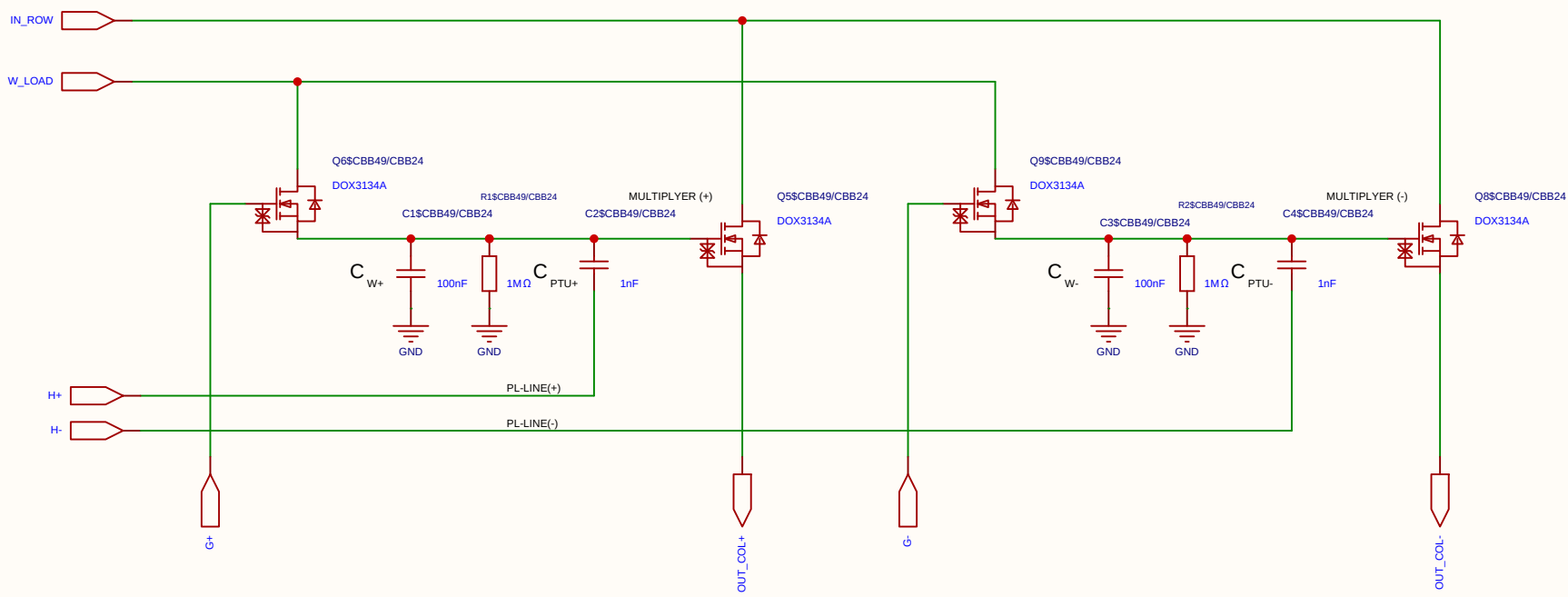
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

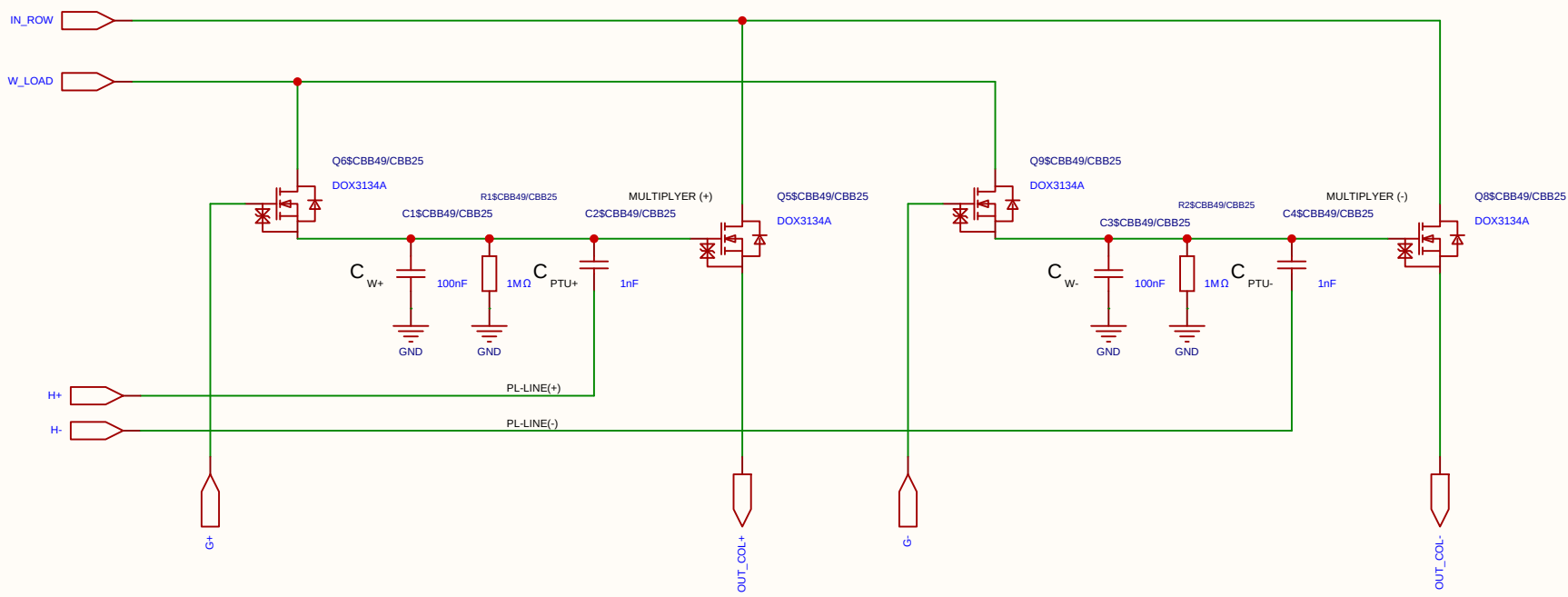
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

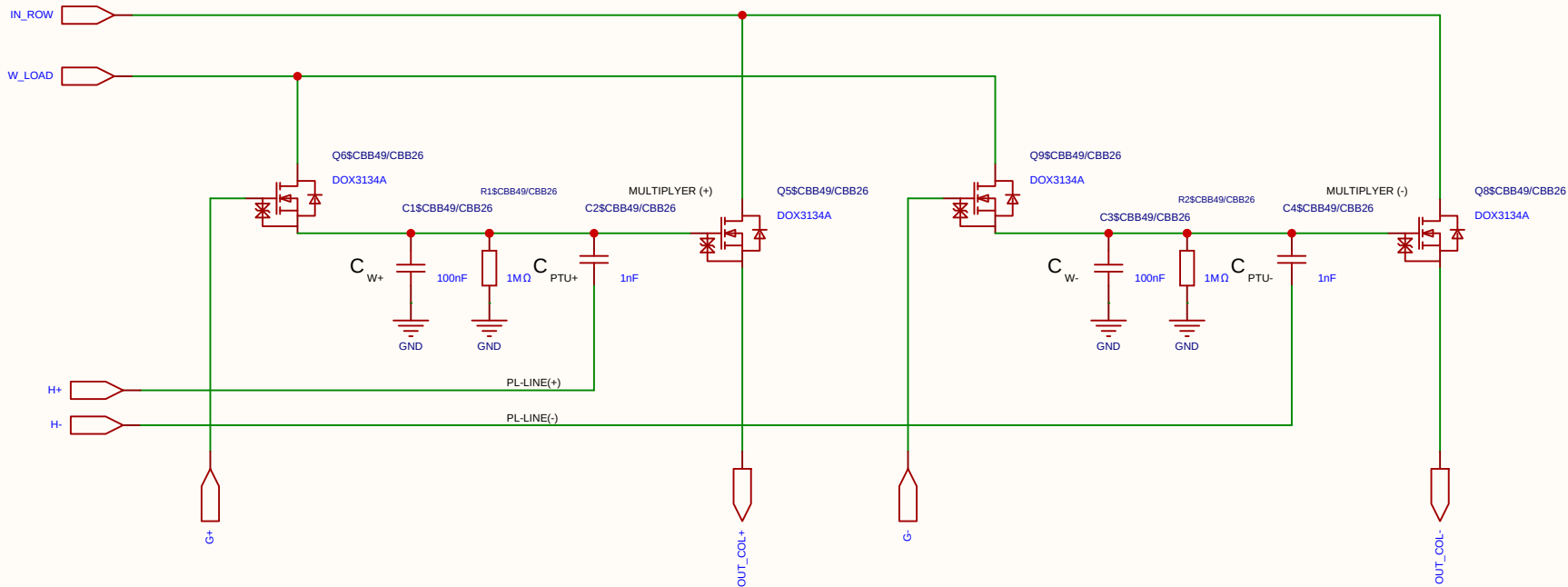
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

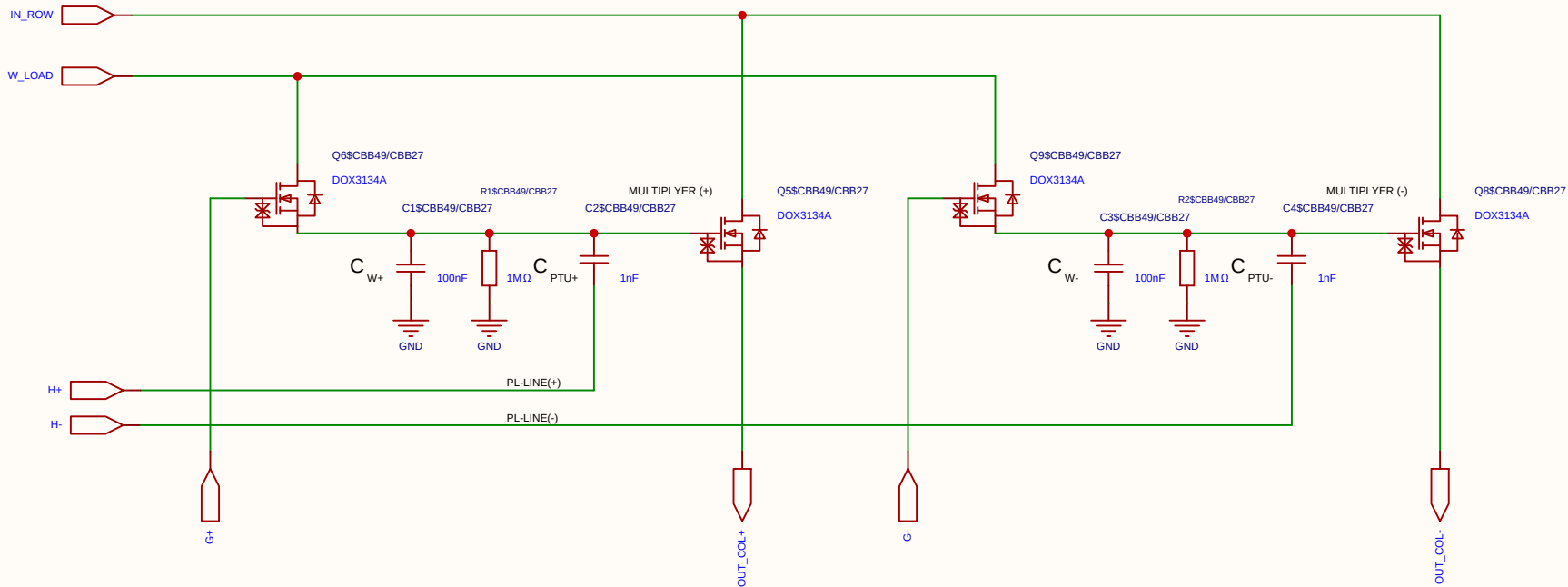
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
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		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

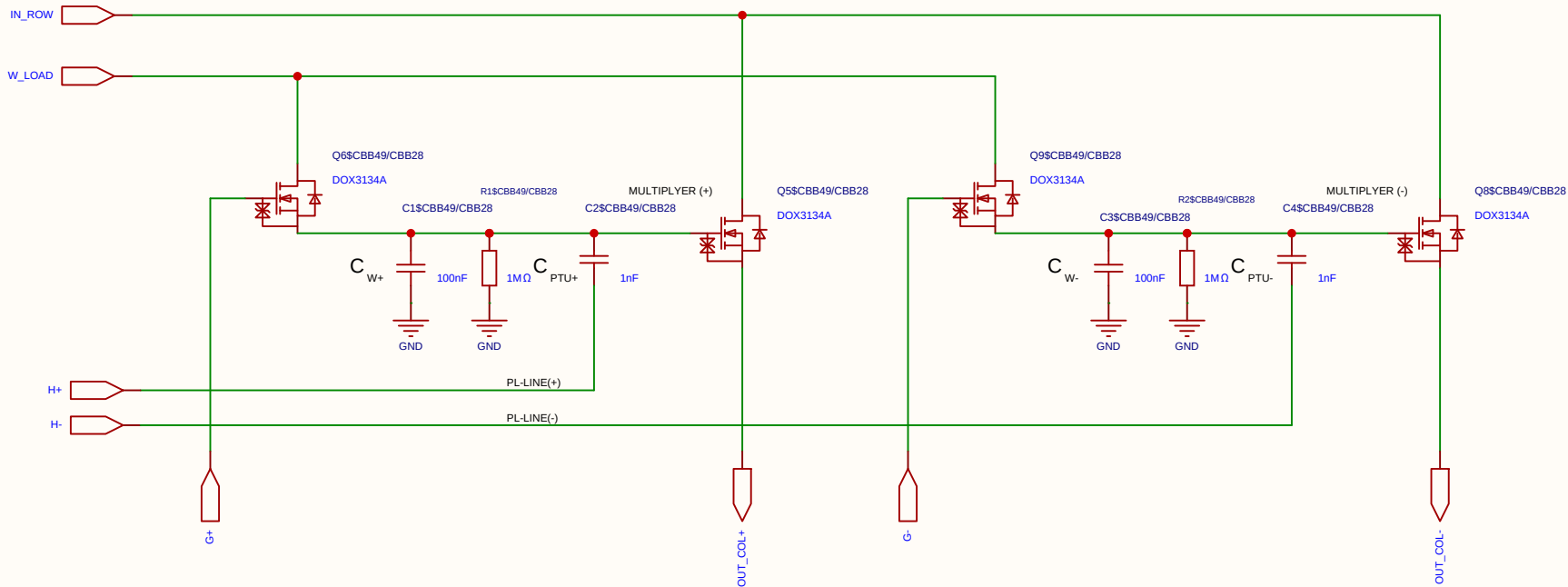
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

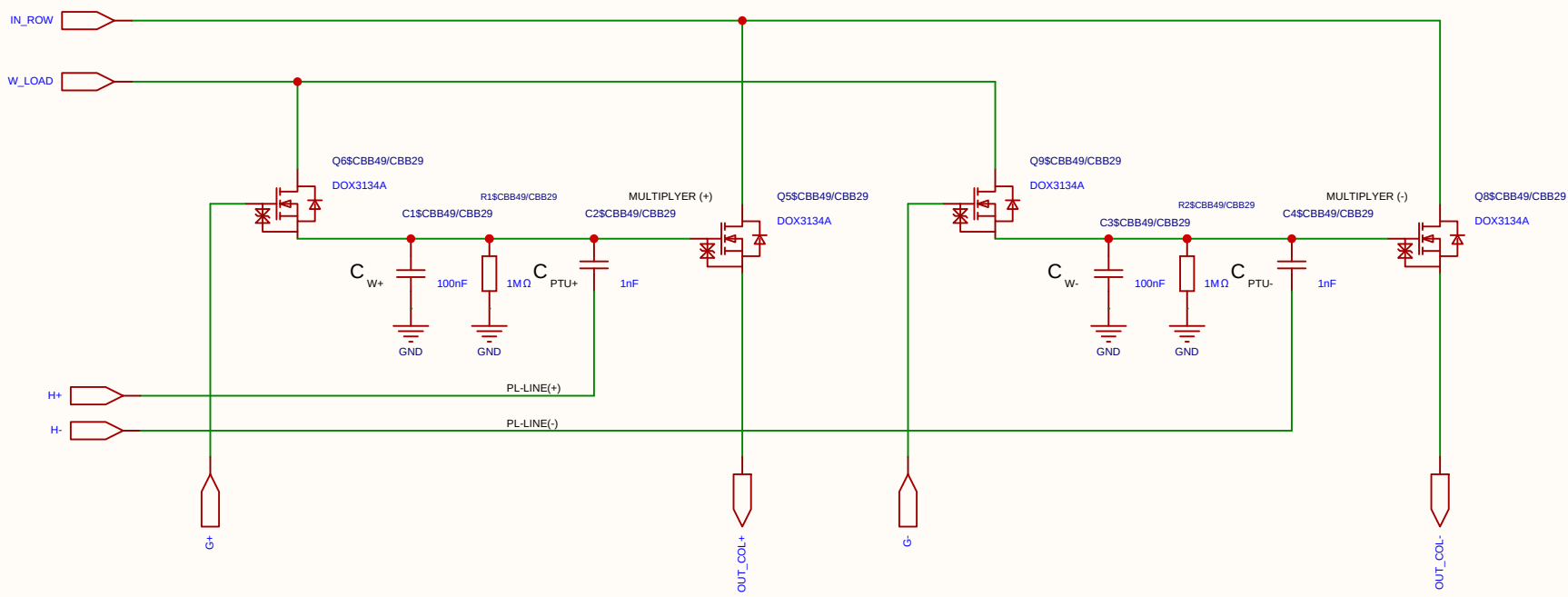
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

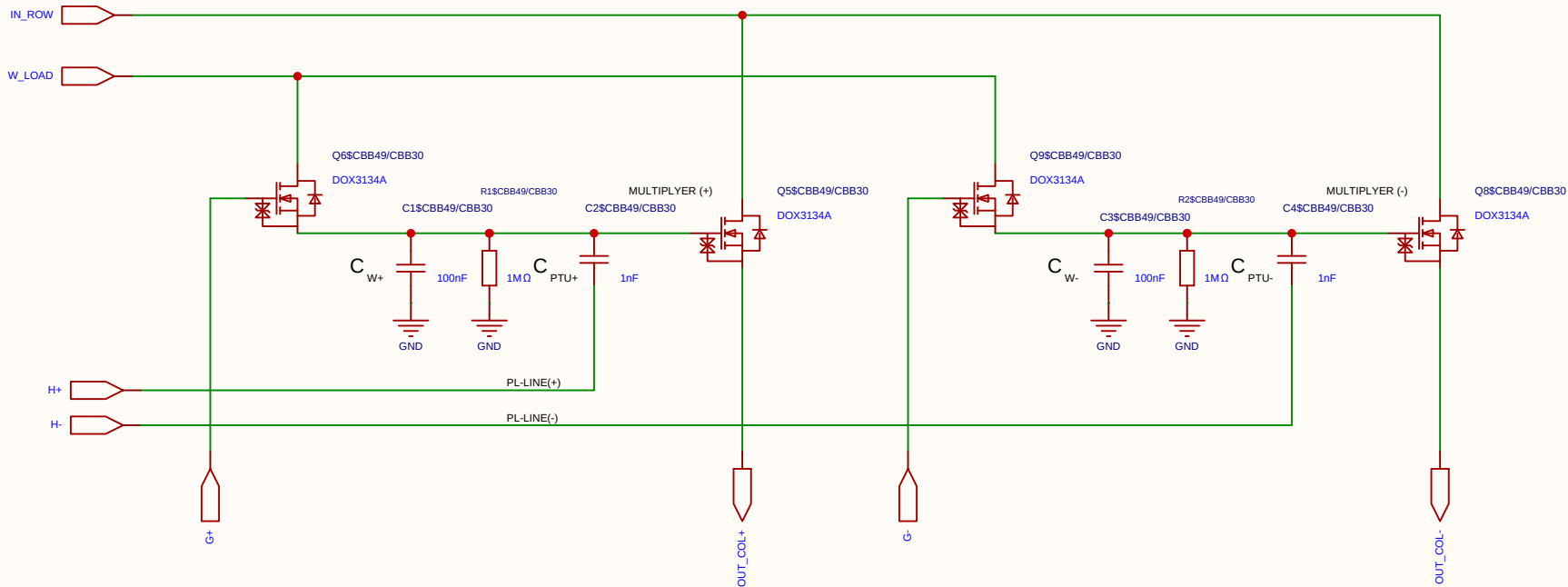
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

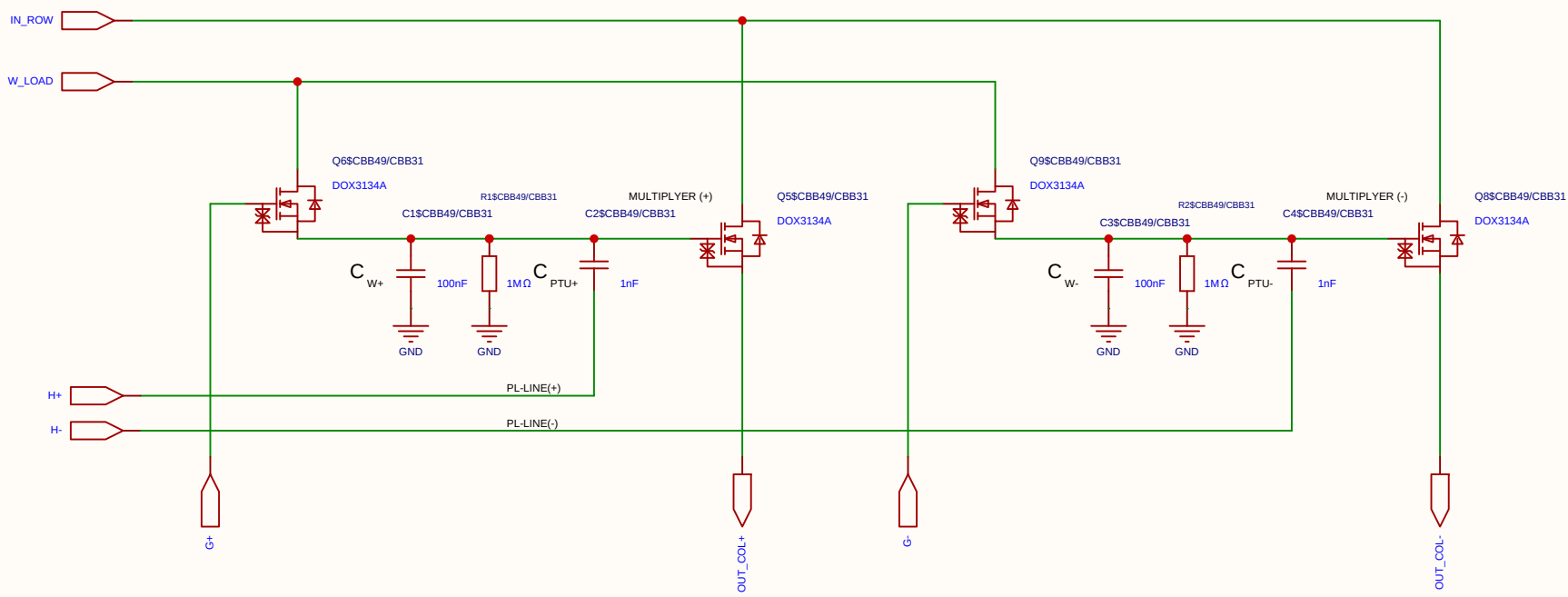
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

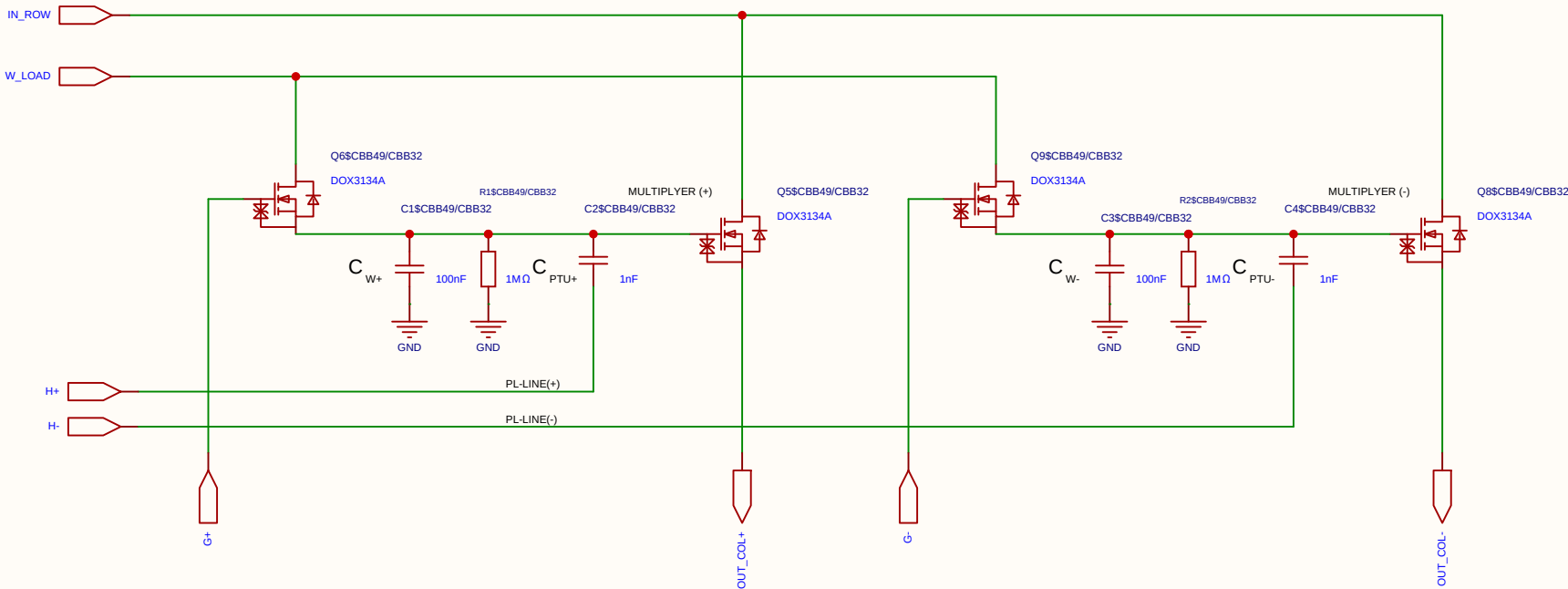
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Board				Page	matrix_3_8_MUL_CELL
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

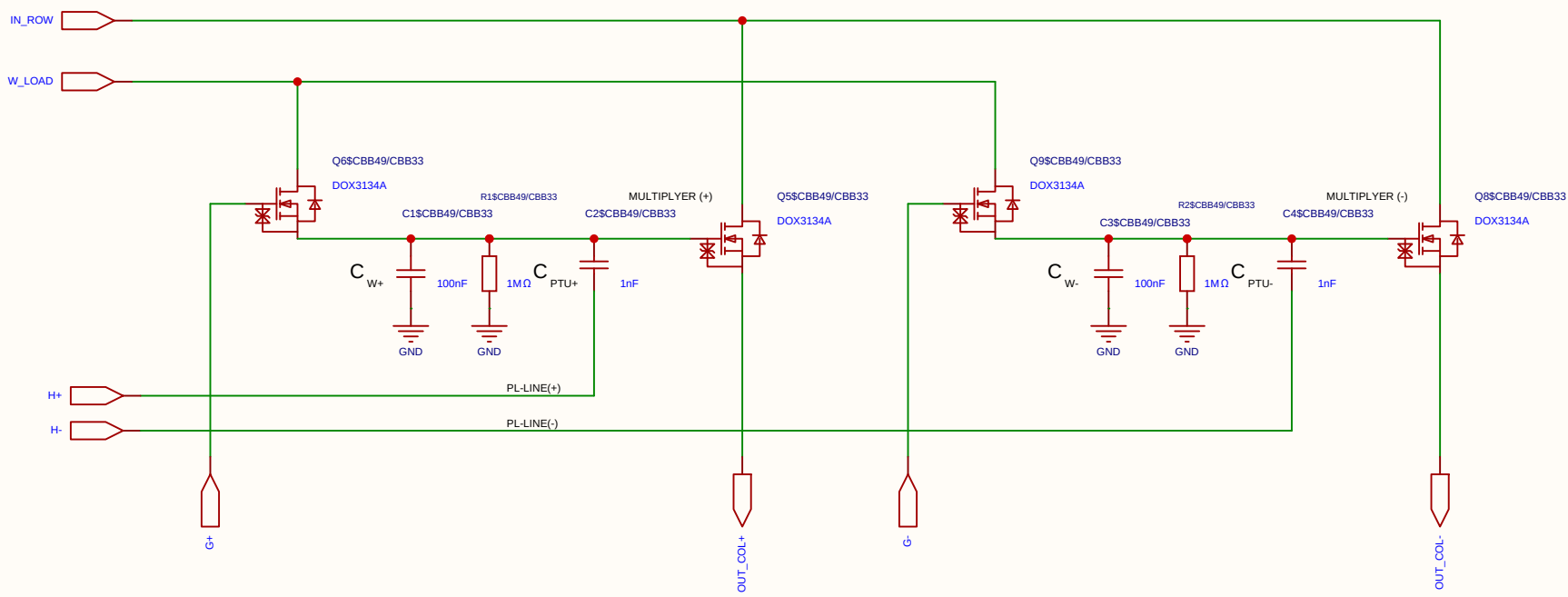
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

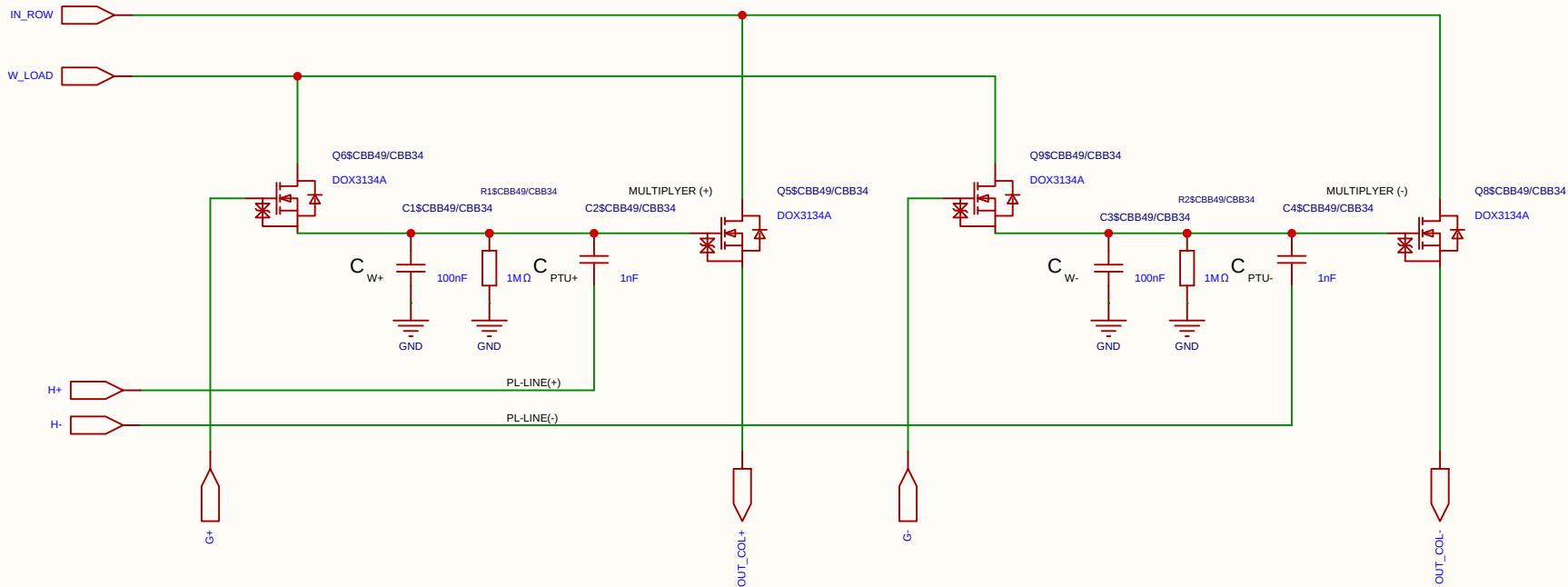
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

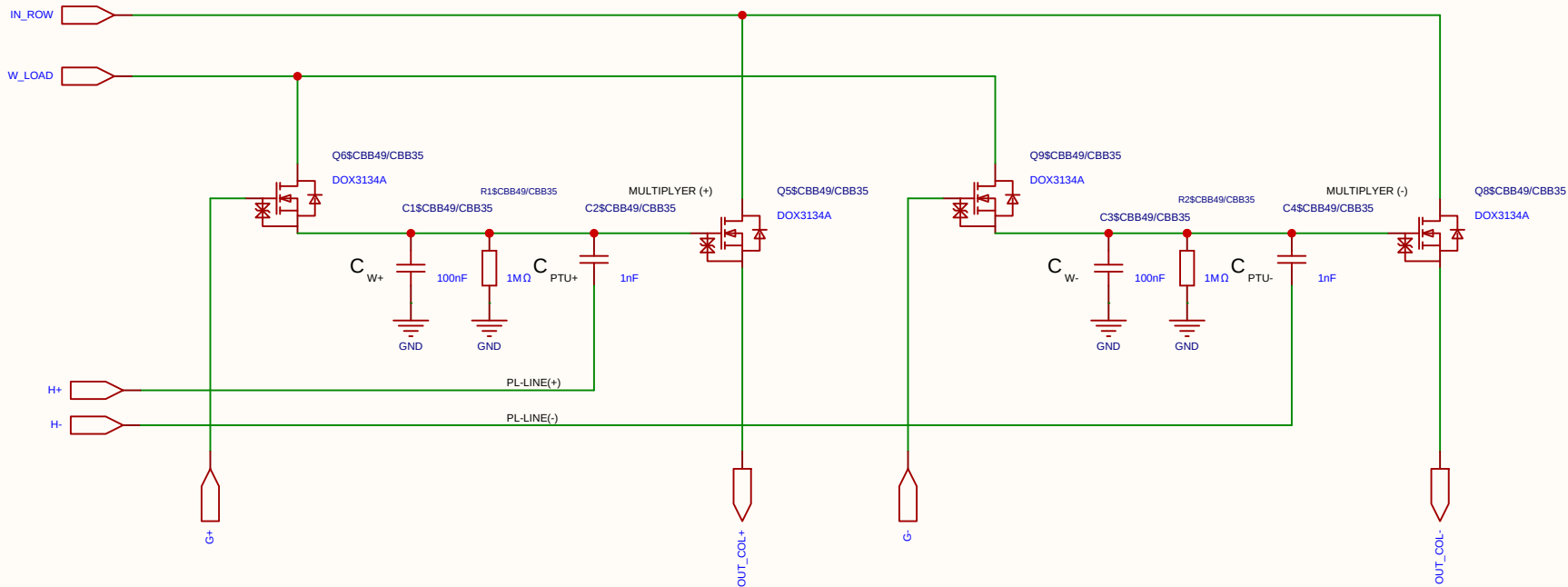
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Drawn	Olle Welin	Inverted_MAML_6x6			
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

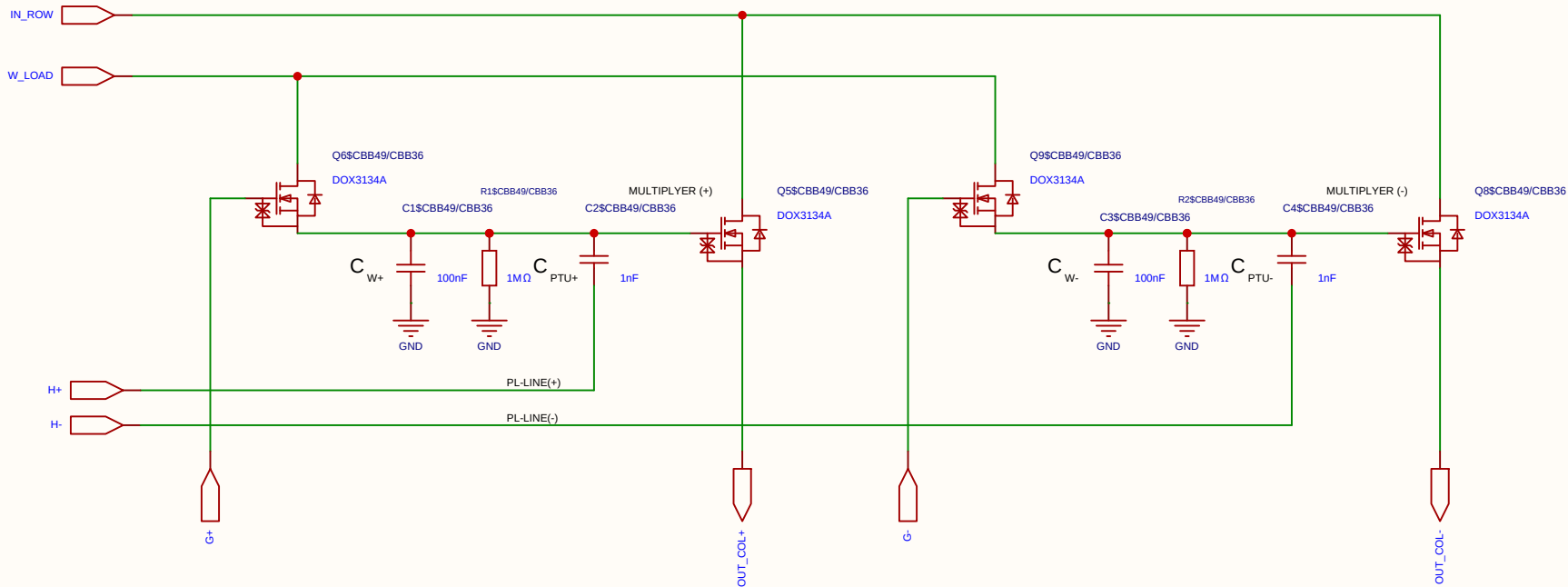
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

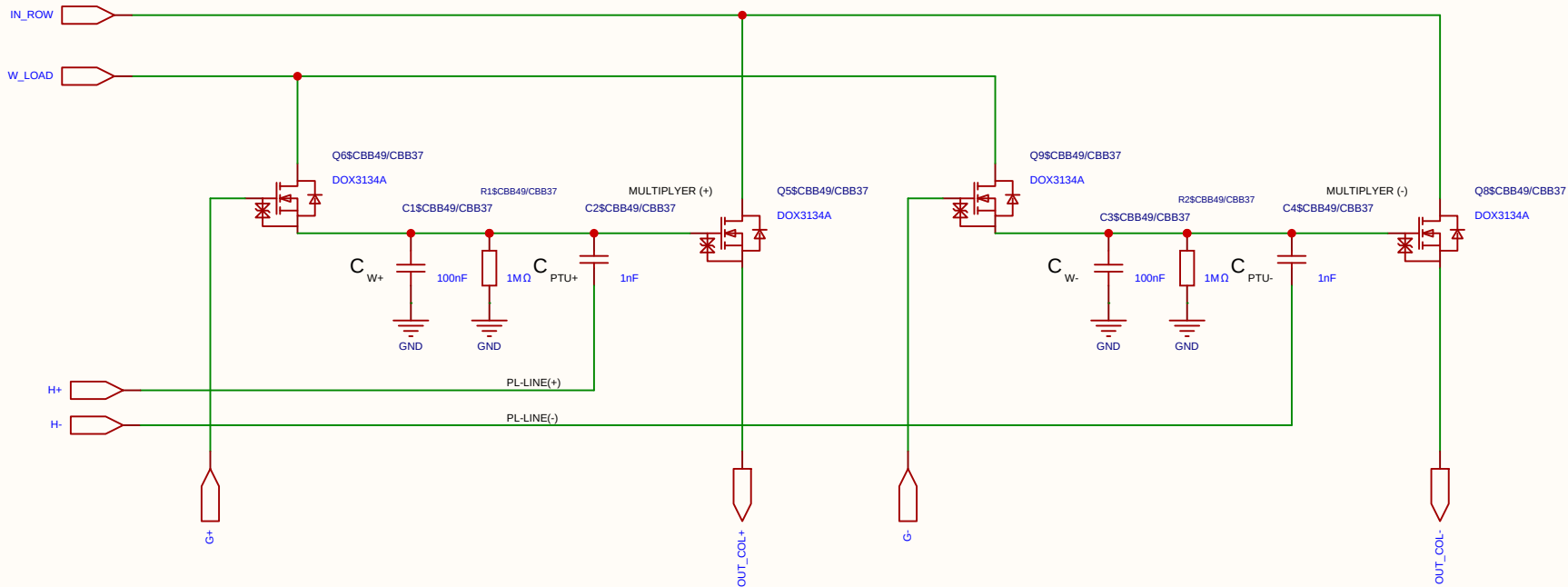
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
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Drawn	Olle Welin	Inverted_MAML_6x6			
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		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

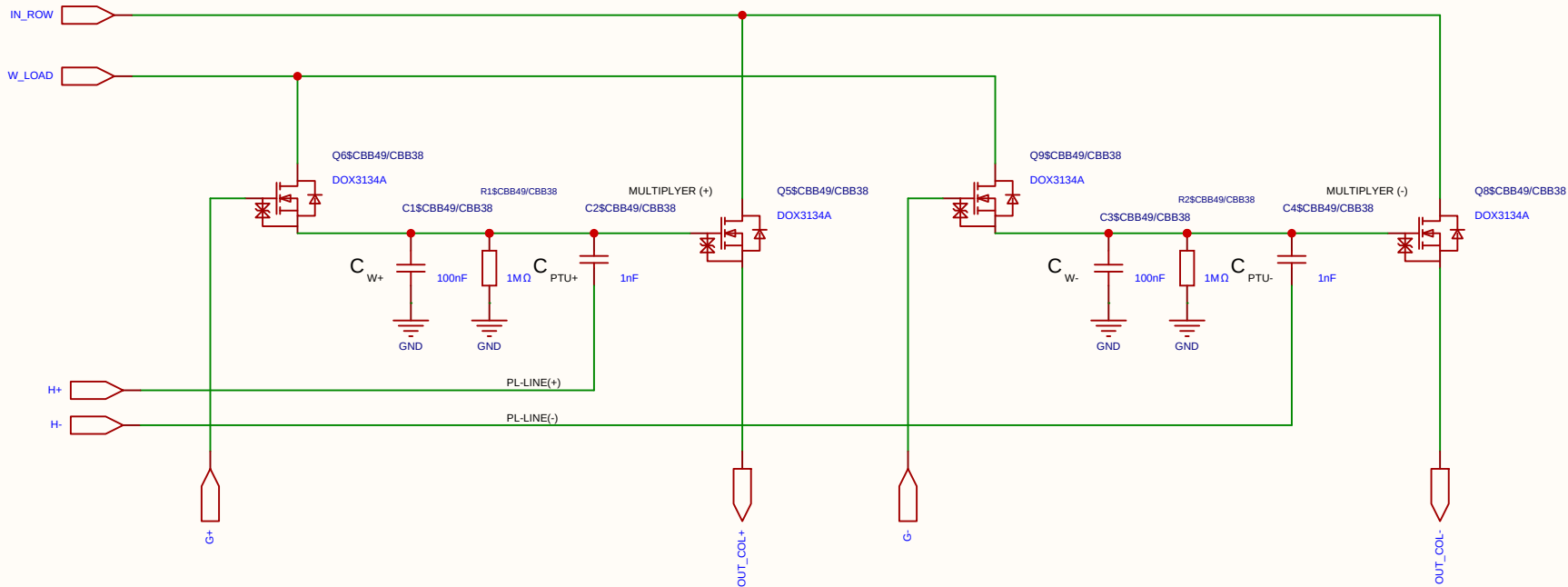
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
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		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

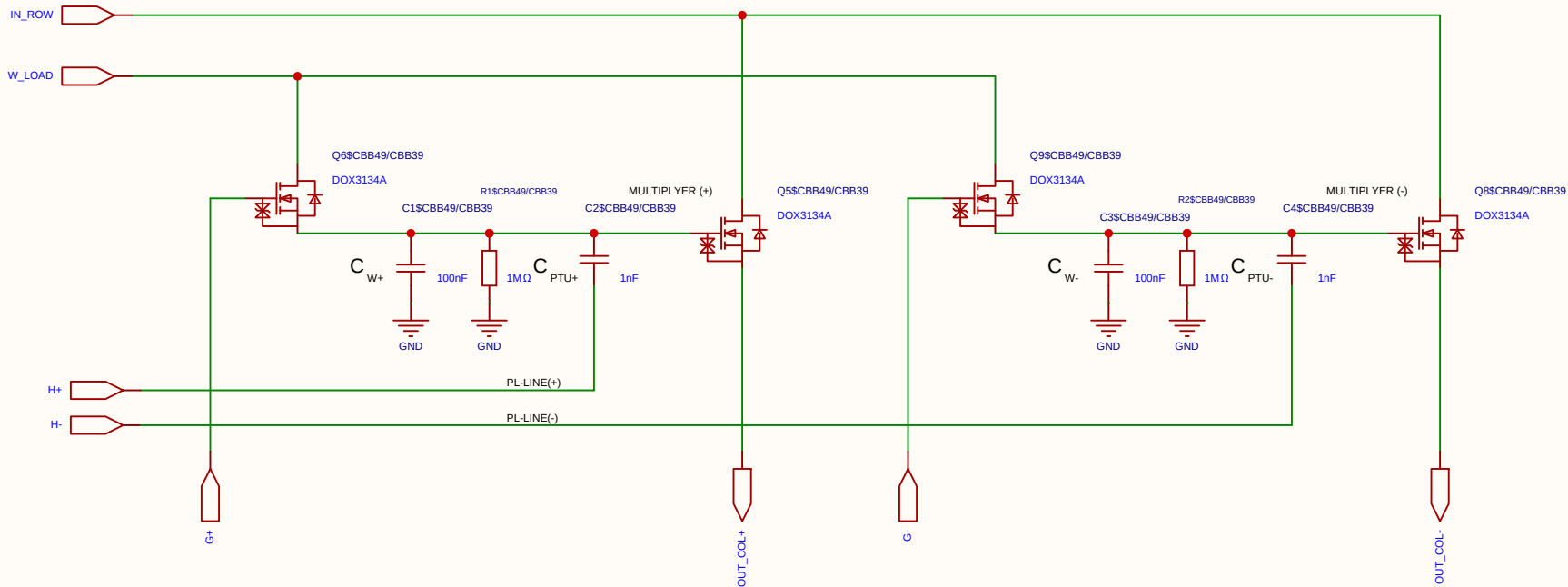
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

Component	Value	Description / Function
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

State	PL Voltage	Perturbation at Gate	Comment
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

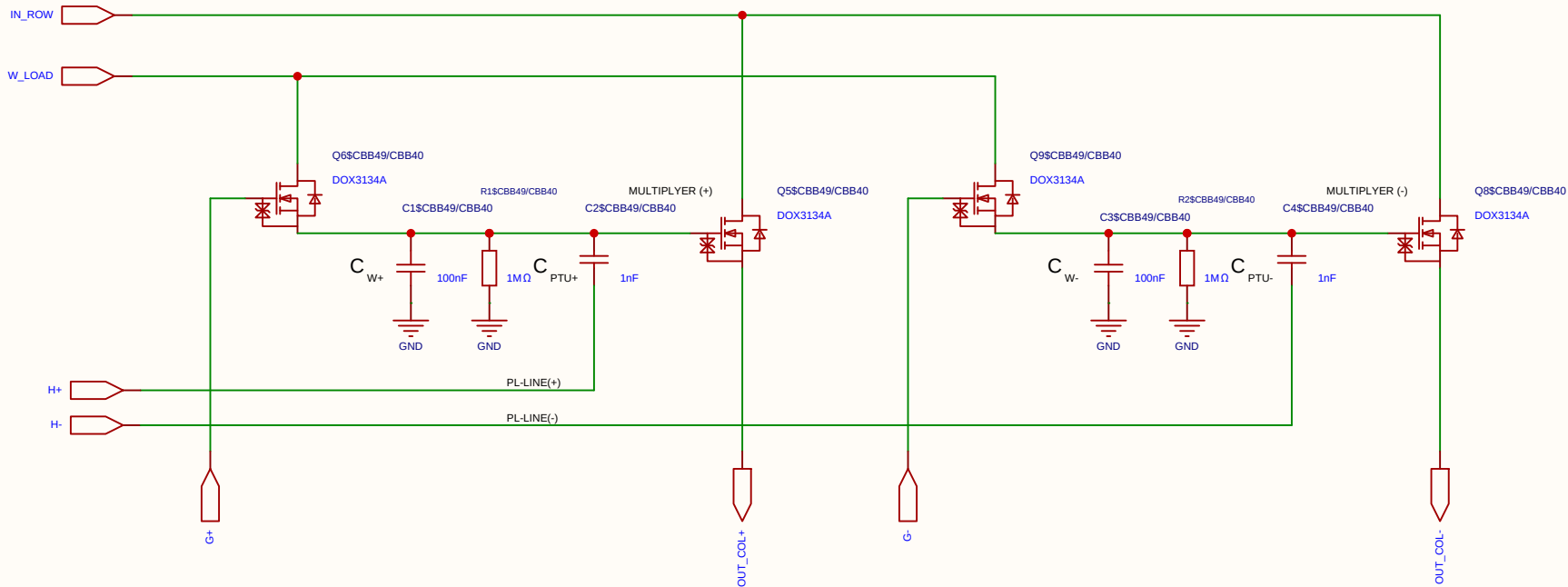
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

Component	Value	Description / Function
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

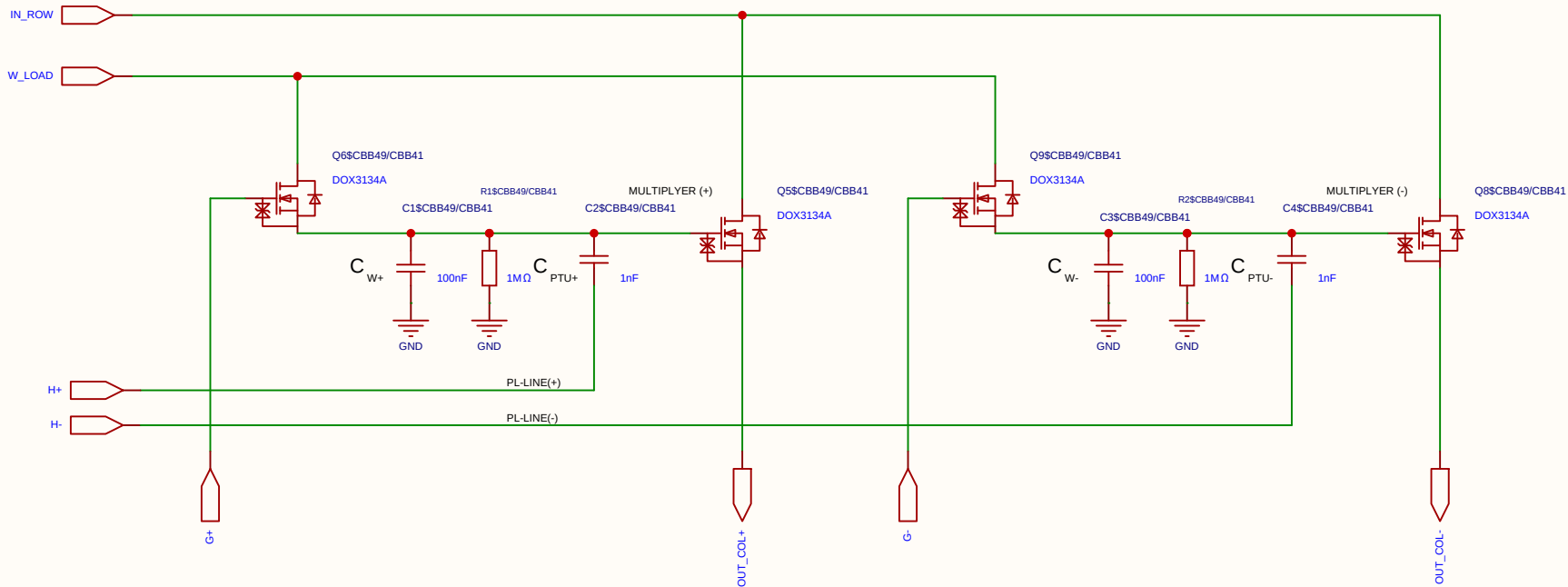
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

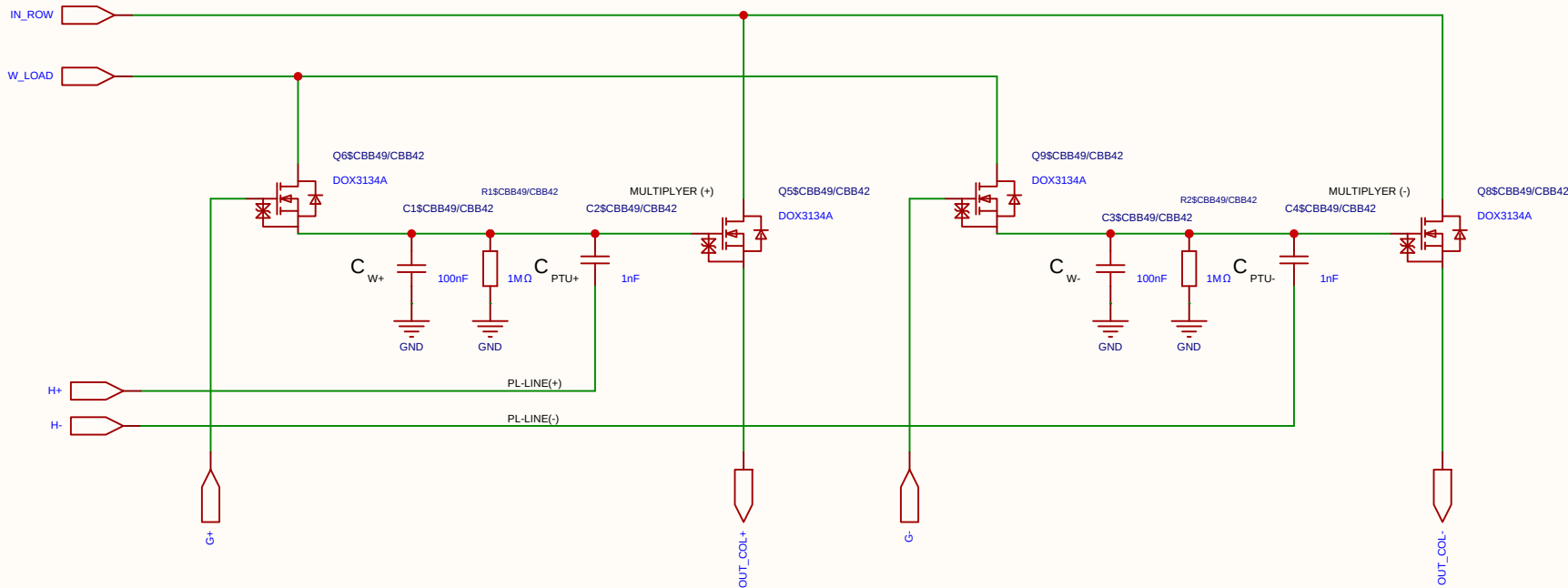
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

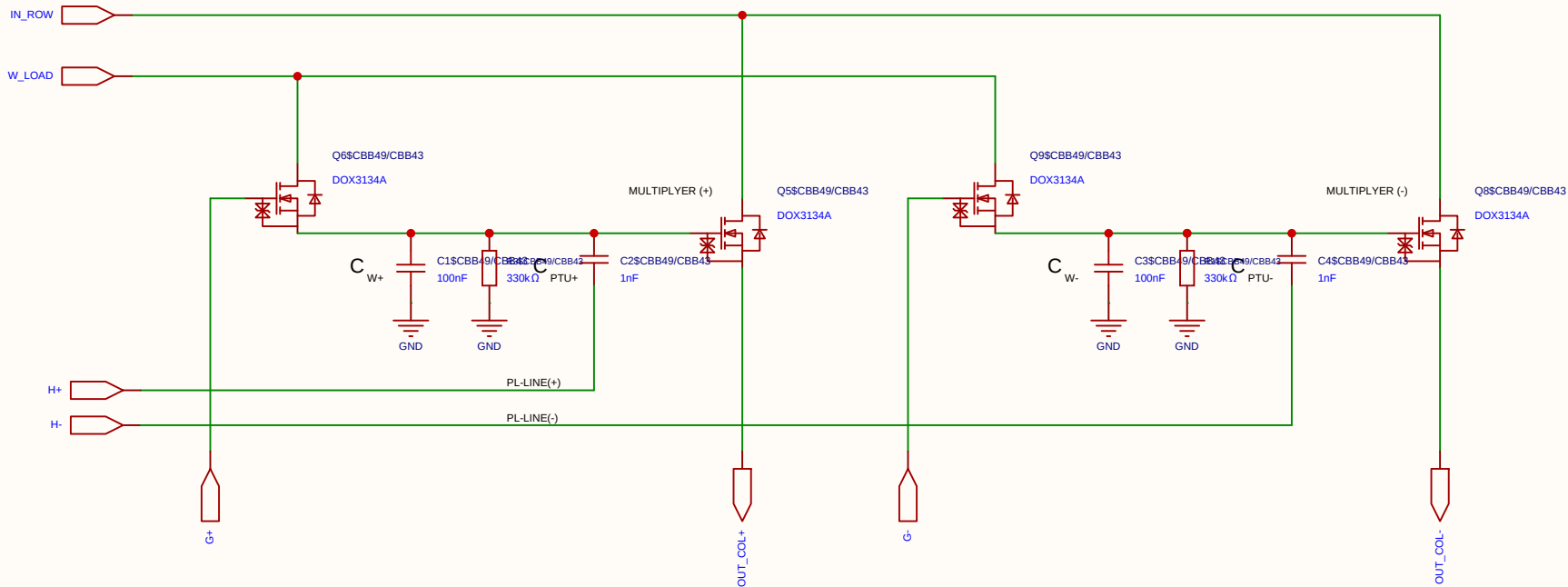
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

Analog REF DISSCHARGE
Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

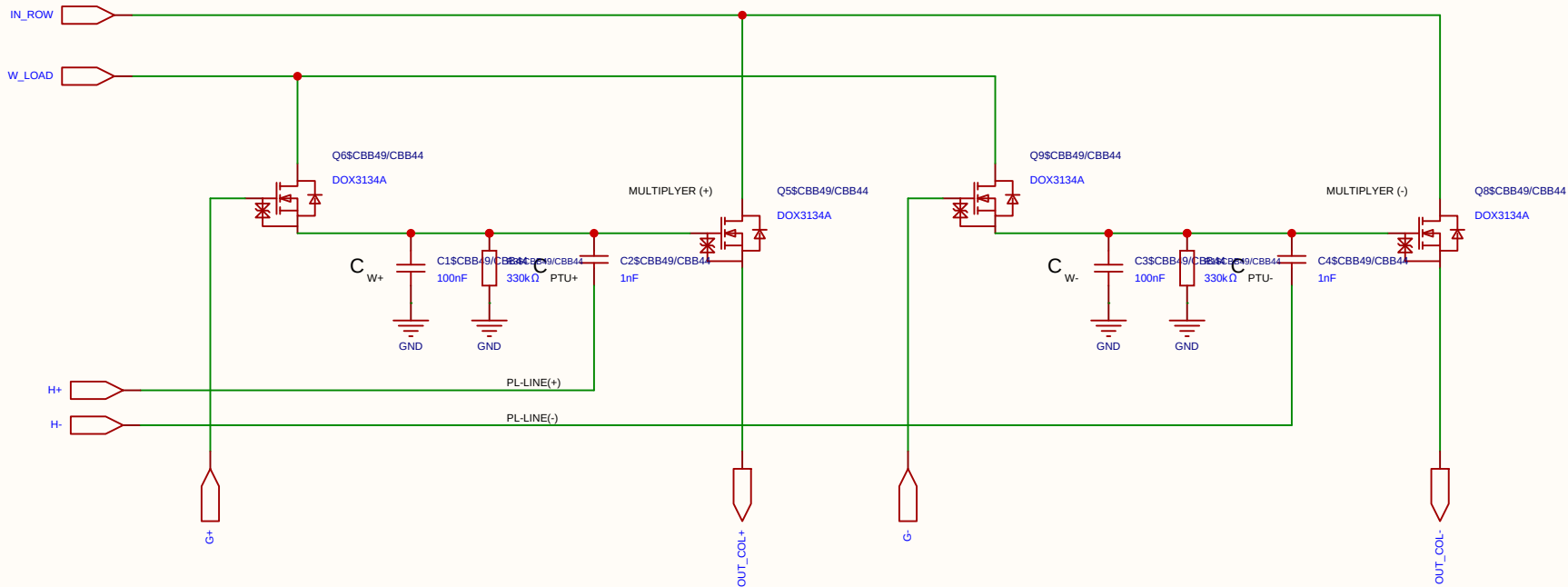
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	REF_CELL_2X_2T1C			Create at	2026-06-11
				Update at	2026-06-19
Board				Page	matrix_3_REF_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog REF DISSCHARGE
Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

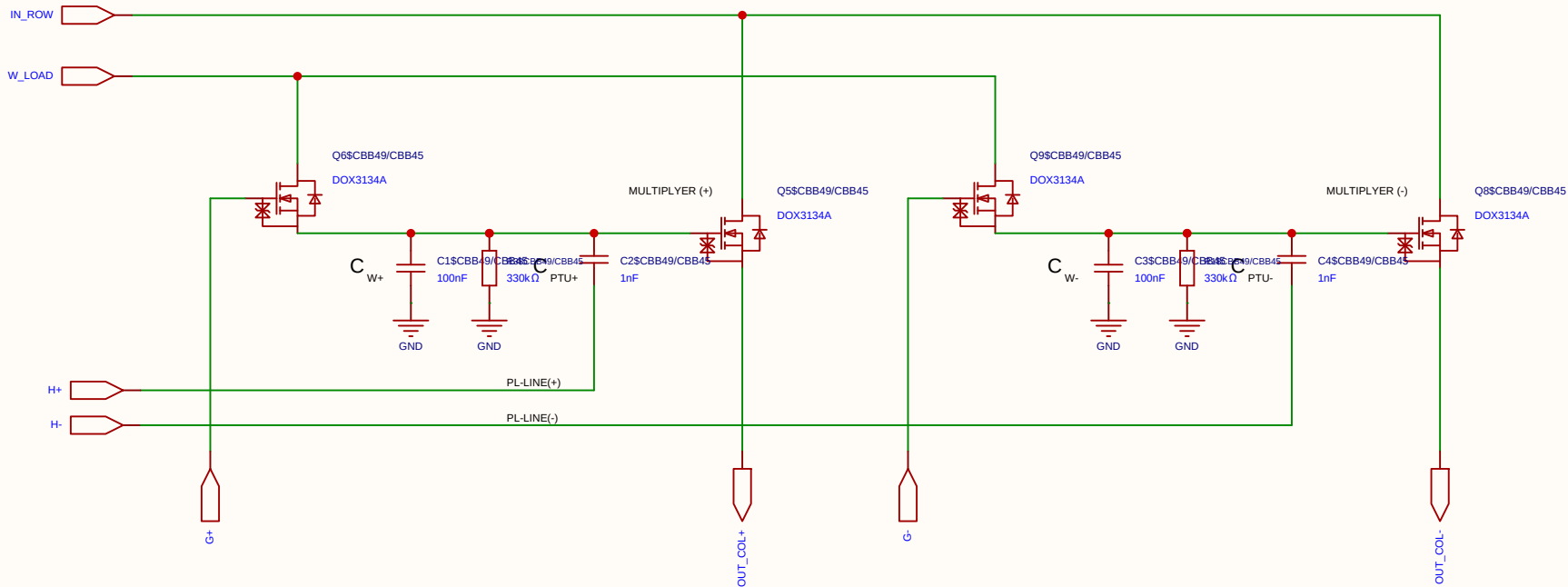
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	REF_CELL_2X_2T1C			Create at	2026-06-11
				Update at	2026-06-19
Board				Page	matrix_3_REF_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog REF DISSCHARGE
Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

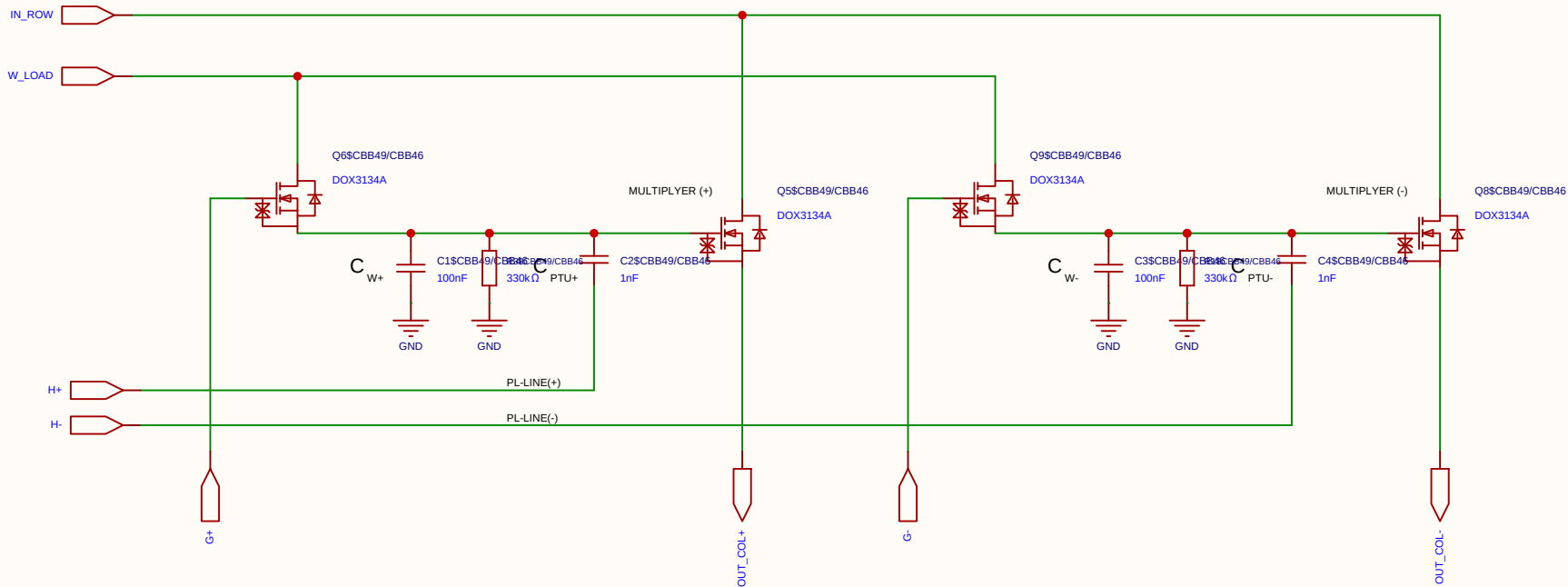
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	REF_CELL_2X_2T1C			Create at	2026-06-11
				Update at	2026-06-19
Board				Page	matrix_3_REF_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog REF DISSCHARGE
Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

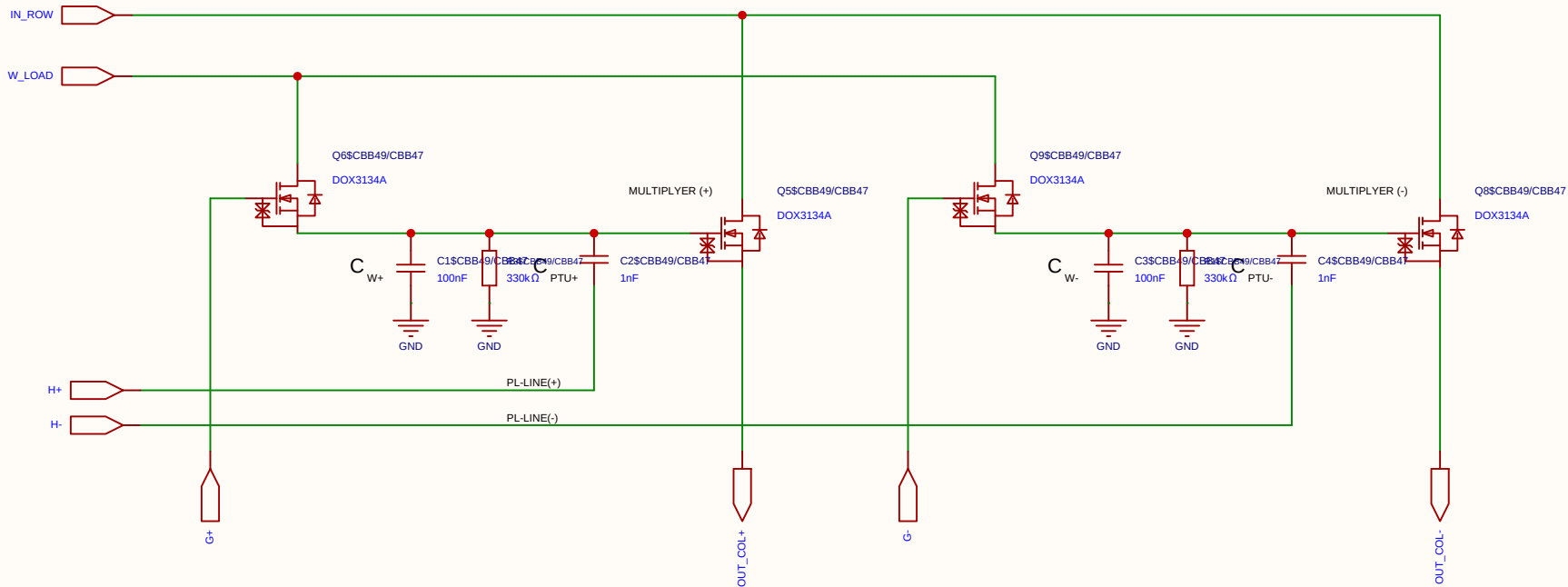
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	REF_CELL_2X_2T1C			Create at	2026-06-11
				Update at	2026-06-19
Board				Page	matrix_3_REF_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog REF DISSCHARGE
Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	REF_CELL_2X_2T1C			Create at	2026-06-11
				Update at	2026-06-19
Board				Page	matrix_3_REF_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

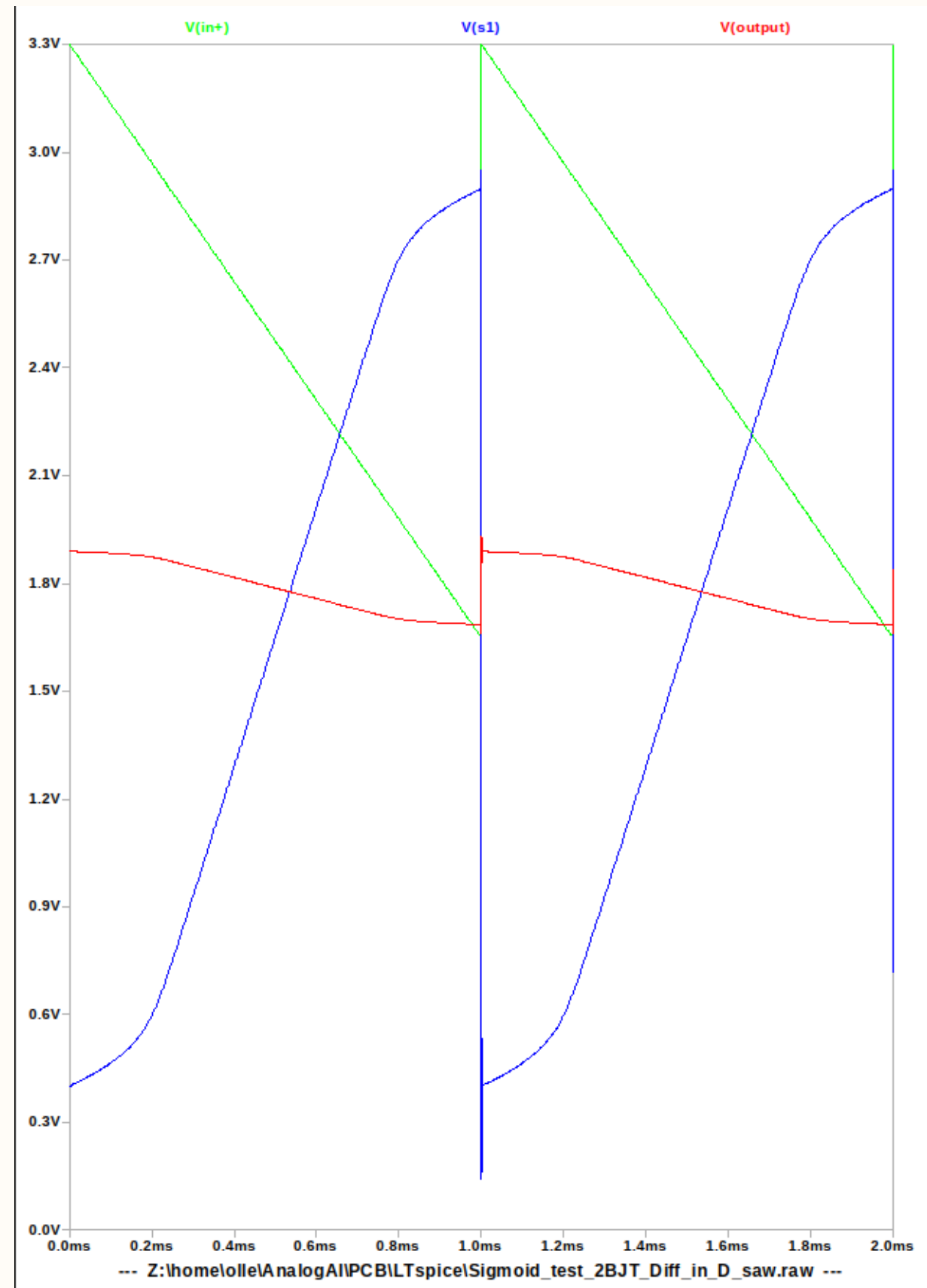
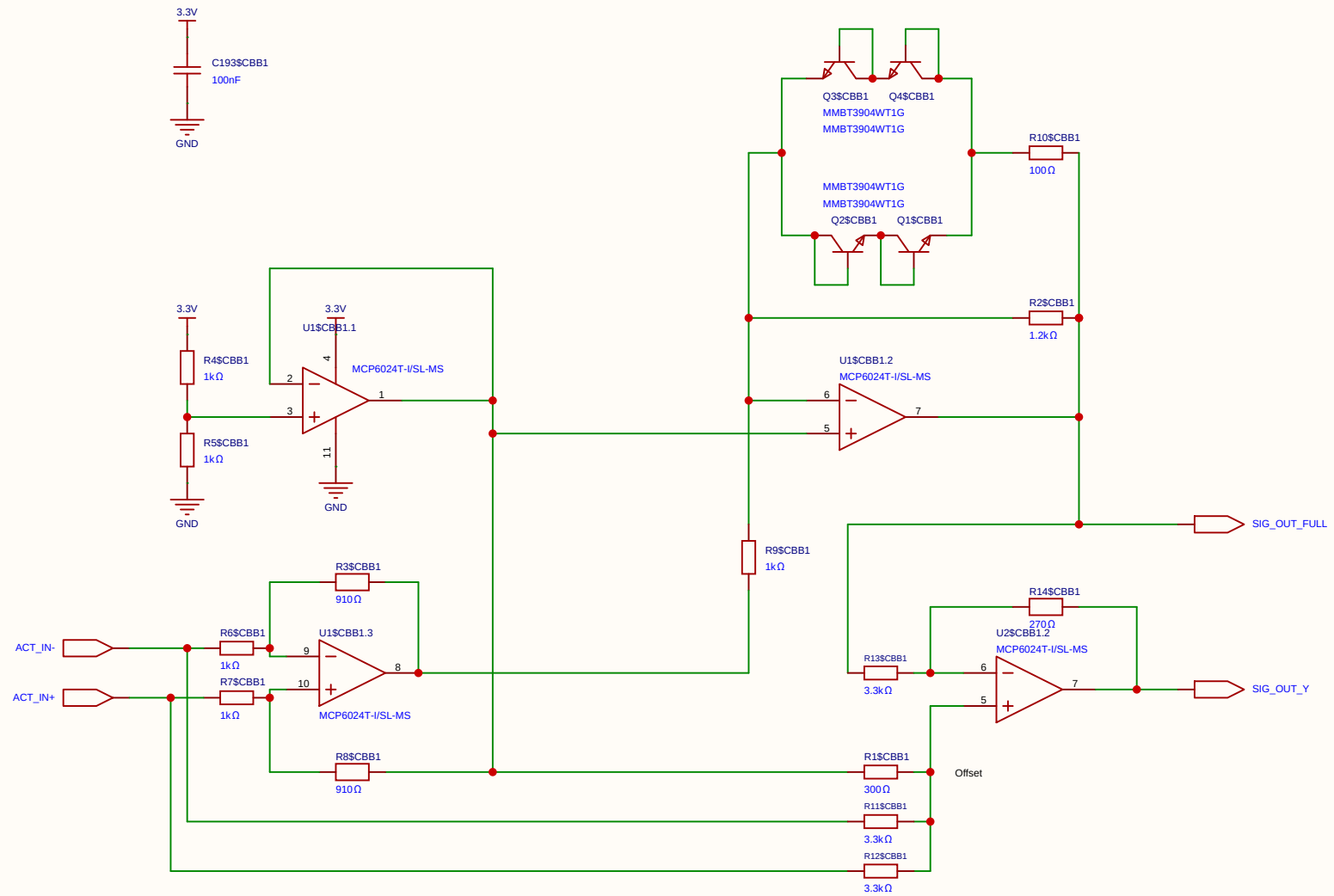
Category	Parameter	Value / Type	Technical Description & Function
Resistors	Discharge resistor (R)	1 M Ω	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

Component	Value	Description / Function
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a $\sim 1/101$ voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

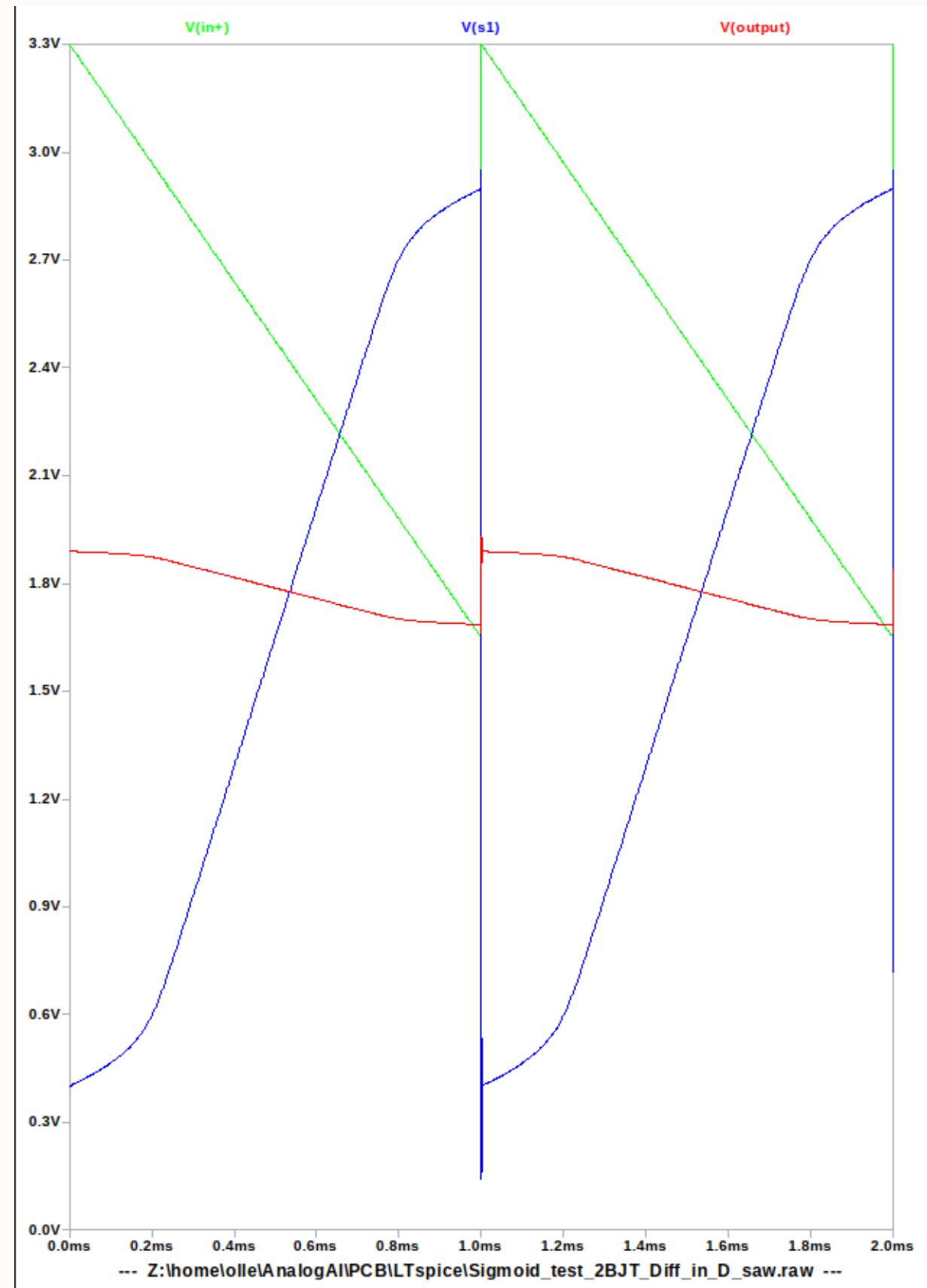
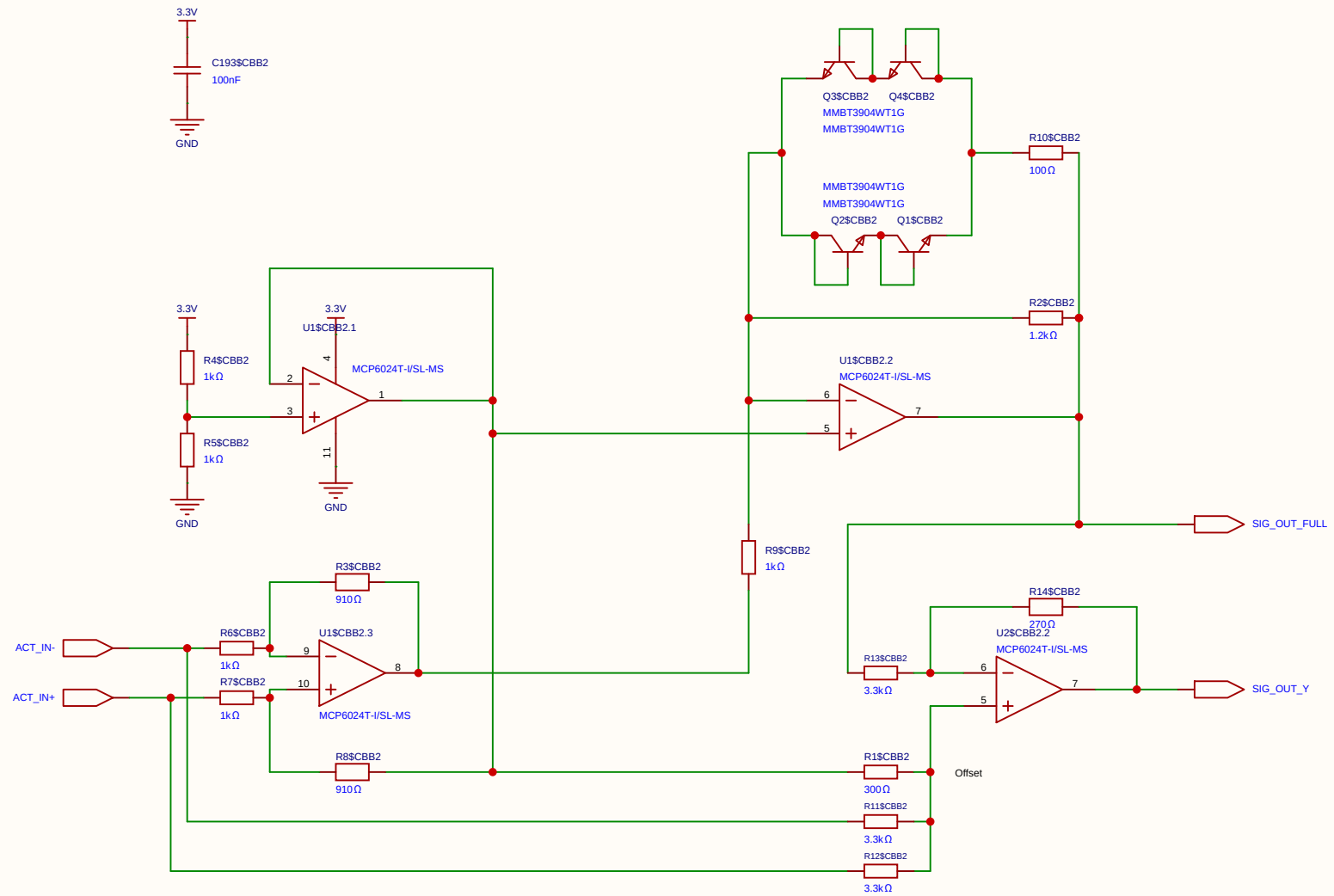
State	PL Voltage	Perturbation at Gate	Comment
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μ s.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

Component	Value	Description / Function
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_I . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_I . Establishes exactly 1.65 V (V_S) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_I . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

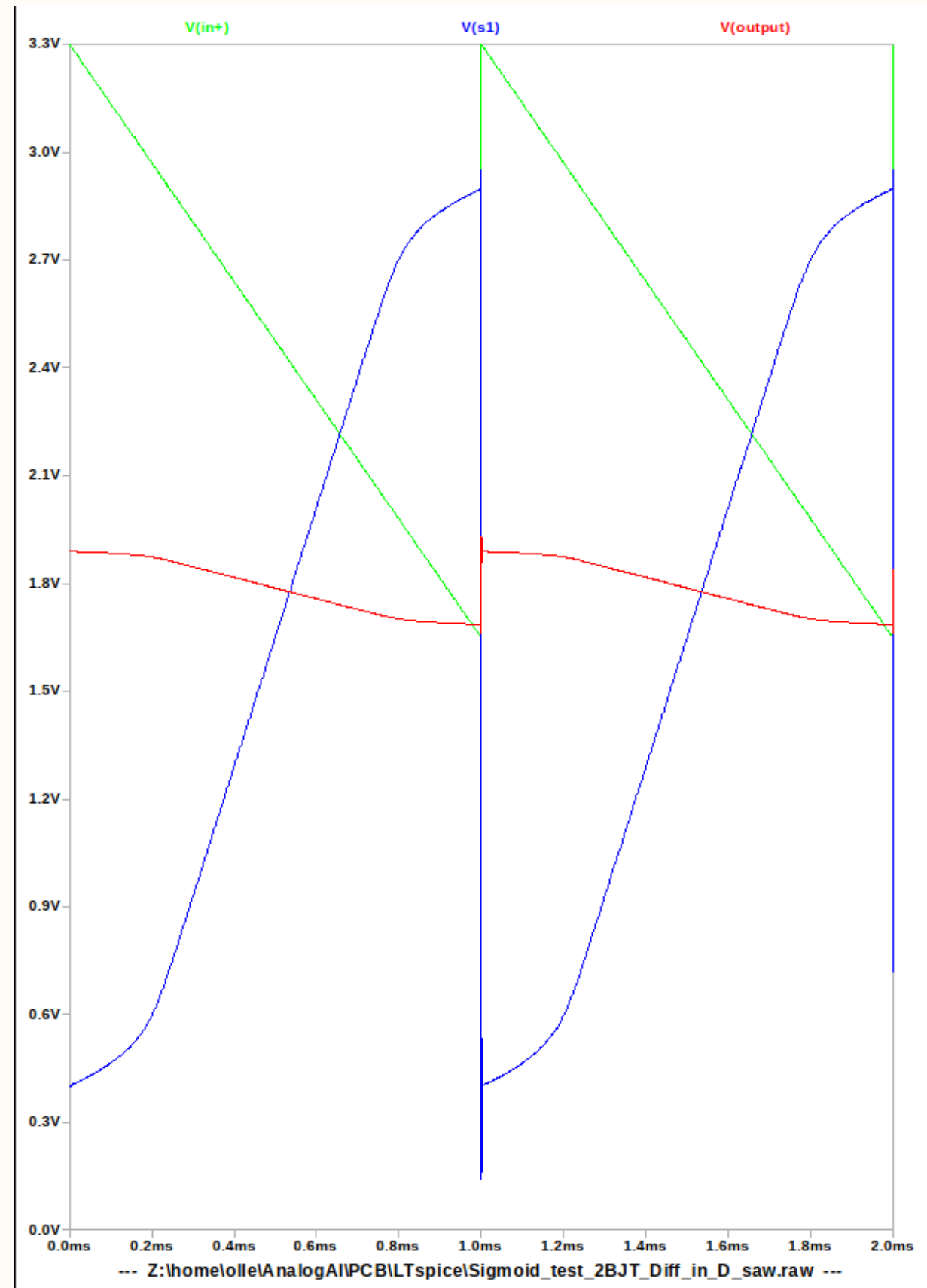
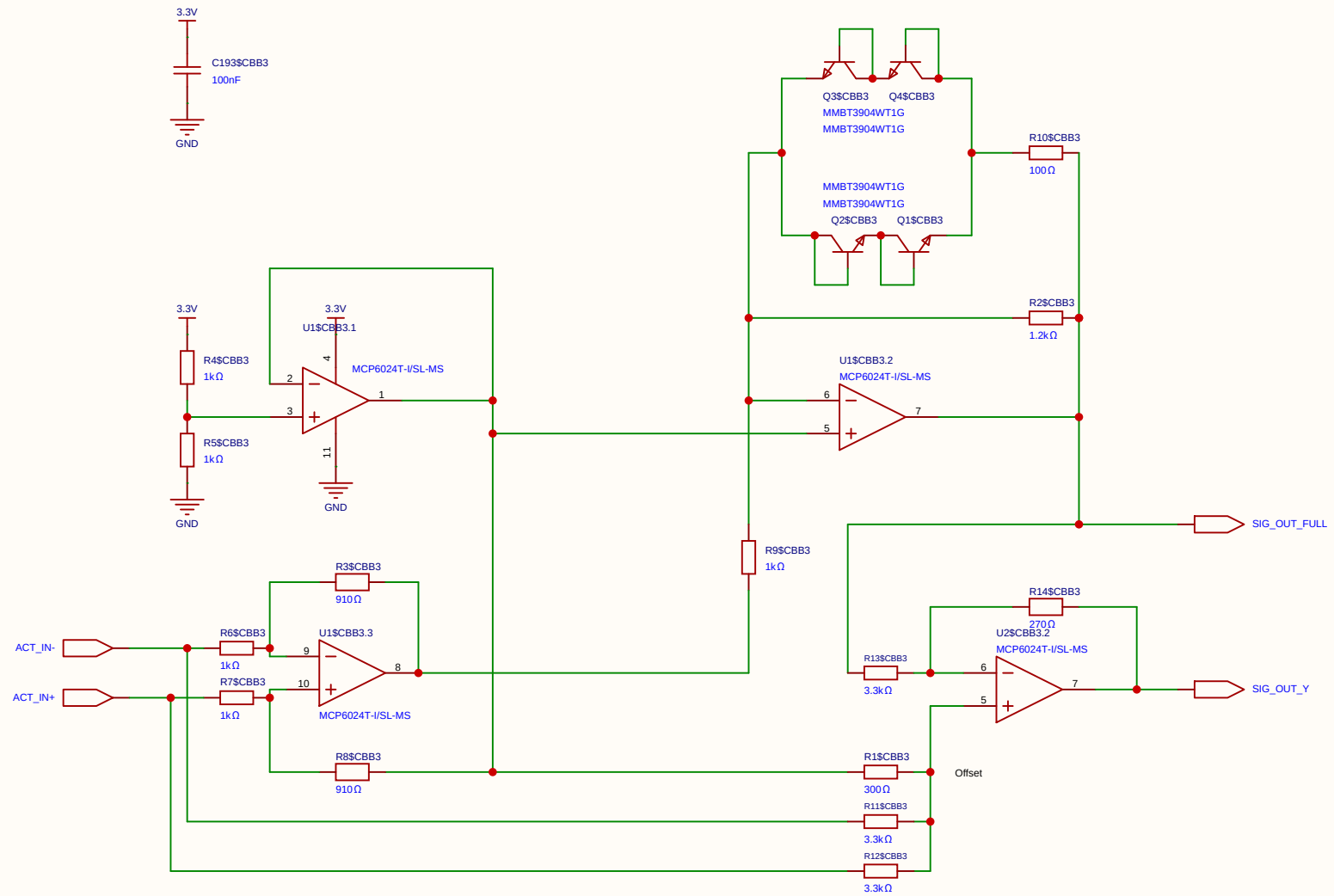




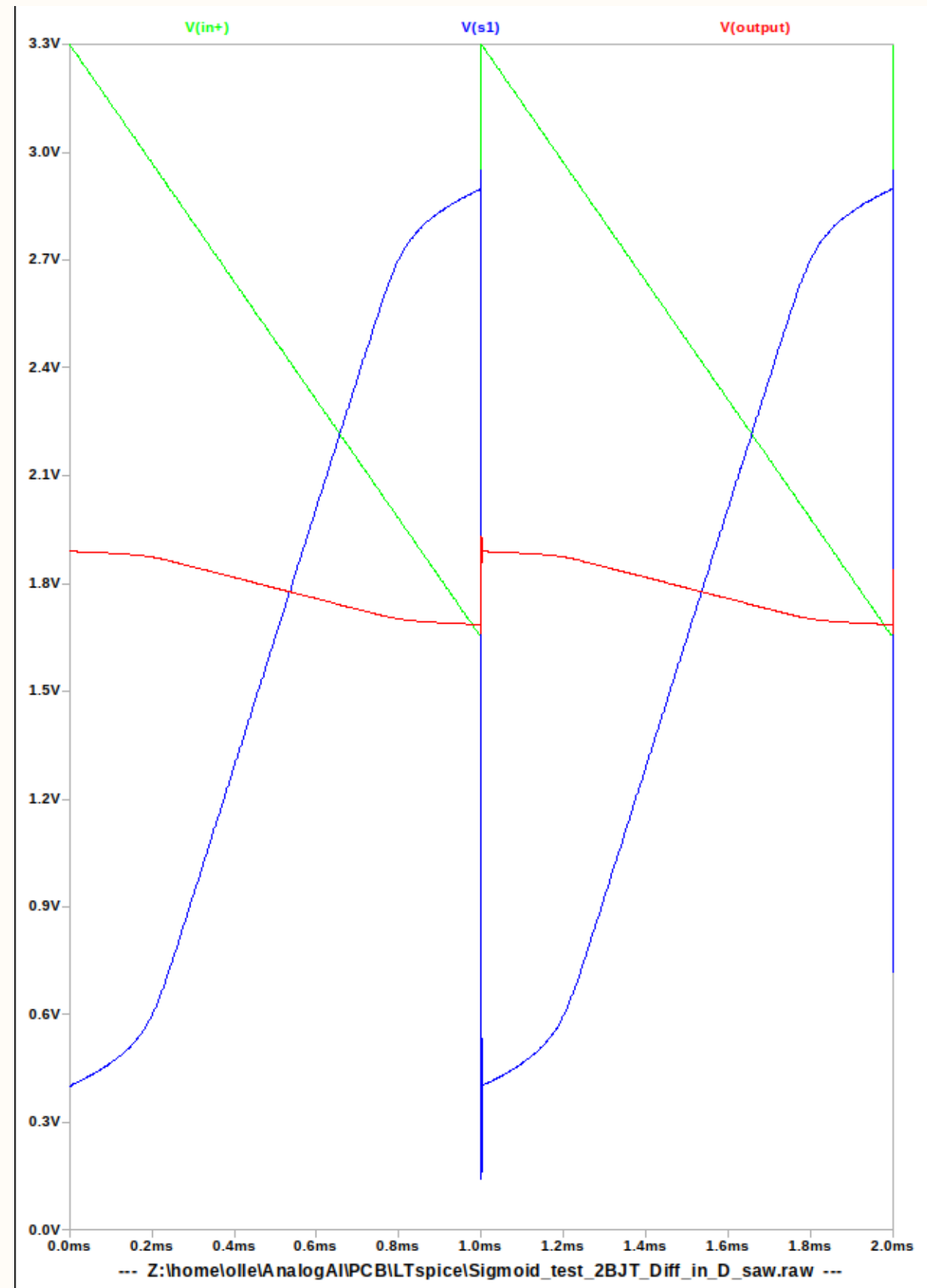
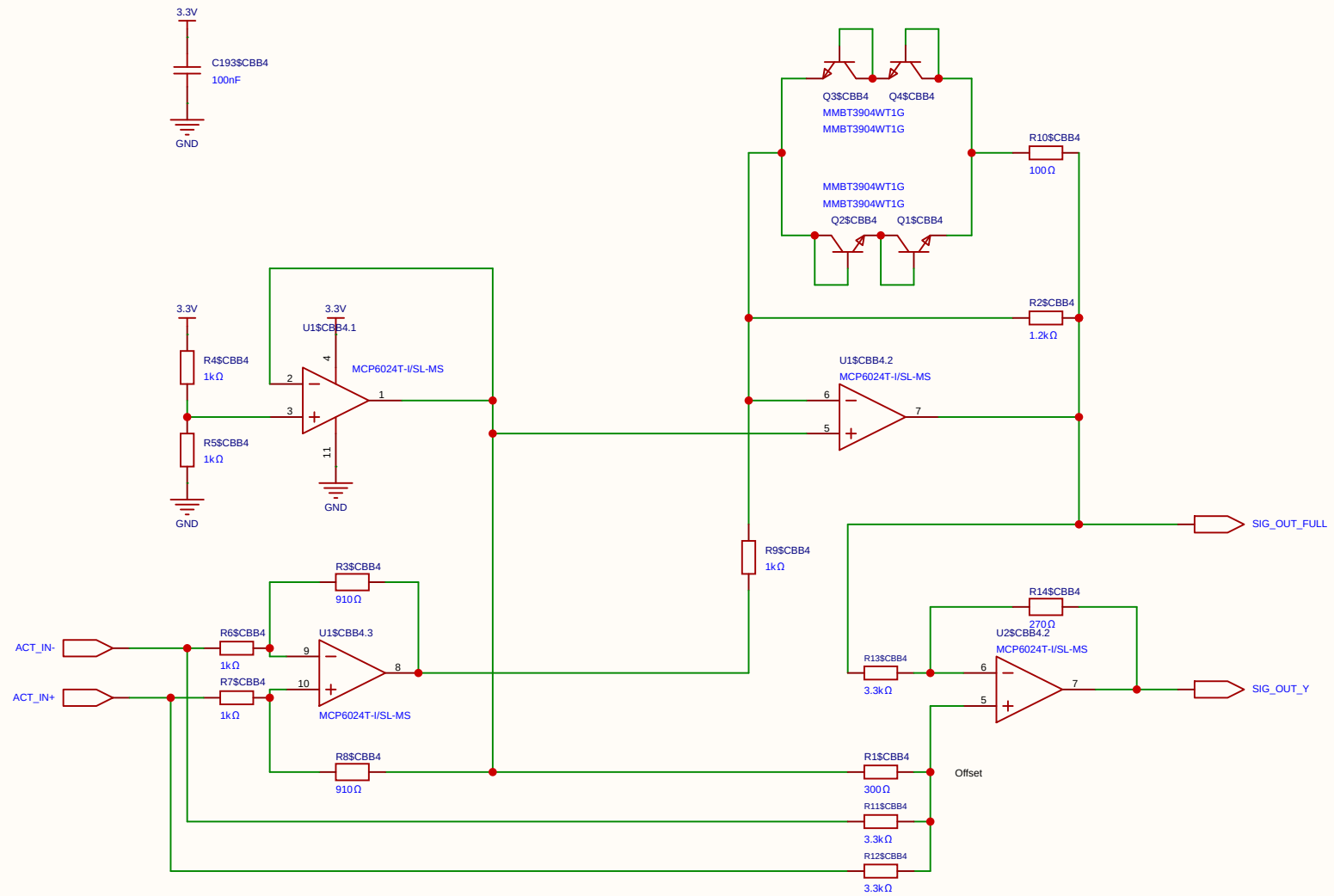
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Board				Page	SIMOID(4)
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Reviewed					
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		V1.0	A4	EasyEDA.com	



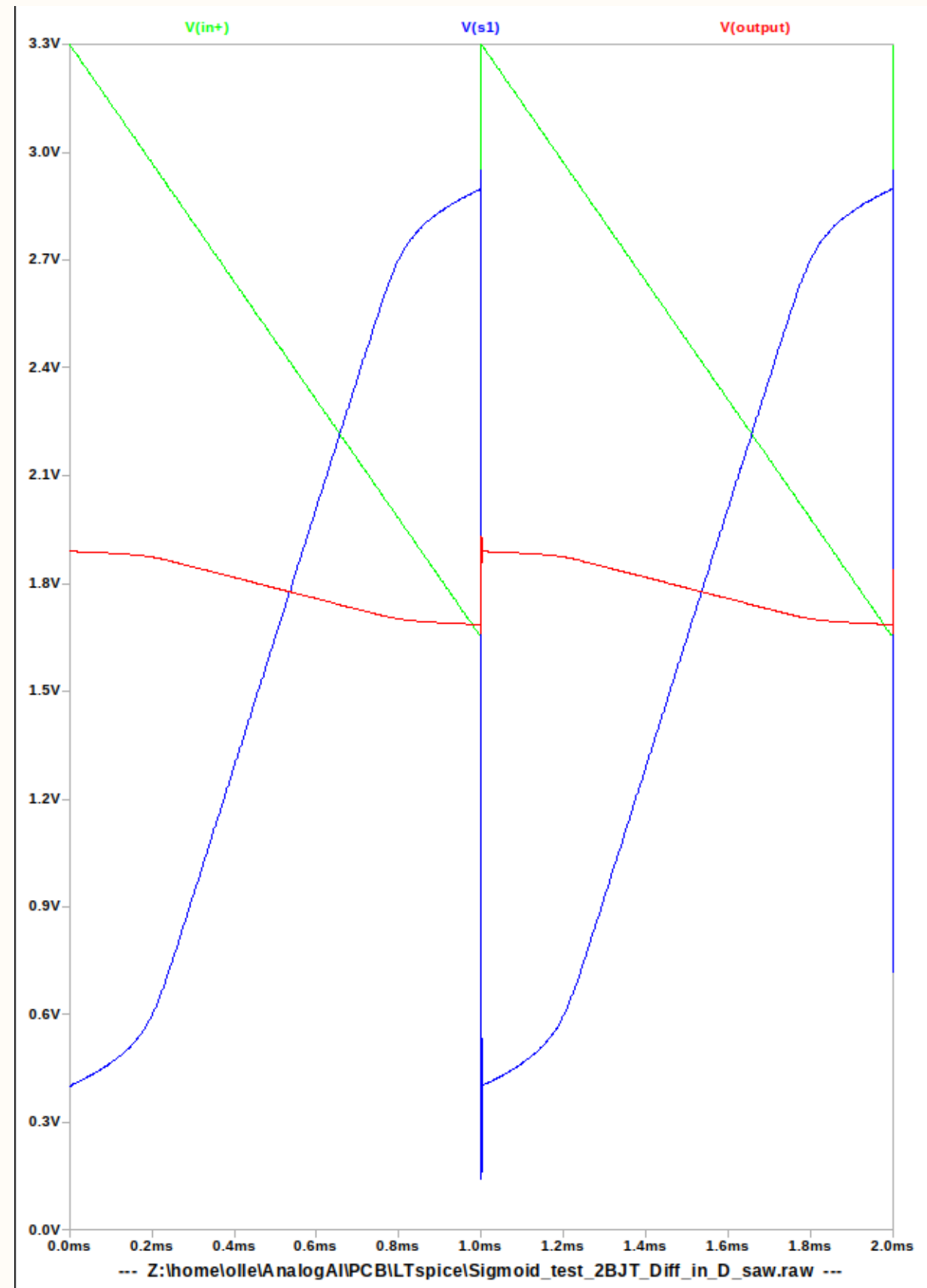
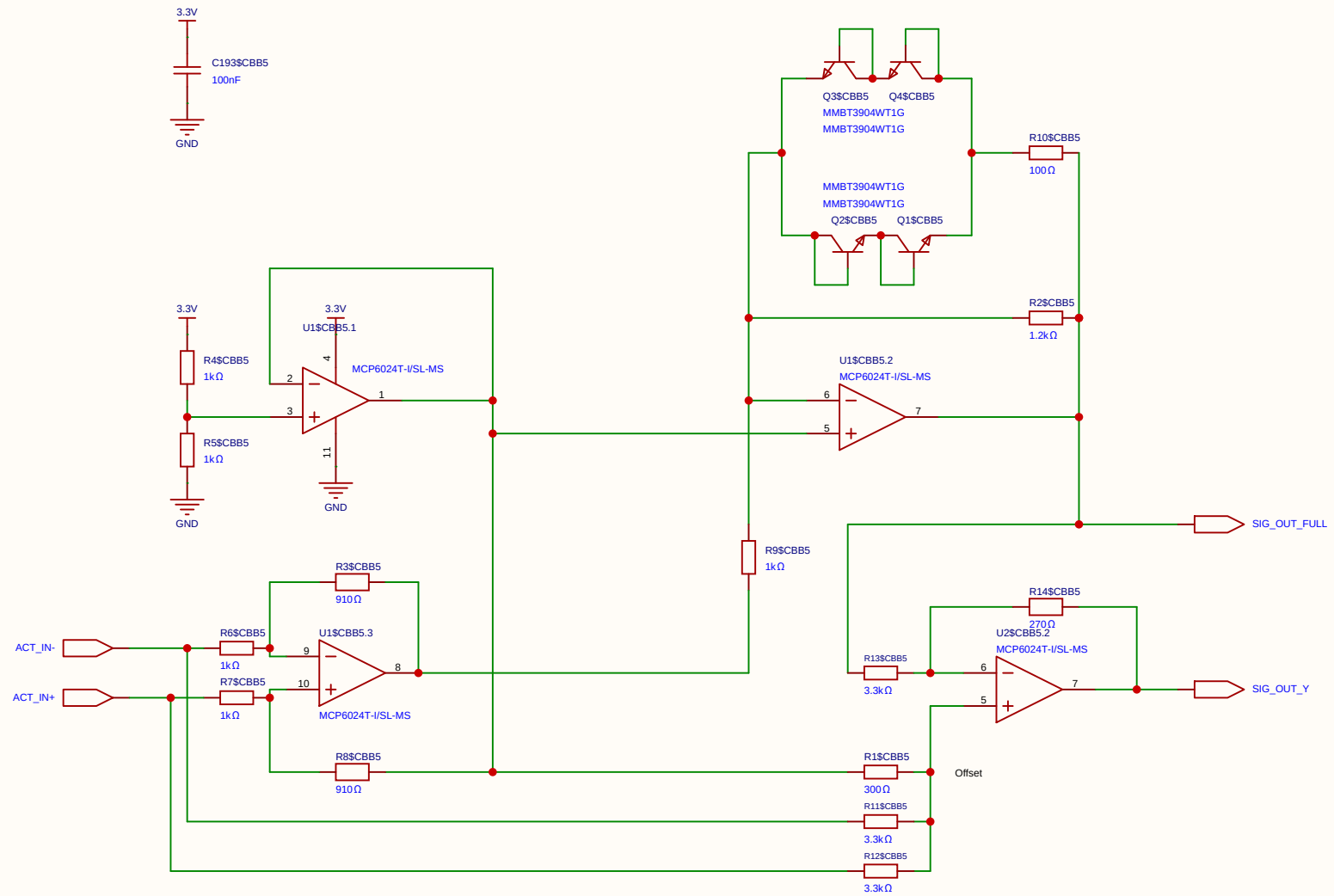
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Board				Page	SIMOID(4)
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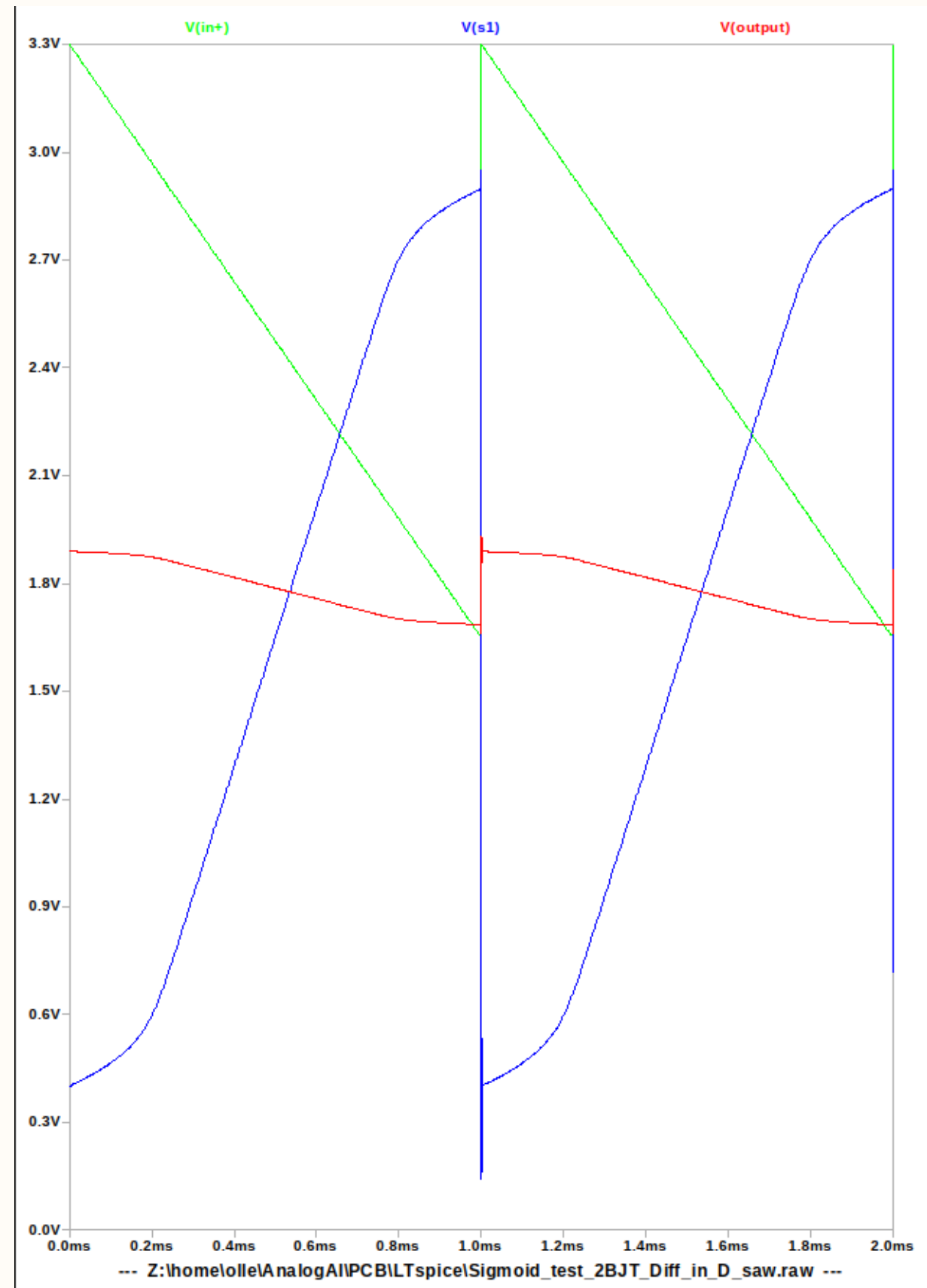
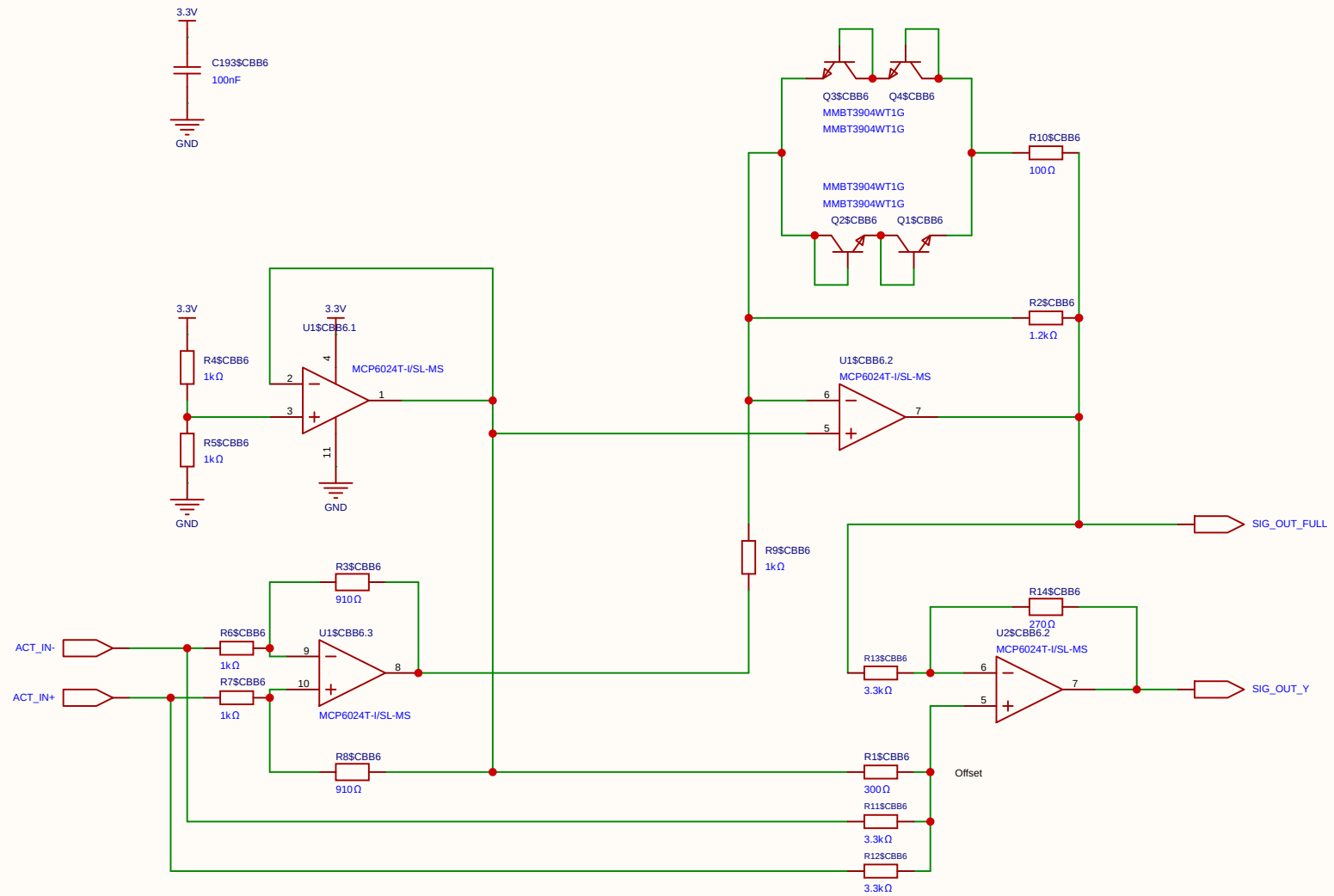
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EasyEDA		V1.0	A4	EasyEDA.com	



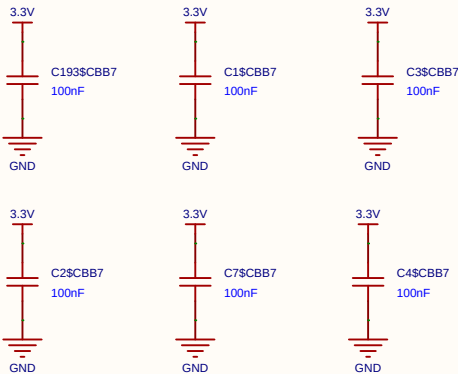
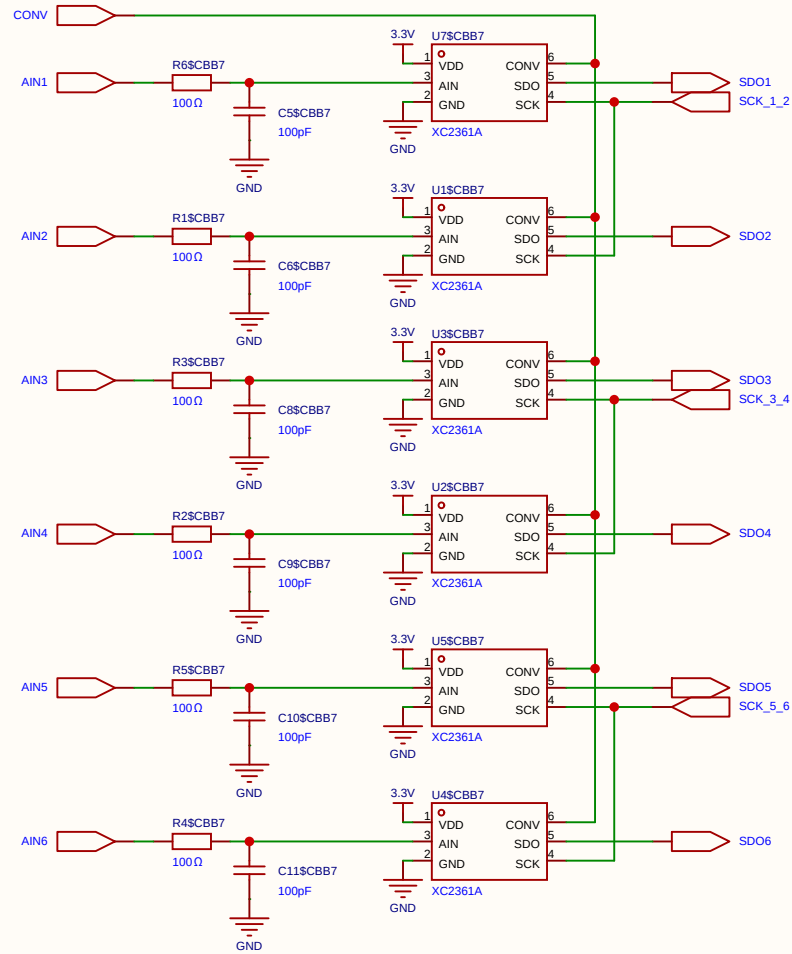
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Board				Page	SIMOID(4)
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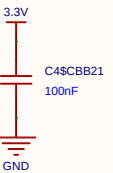
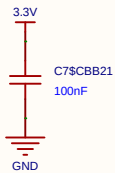
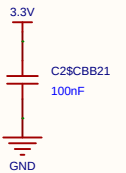
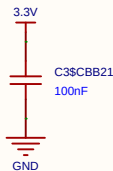
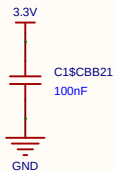
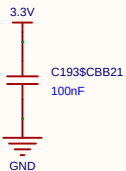
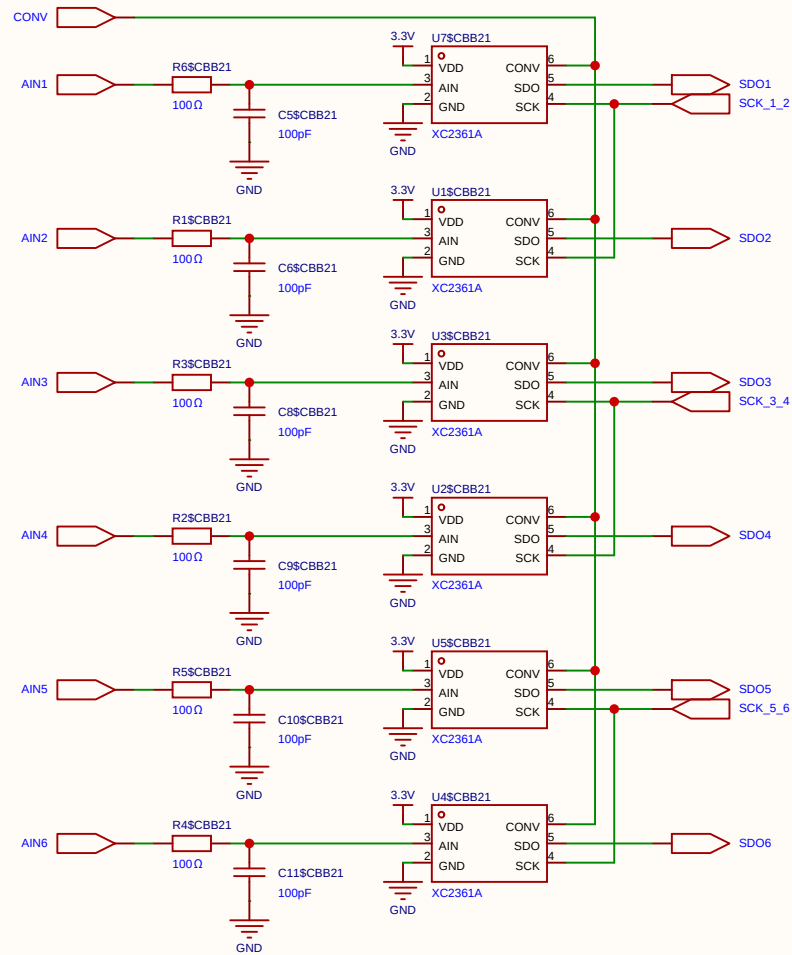
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Board				Page	SIMOID(4)
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EasyEDA		V1.0	A4	EasyEDA.com	



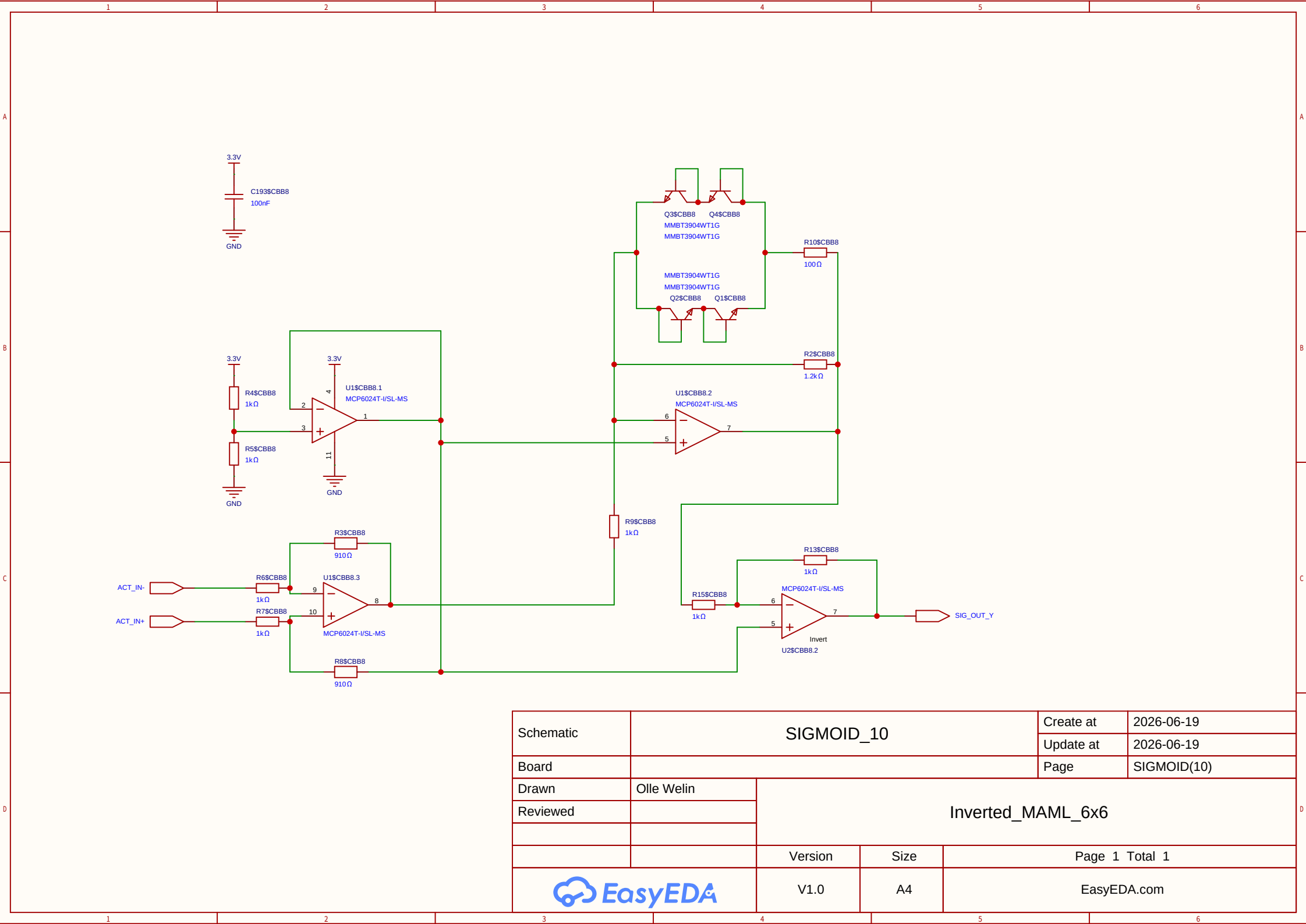
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Board				Page	SIMOID(4)
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Reviewed					
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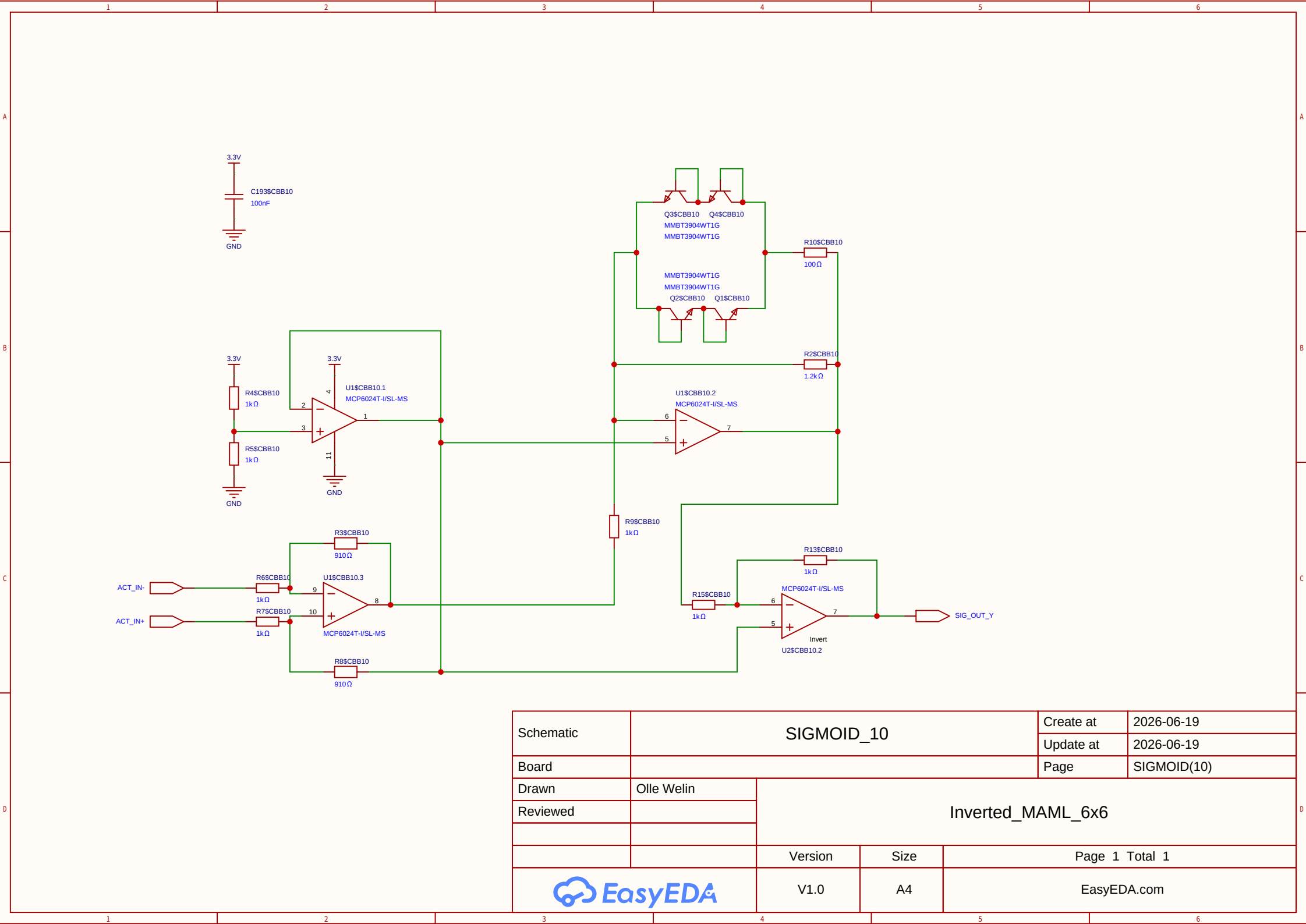
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Board				Page	ADC_BANK
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Reviewed					
			Version	Size	Page 1 Total 1
EasyEDA			V1.0	A4	EasyEDA.com



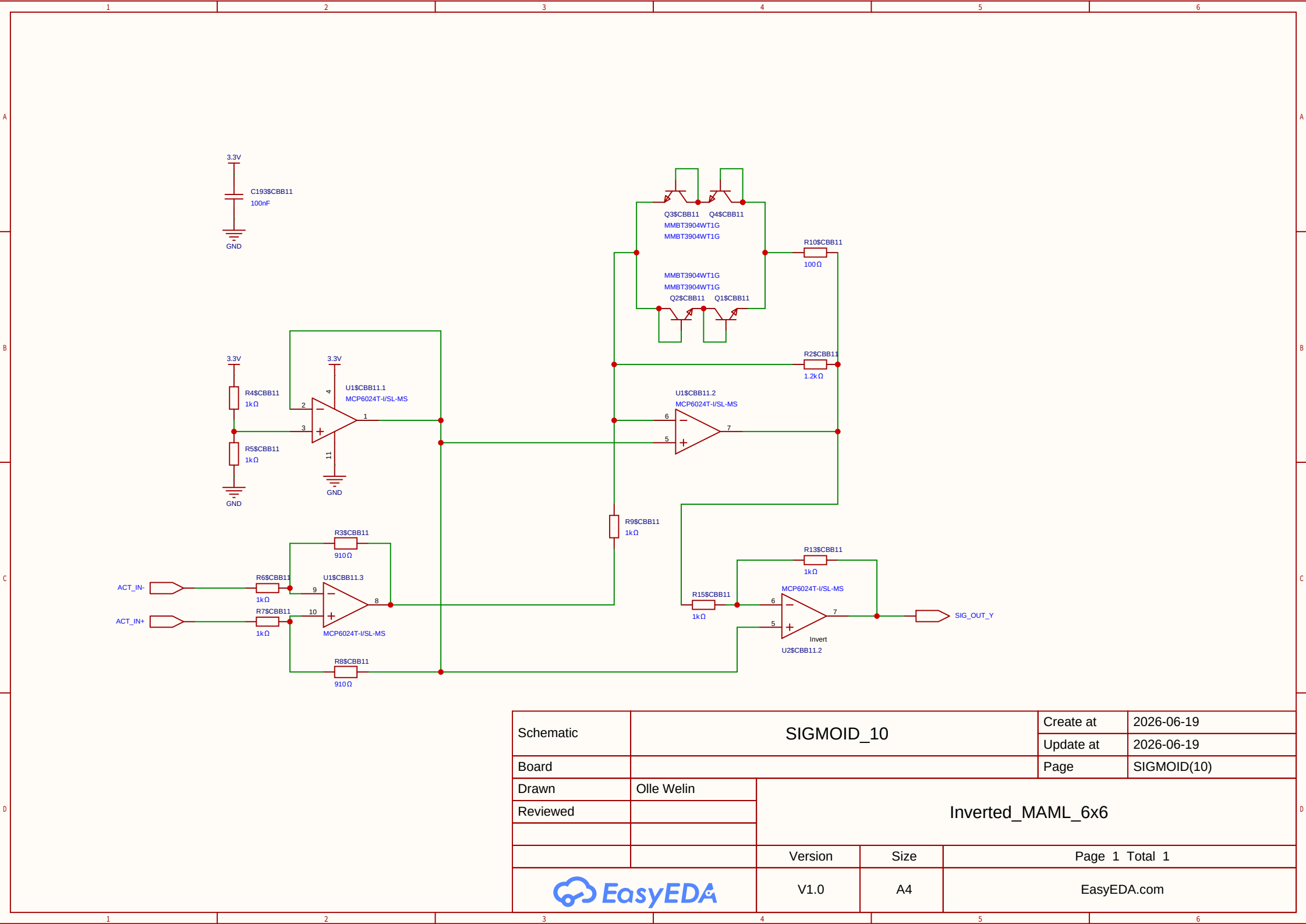
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Board				Page	ADC_BANK
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			Version	Size	Page 1 Total 1
EasyEDA			V1.0	A4	EasyEDA.com



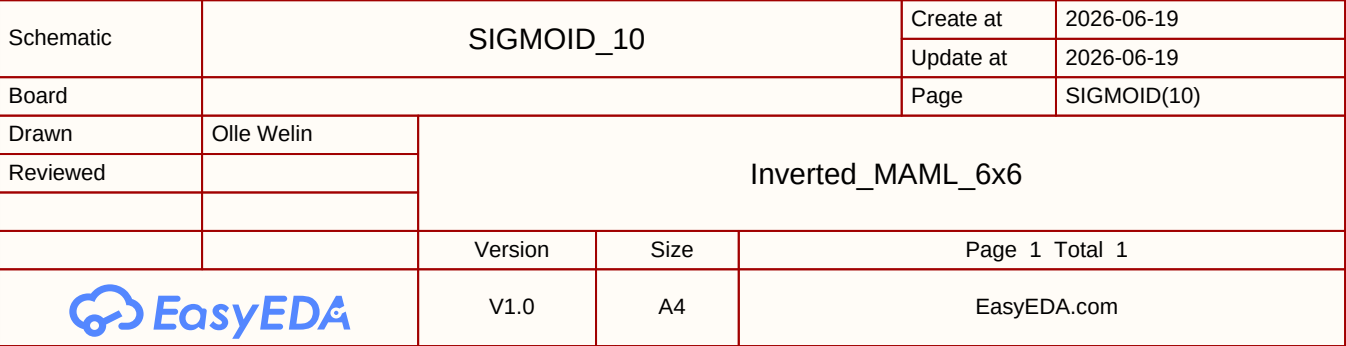
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Board				Page	SIGMOID(10)
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

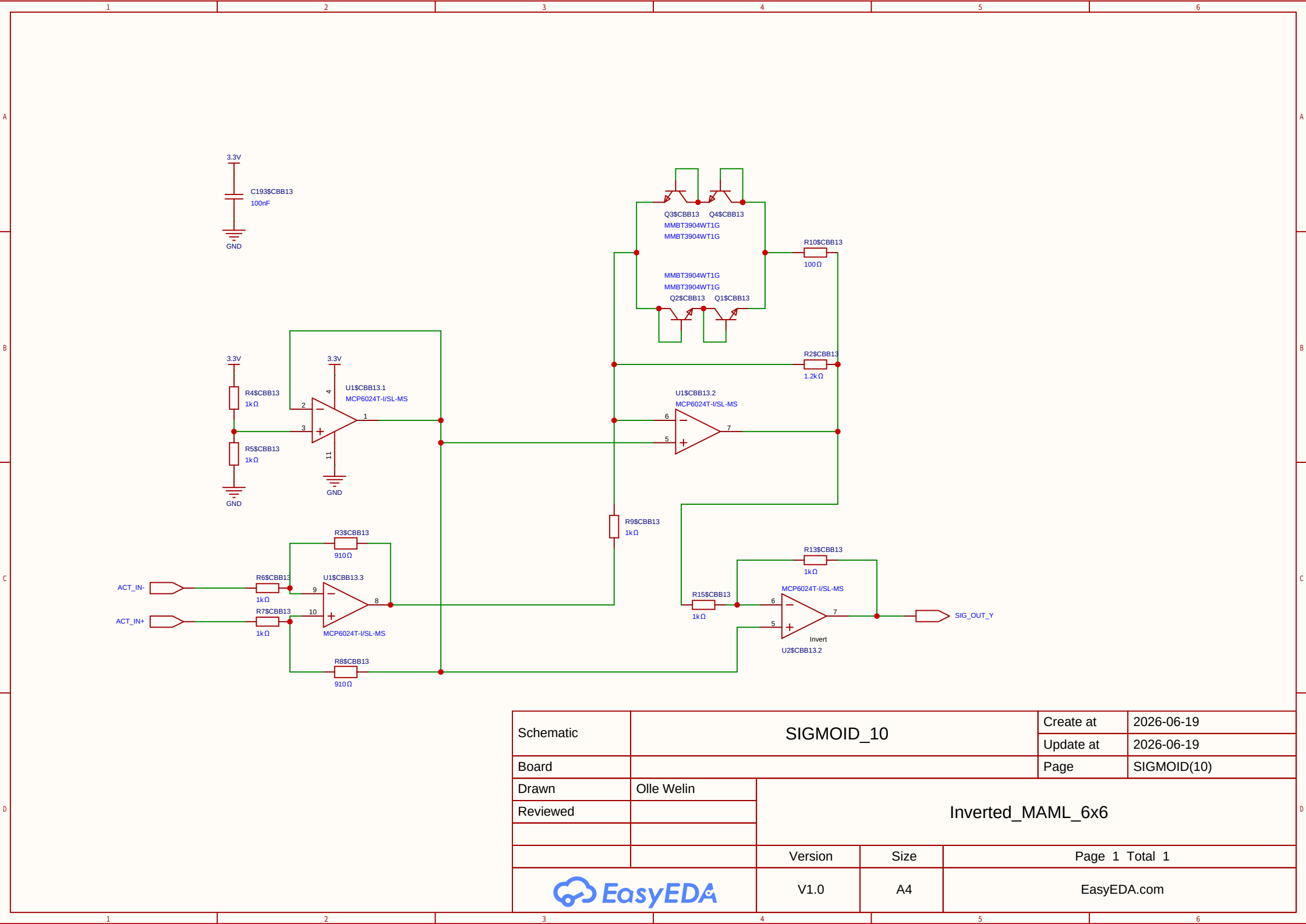


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Board				Page	SIGMOID(10)
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Reviewed					
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EasyEDA		V1.0	A4	EasyEDA.com	

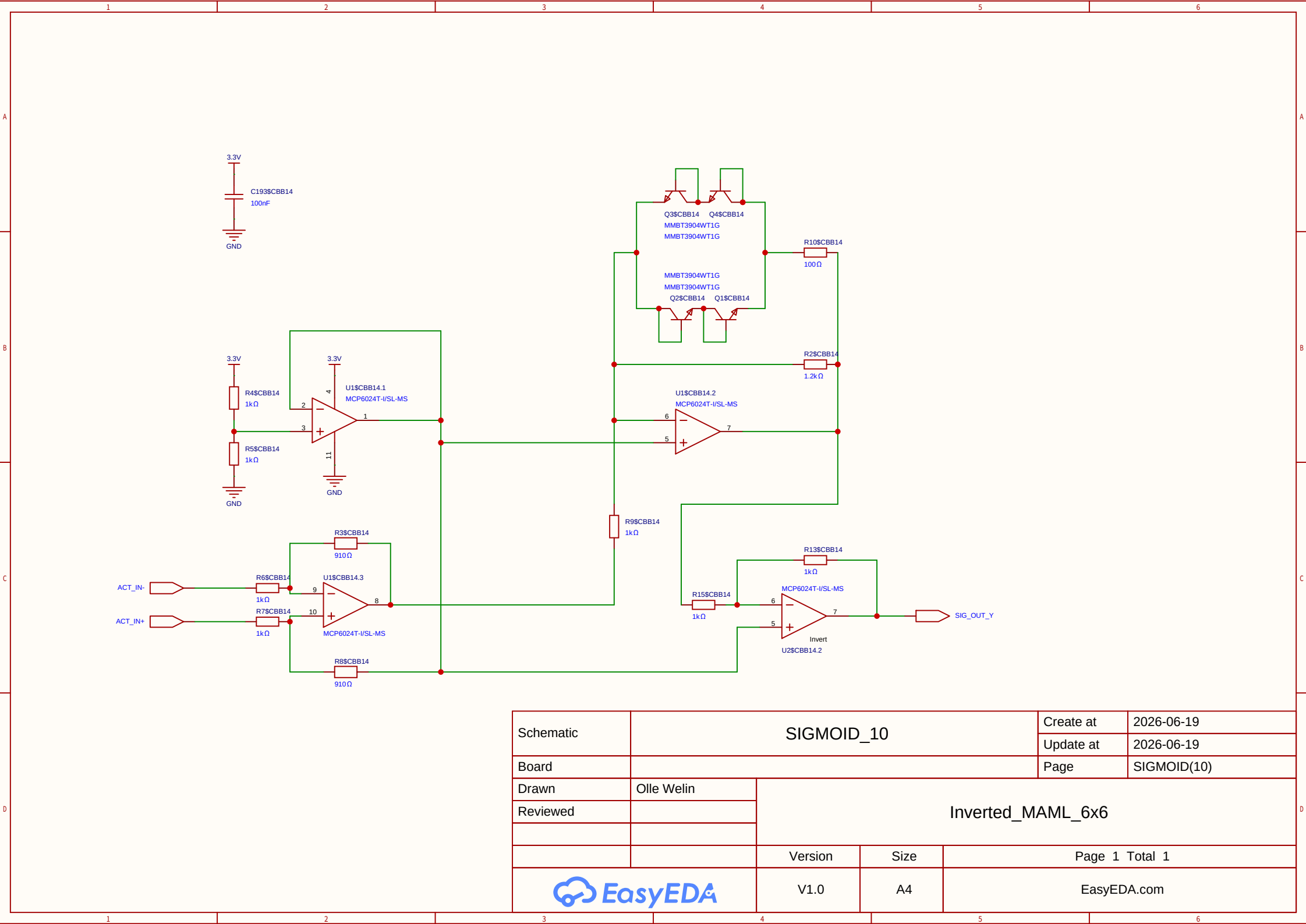


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Board				Page	SIGMOID(10)
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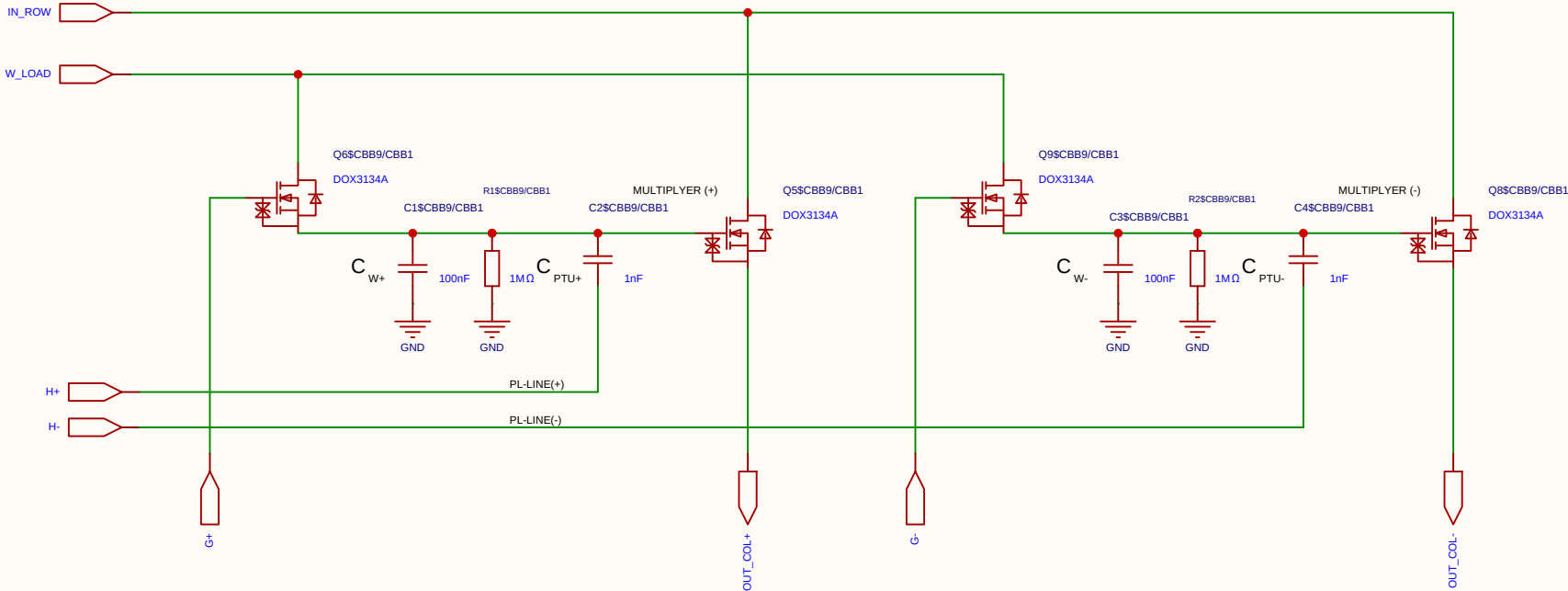


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Board				Page	SIGMOID(10)
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Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	



Schematic	SIGMOID_10			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	SIGMOID(10)
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

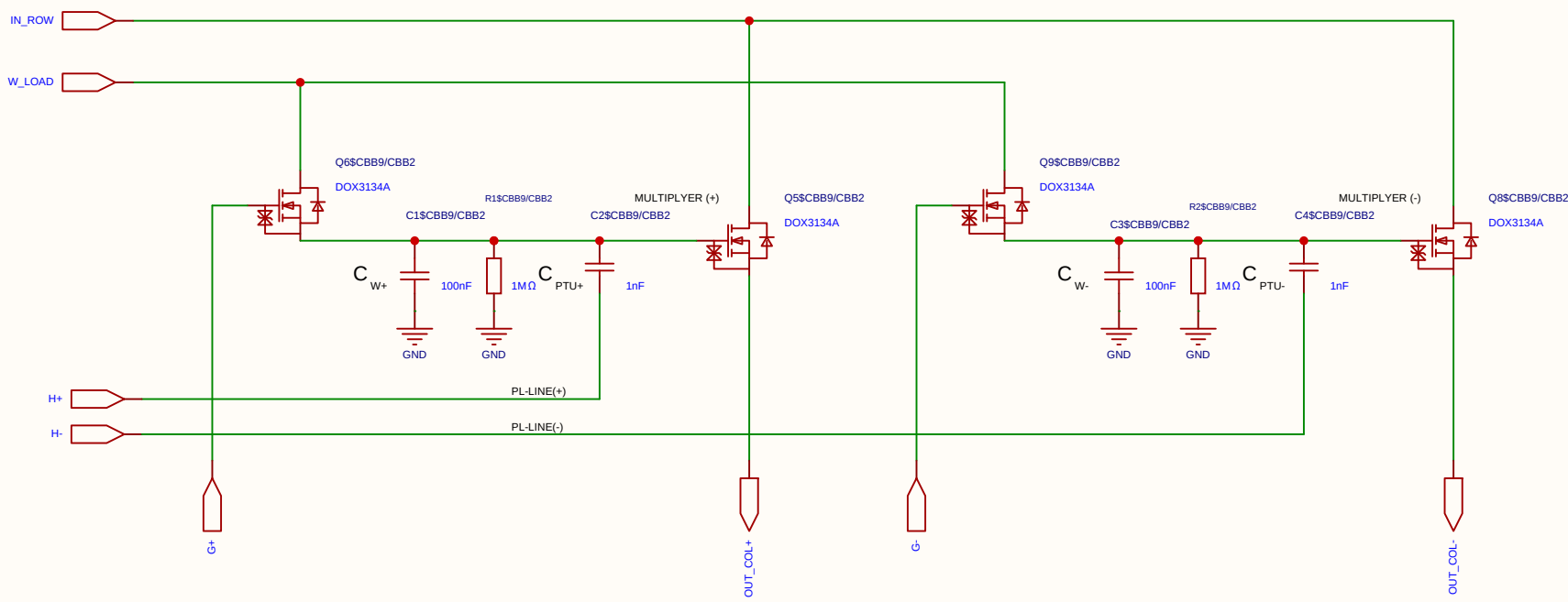
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

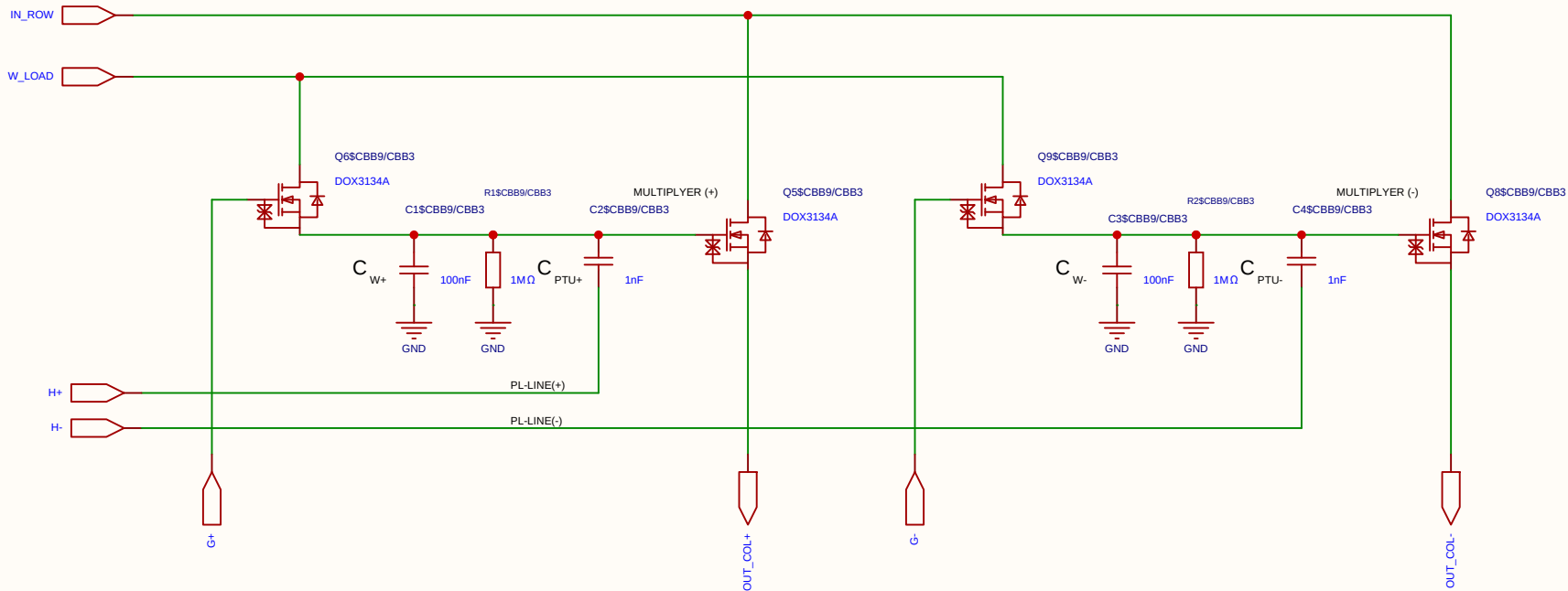
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

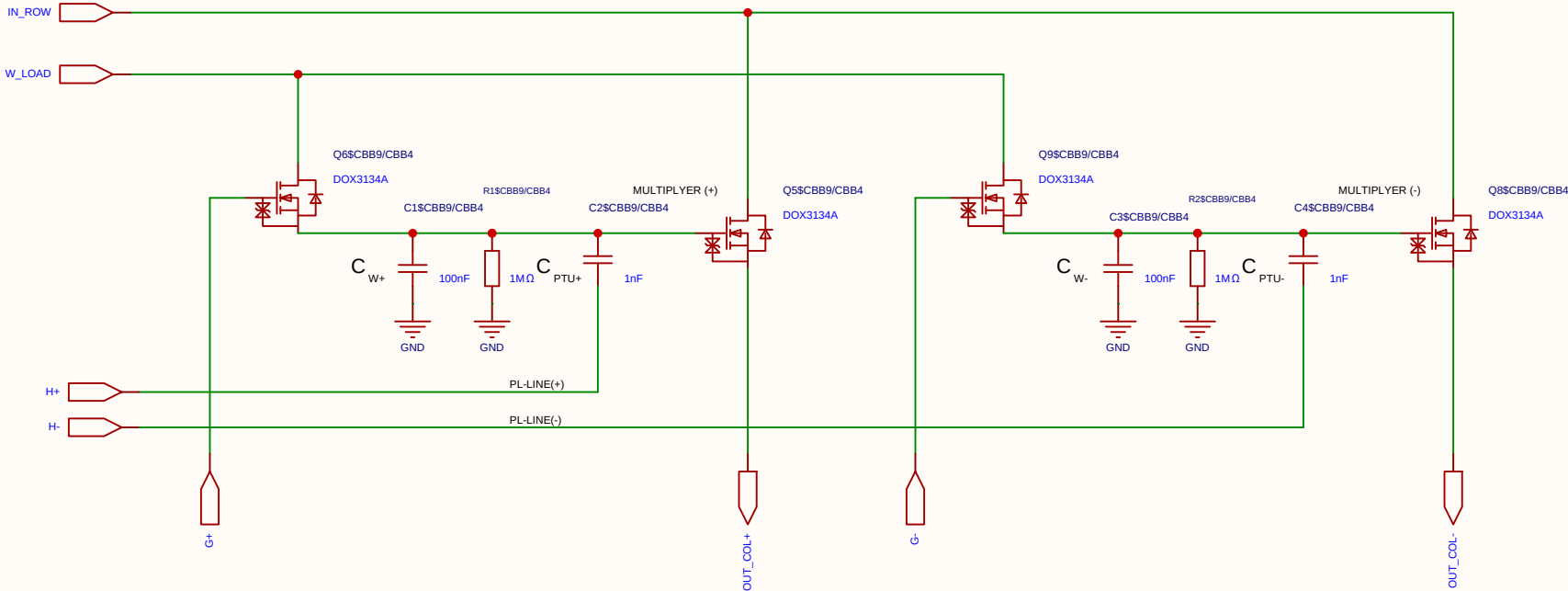
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

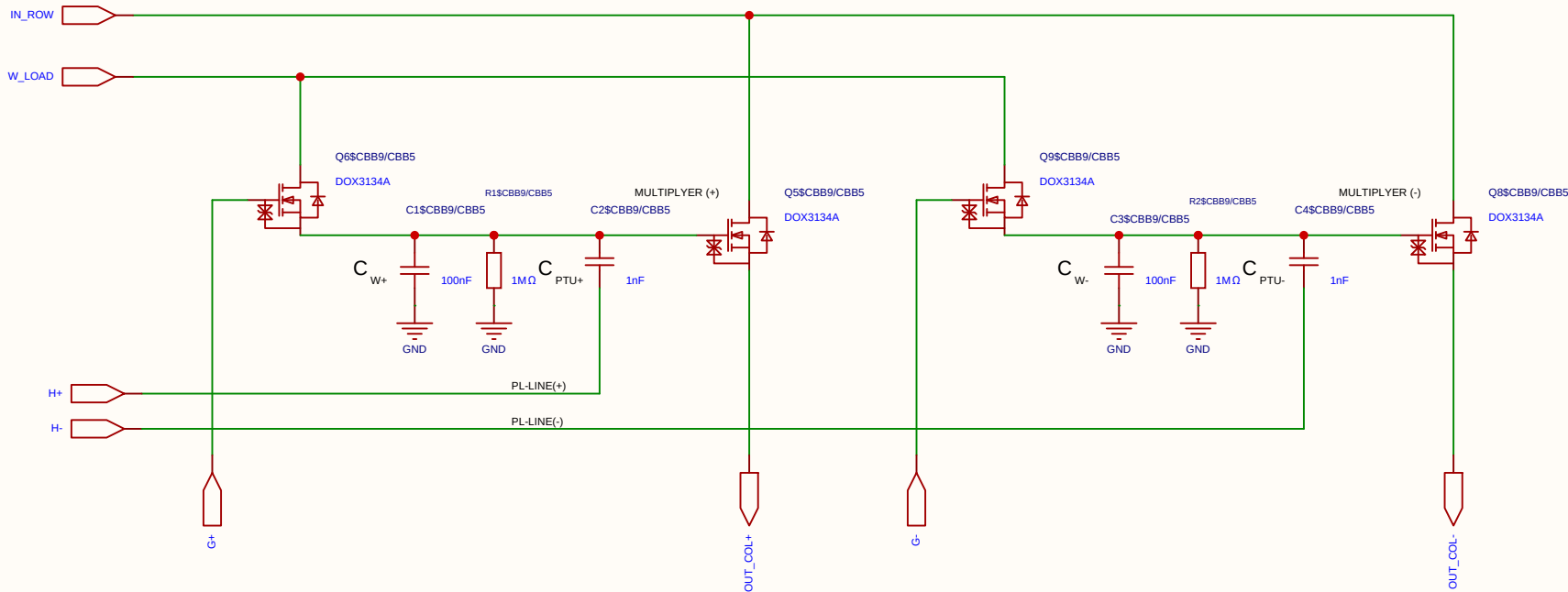
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

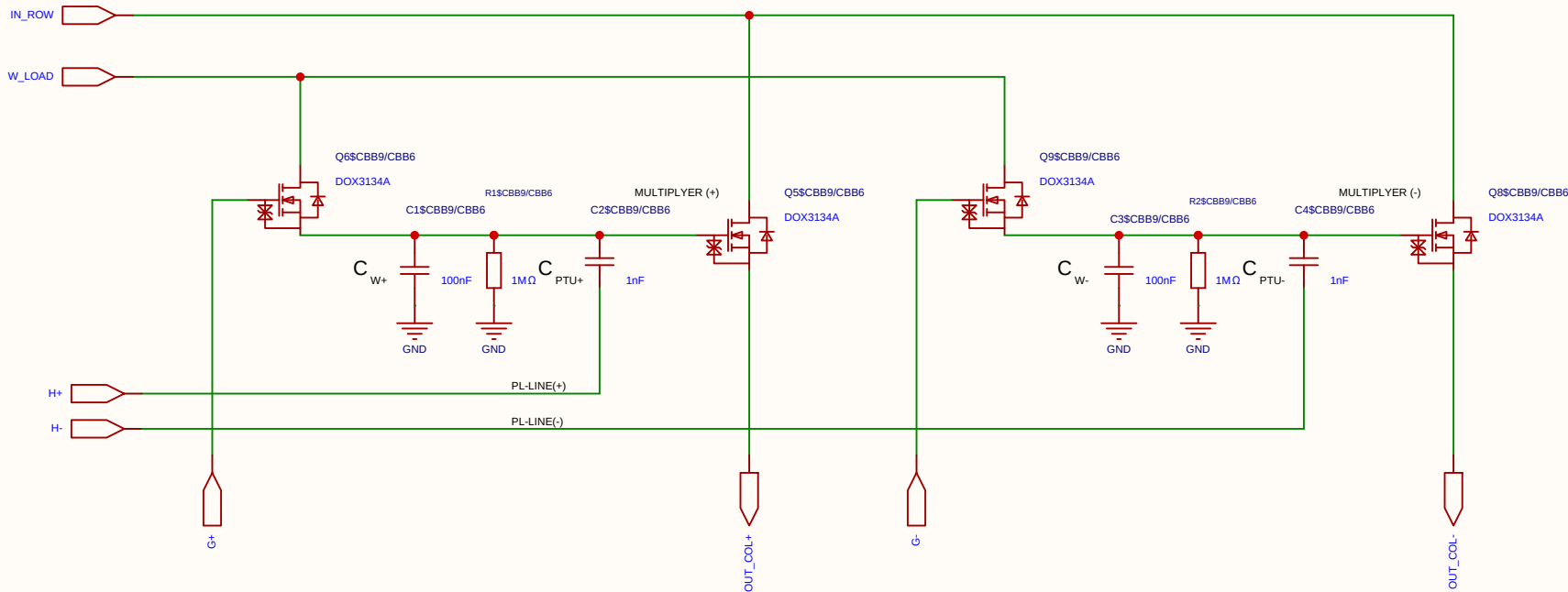
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

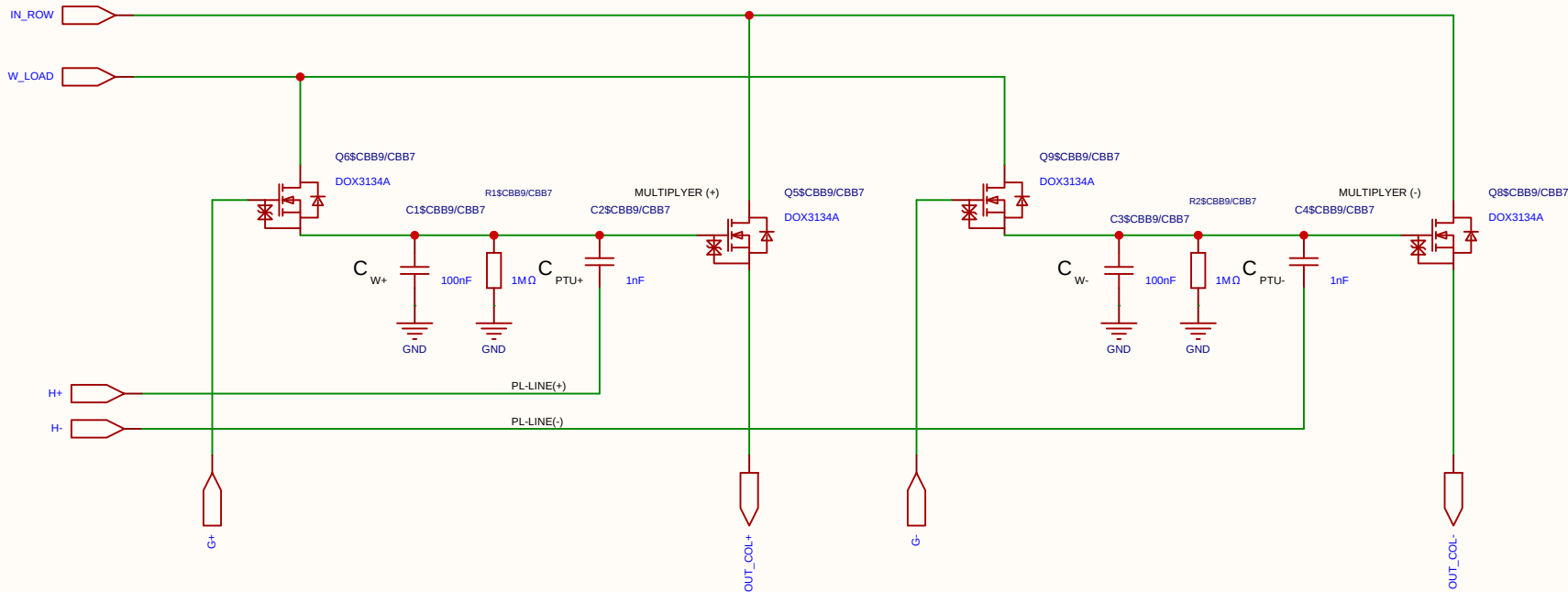
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Drawn	Olle Welin	Inverted_MAML_6x6			
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		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Category	Parameter	Value / Type	Technical Description & Function
Resistors	Discharge resistor (R)	1 M Ω	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

Component	Value	Description / Function
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a $\sim 1/101$ voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μ s.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

Component	Value	Description / Function
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_I . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_I . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_I . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
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The schematic diagram illustrates a 2x2 matrix multiplier circuit. It features four primary inputs: **IN_ROW**, **W_LOAD**, **H+**, and **H-**. The circuit is powered by a 5V supply and ground. The output is split into two channels, **OUT_COL+** and **OUT_COL-**, each passing through a 50Ω load.

The circuit is composed of two main multiplier blocks, **MULTIPLYER (+)** and **MULTIPLYER (-)**, each implemented with a differential pair of transistors (Q6/Q5 and Q9/Q8) and a current mirror (Q8). The multiplier blocks are connected to the **IN_ROW** and **W_LOAD** inputs. The **H+** and **H-** inputs are connected to the **OUT_COL+** and **OUT_COL-** outputs, respectively, through a 50Ω load.

The multiplier blocks are biased by a 5V supply and ground. The biasing network includes a 100nF capacitor (C1) connected to the **W_LOAD** input, a 1MΩ resistor (R1) connected to the **W_LOAD** input, and a 1nF capacitor (C2) connected to the **W_LOAD** input. The multiplier blocks are also biased by a 5V supply and ground.

Category	Parameter	Value / Type	Technical Description & Function
Resistors	Discharge resistor (R)	1 M Ω	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

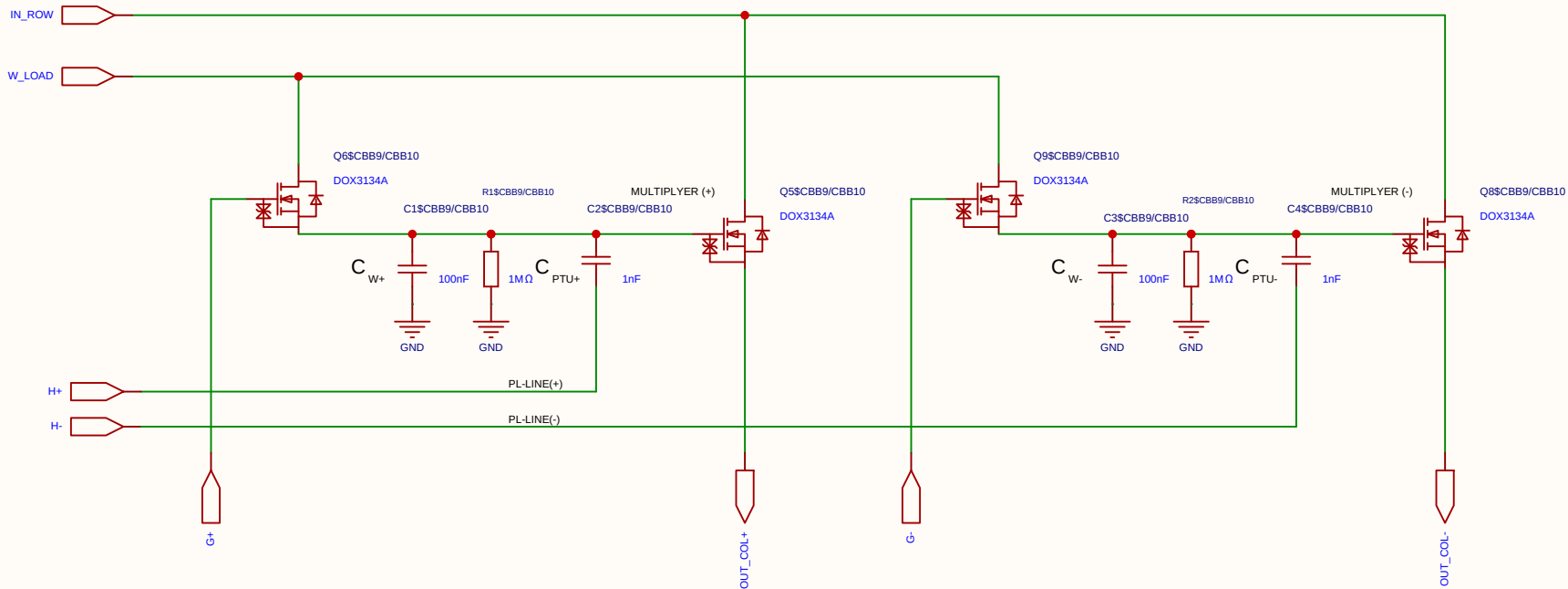
Component	Value	Description / Function
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a $\sim 1/101$ voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

State	PL Voltage	Perturbation at Gate	Comment
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μ s.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

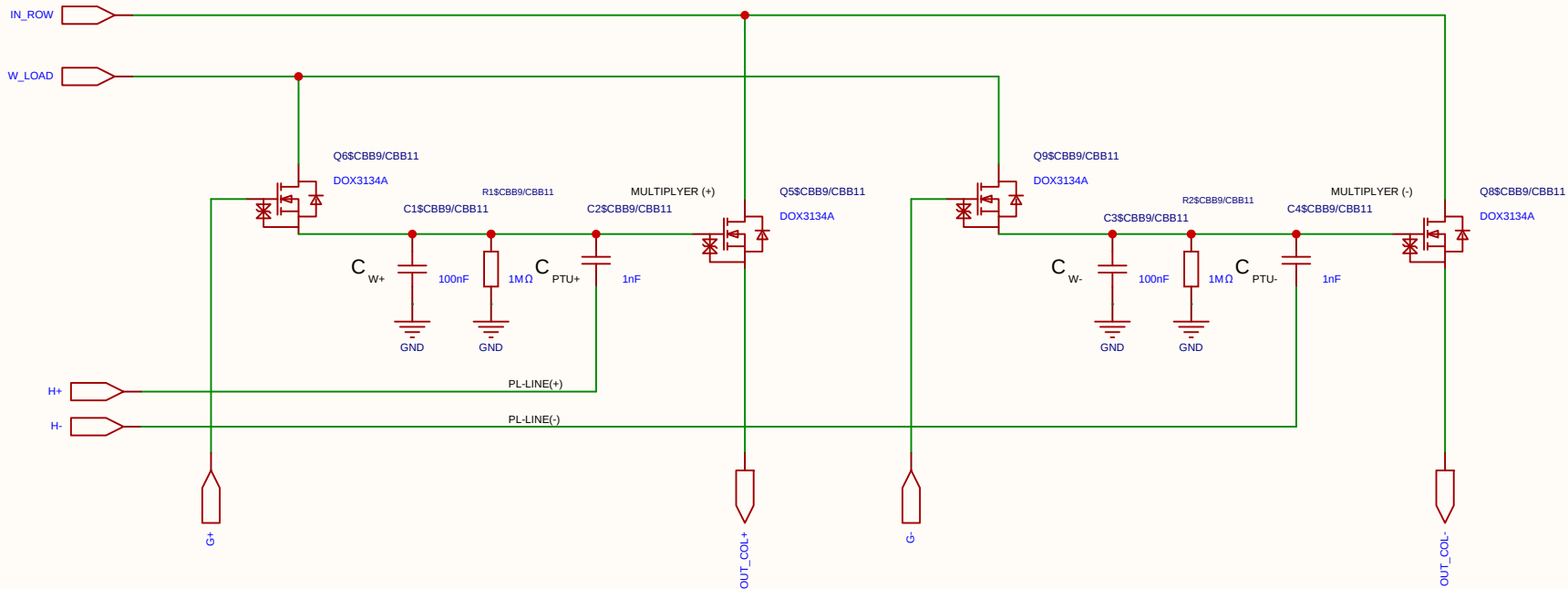
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

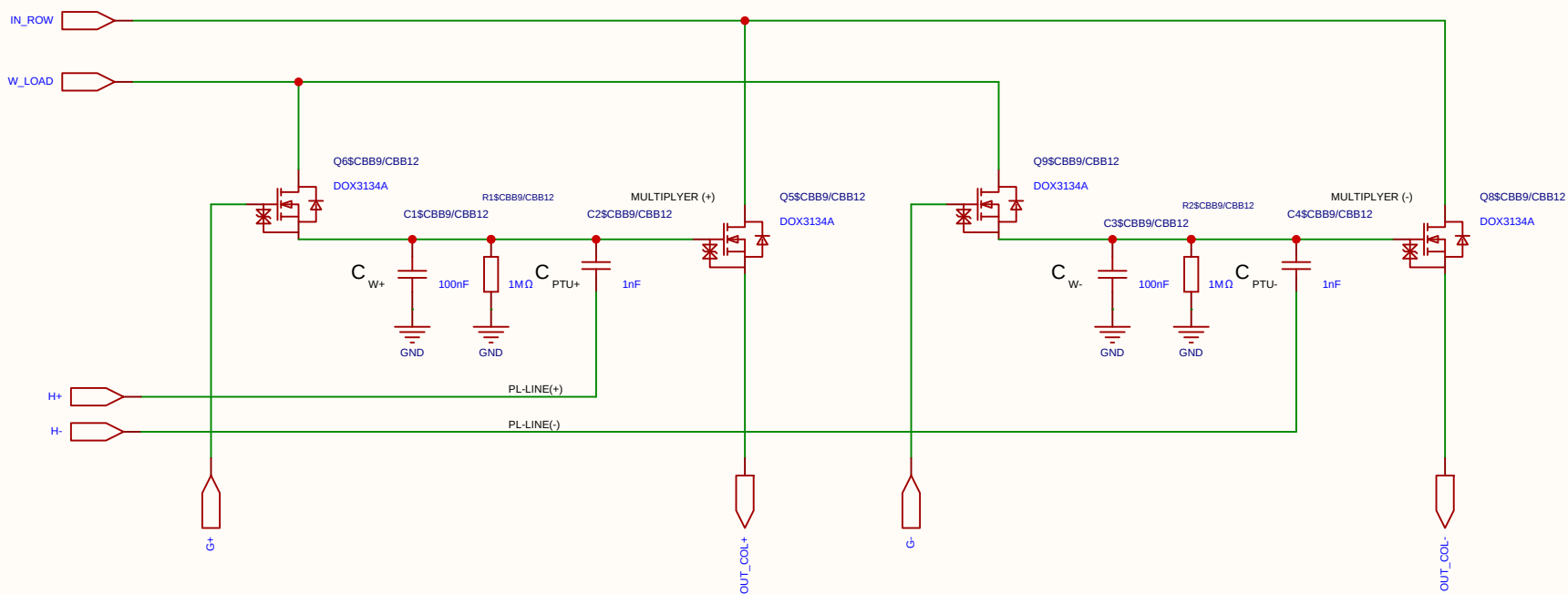
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

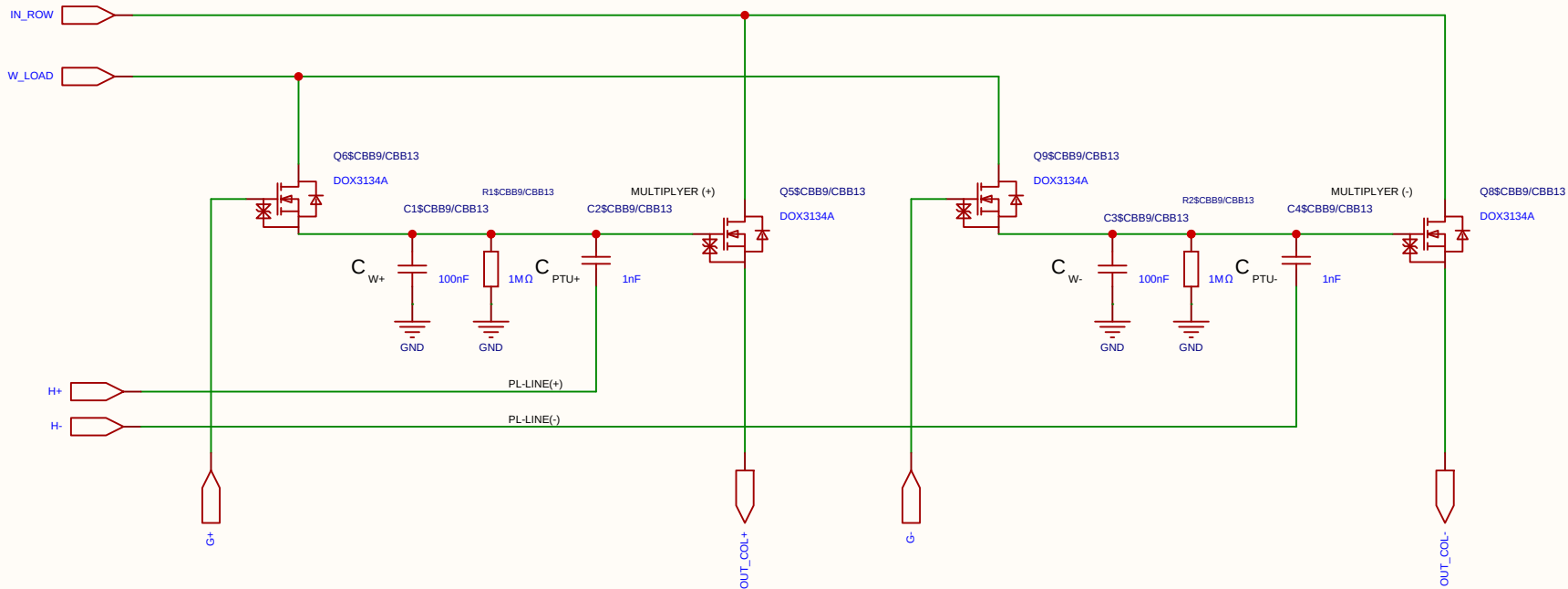
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

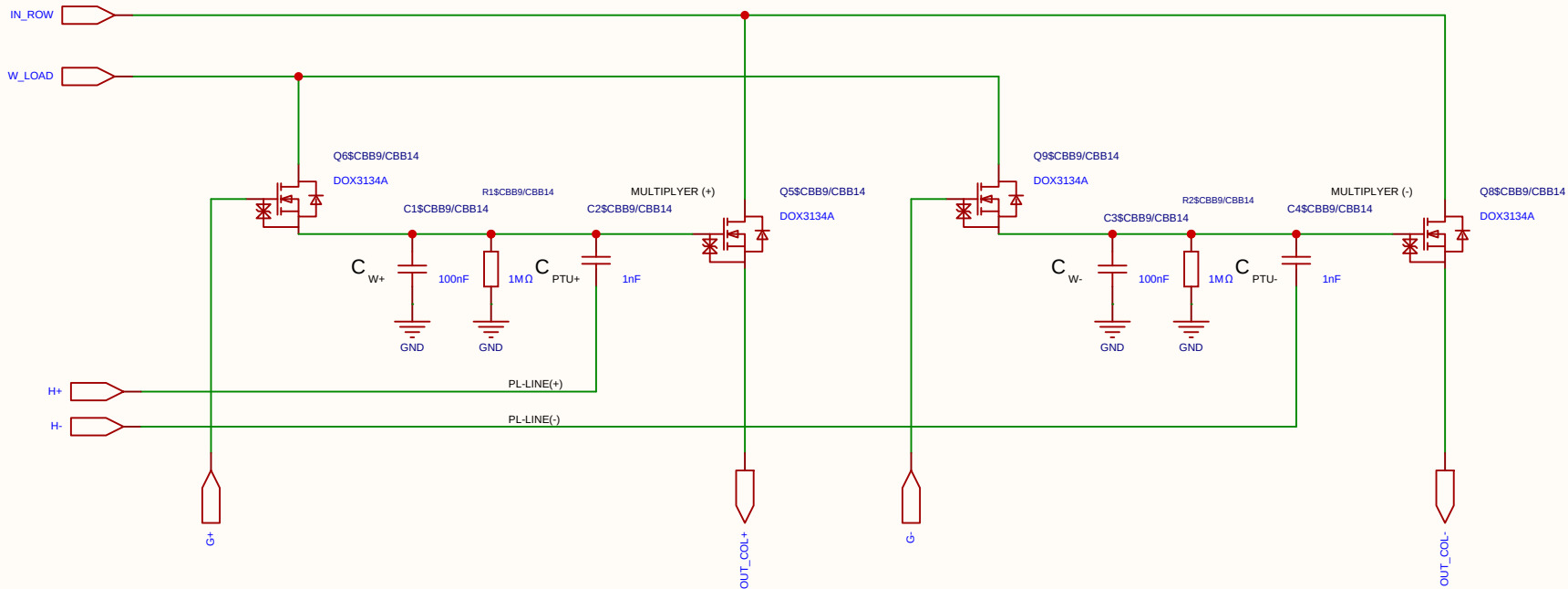
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

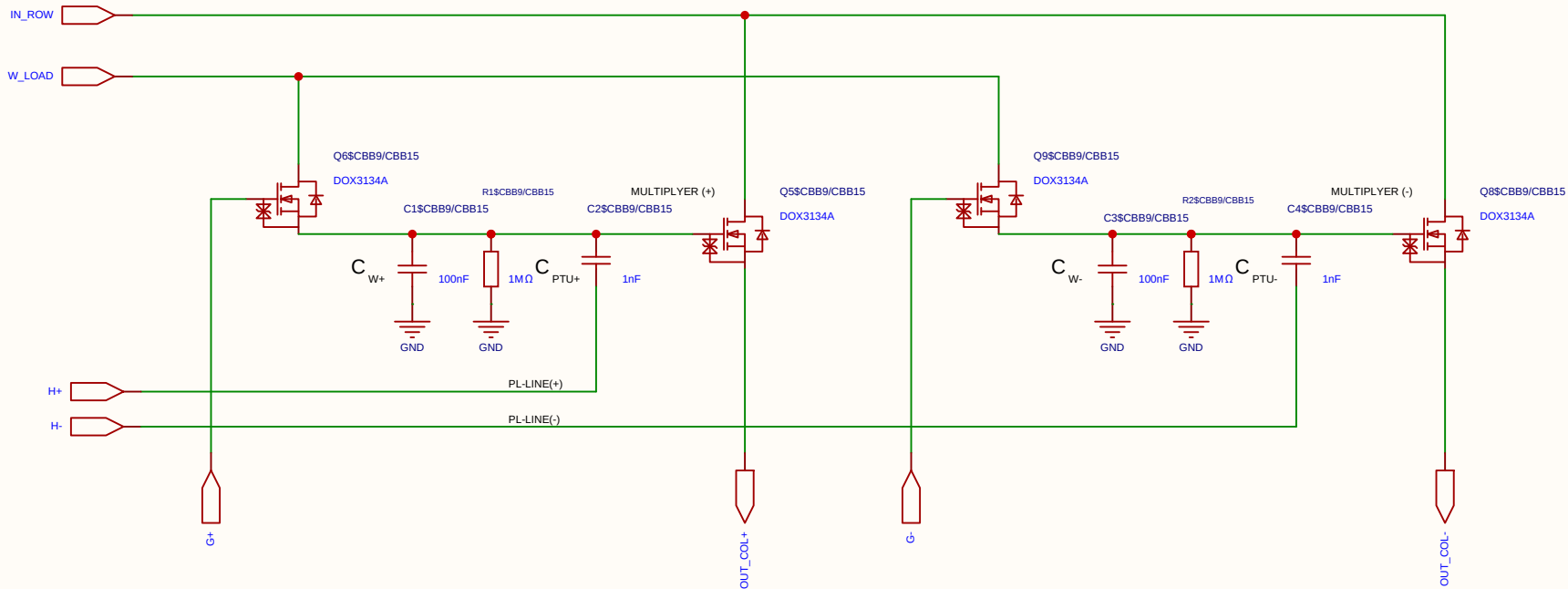
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

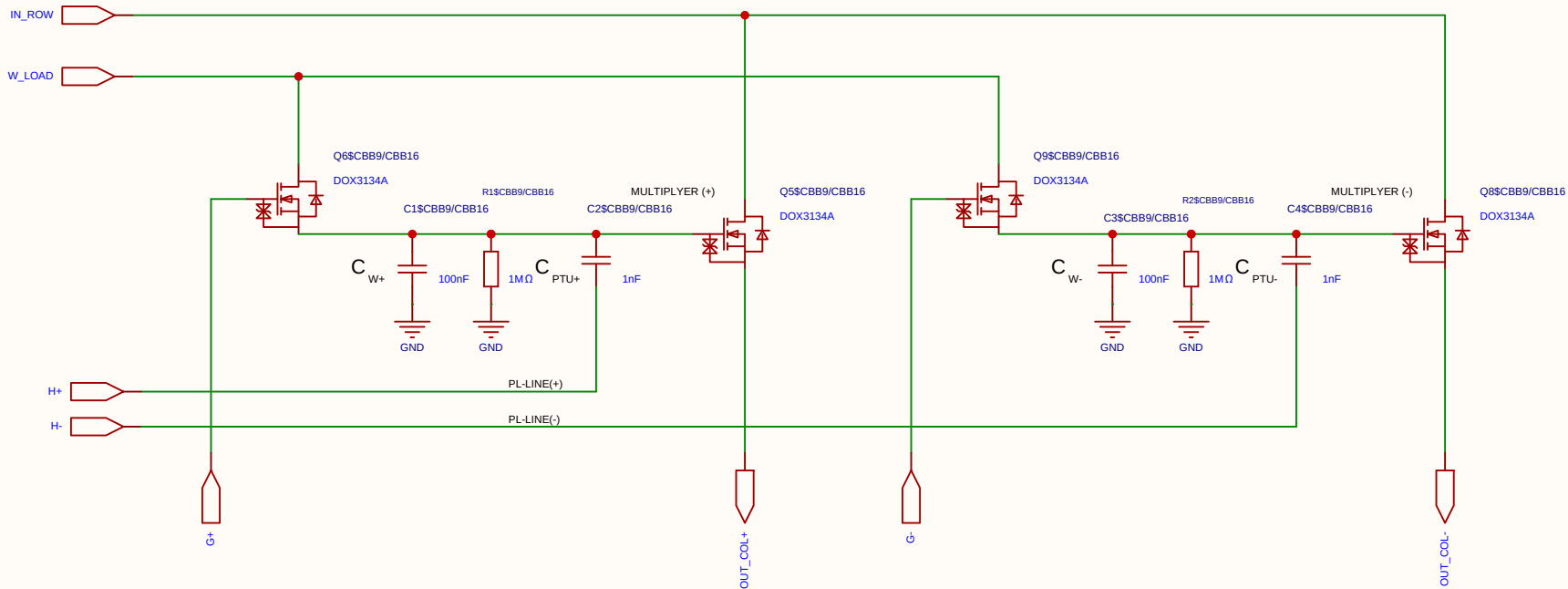
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
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		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

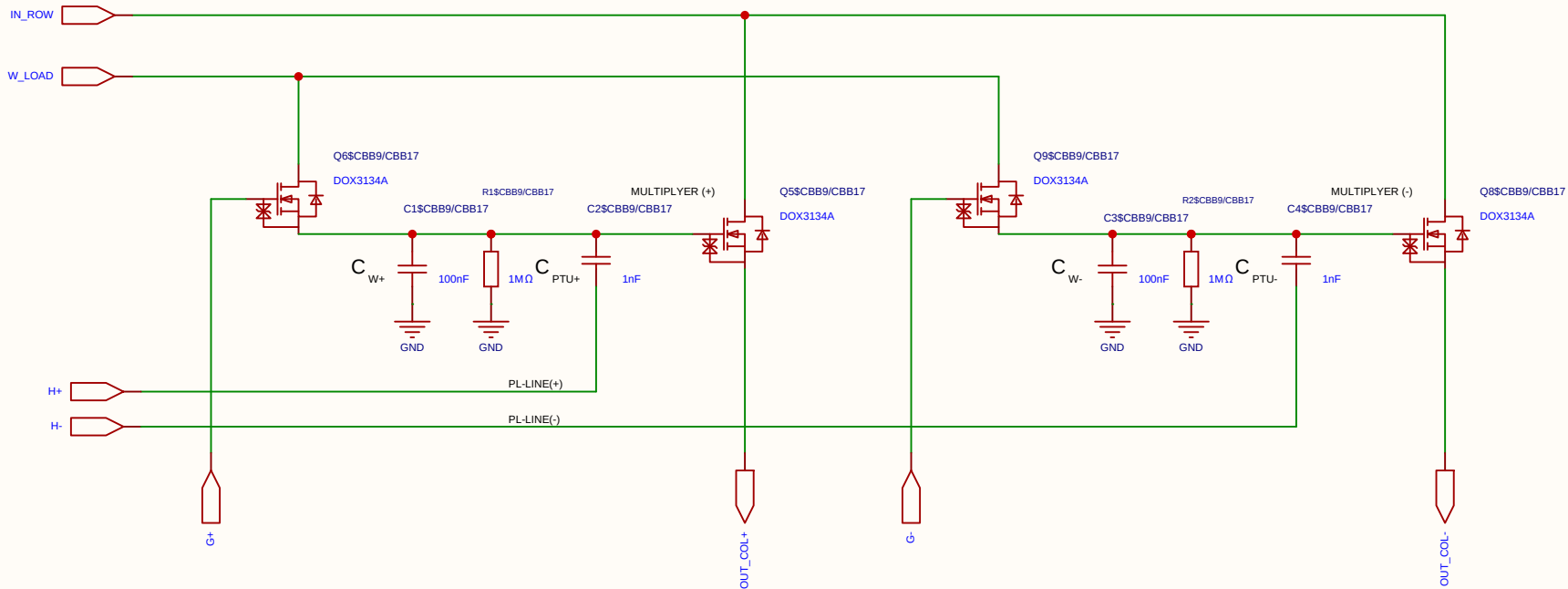
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

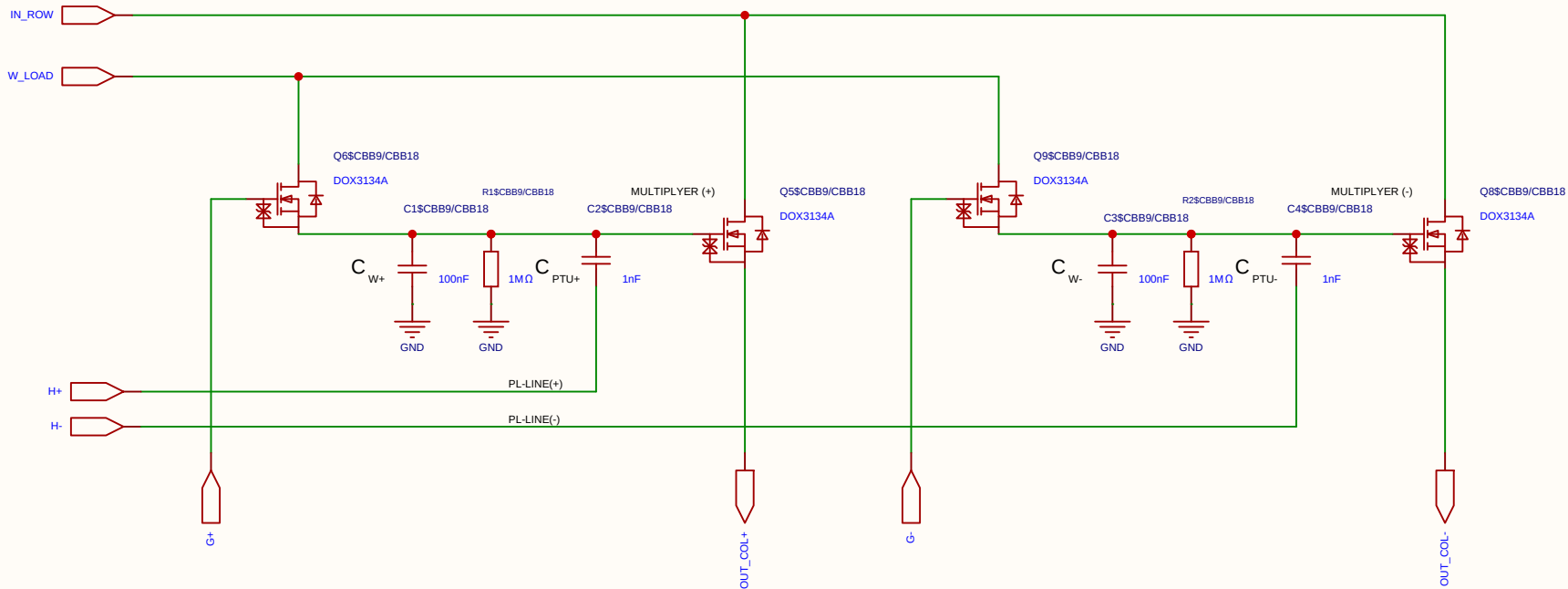
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

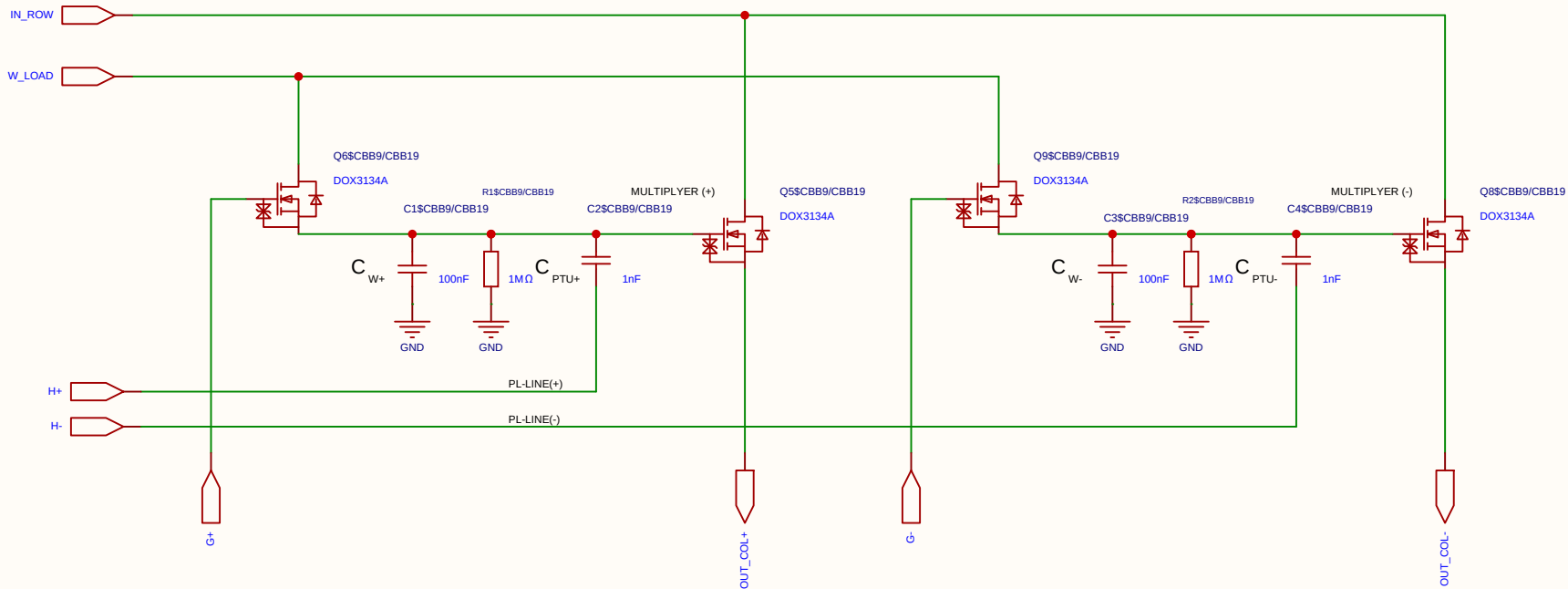
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
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		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
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Reviewed					
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Category	Parameter	Value / Type	Technical Description & Function
Resistors	Discharge resistor (R)	1 M Ω	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

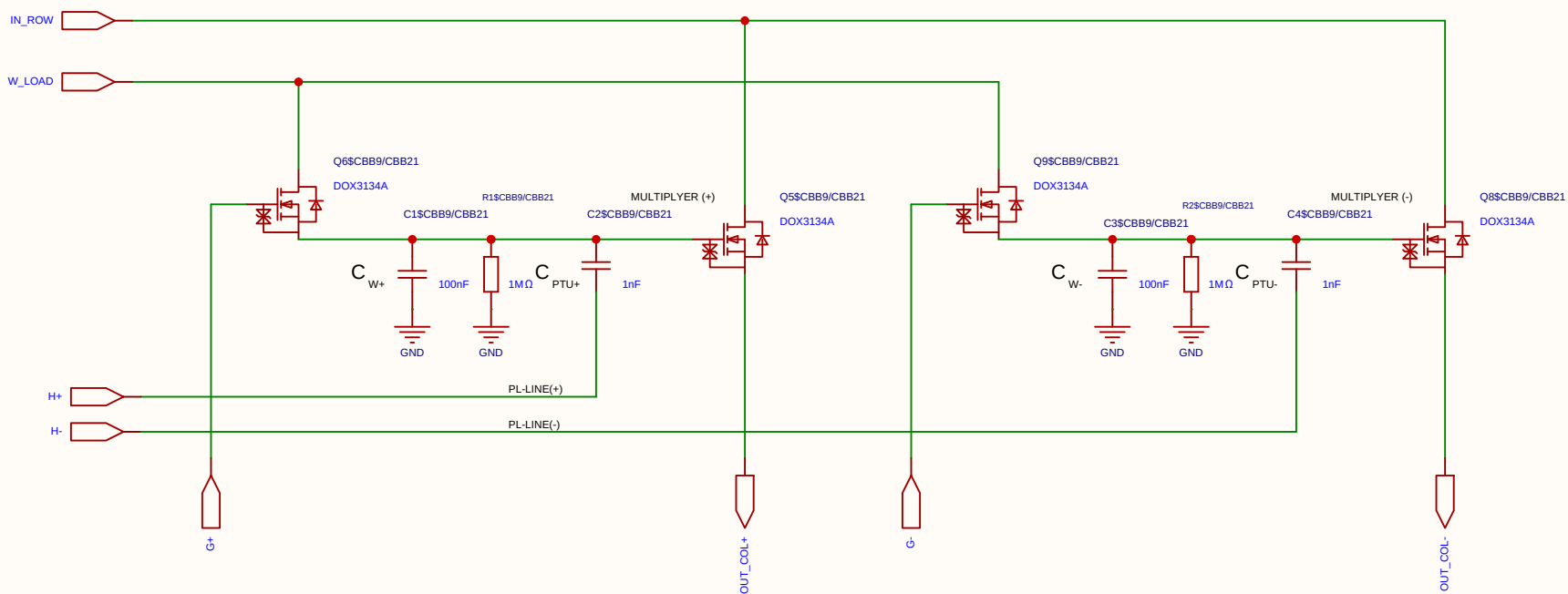
Component	Value	Description / Function
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a $\sim 1/101$ voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

Component	Value	Description / Function
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_I . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_I . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_I . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

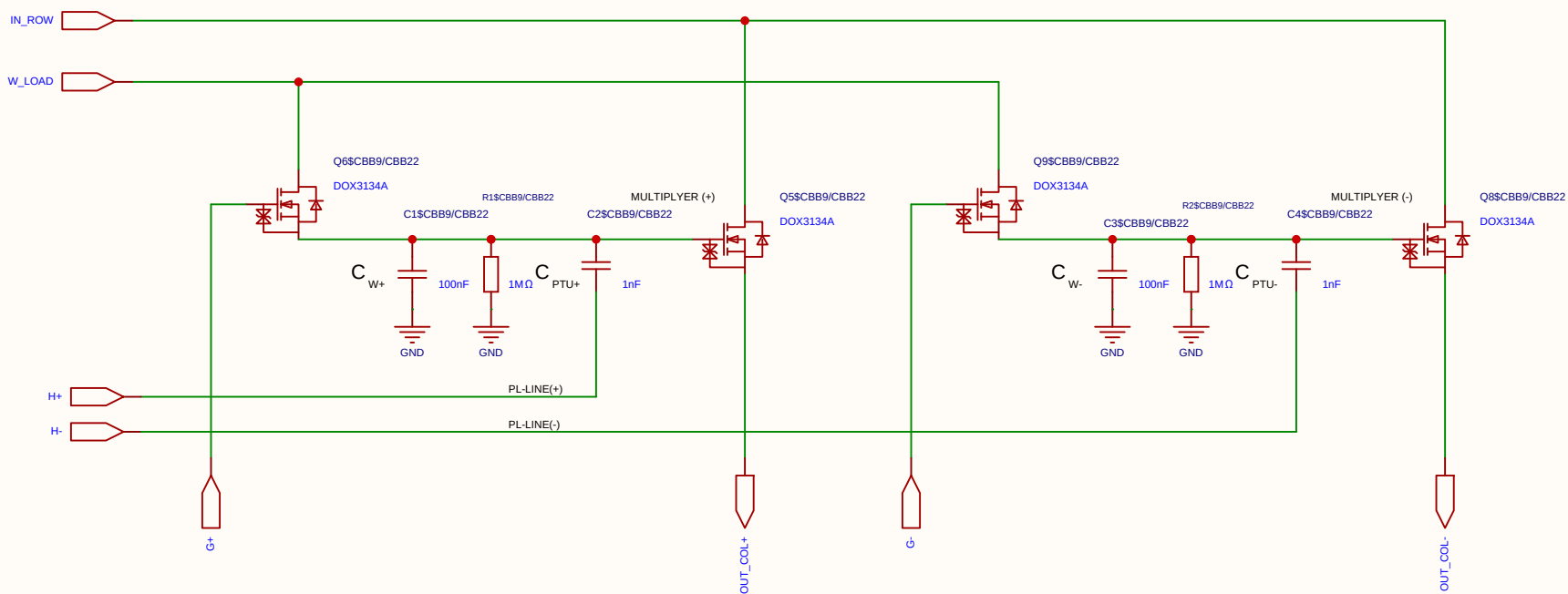
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Board				Page	matrix_3_8_MUL_CELL
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

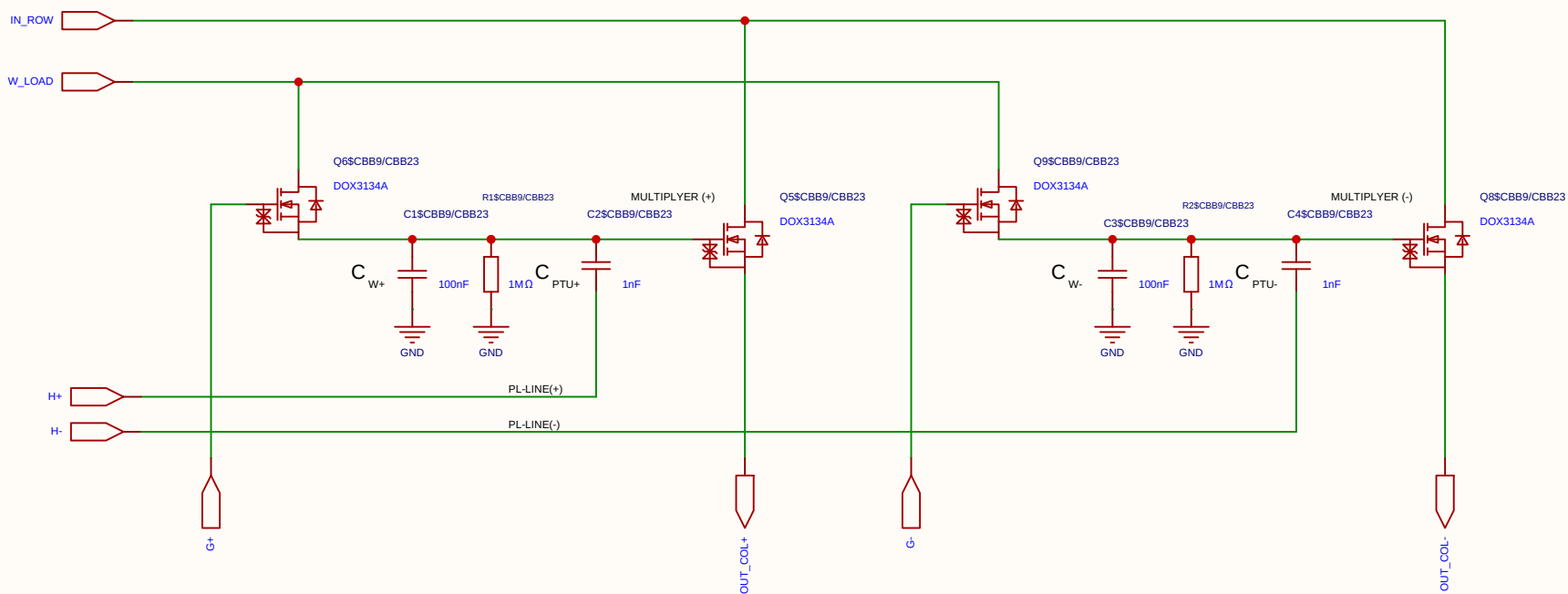
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Drawn	Olle Welin	Inverted_MAML_6x6			
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

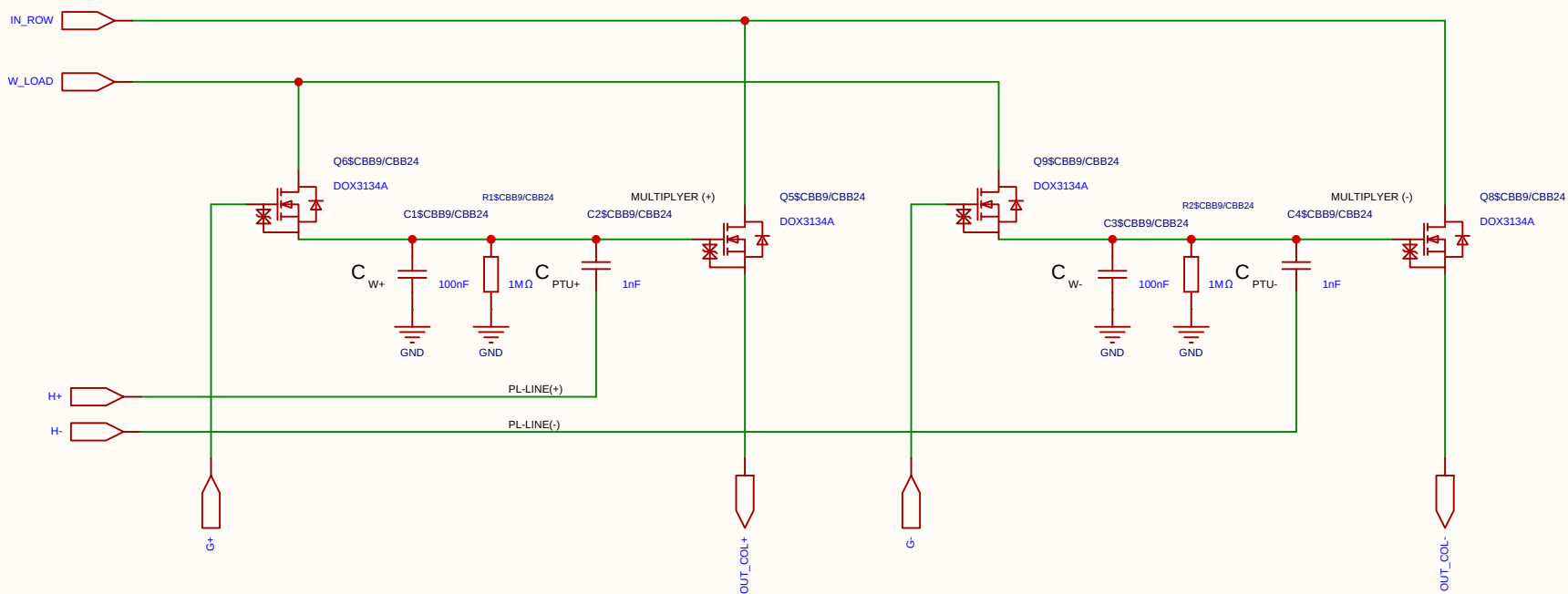
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

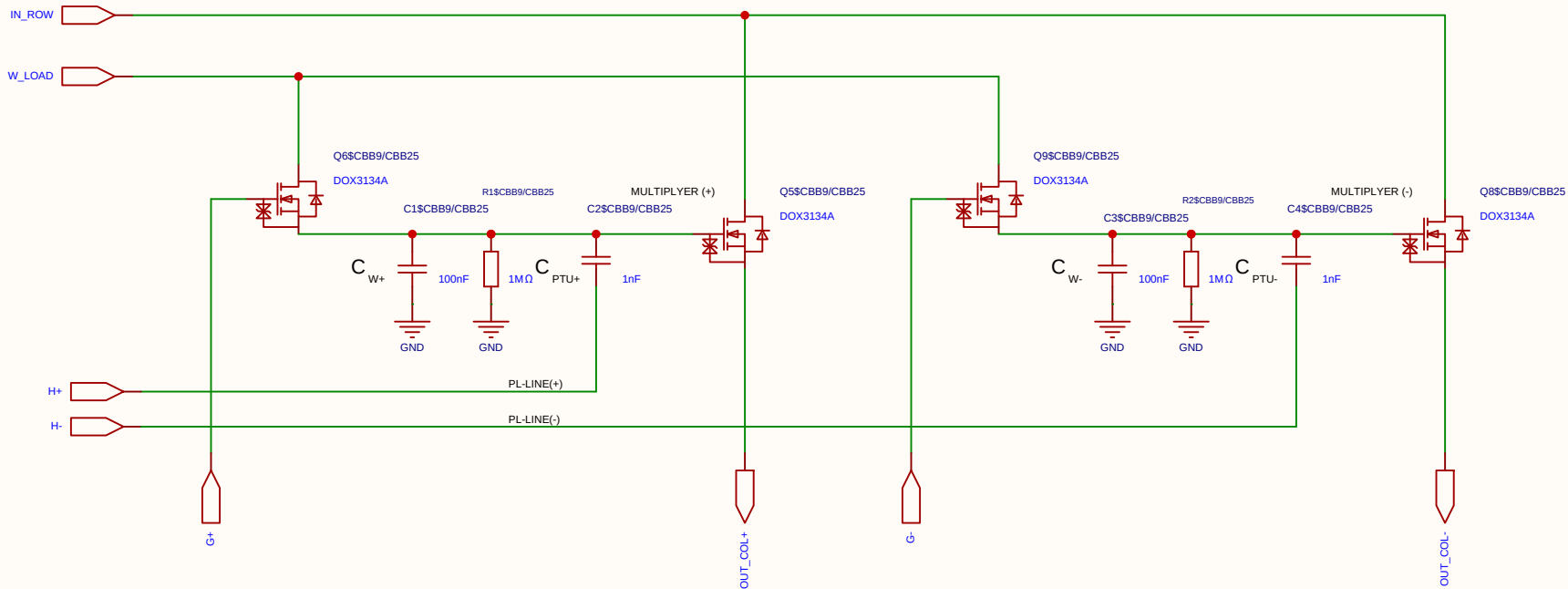
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

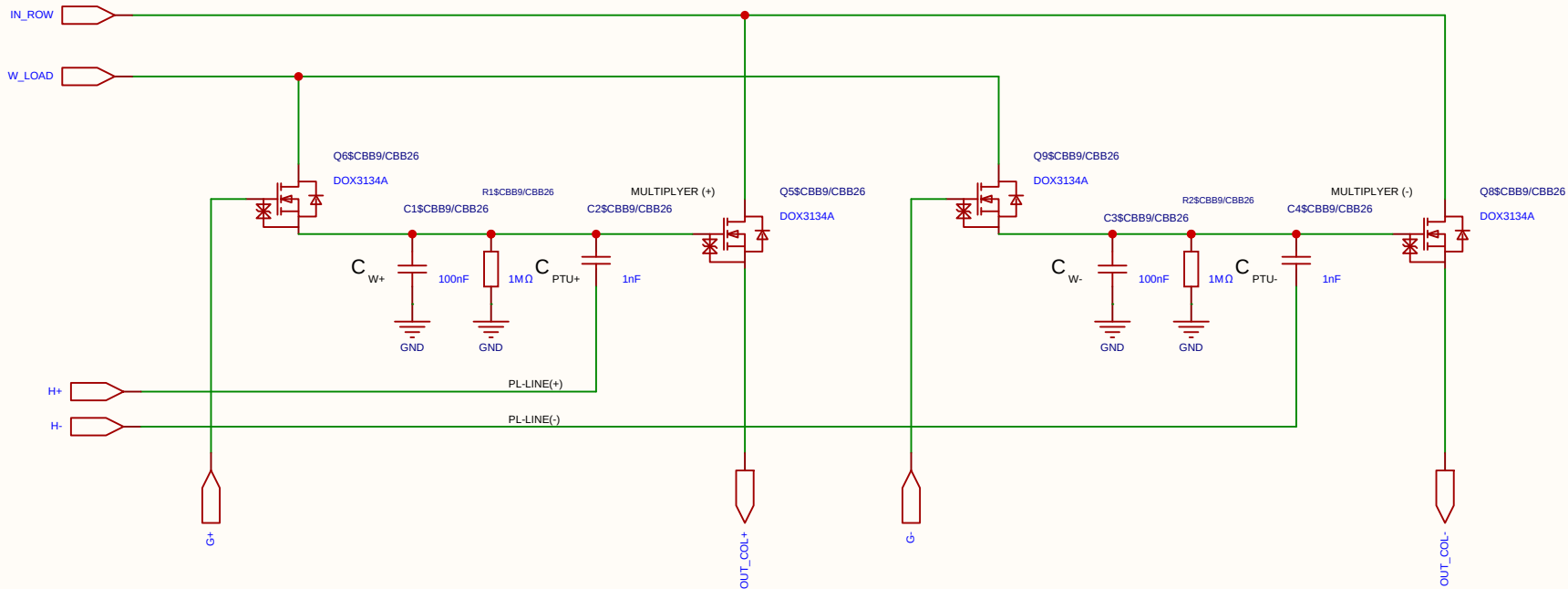
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

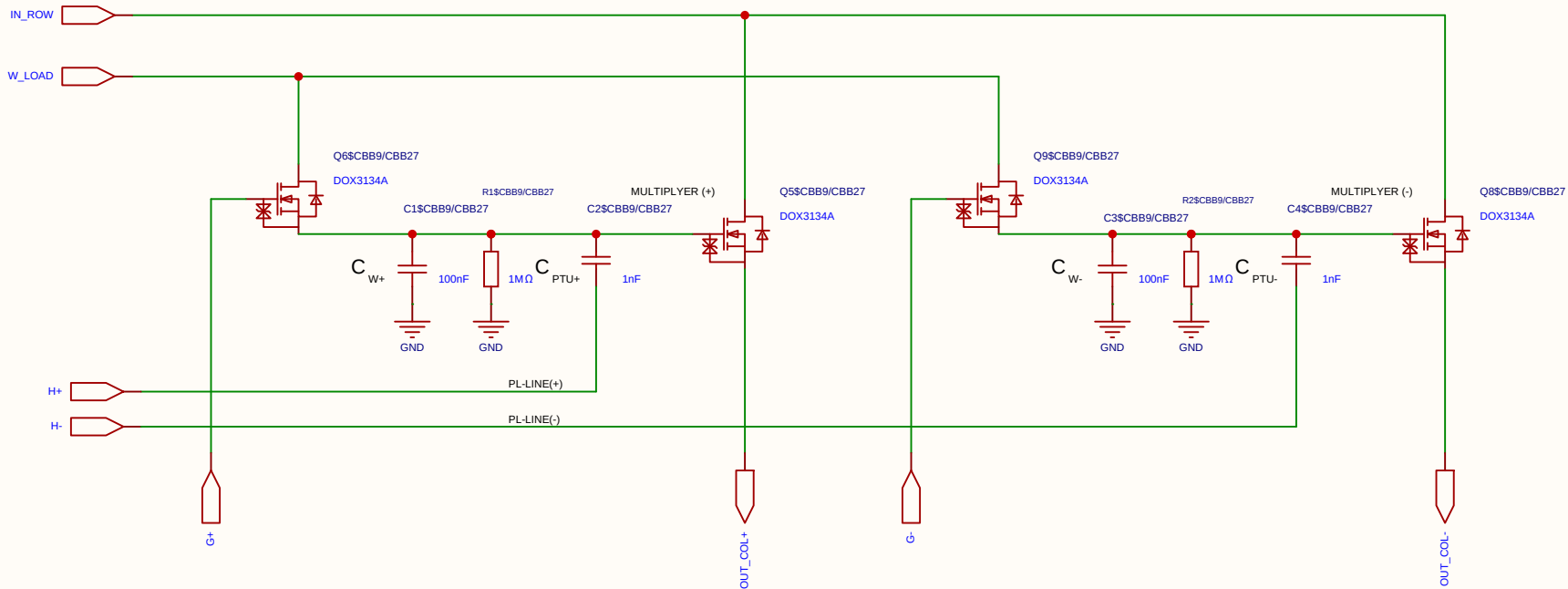
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

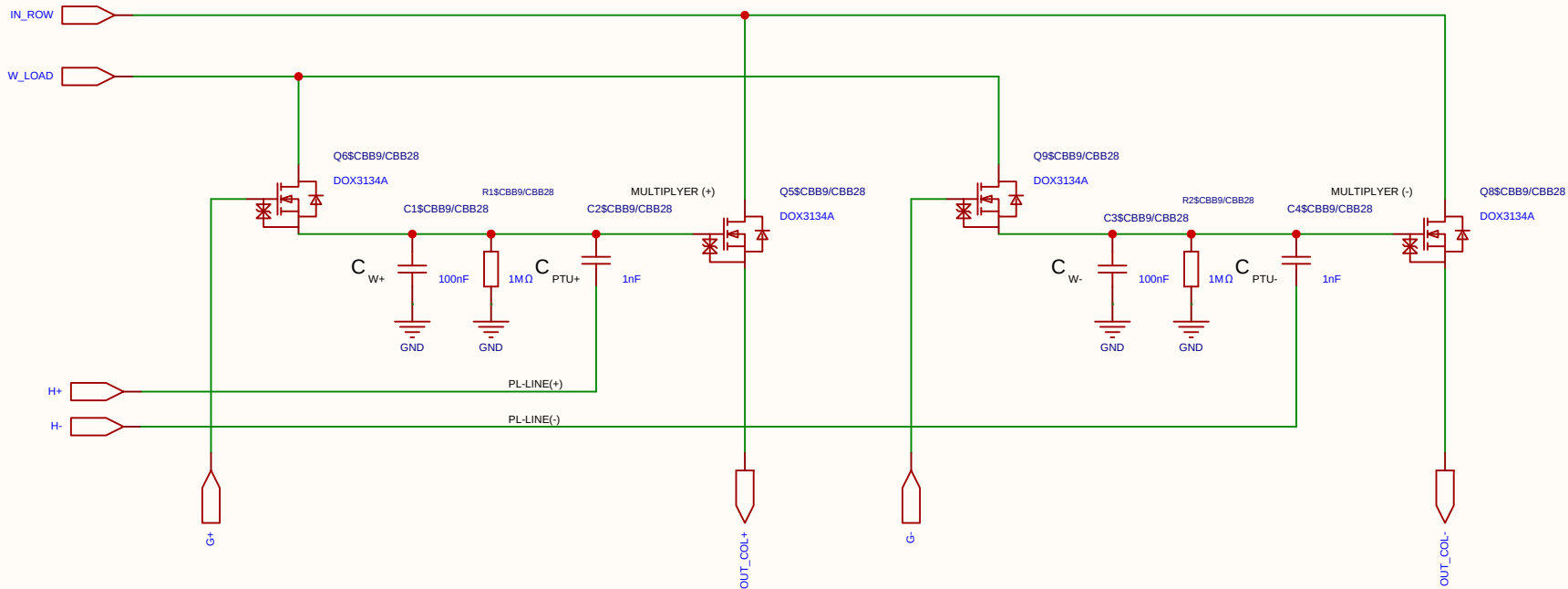
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Board				Page	matrix_3_8_MUL_CELL
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

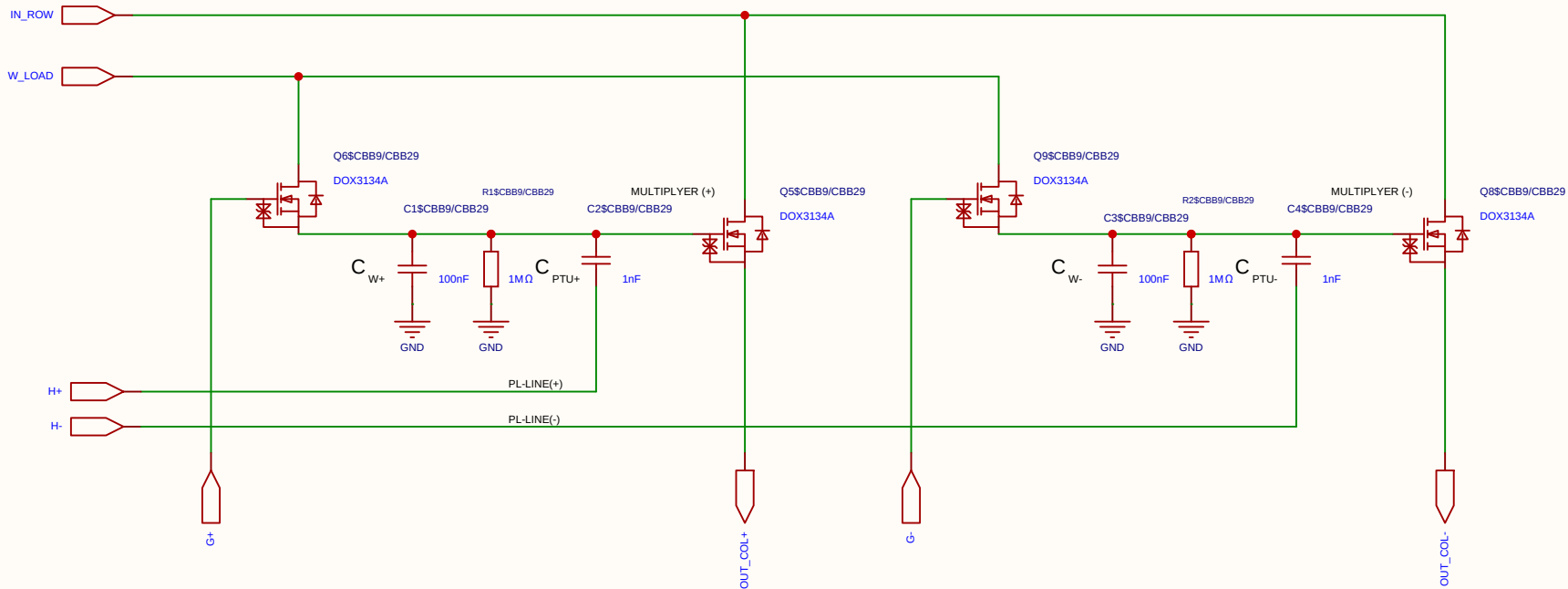
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

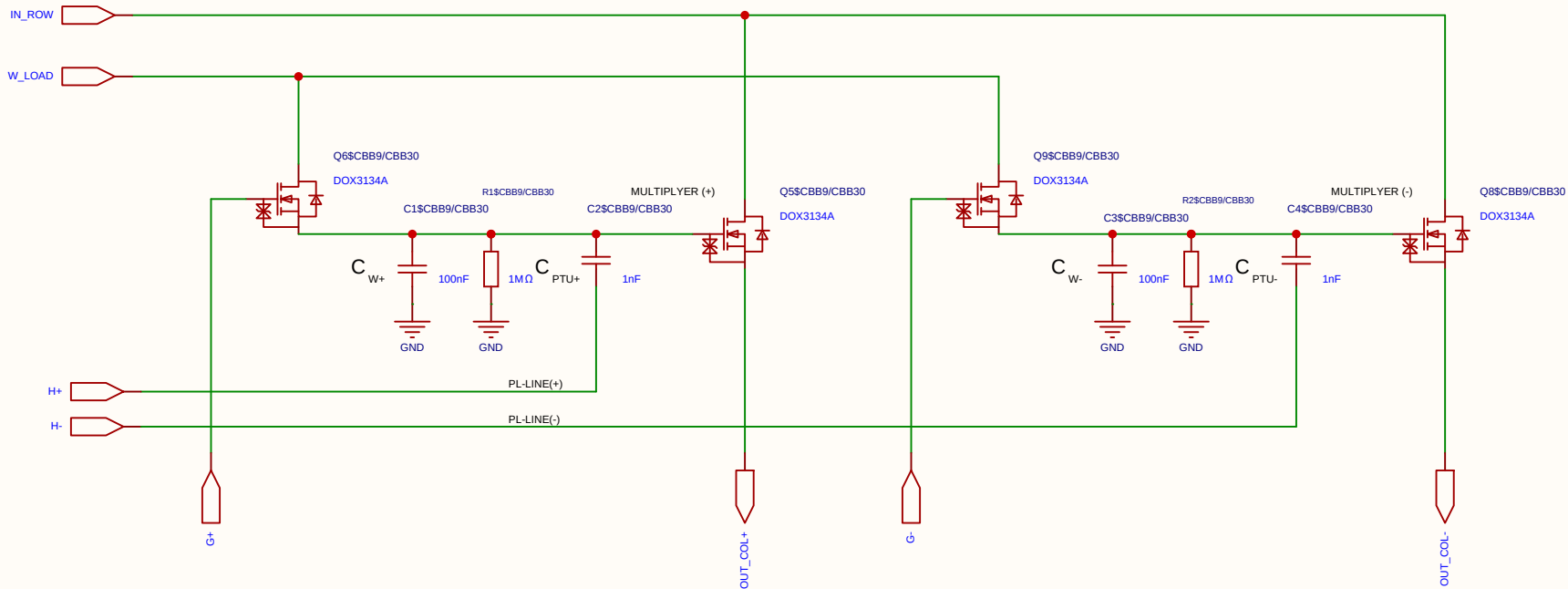
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

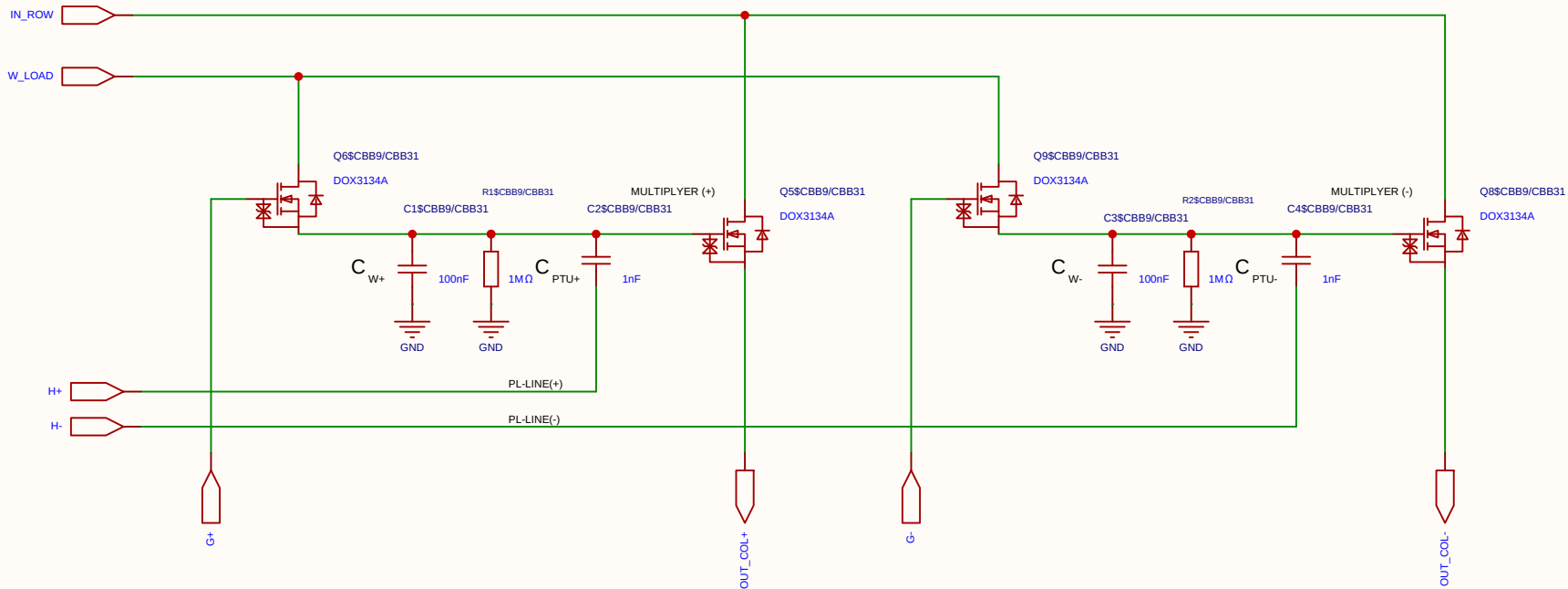
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

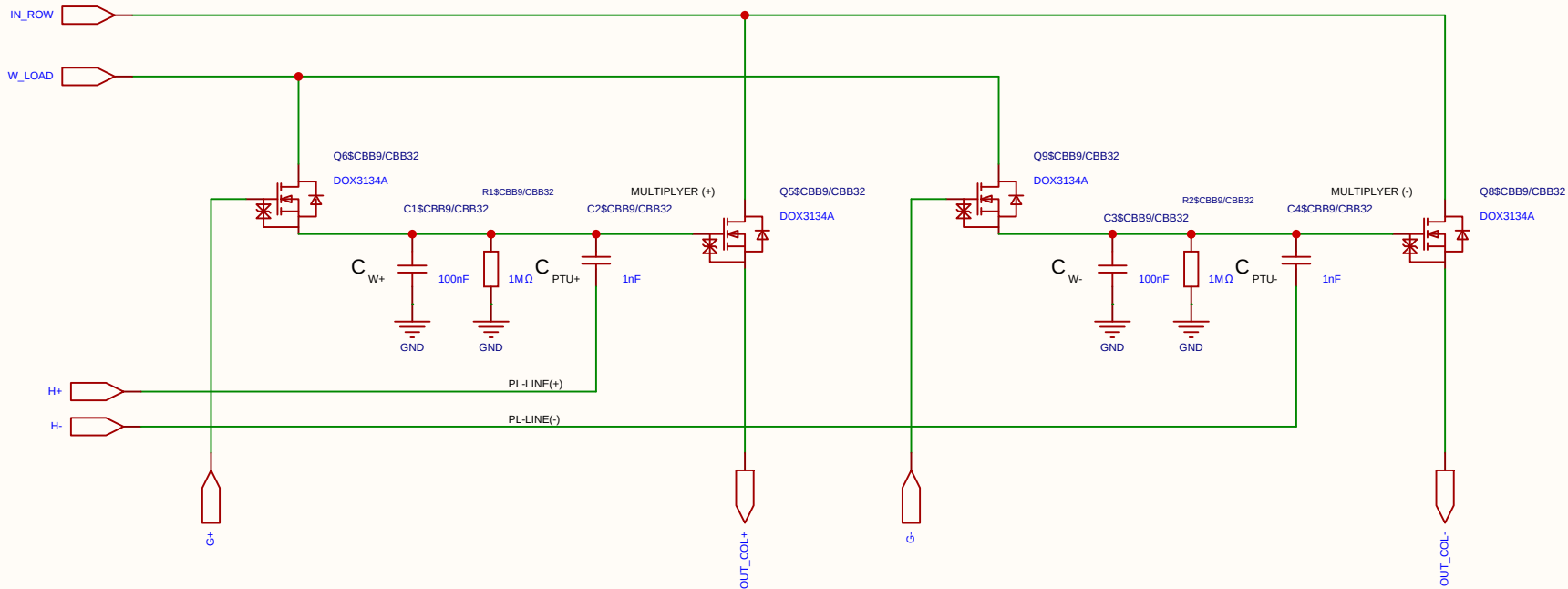
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Category	Parameter	Value / Type	Technical Description & Function
Resistors	Discharge resistor (R)	1 M Ω	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

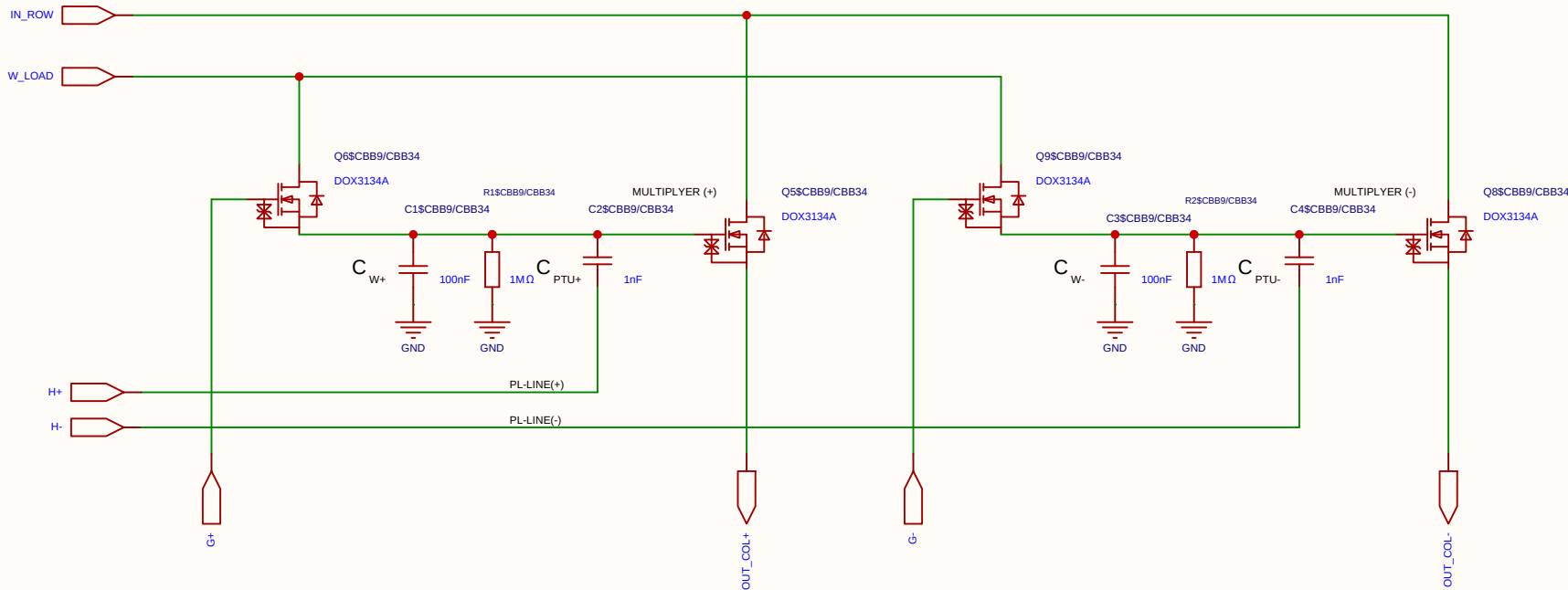
Component	Value	Description / Function
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a $\sim 1/101$ voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μ s.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

Component	Value	Description / Function
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_I . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_I . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_I . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
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		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

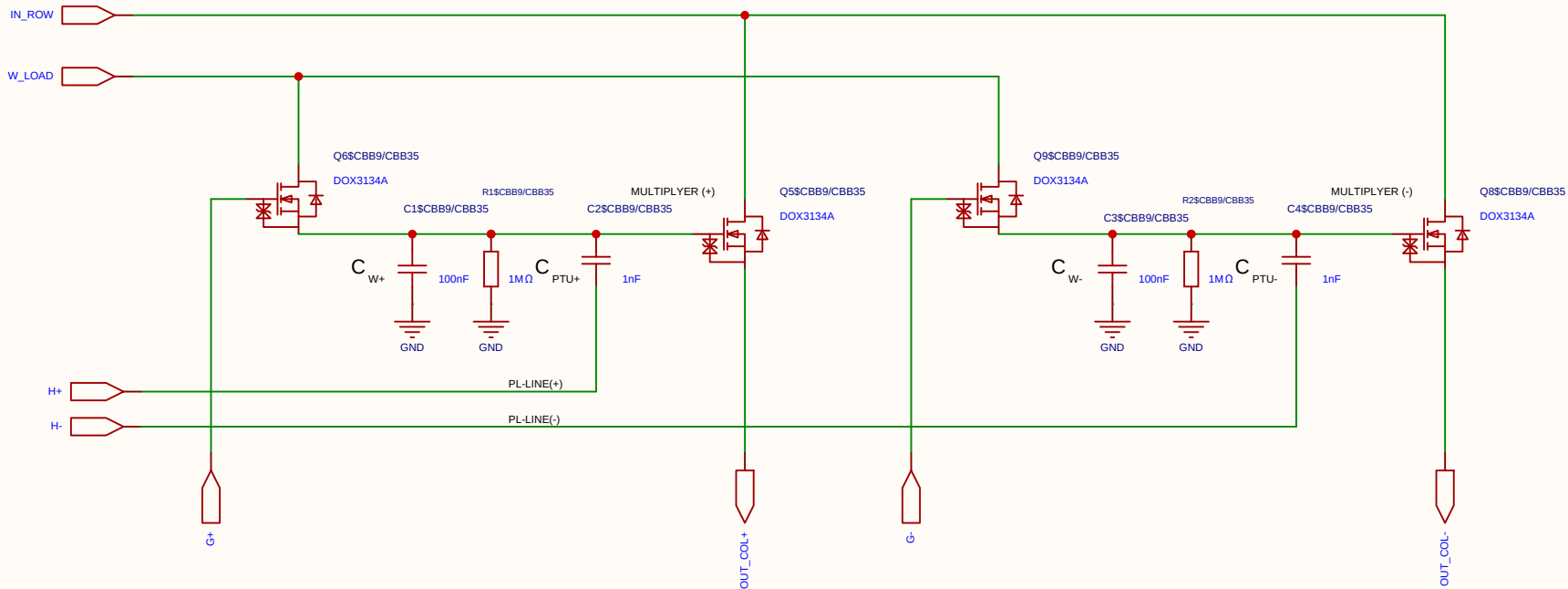
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
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Drawn	Olle Welin	Inverted_MAML_6x6			
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		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

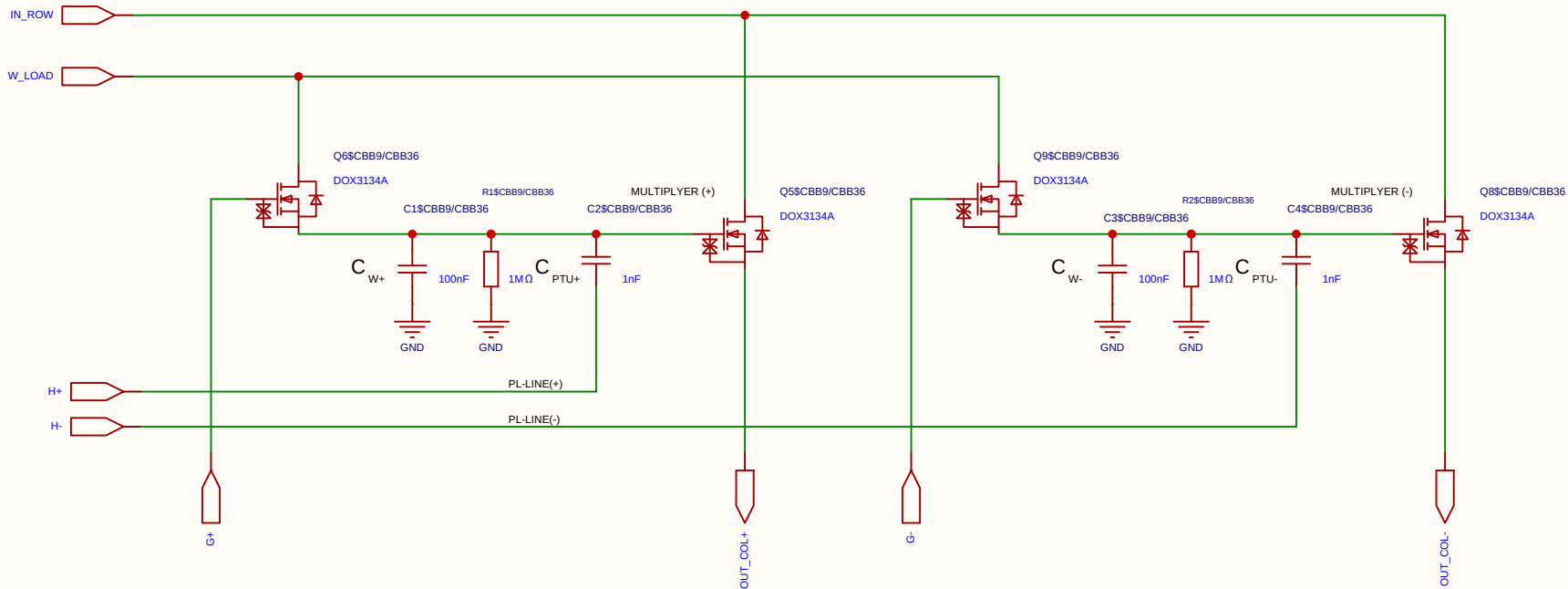
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

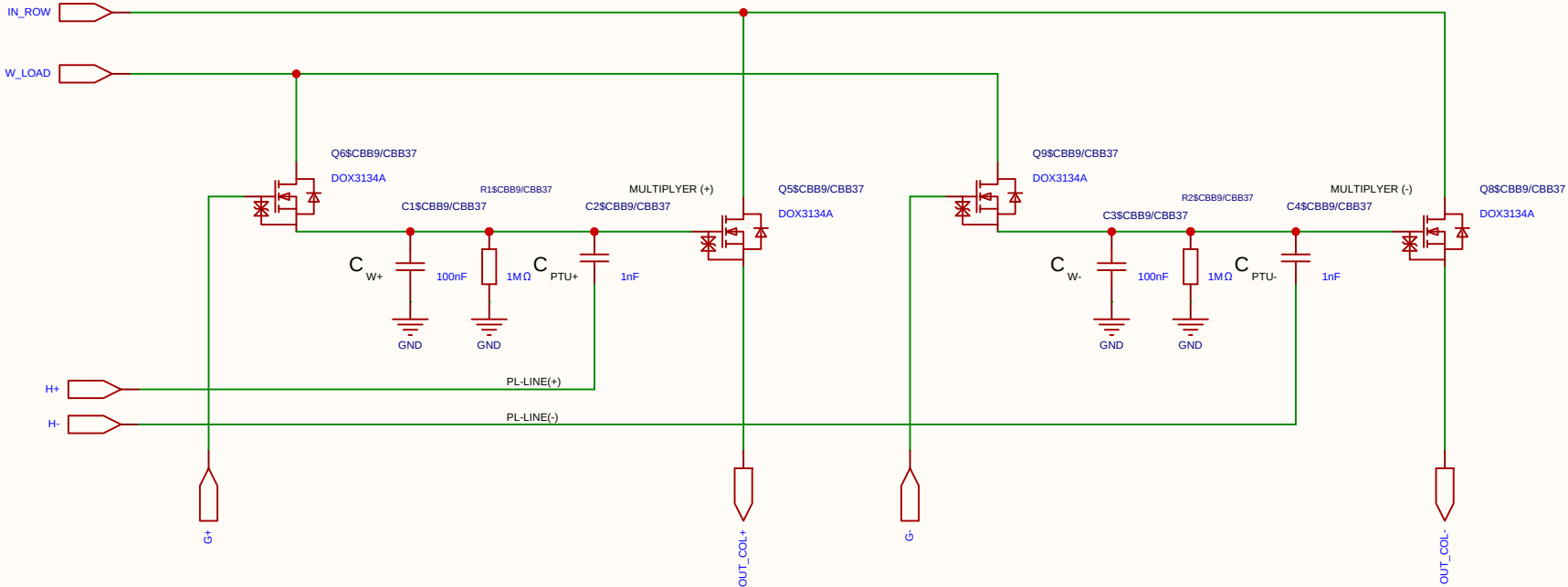
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

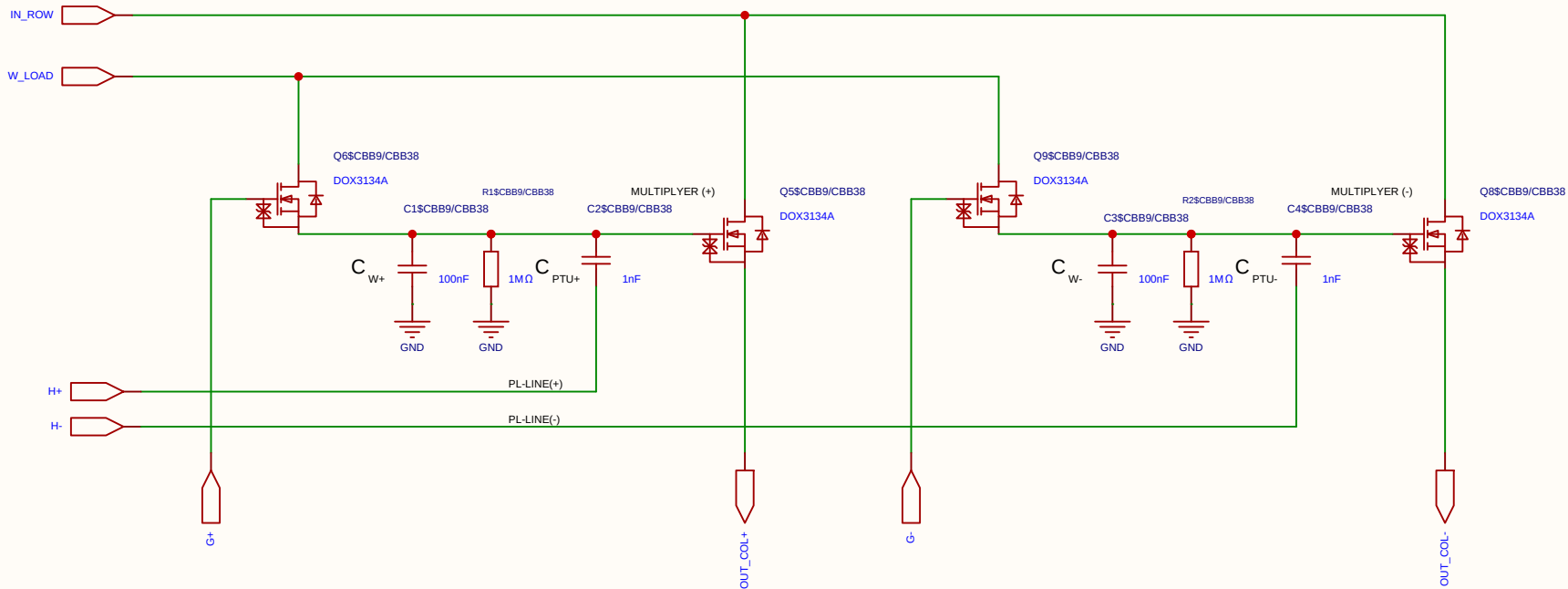
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

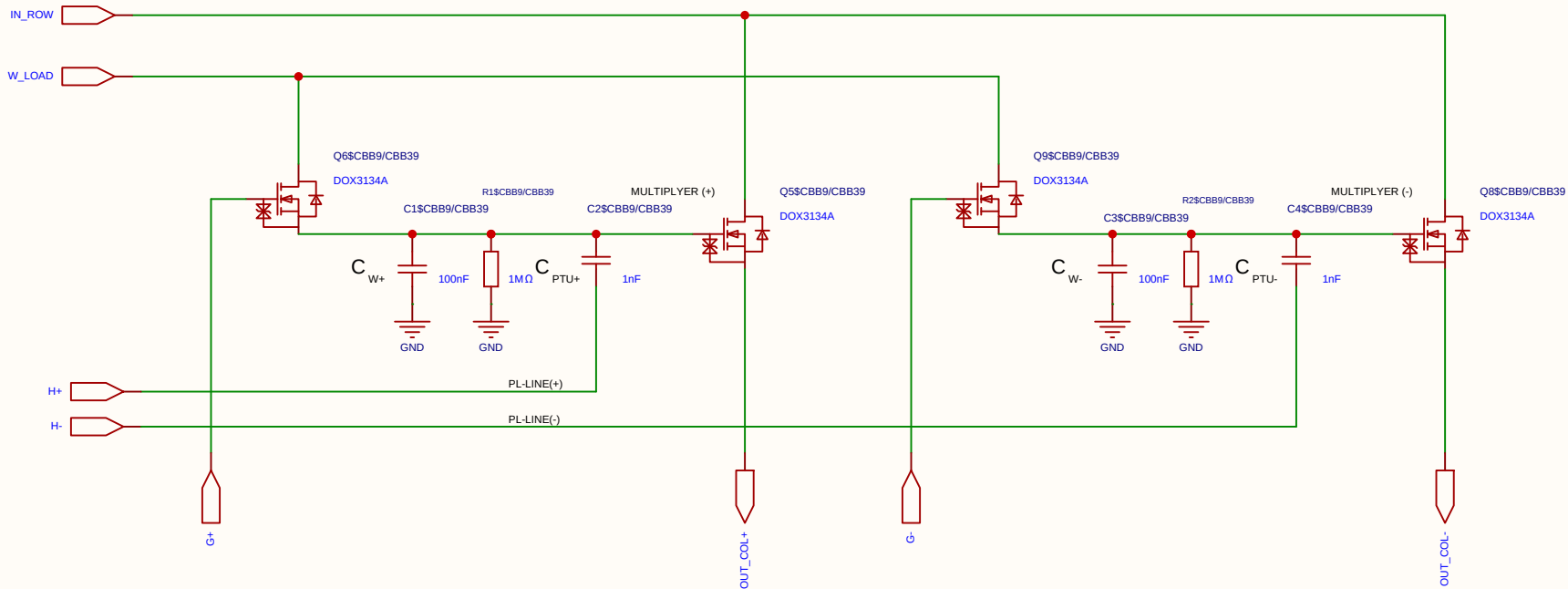
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

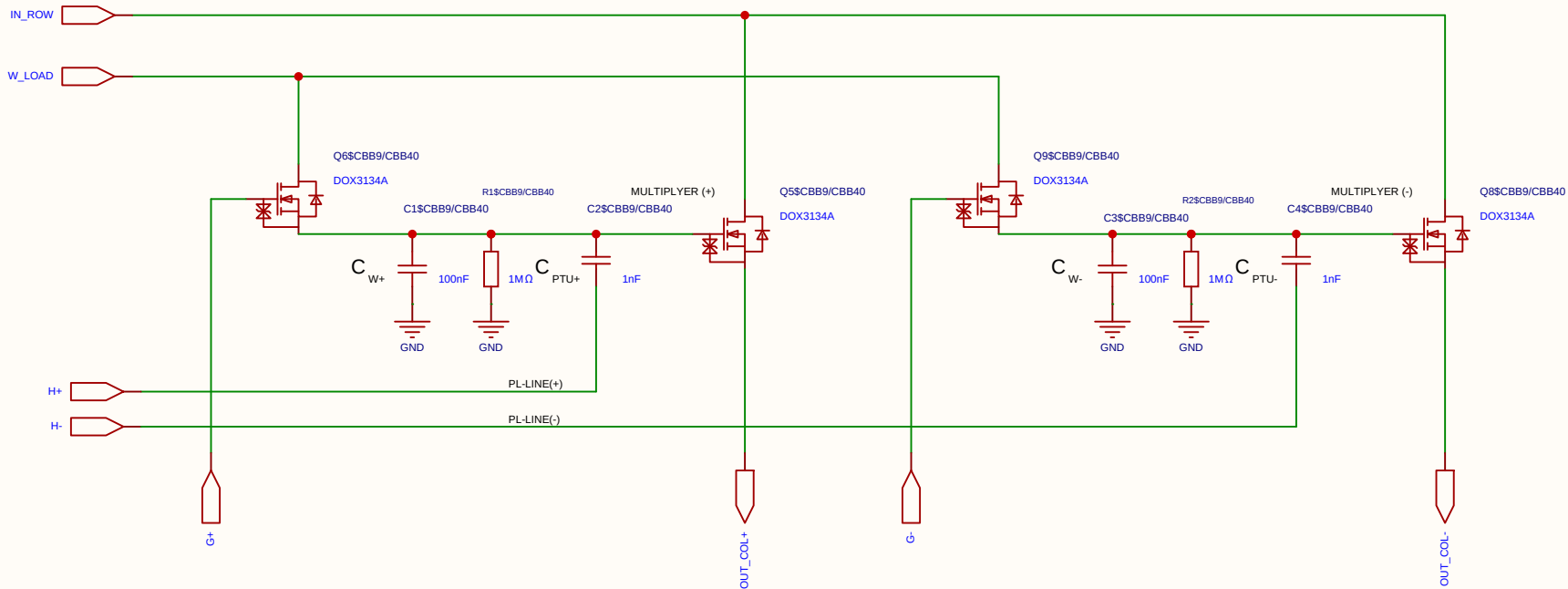
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

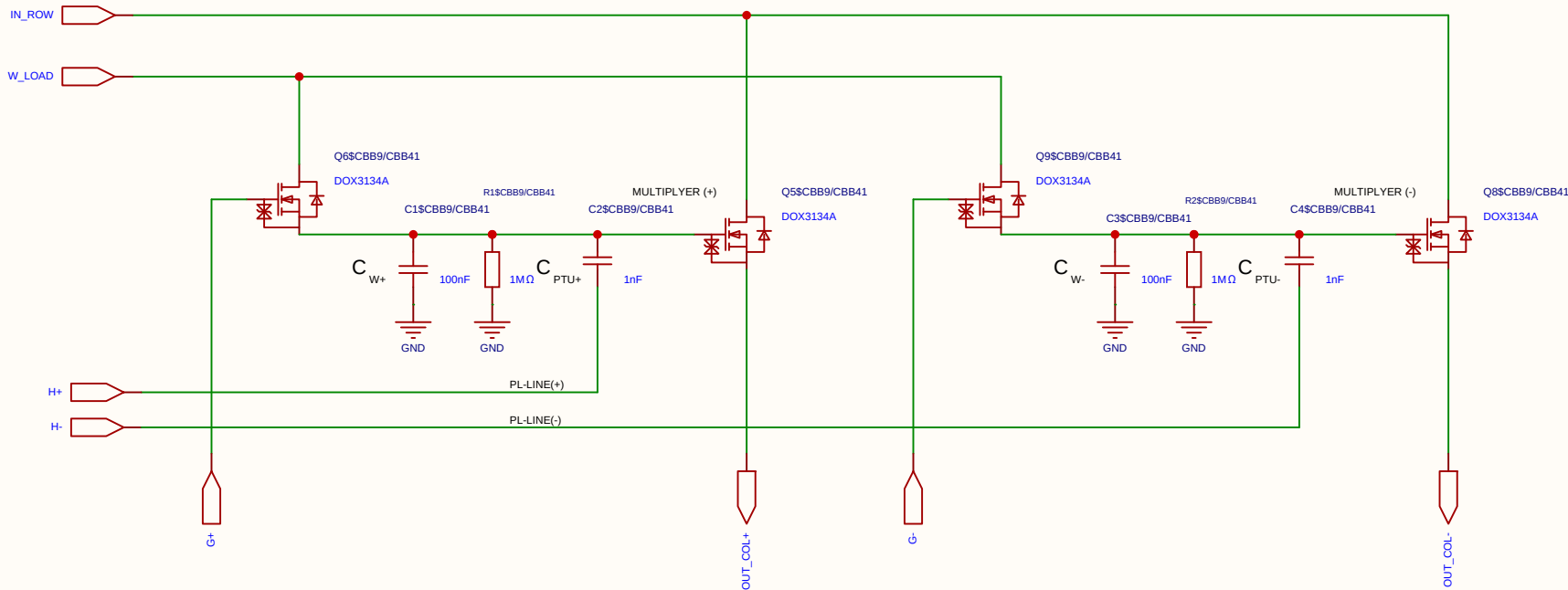
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

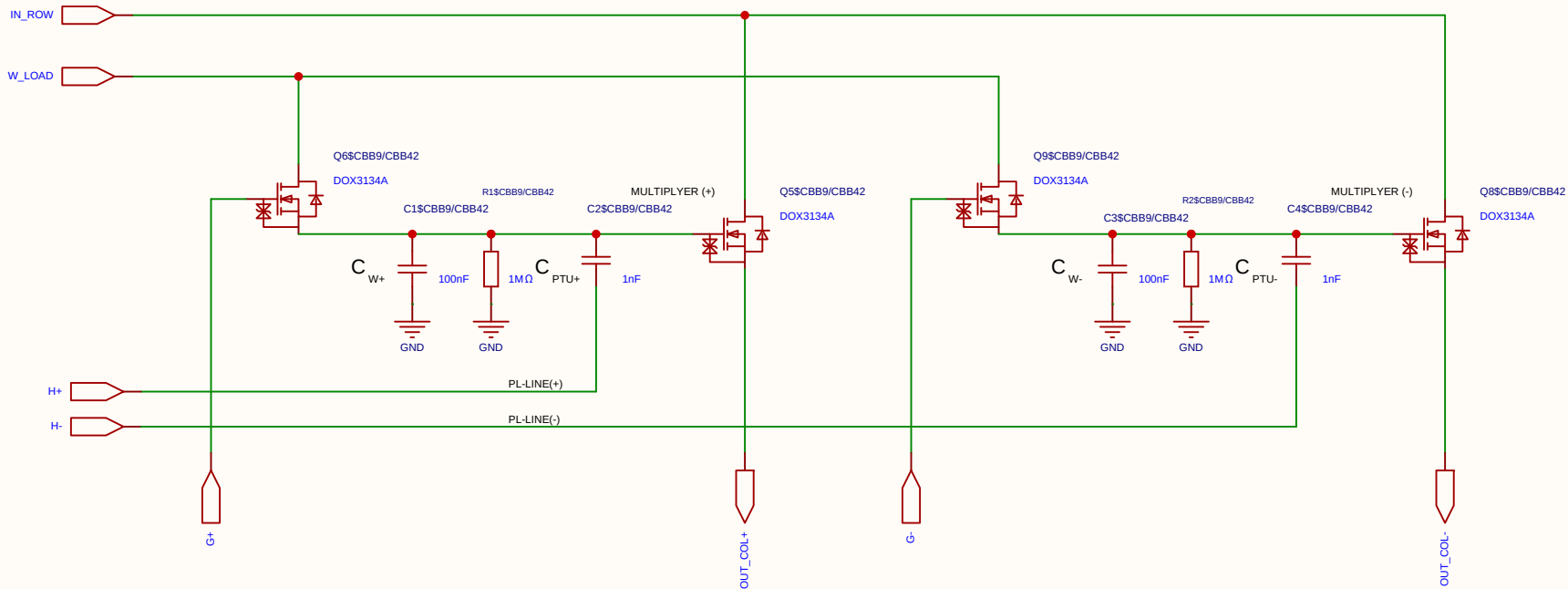
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

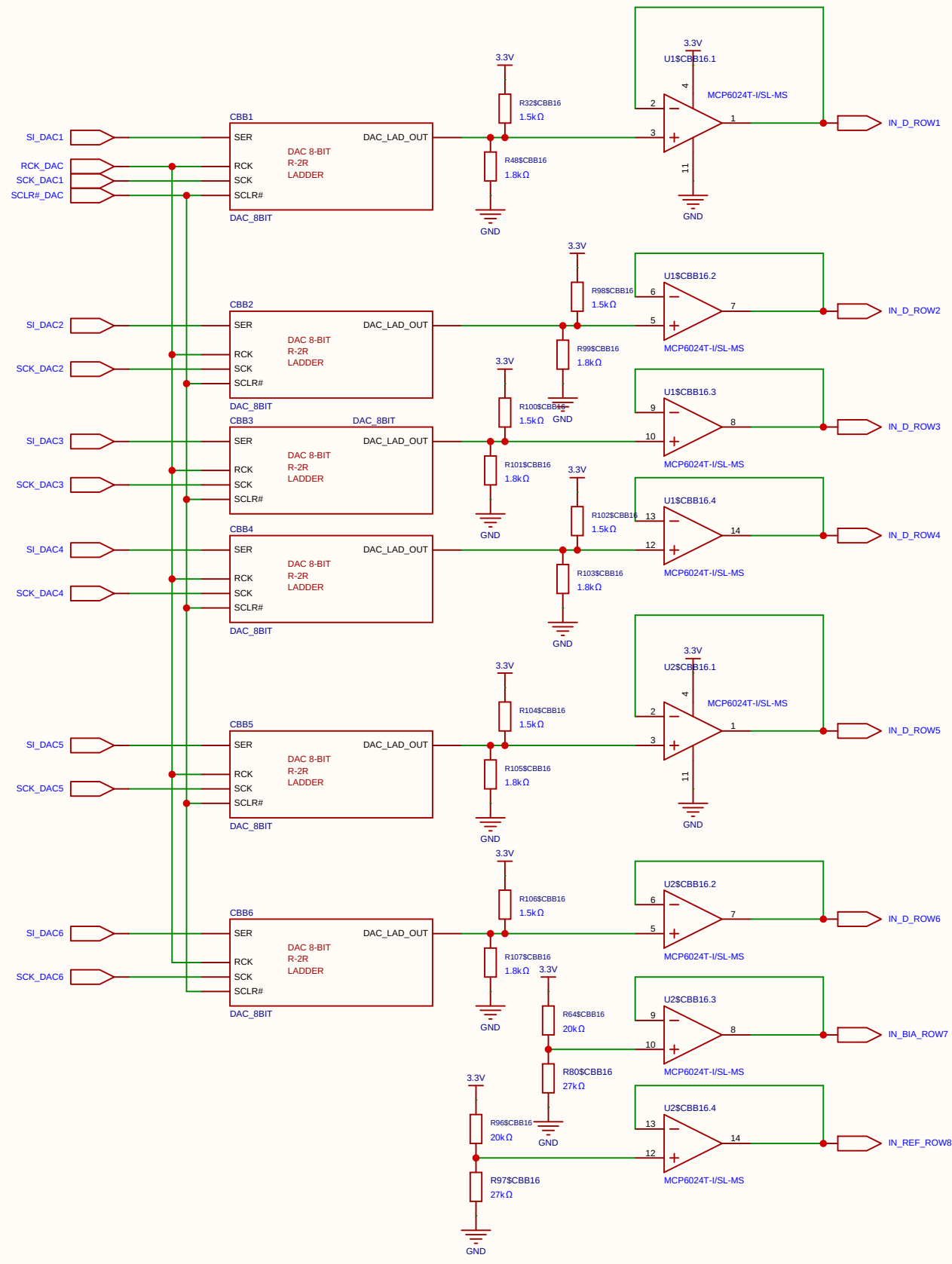
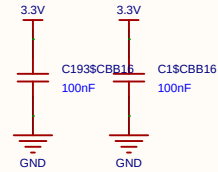
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

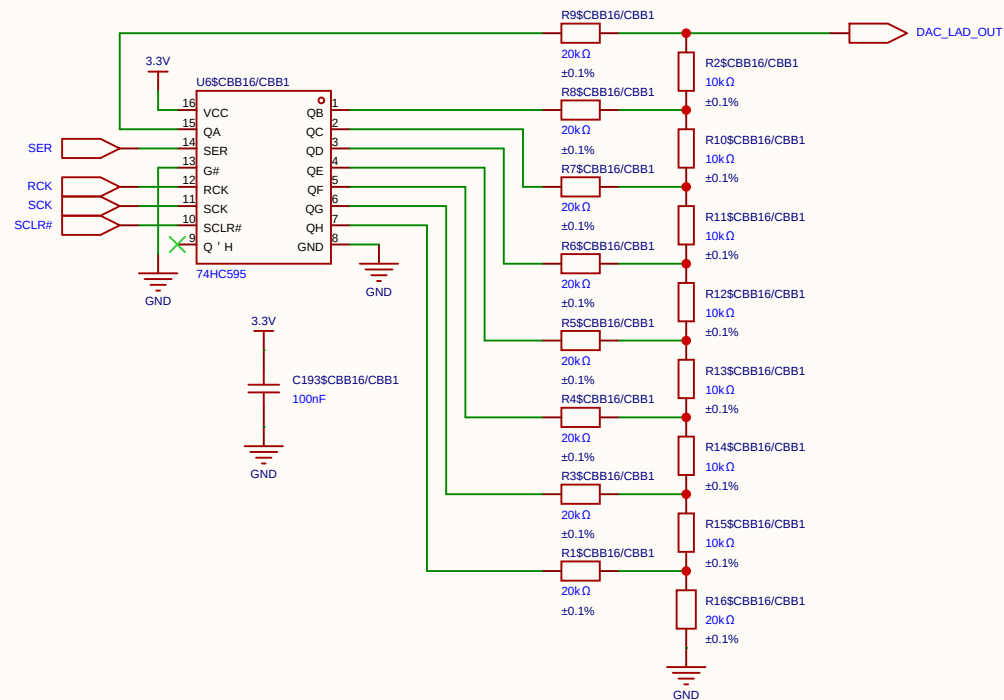
COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
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Reviewed					
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EasyEDA		V1.0	A4	EasyEDA.com	

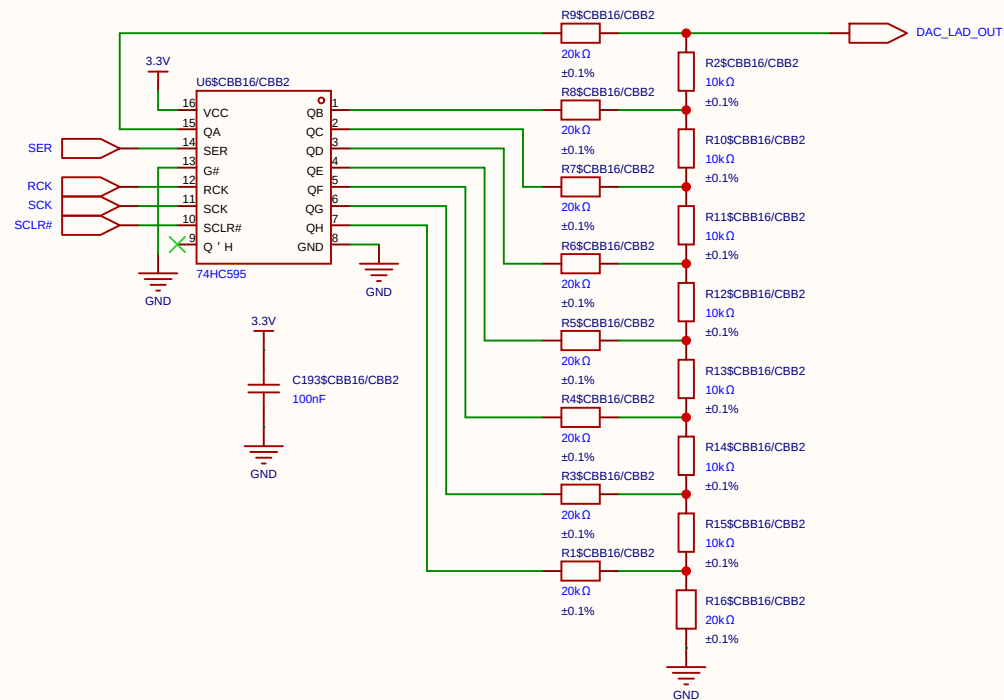
Voltages Data swing (Row / V_{ds}) 250 mV Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.



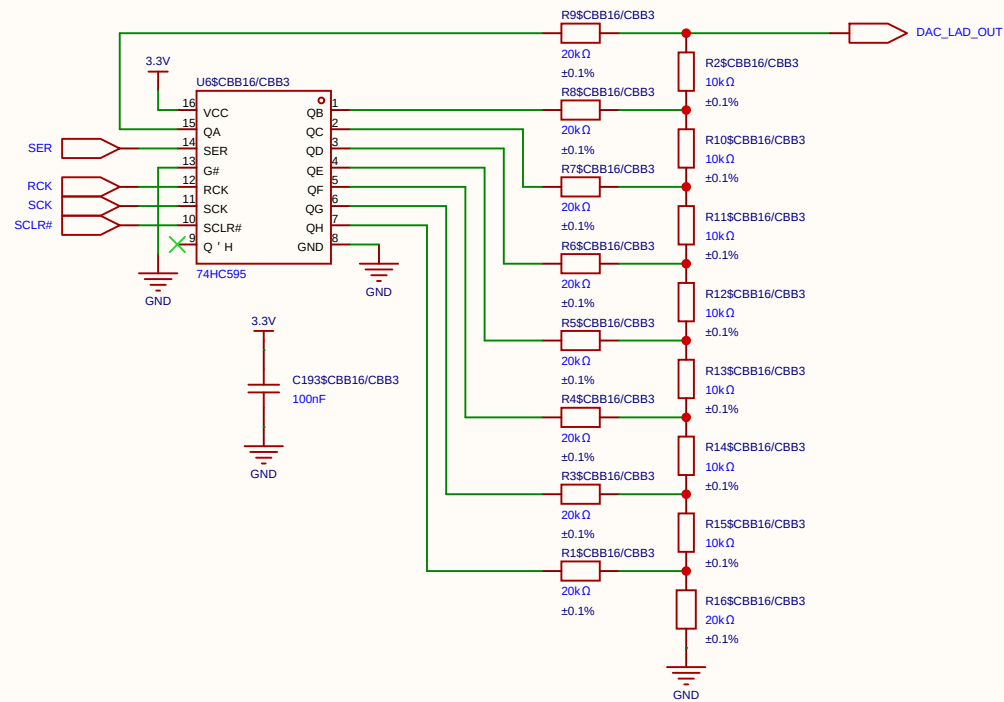
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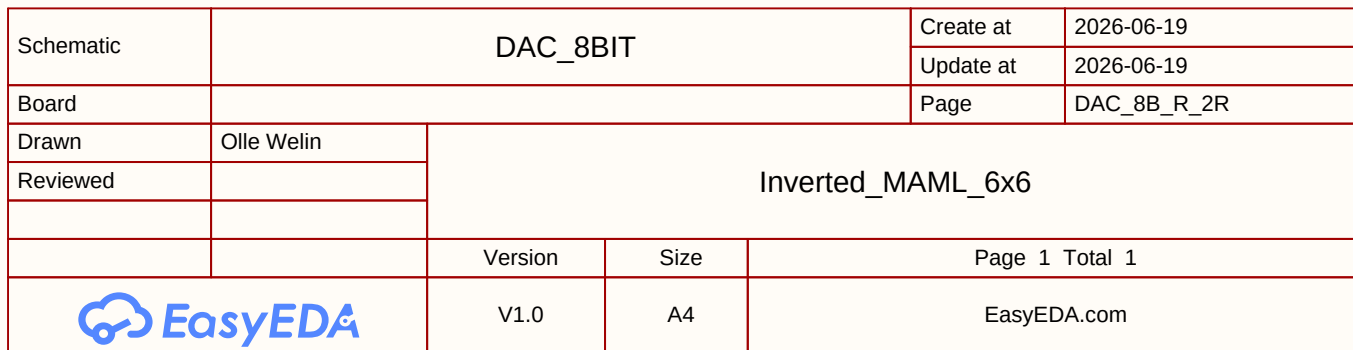
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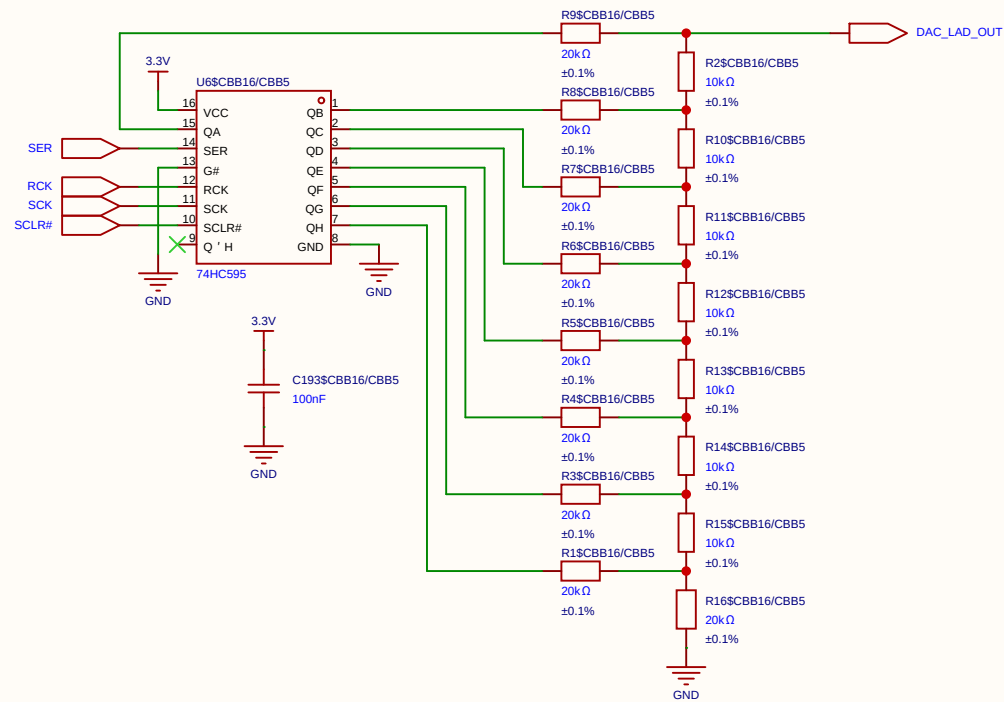


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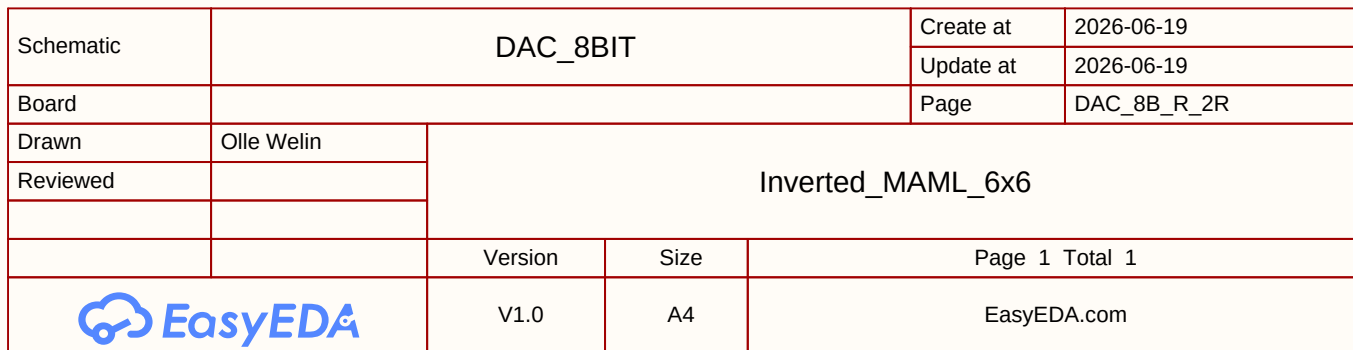


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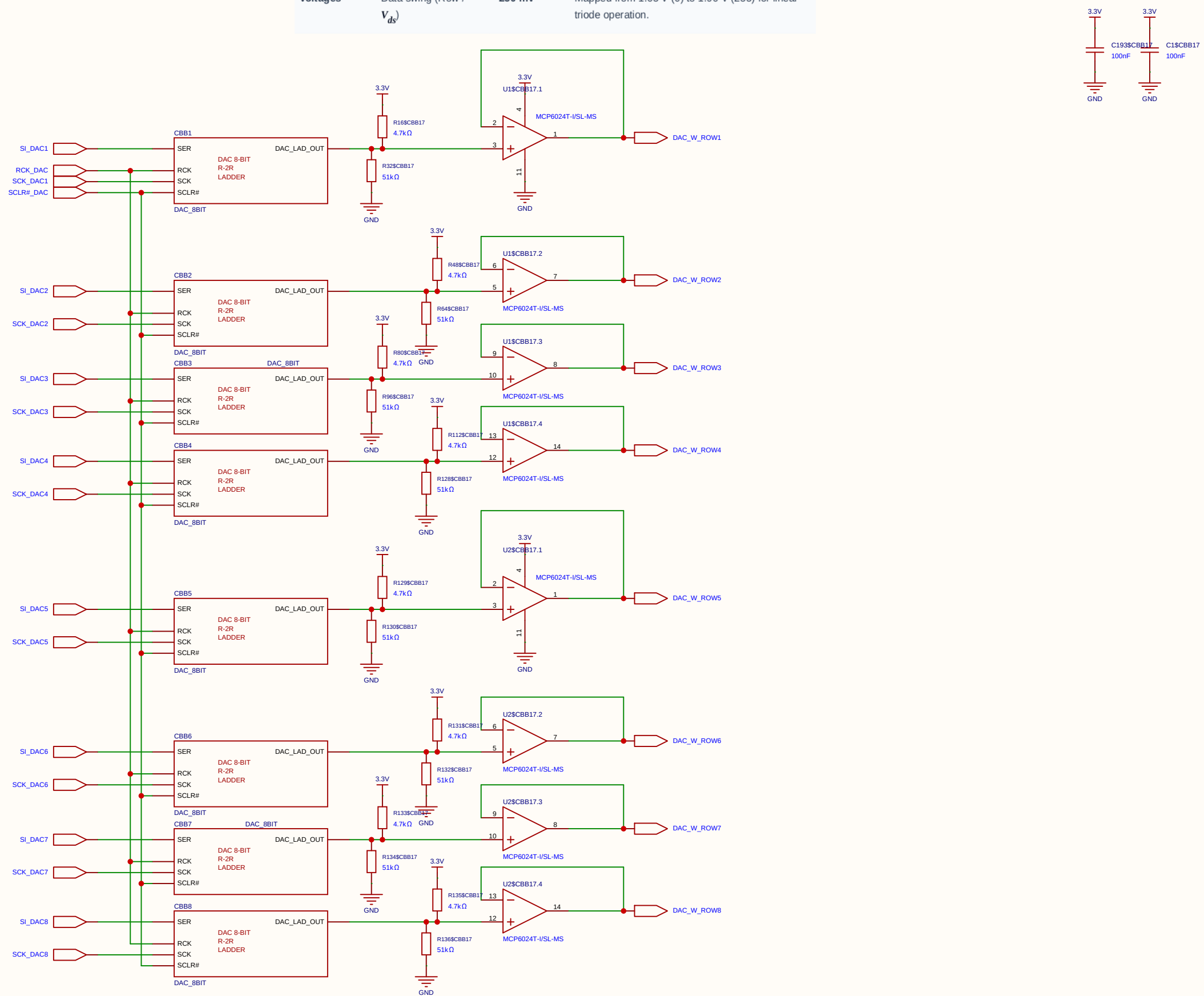




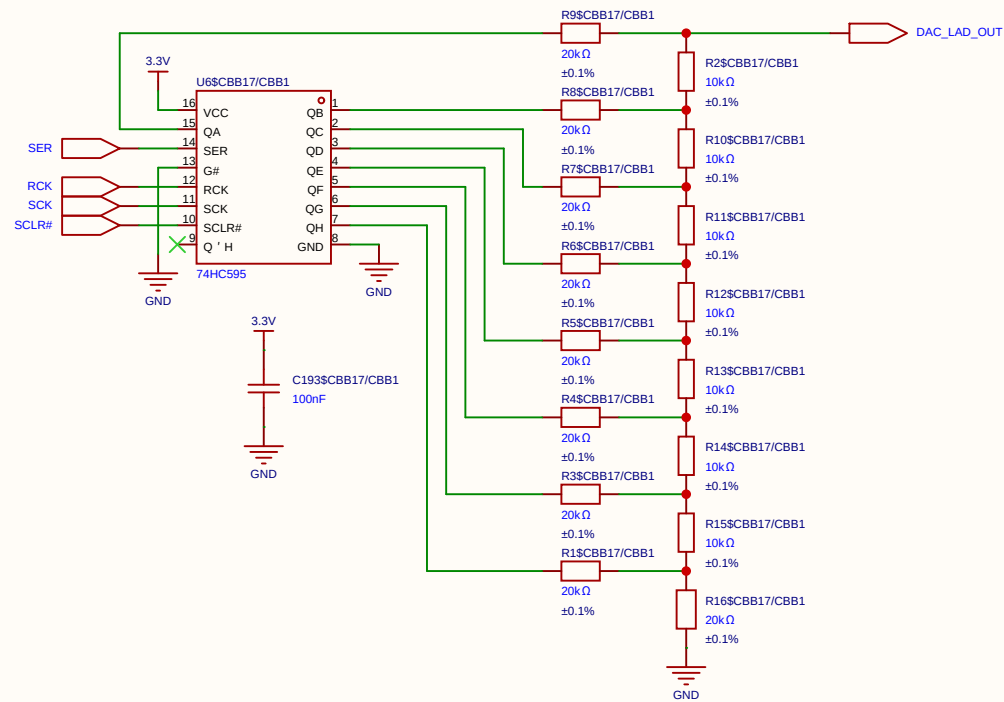
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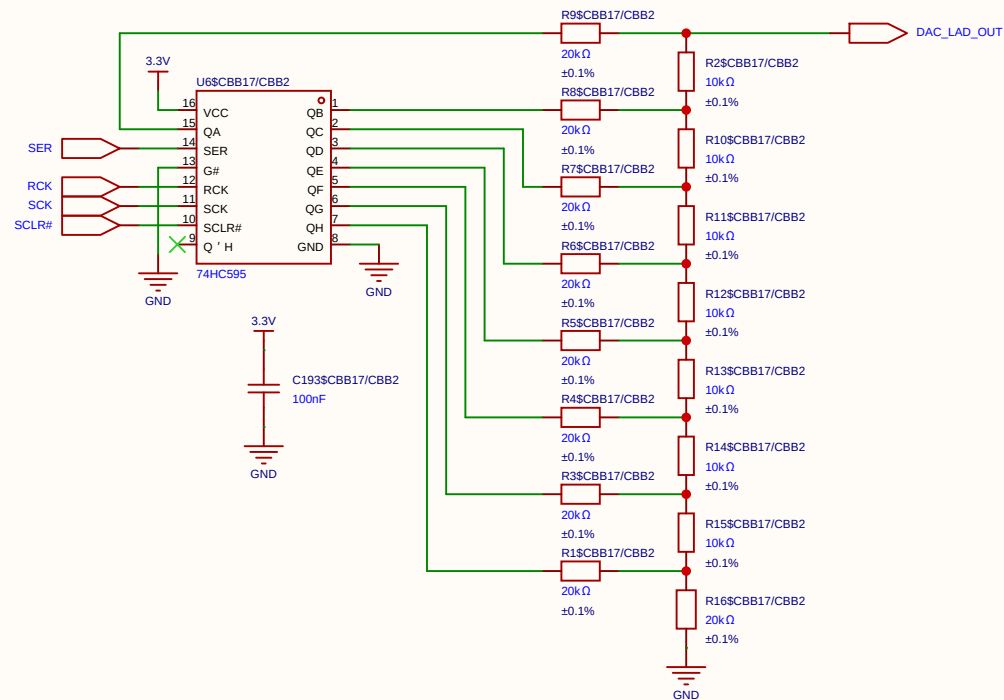
Voltages Data swing (Row / V_{ds}) 250 mV Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.



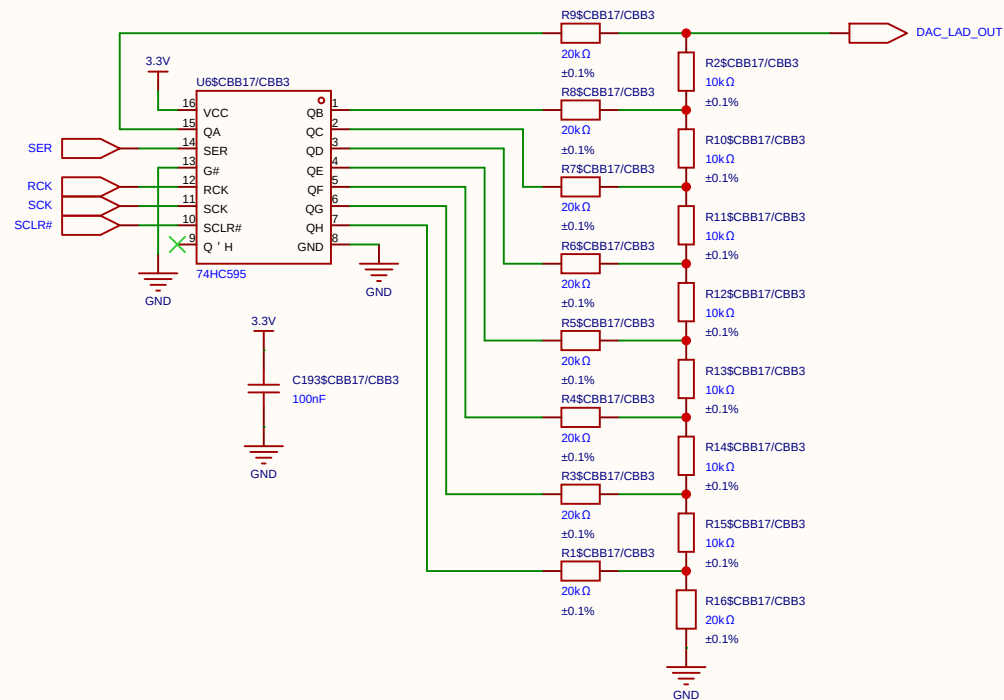
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EasyEDA		V1.0	A4	EasyEDA.com	



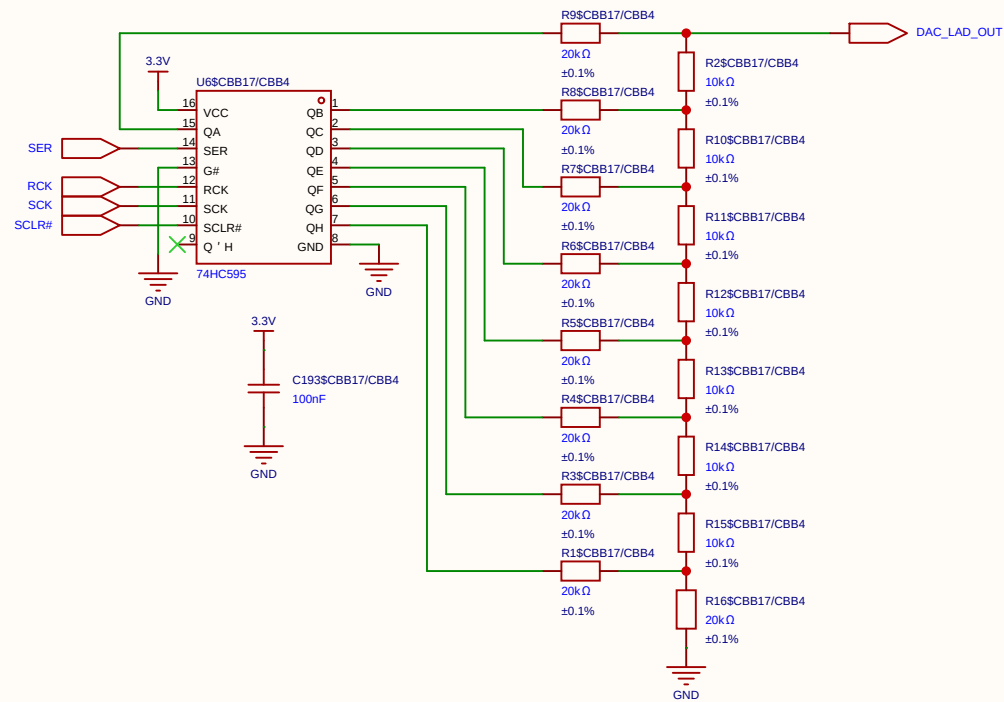
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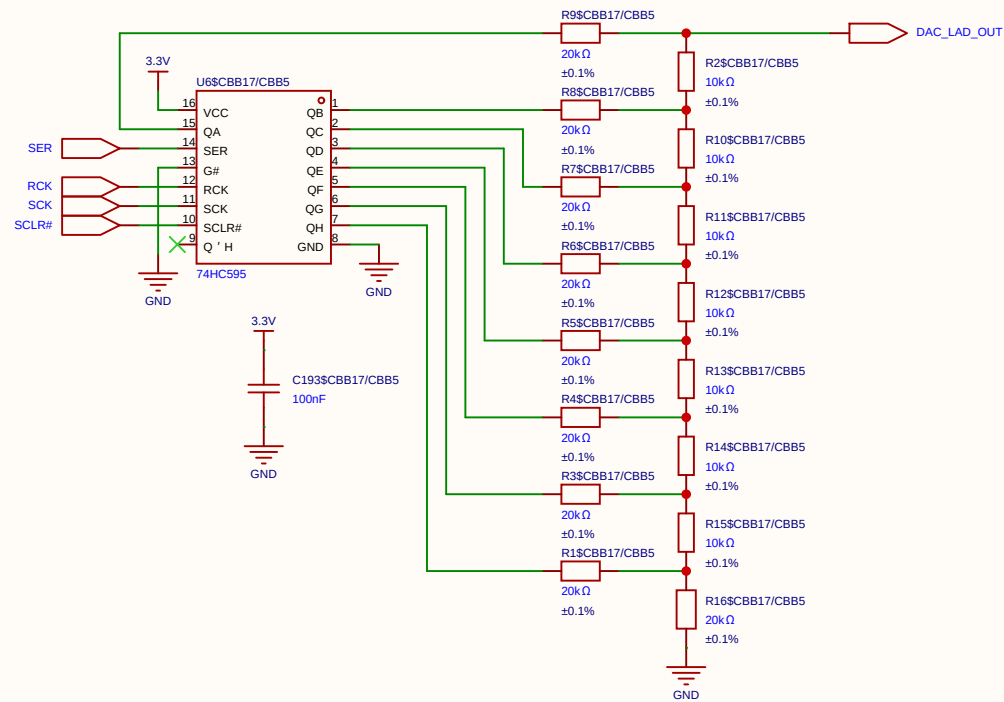
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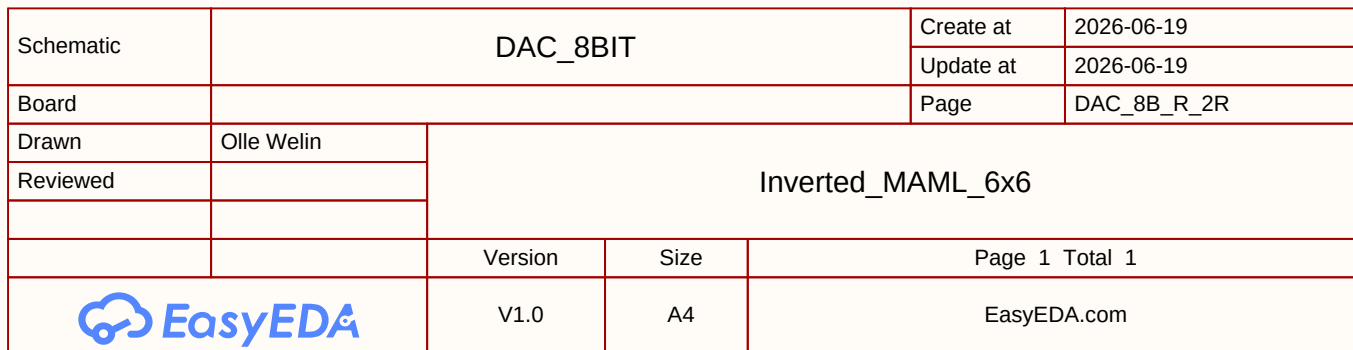
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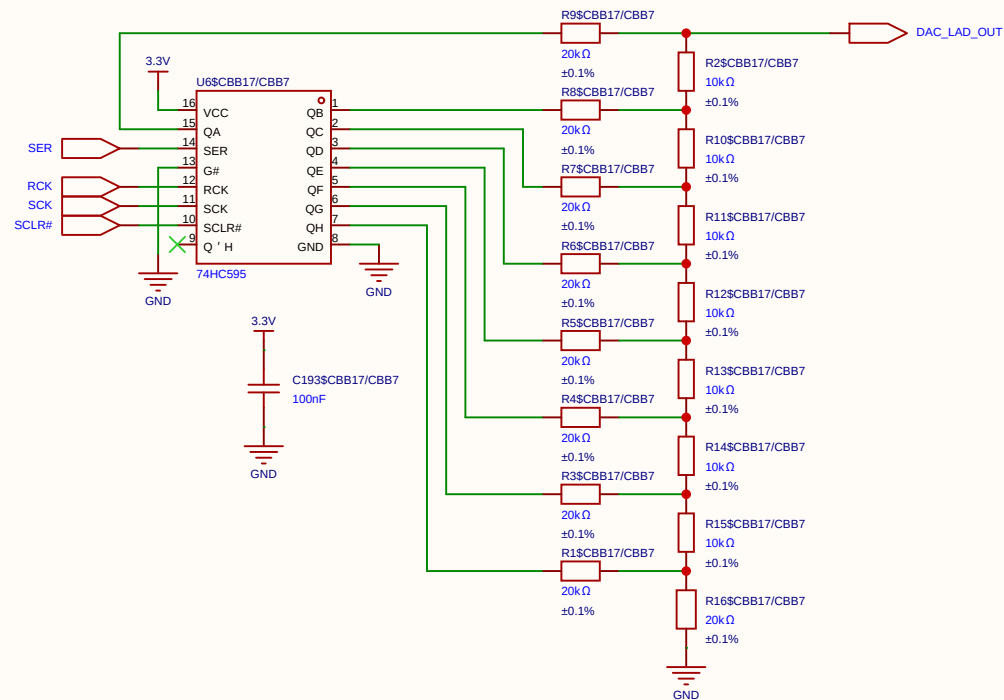


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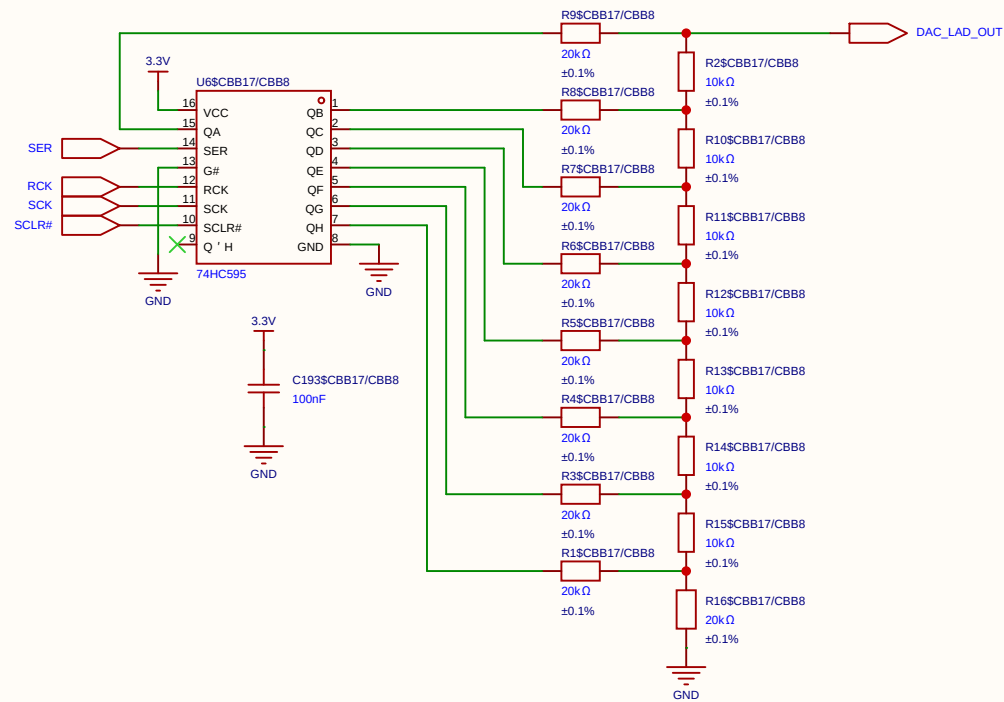


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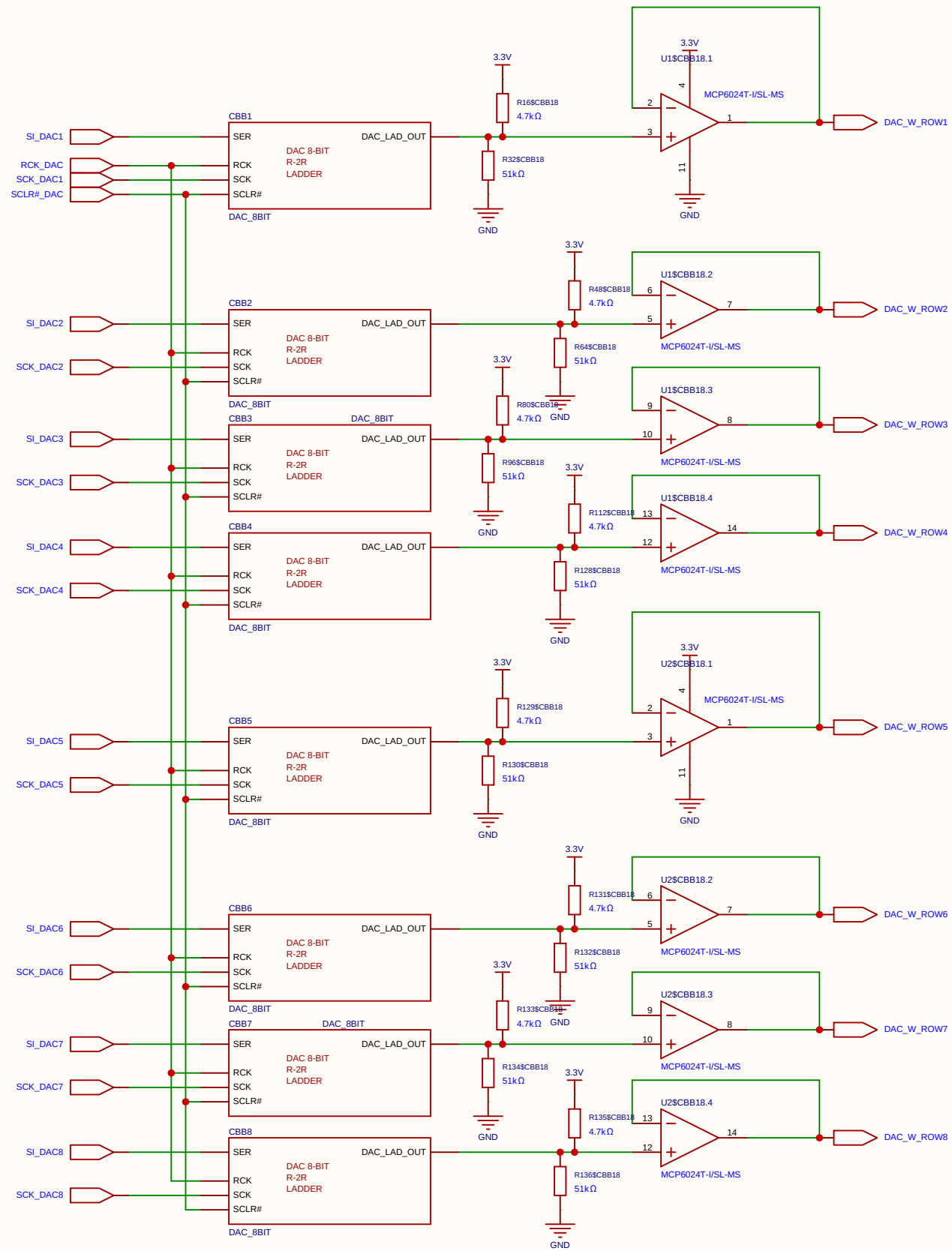


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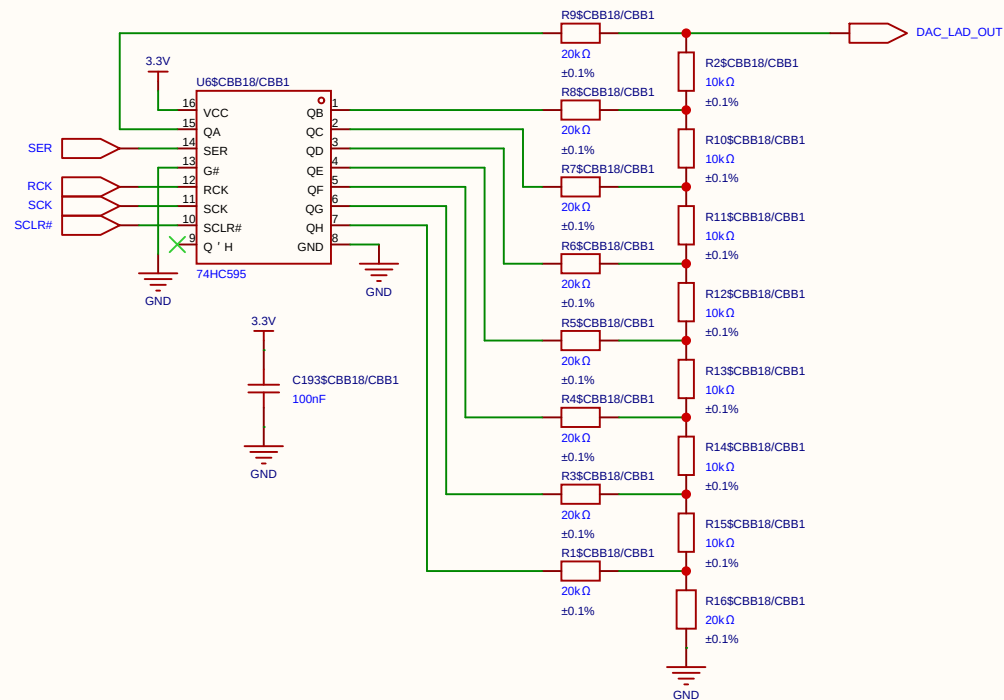


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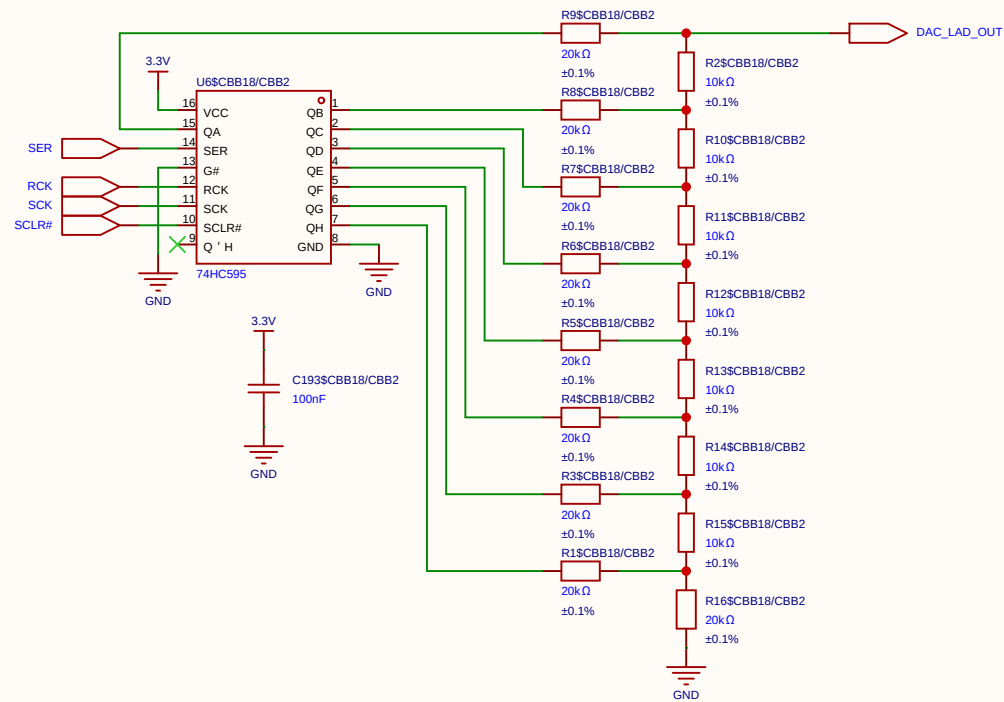
Voltages Data swing (Row / V_{ds}) 250 mV Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.



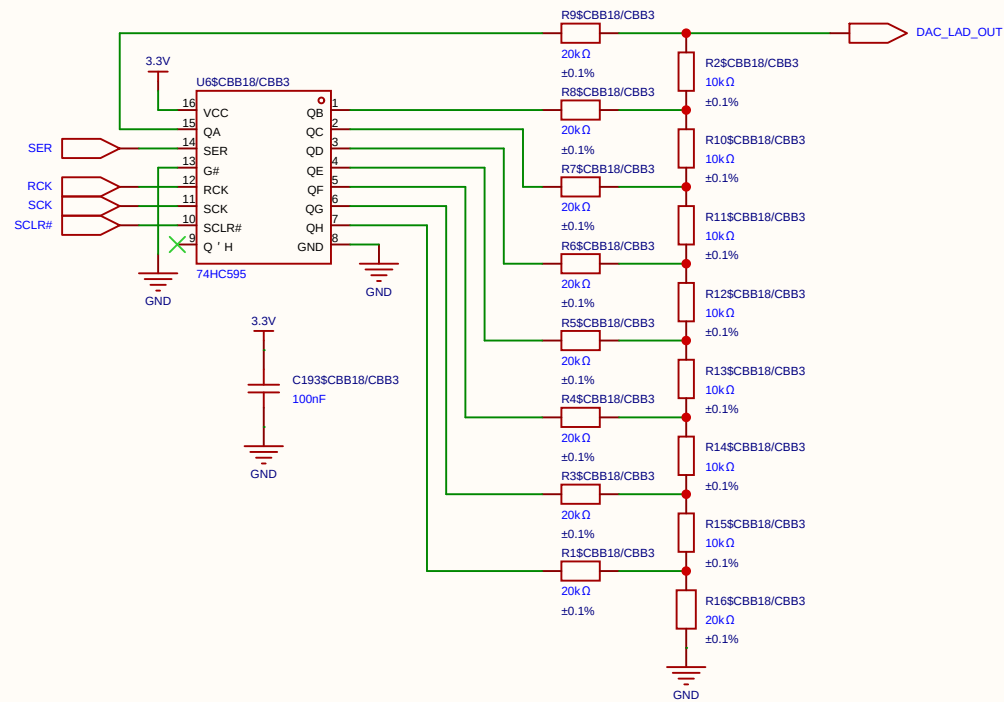
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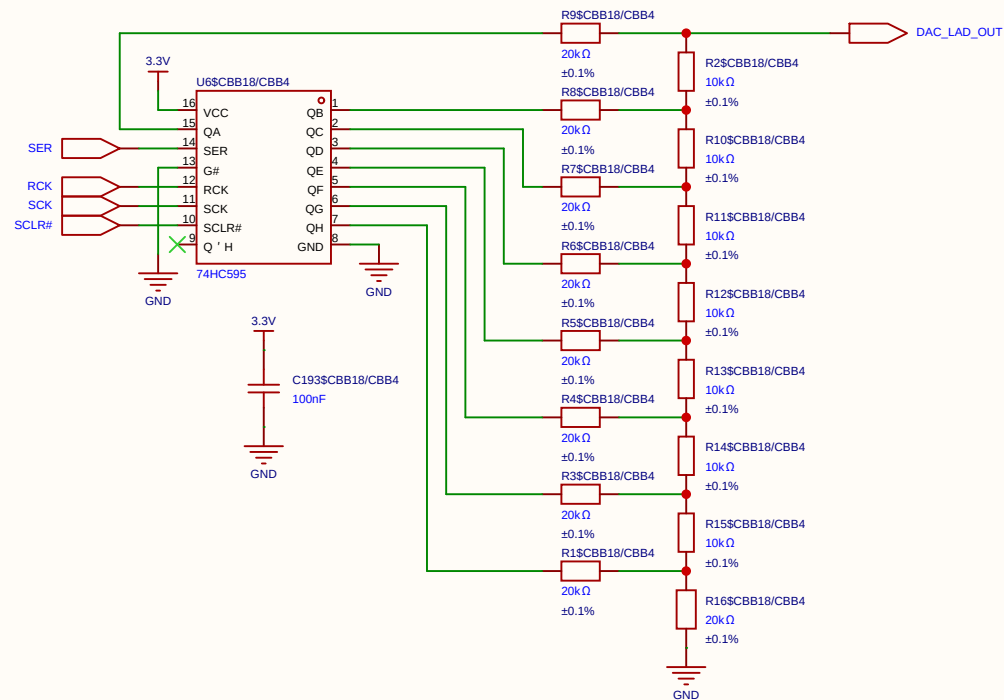
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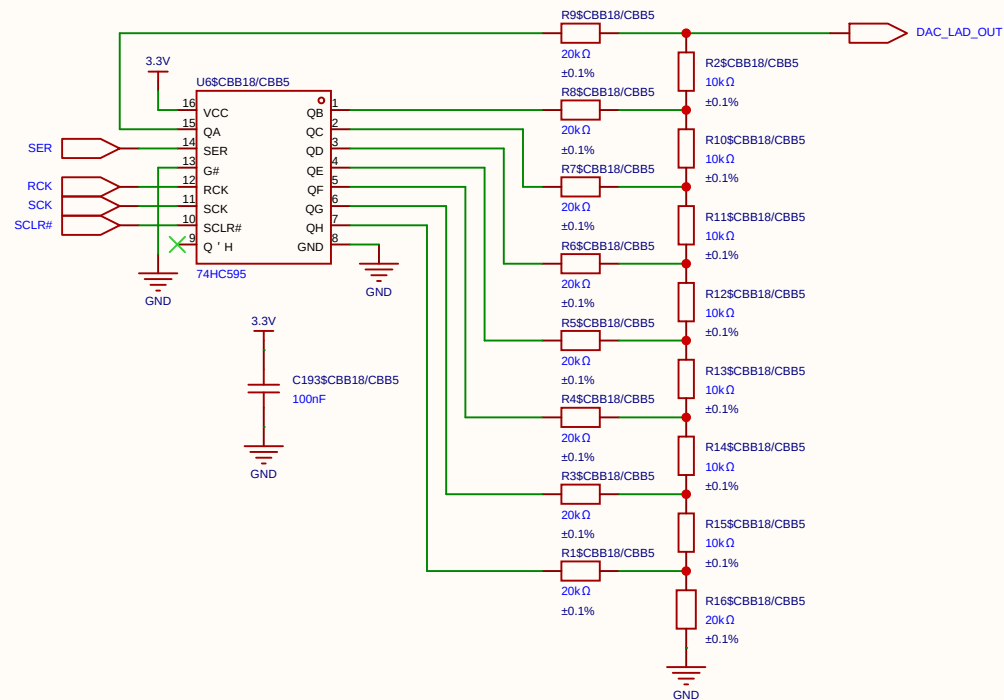
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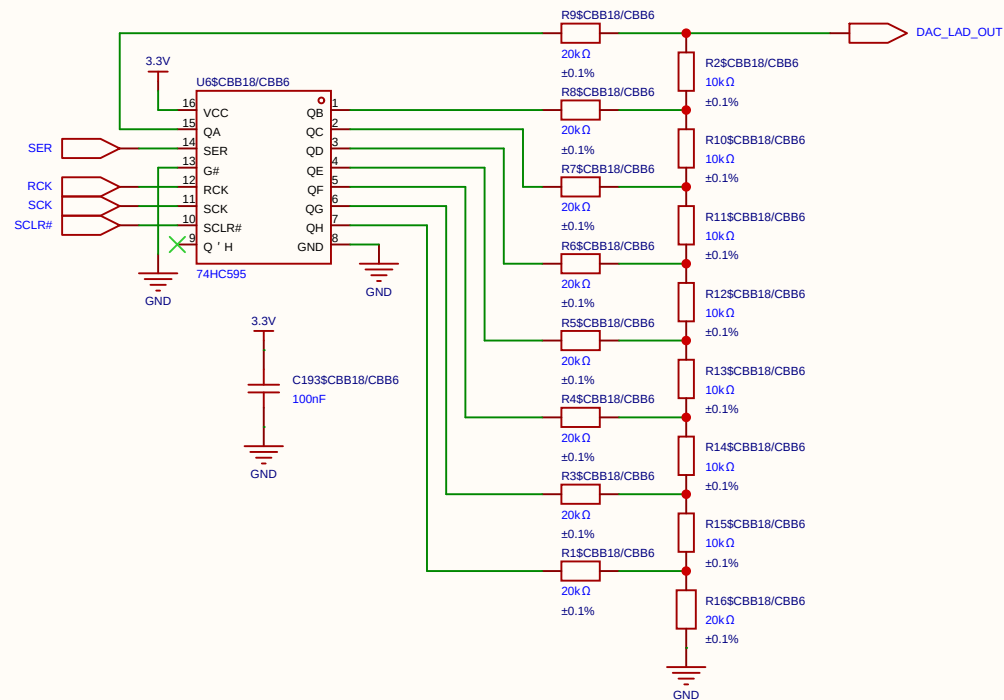
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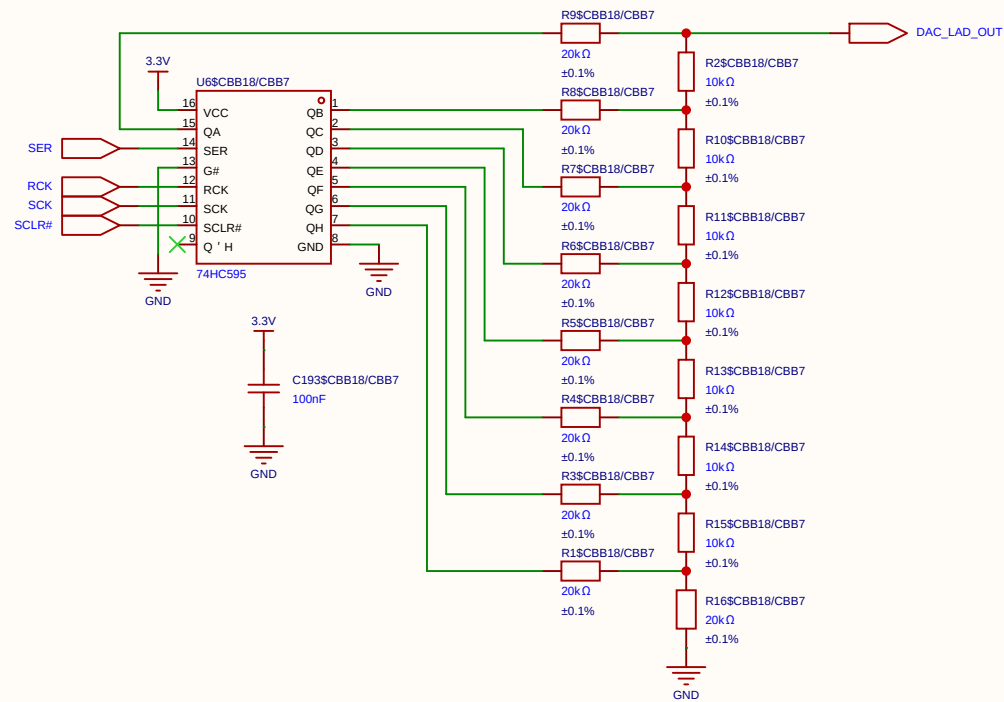
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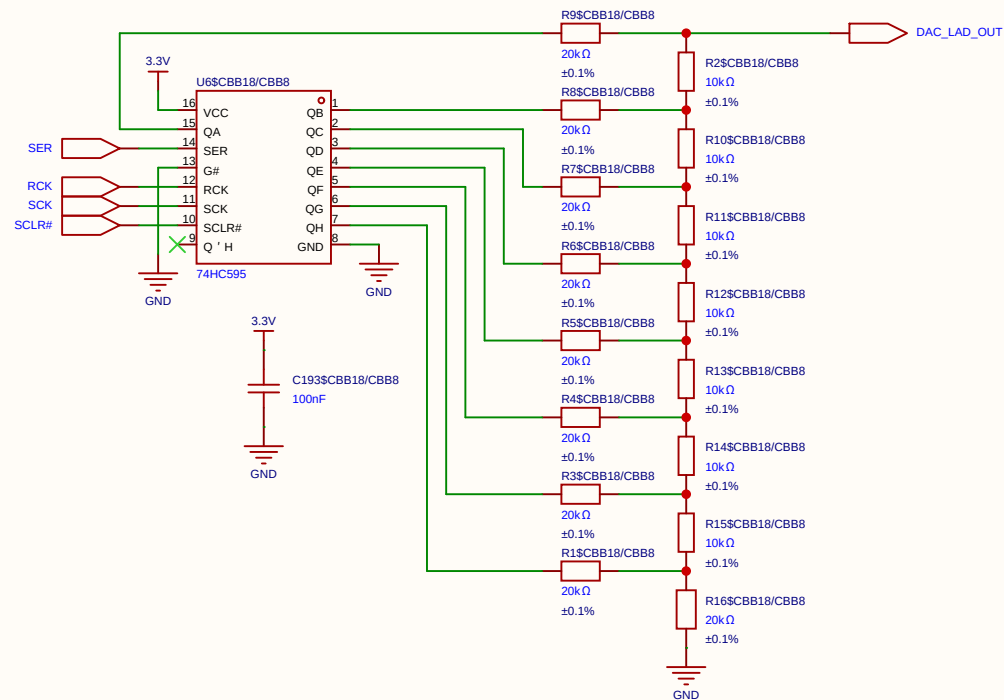
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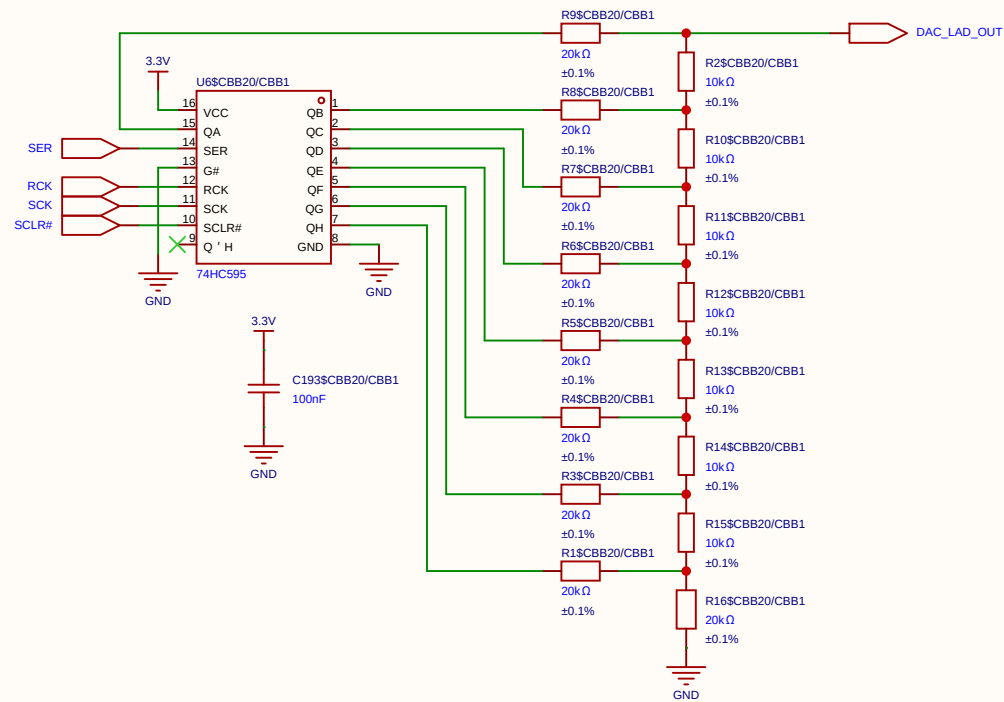
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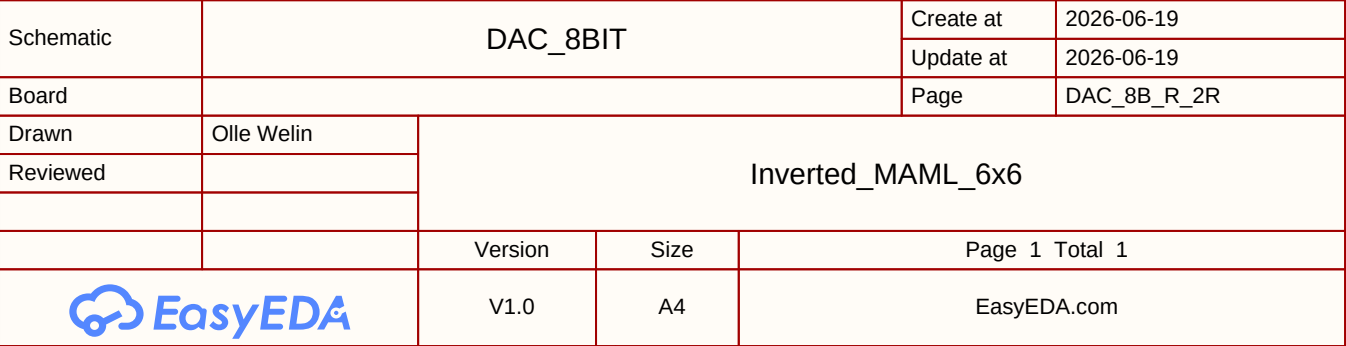
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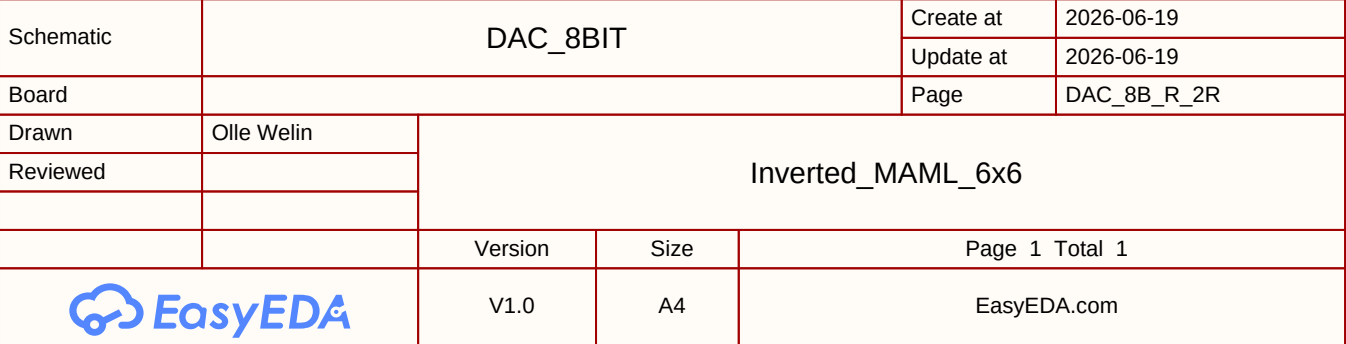


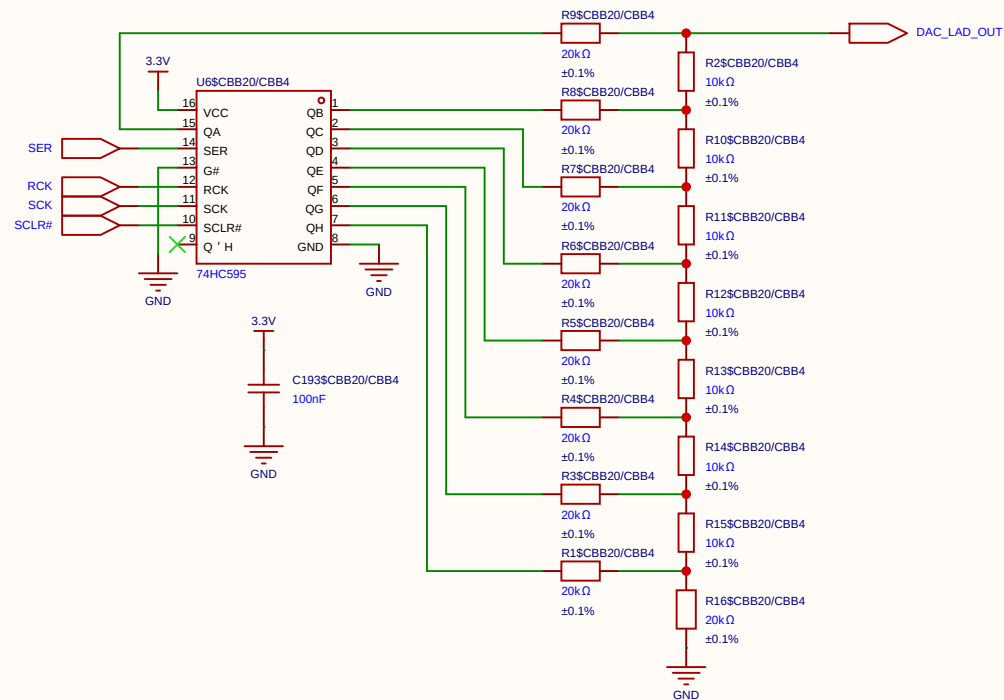
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EasyEDA		V1.0	A4	EasyEDA.com	



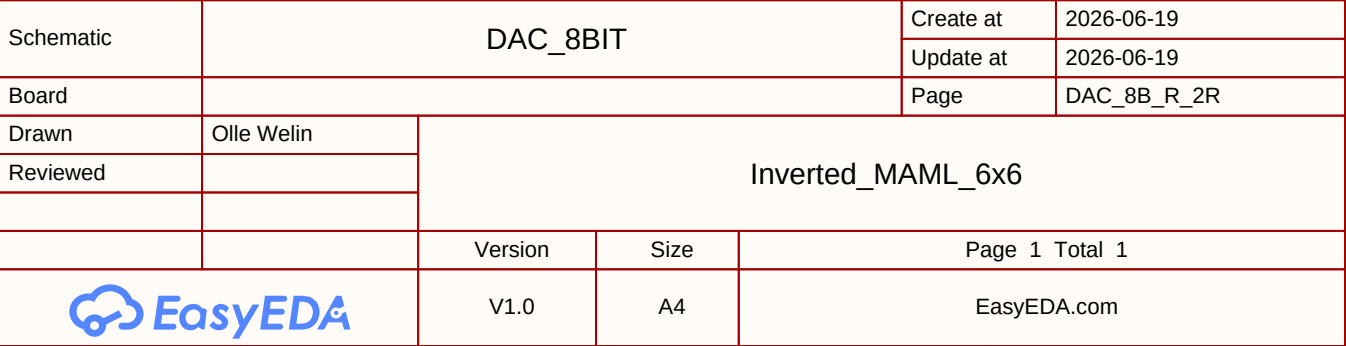
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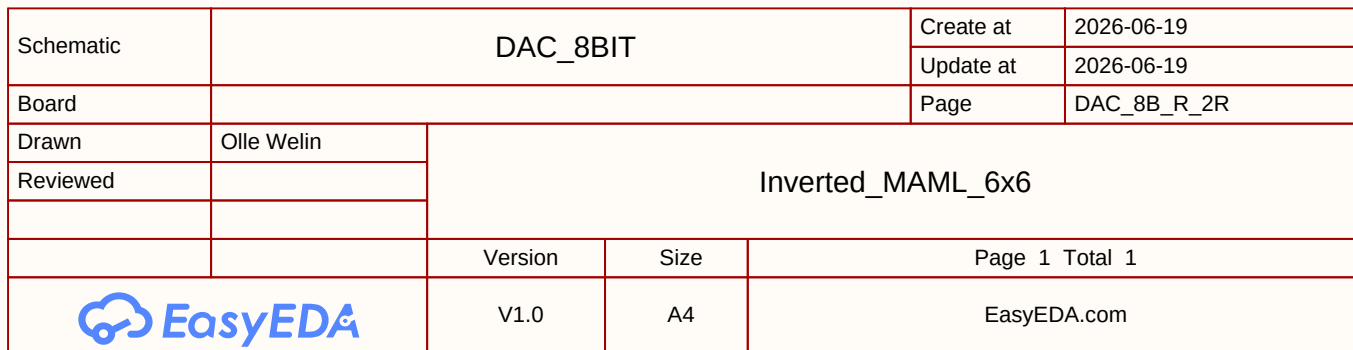


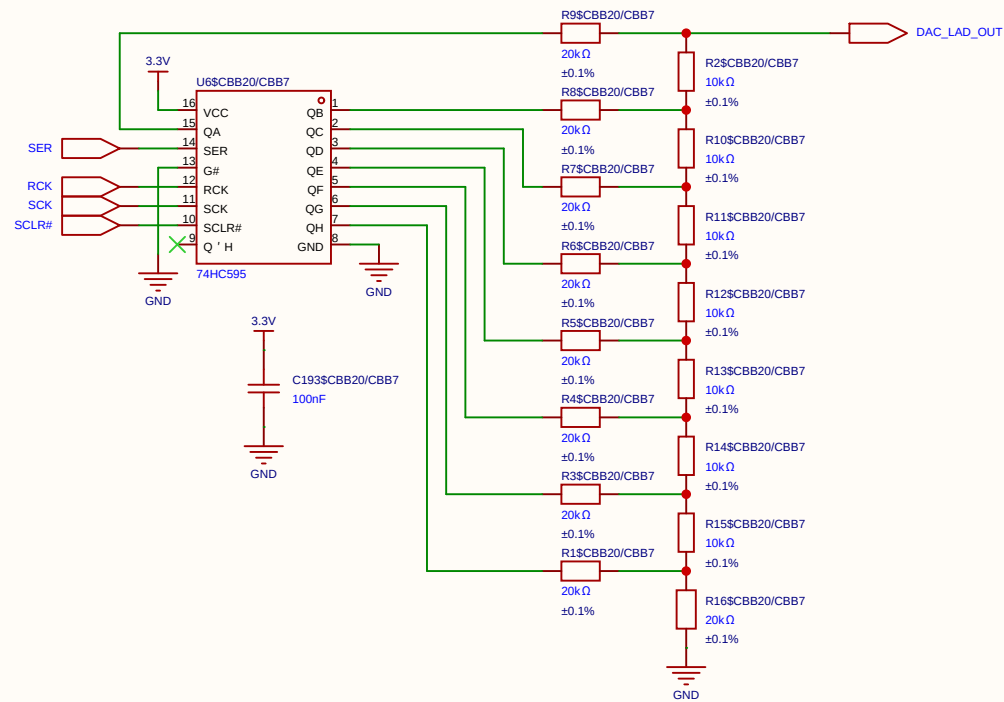




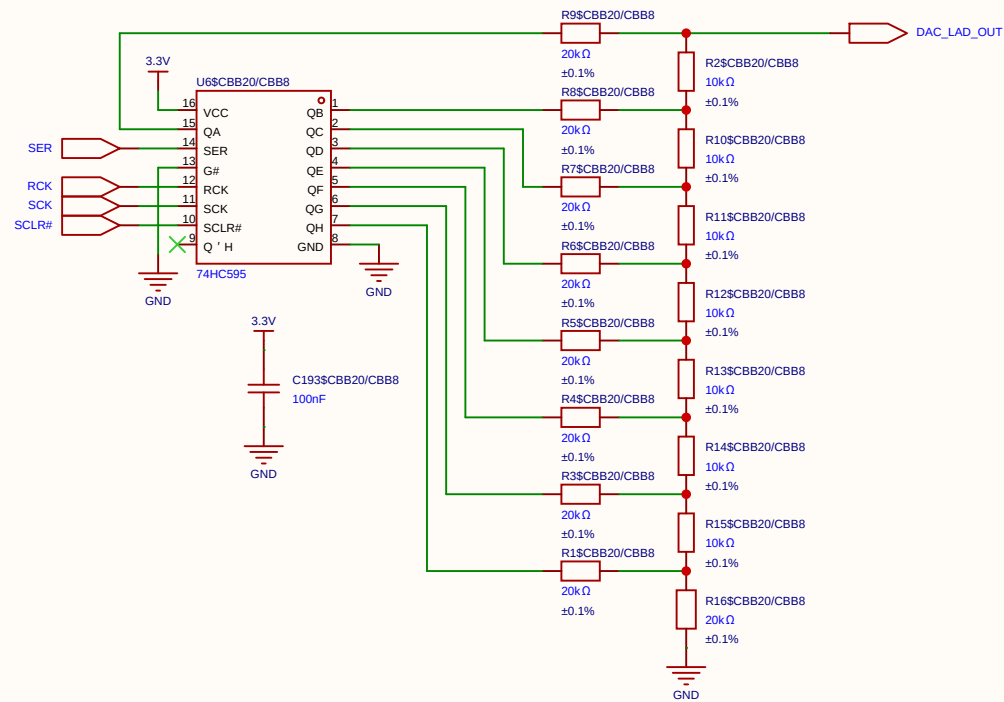
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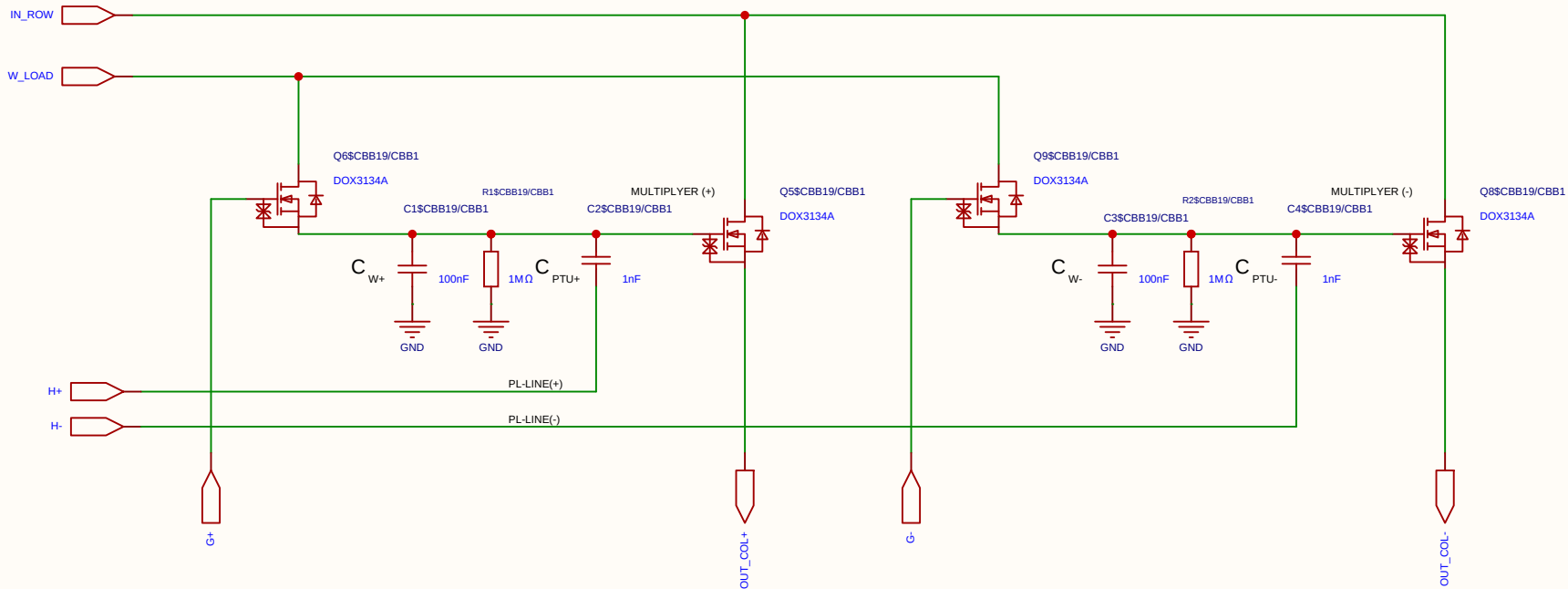


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Schematic	DAC_8BIT			Create at	2026-06-19
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Board				Page	DAC_8B_R_2R
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

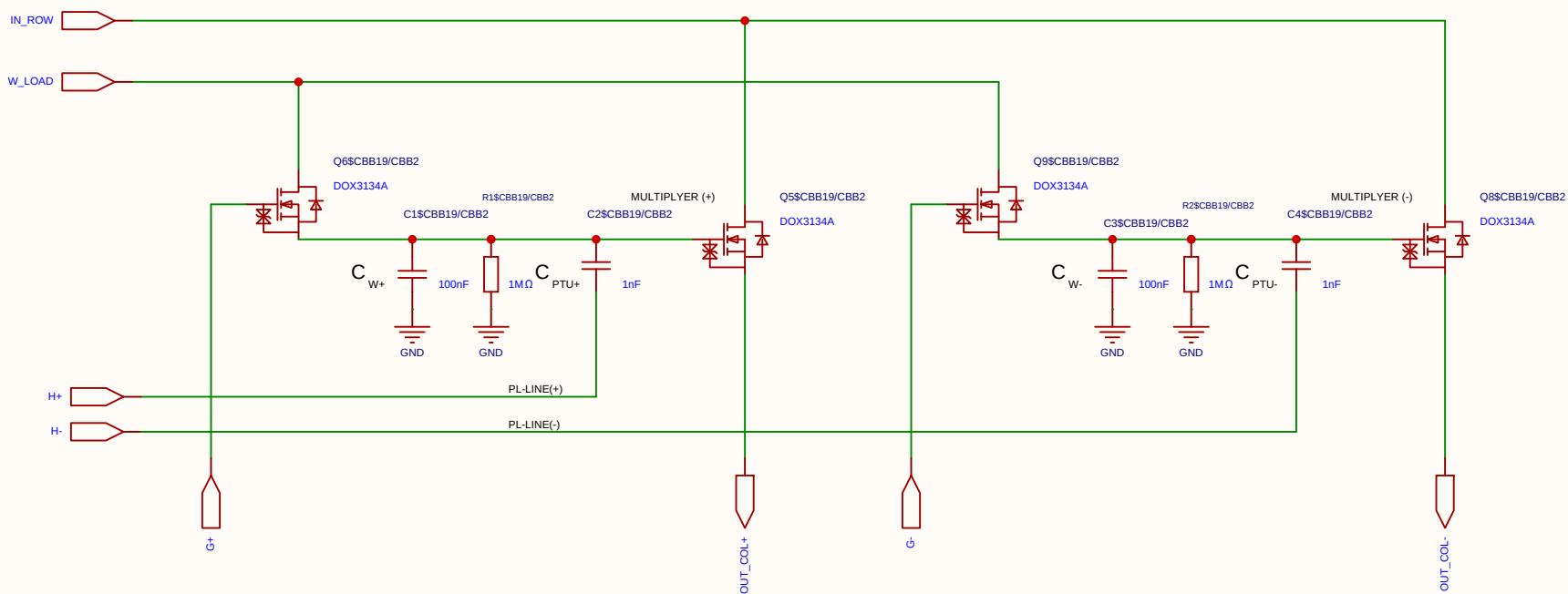
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

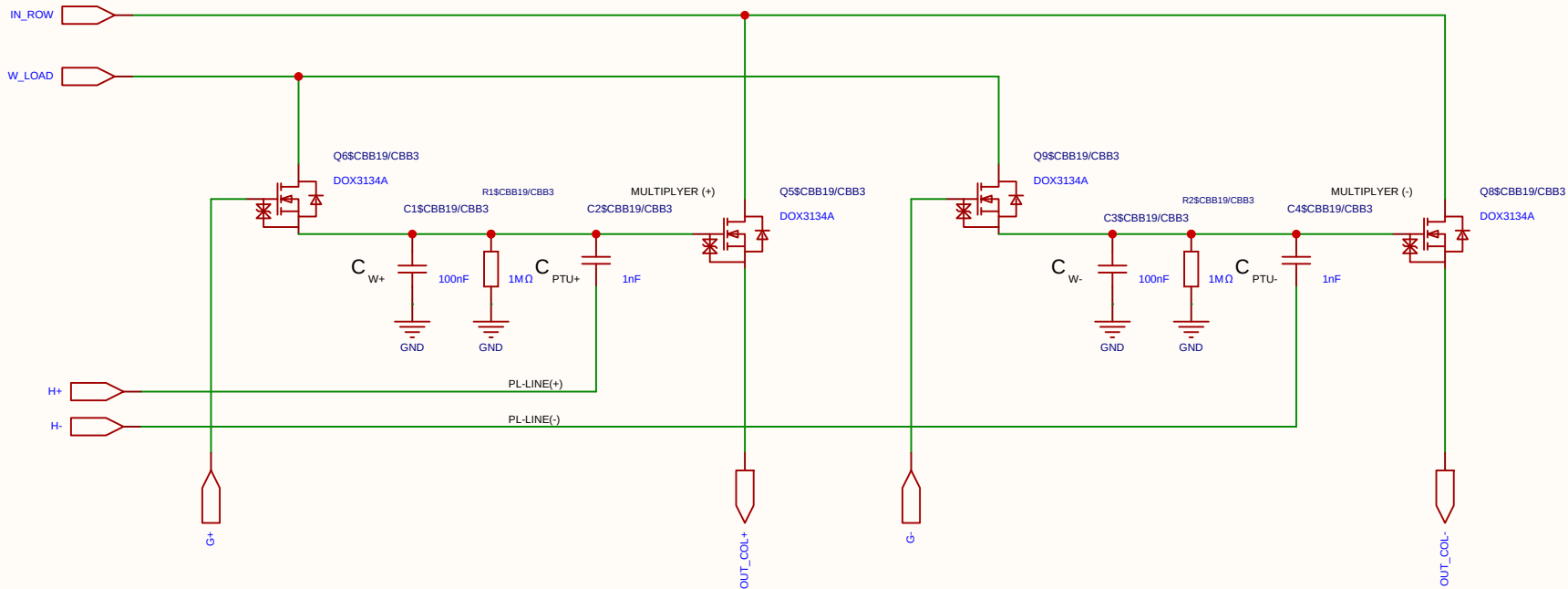
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

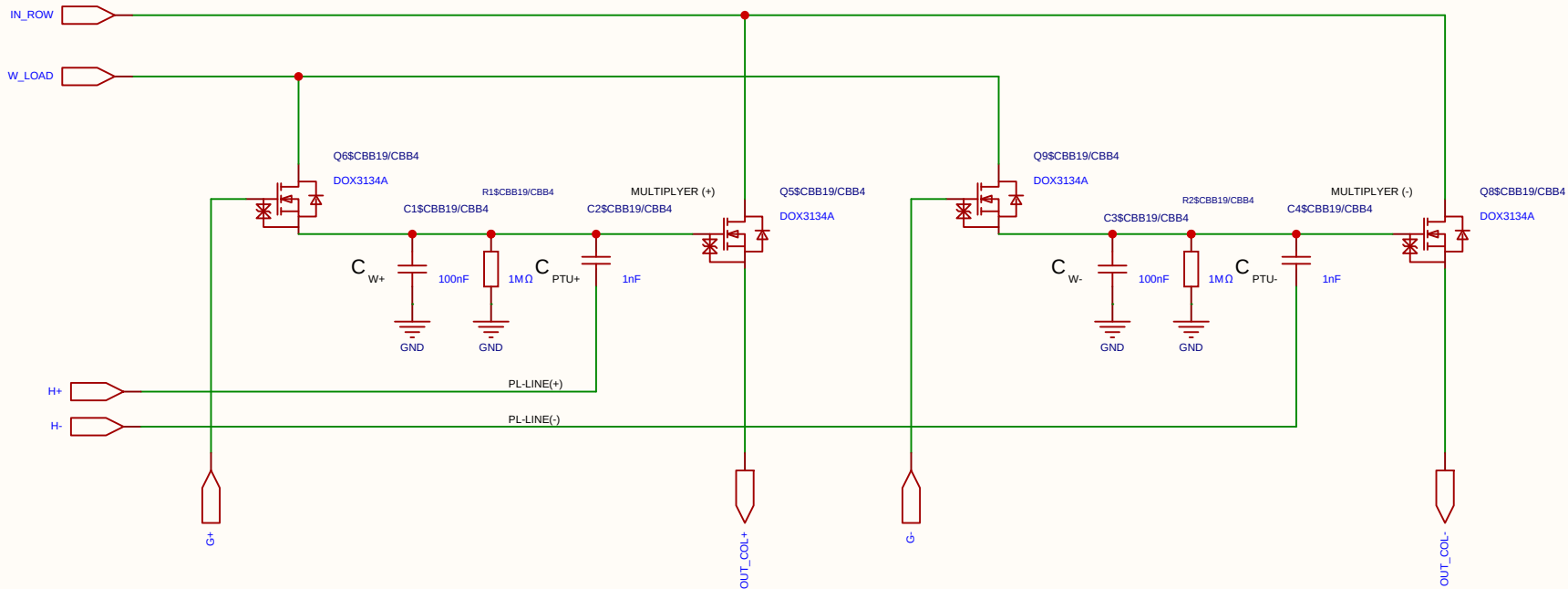
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

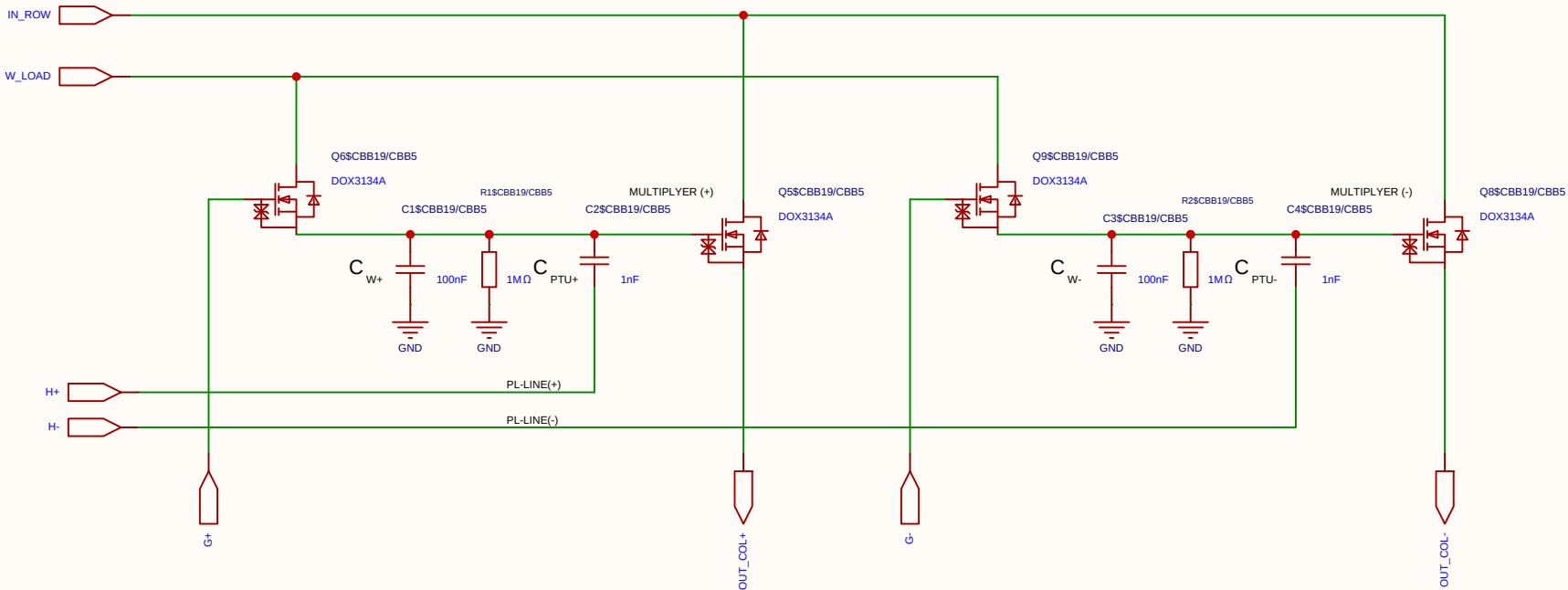
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

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The schematic diagram illustrates a 2-4 bit multiplier circuit. It features two 4-bit multipliers, MULTIPLIER (+) and MULTIPLIER (-), which are used to calculate the product of two 4-bit numbers. The inputs are IN_ROW and W_LOAD. The outputs are OUT_COL+ and OUT_COL-. The circuit includes various components like transistors (Q6, Q5, Q9, Q8), capacitors (C1, C2, C3, C4), resistors (R1, R2), and a 100nF capacitor. The circuit is powered by a 5V supply and ground.

Category	Parameter	Value / Type	Technical Description & Function
Resistors	Discharge resistor (R)	1 M Ω	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

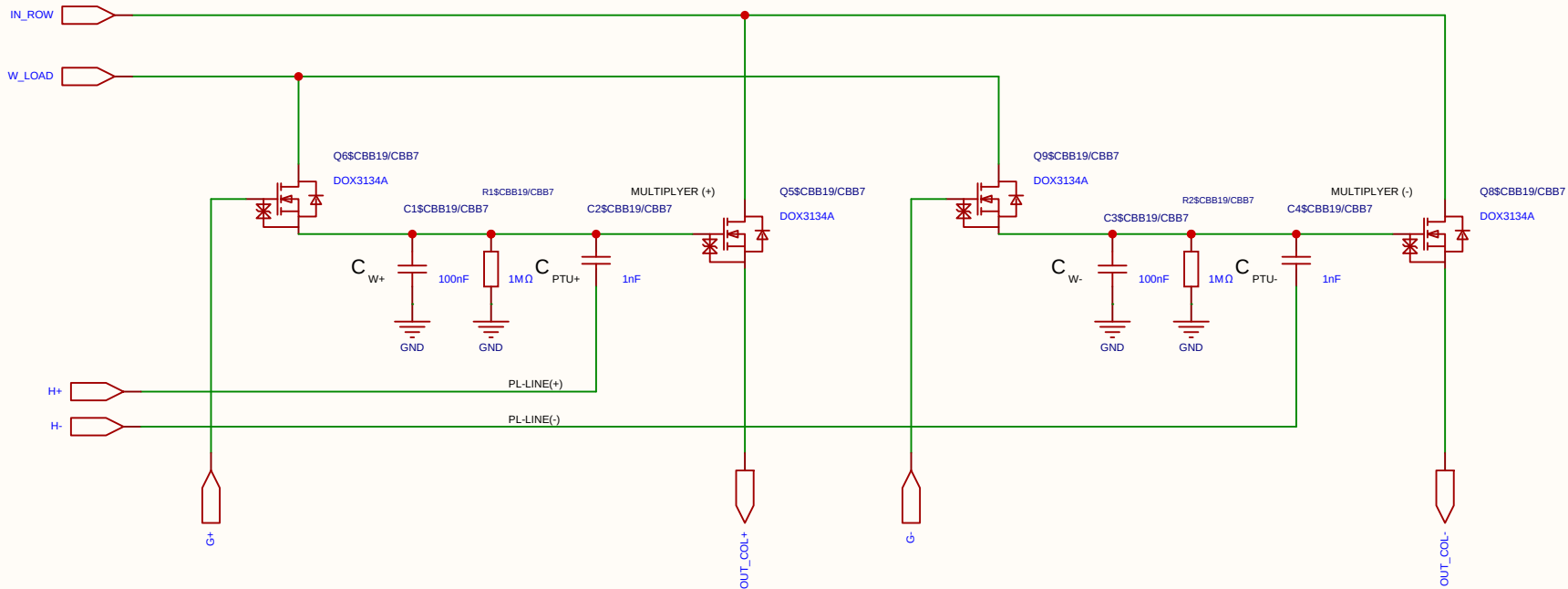
Component	Value	Description / Function
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a $\sim 1/101$ voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μ s.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

Component	Value	Description / Function
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_I . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_I . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_I . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.



Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
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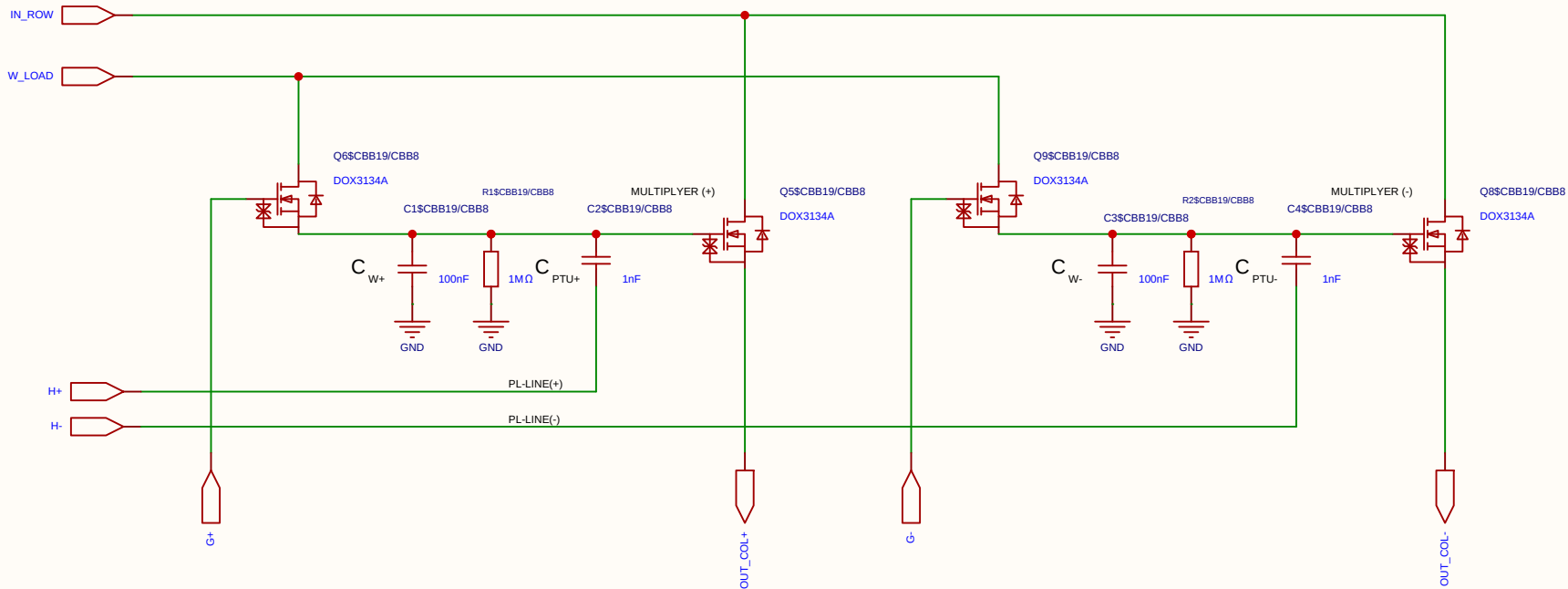
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
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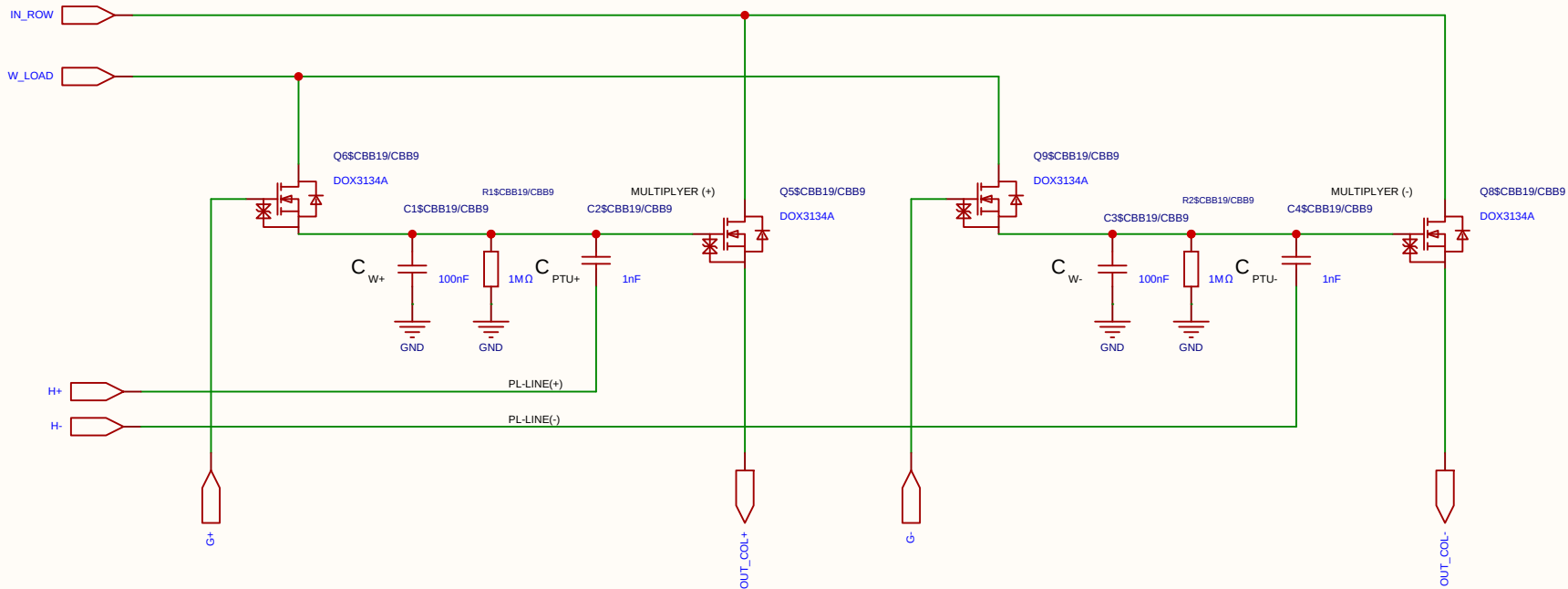
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
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Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

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Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
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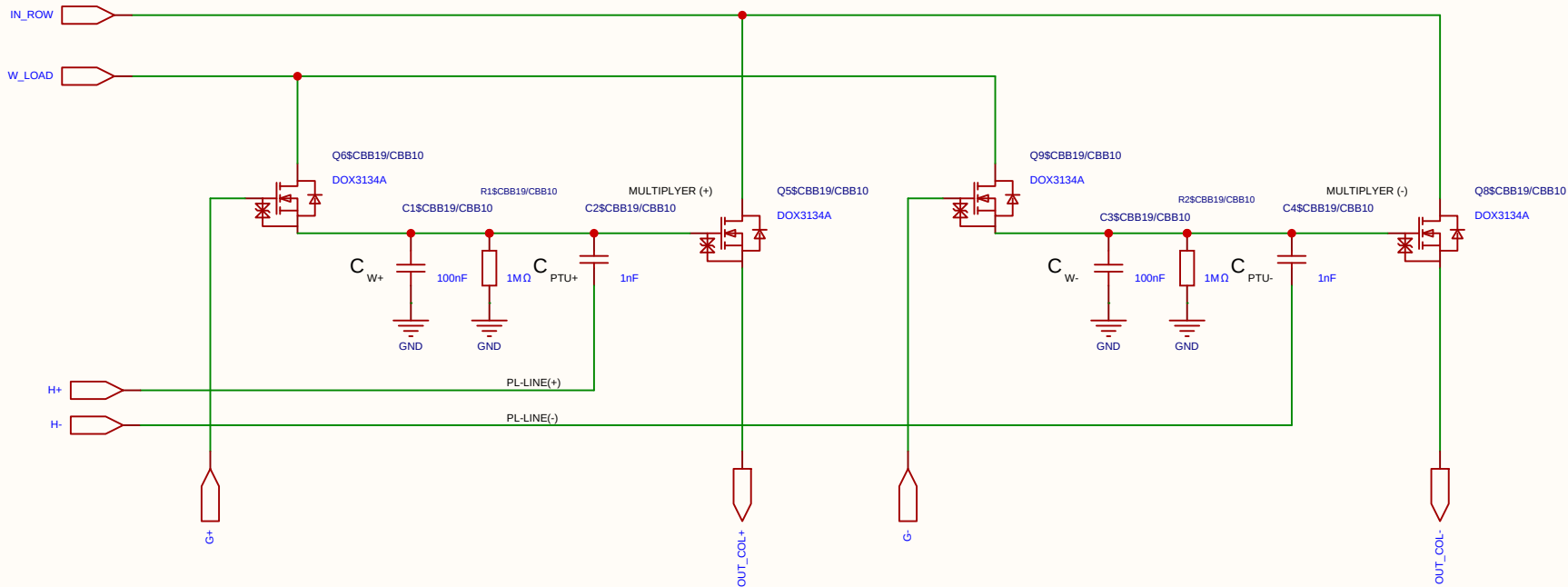
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This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
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	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
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4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
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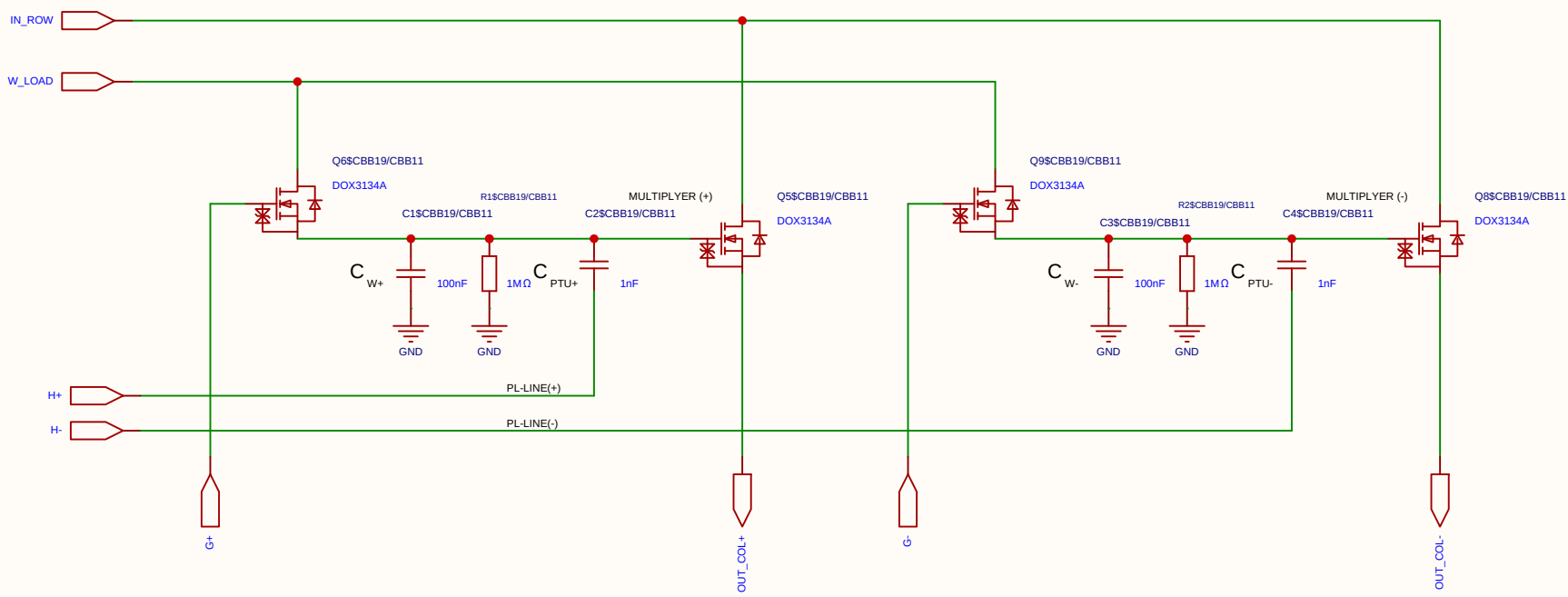
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
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Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

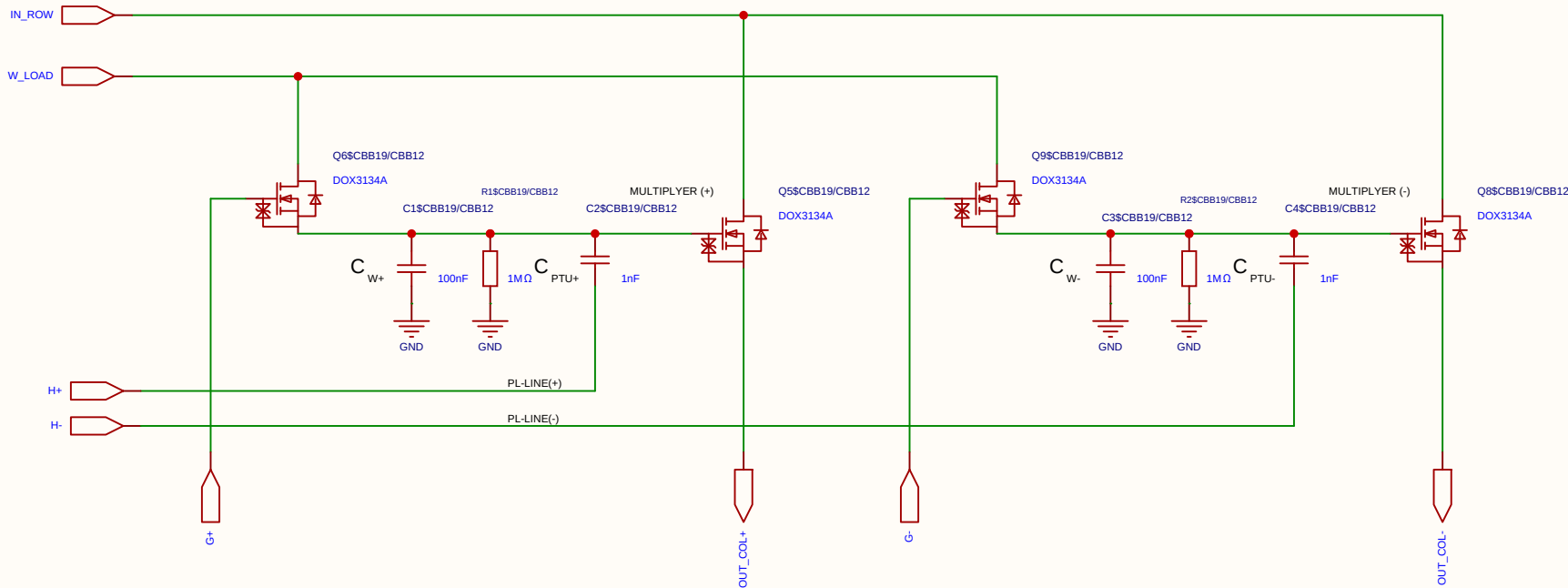
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Board				Page	matrix_3_8_MUL_CELL
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Reviewed					
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

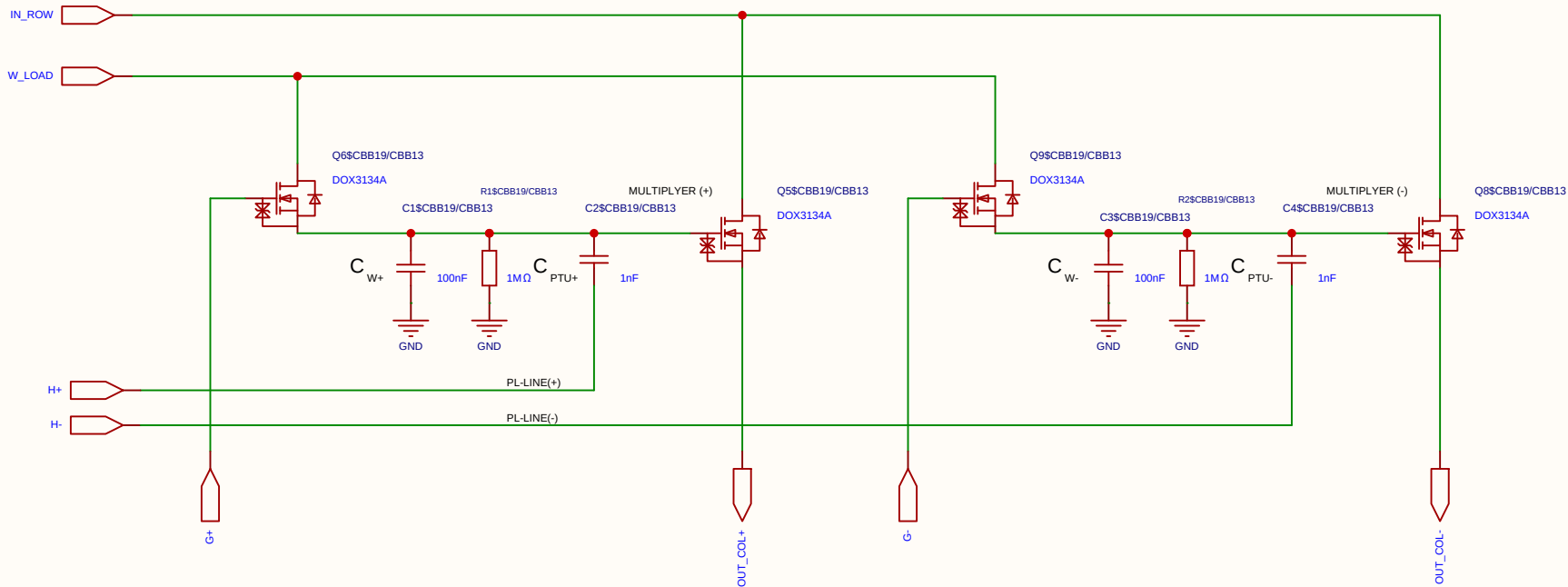
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Drawn	Olle Welin	Inverted_MAML_6x6			
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

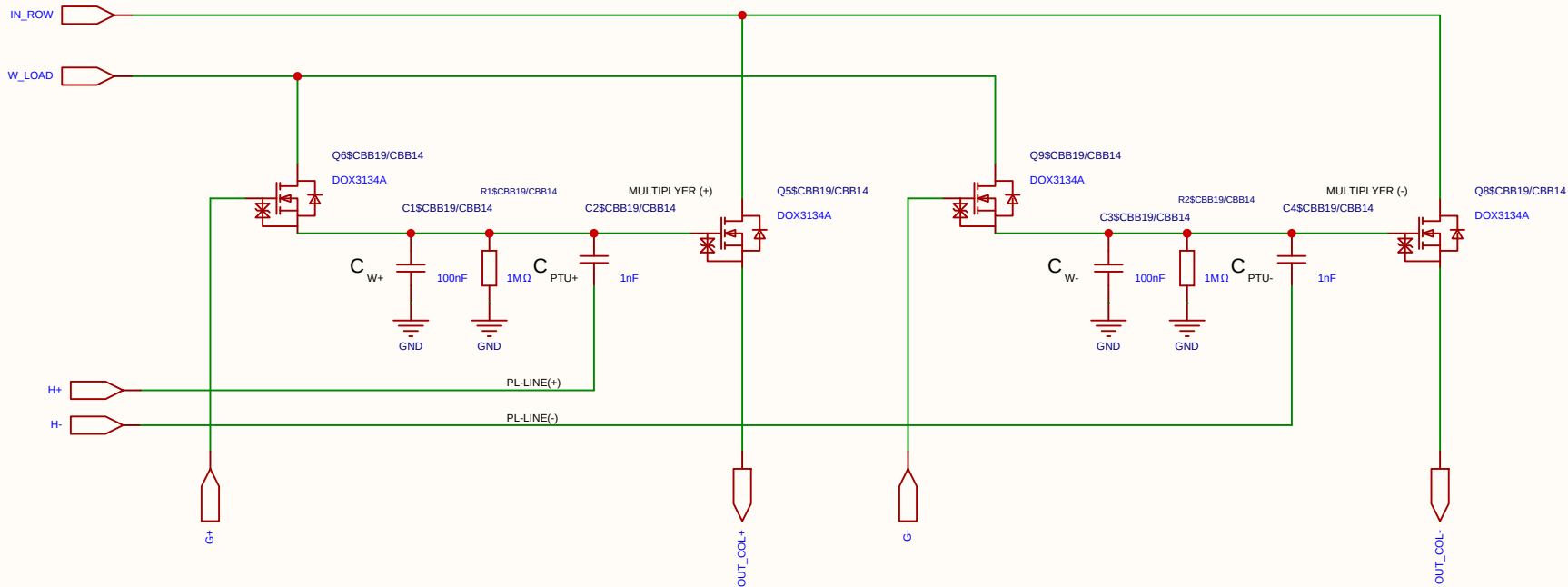
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Drawn	Olle Welin	Inverted_MAML_6x6			
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

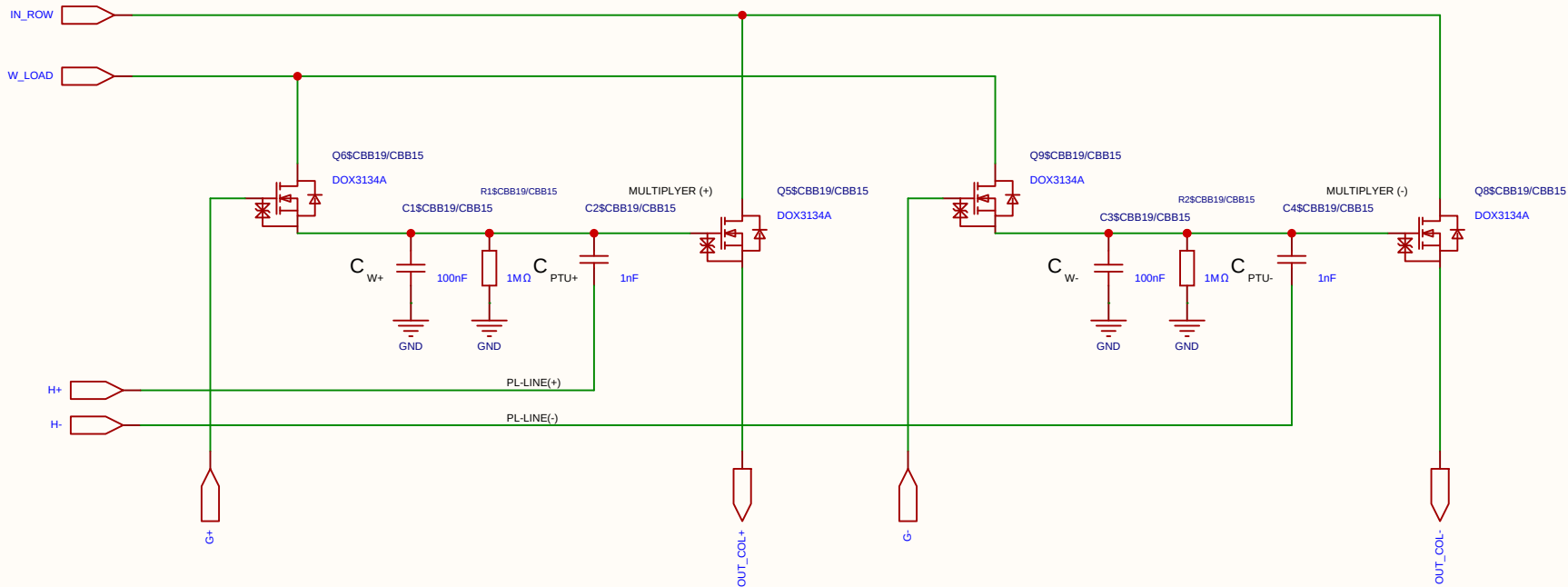
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Drawn	Olle Welin	Inverted_MAML_6x6			
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

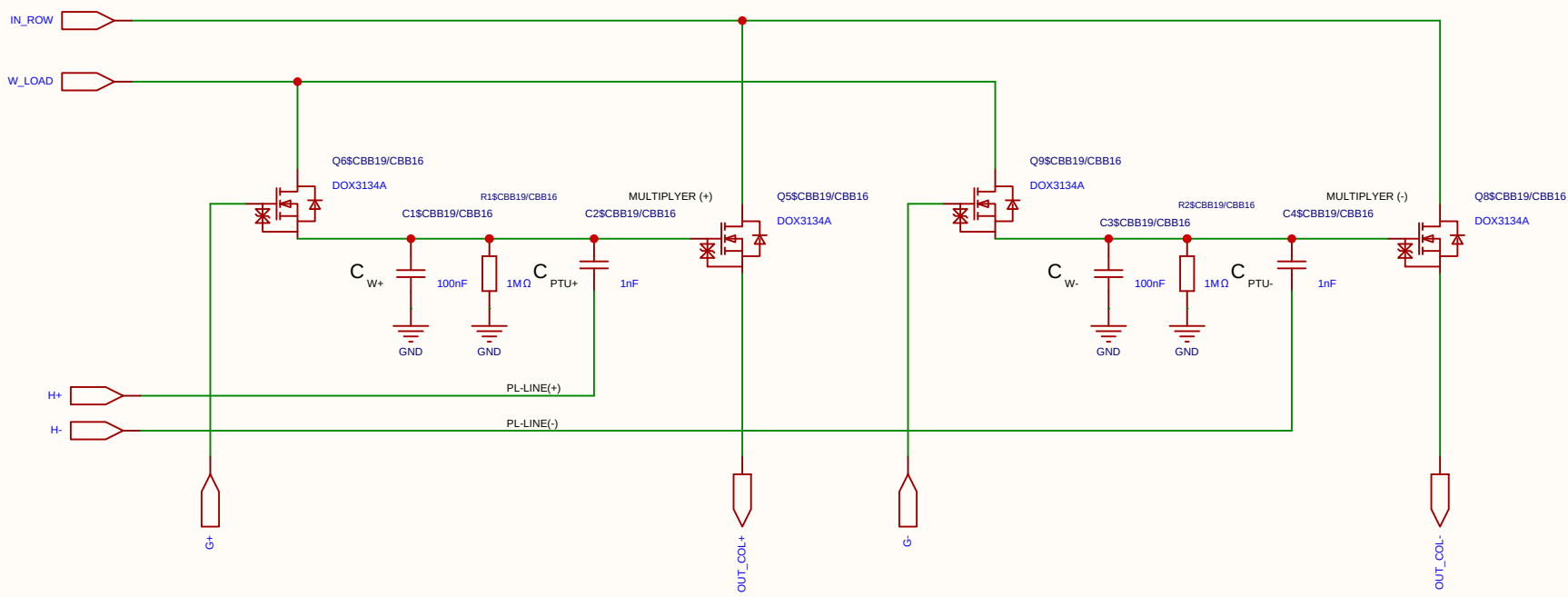
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Drawn	Olle Welin	Inverted_MAML_6x6			
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

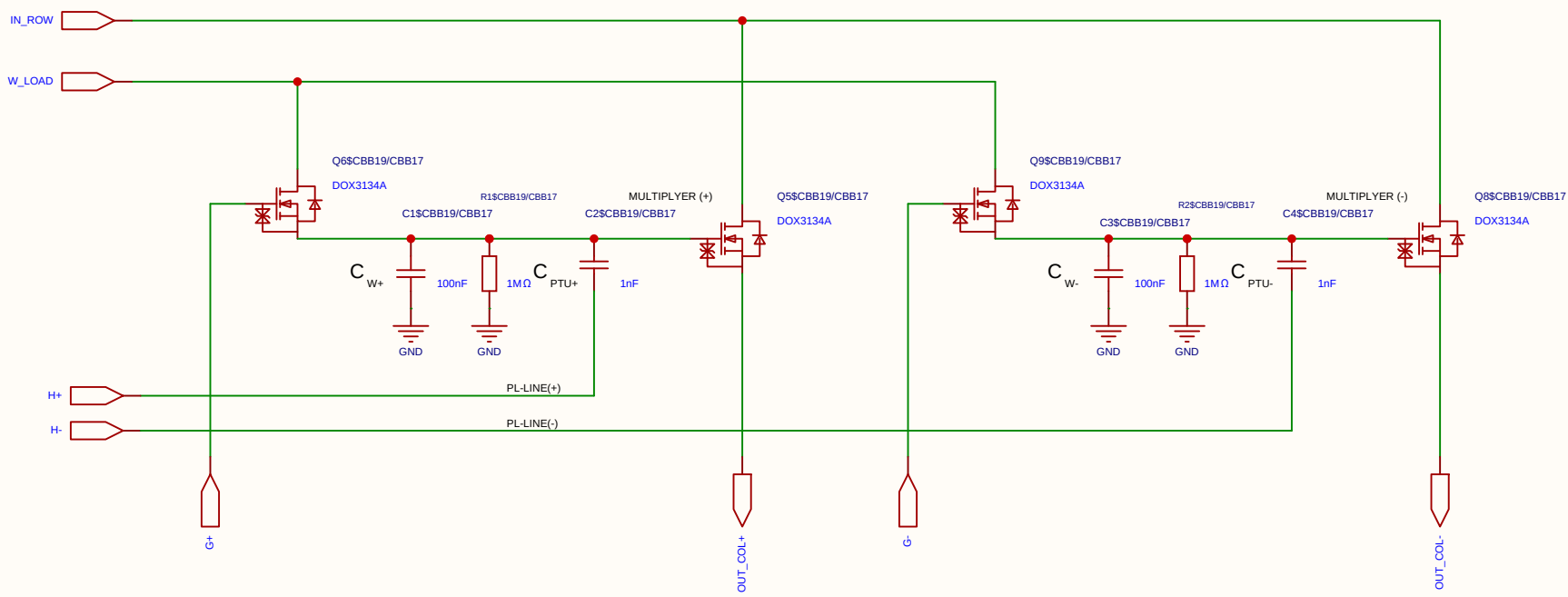
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
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Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

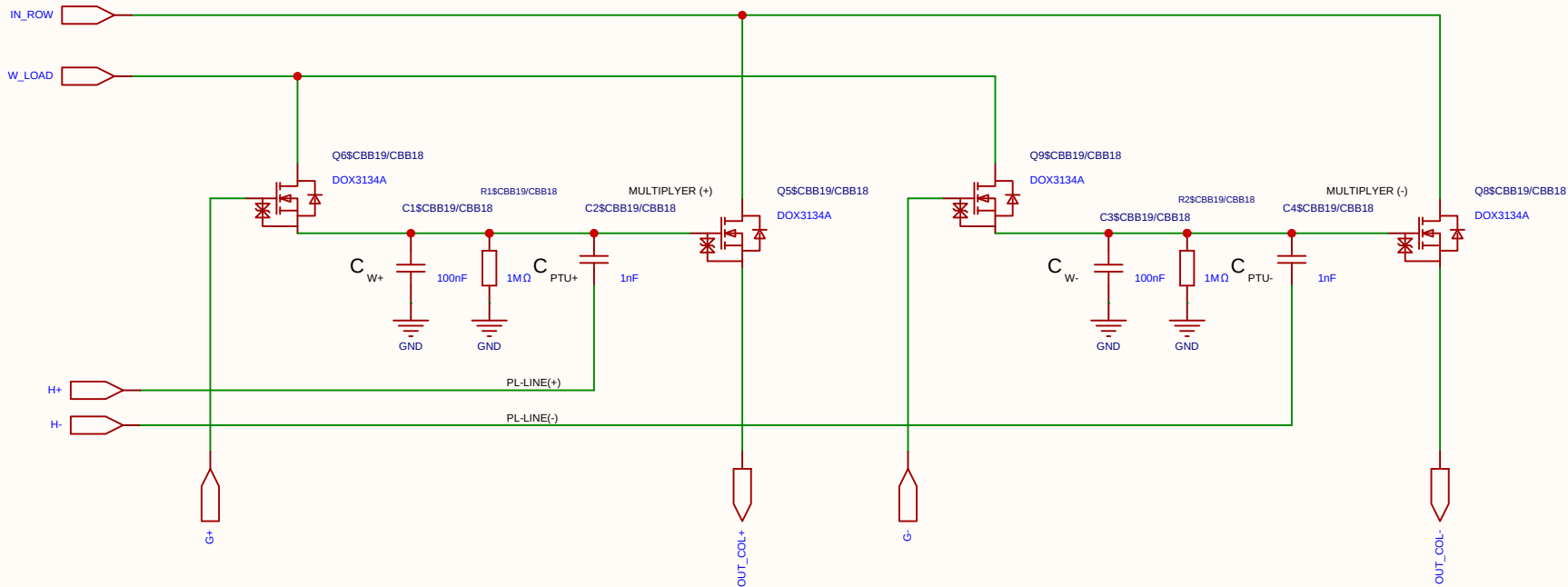
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
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Drawn	Olle Welin	Inverted_MAML_6x6			
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Board				Page	matrix_3_8_MUL_CELL
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Reviewed					
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Category	Parameter	Value / Type	Technical Description & Function
Resistors	Discharge resistor (R)	1 M Ω	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

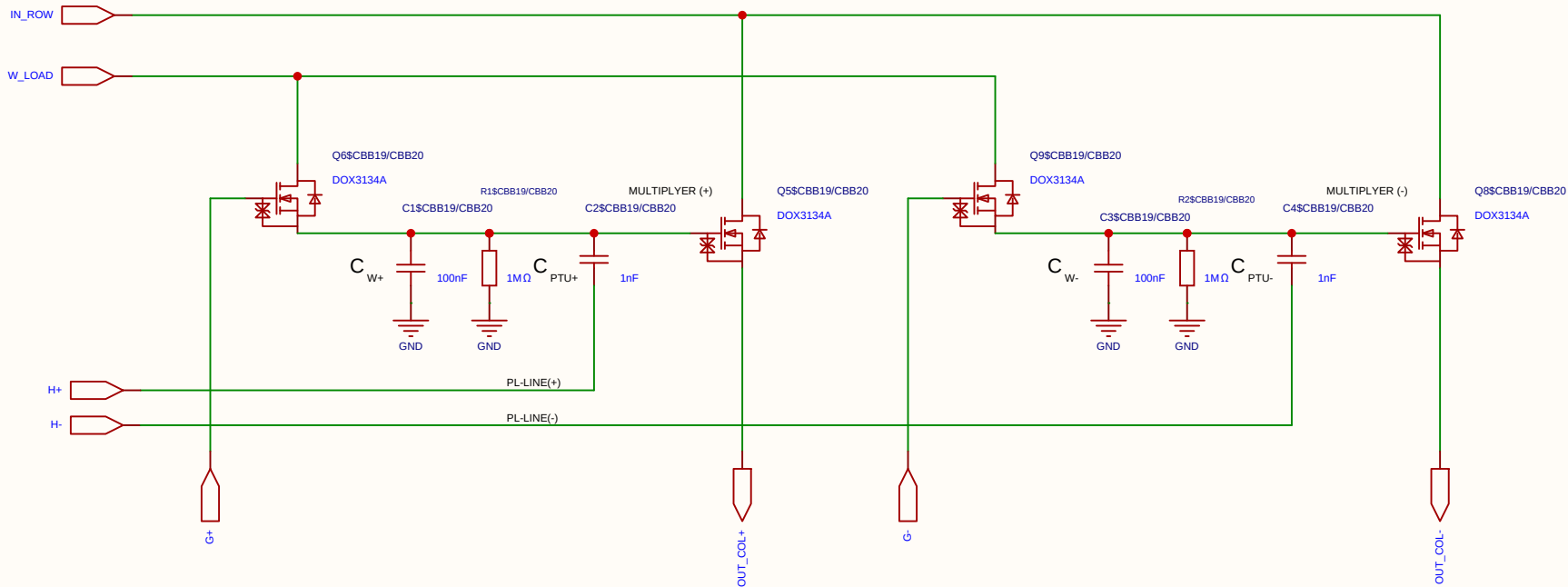
Component	Value	Description / Function
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a $\sim 1/101$ voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μ s.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

Component	Value	Description / Function
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_I . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_I . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_I . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.



Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

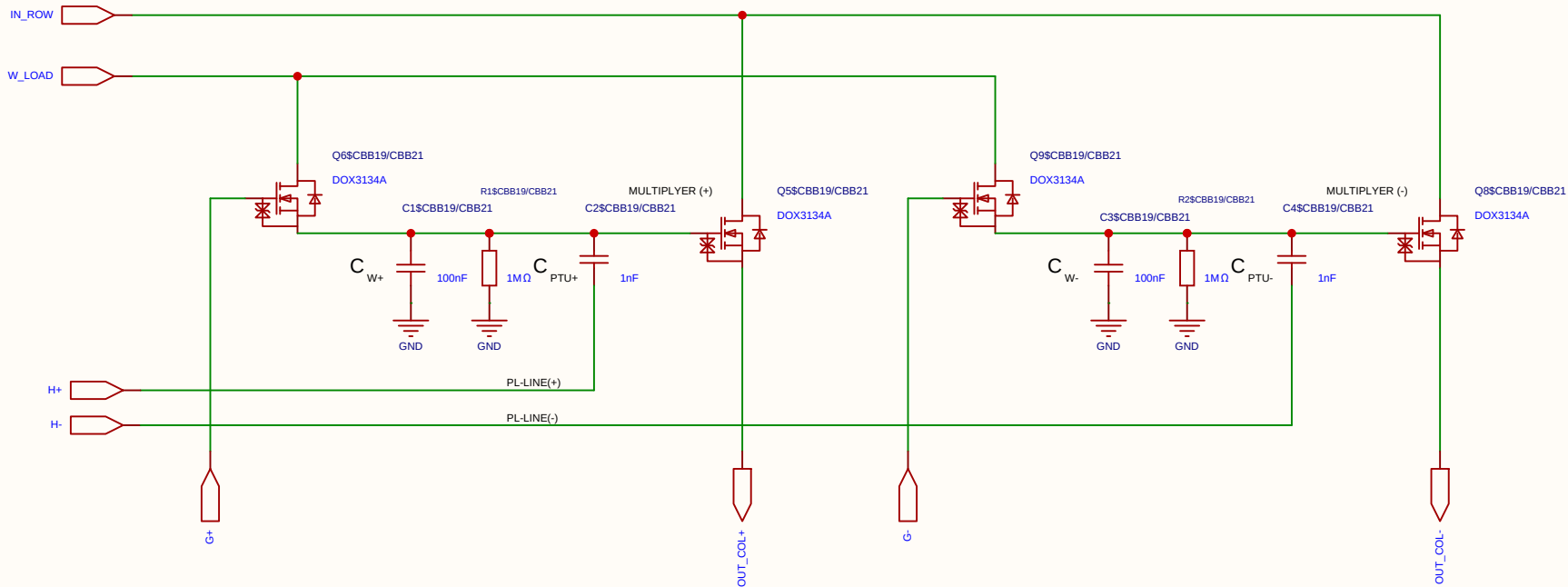
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

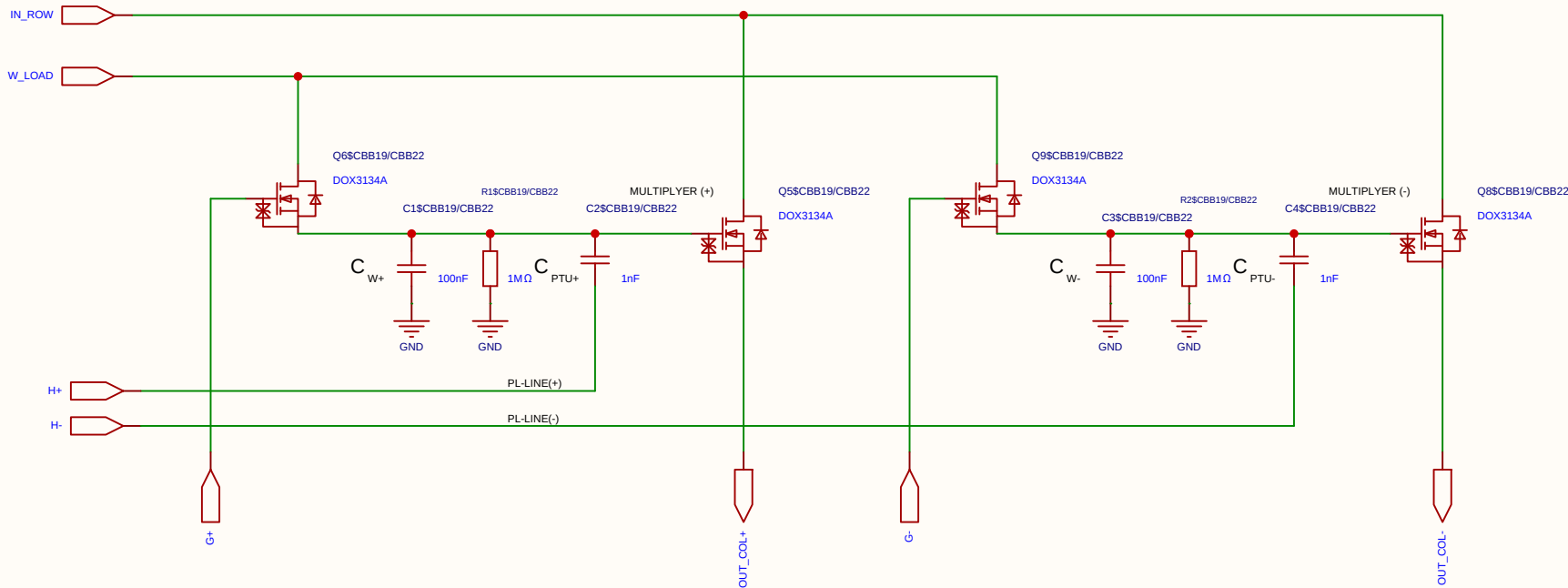
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

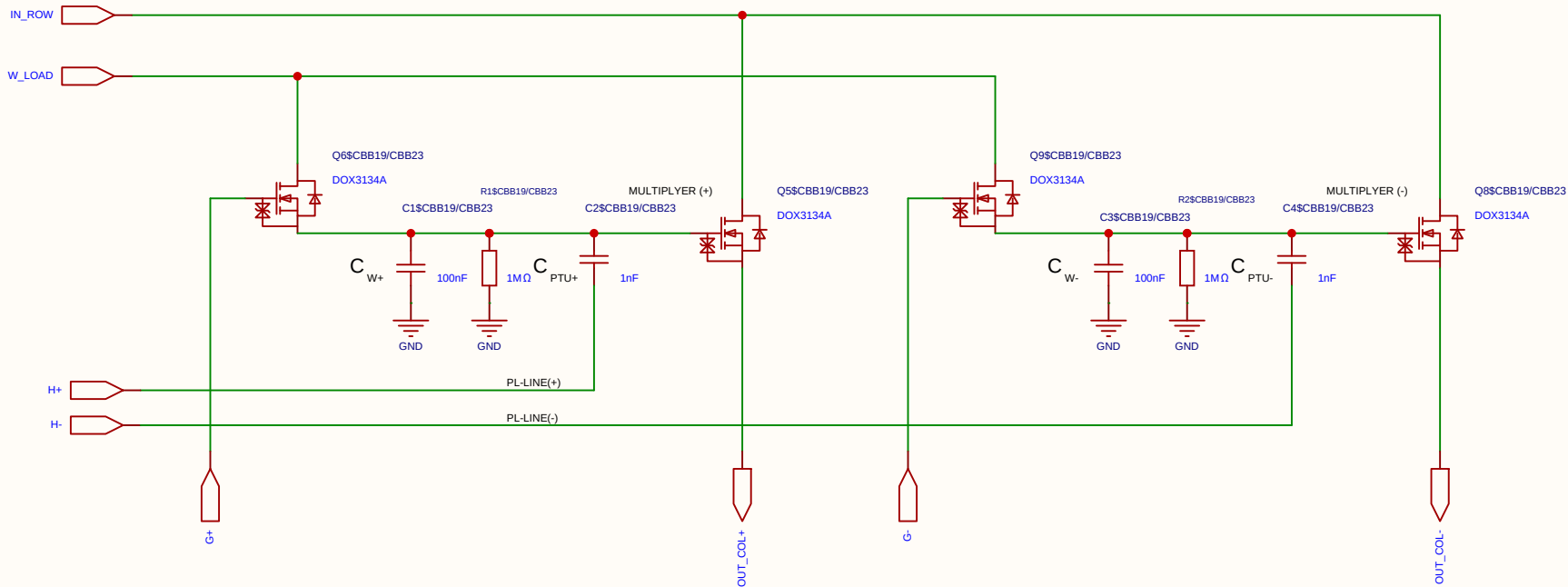
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

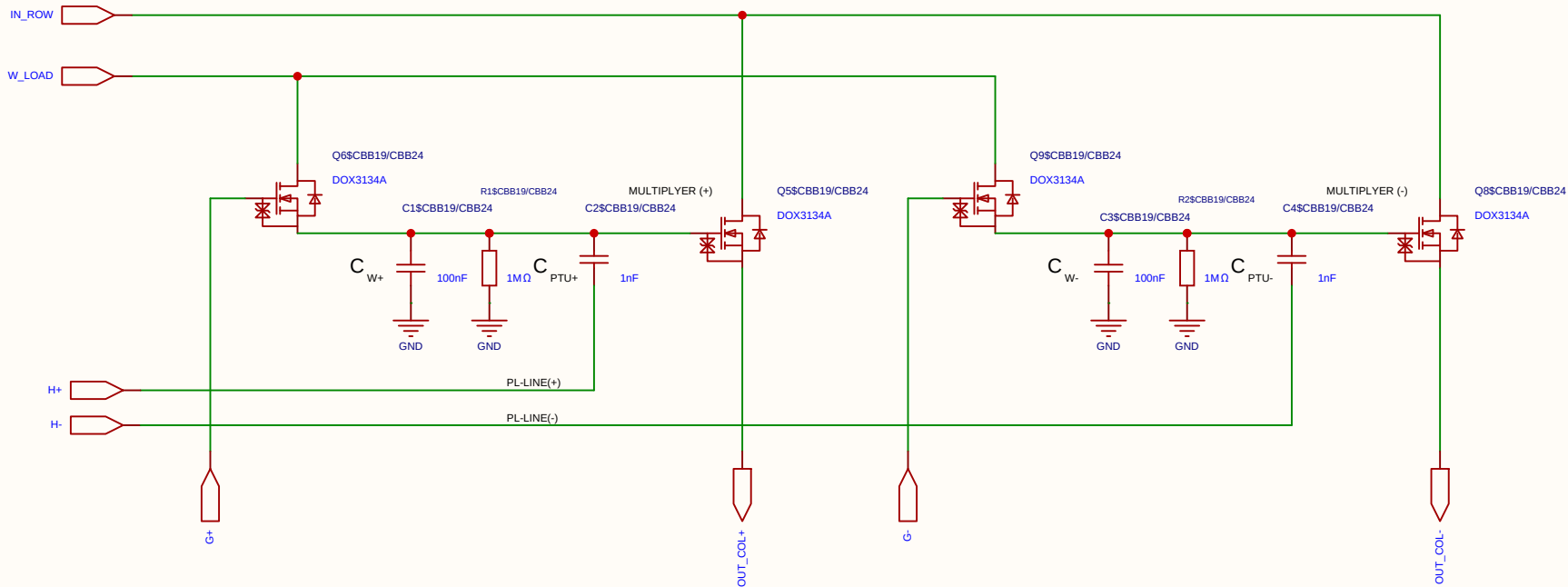
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Drawn	Olle Welin	Inverted_MAML_6x6			
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

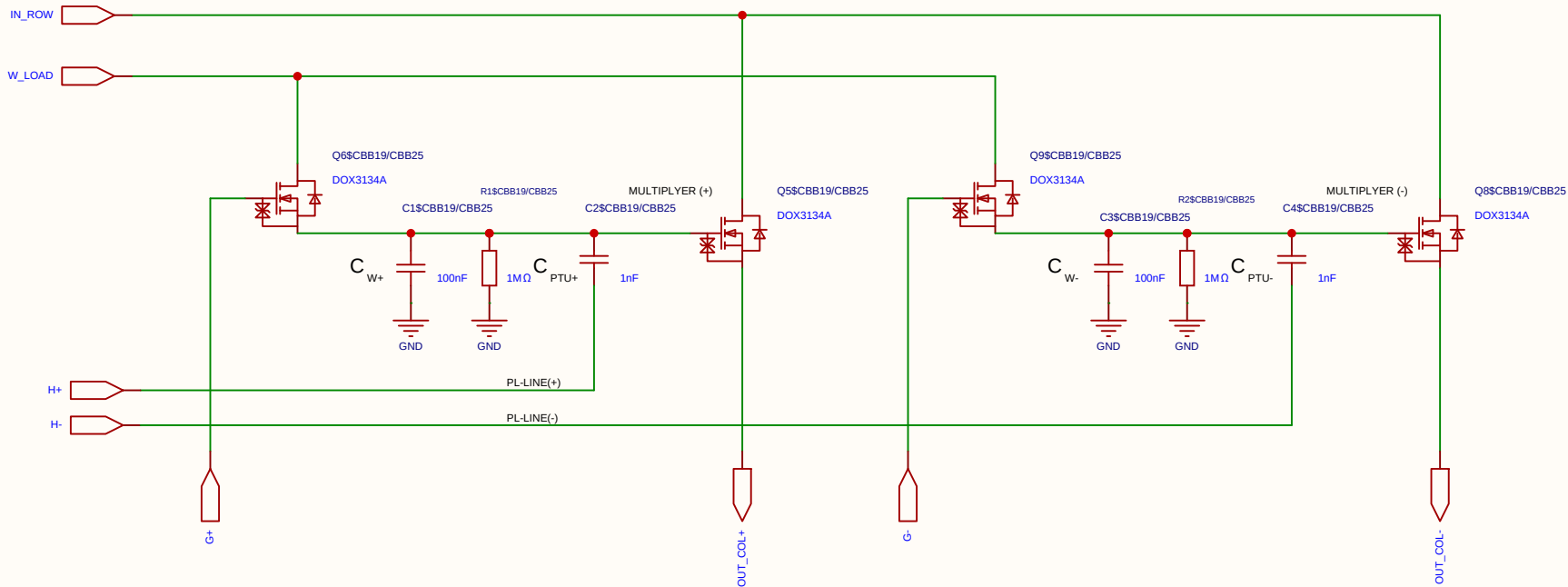
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

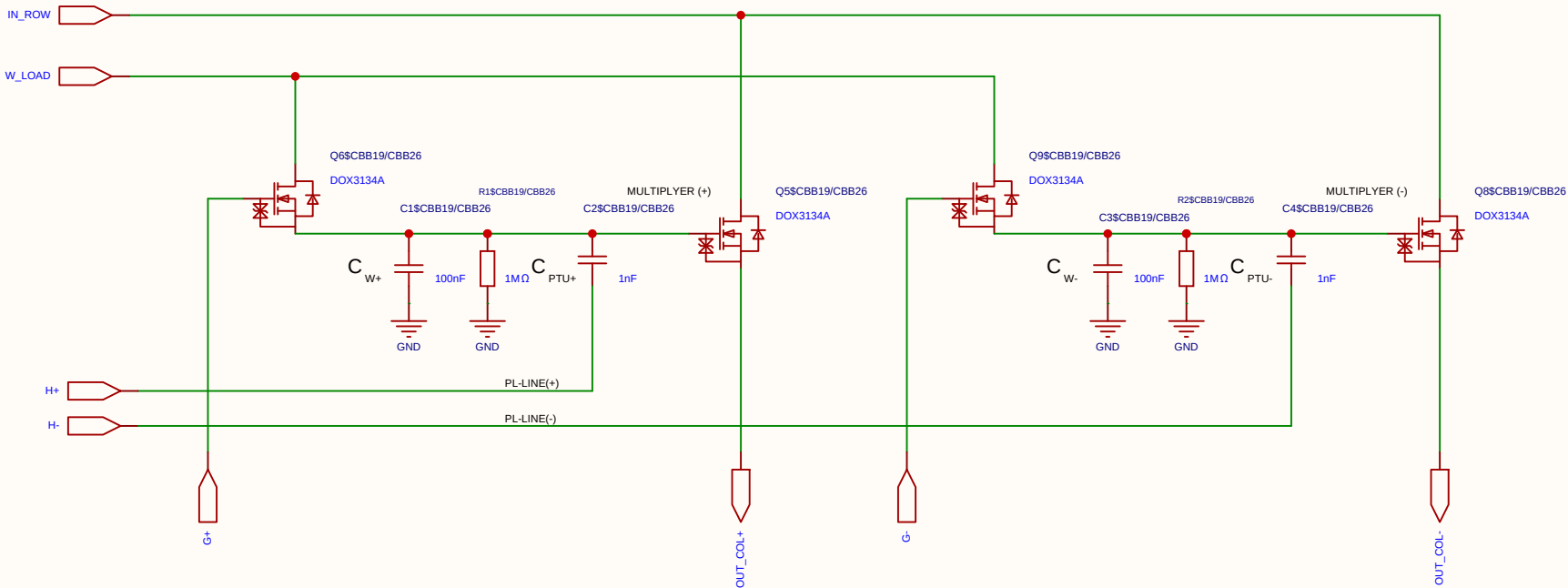
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
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Drawn	Olle Welin	Inverted_MAML_6x6			
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		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

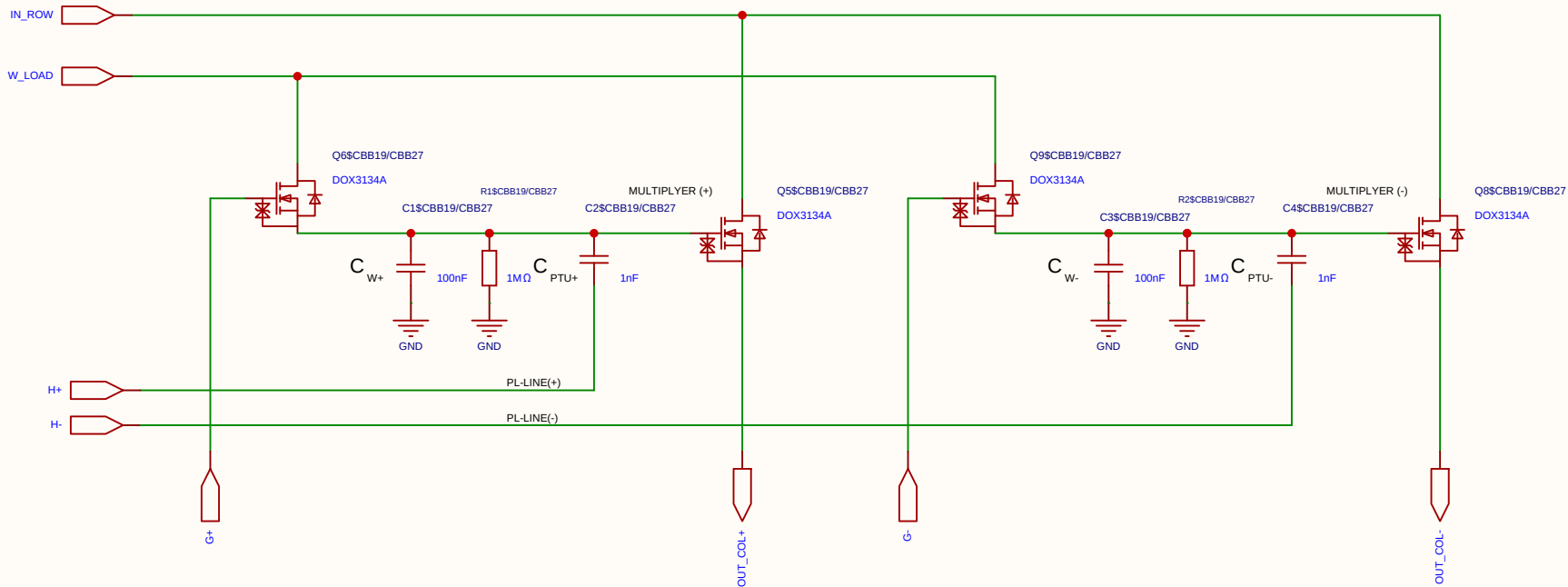
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

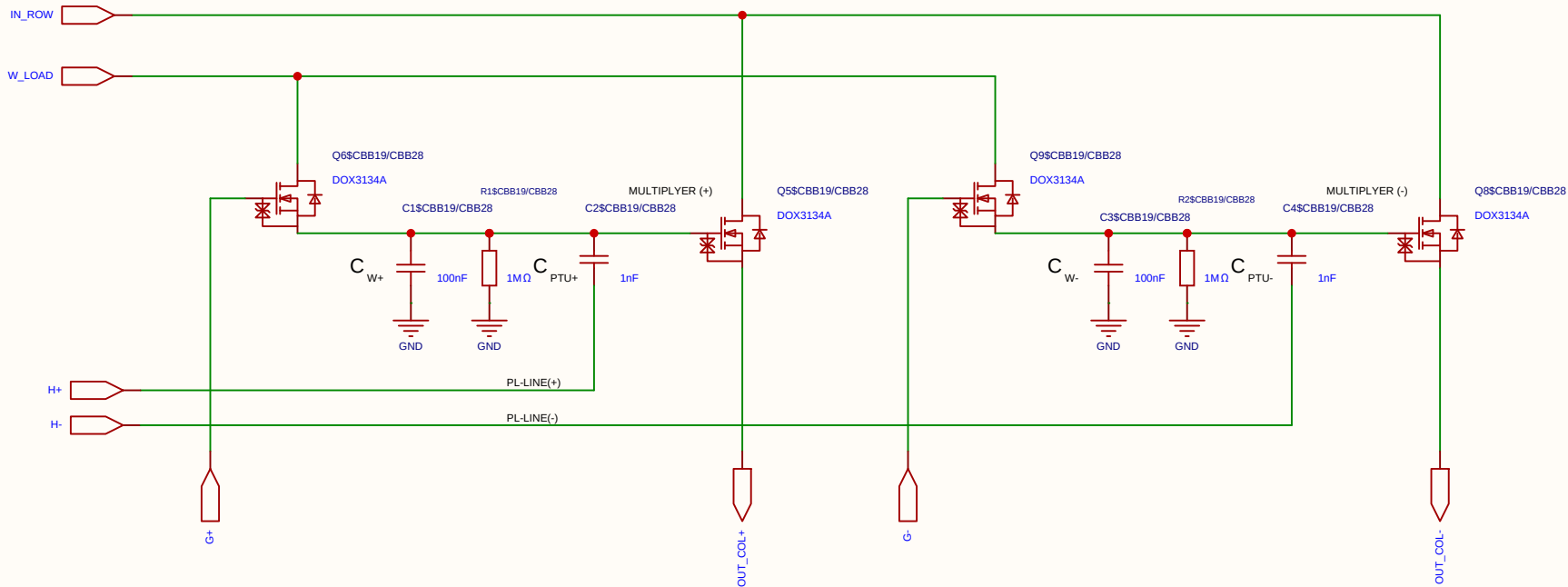
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

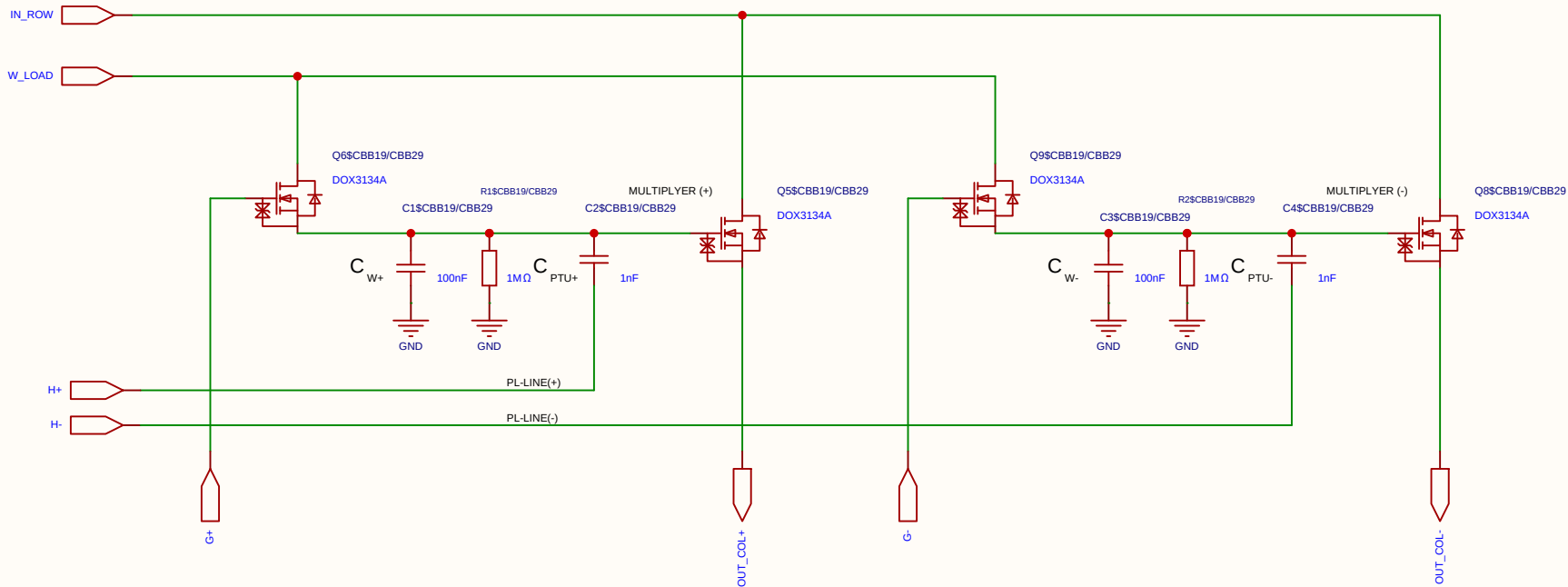
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

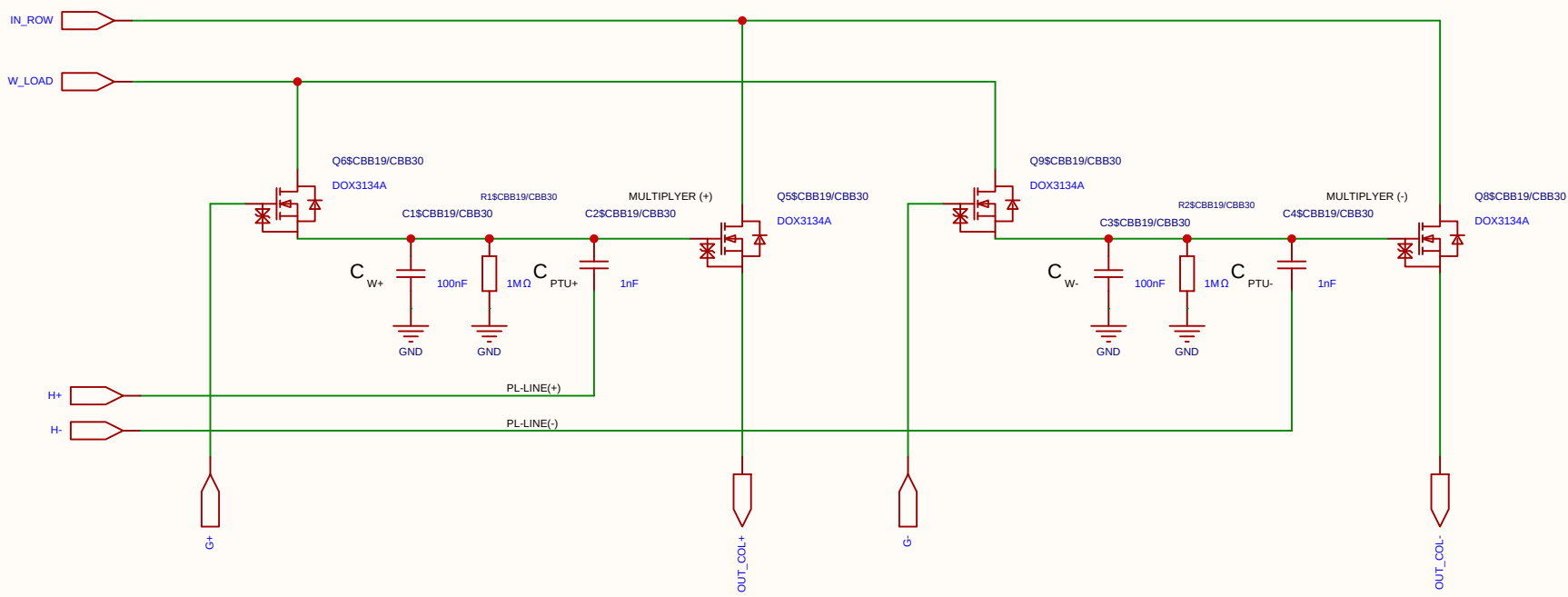
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

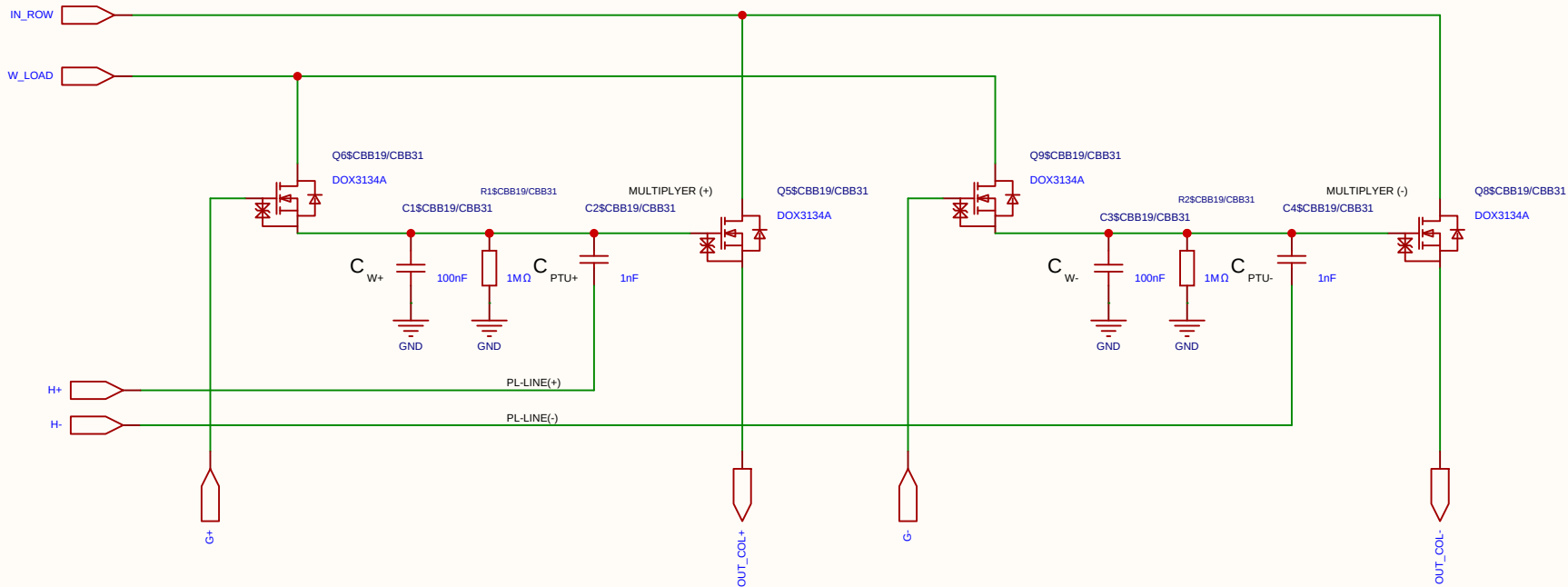
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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Drawn	Olle Welin	Inverted_MAML_6x6			
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Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

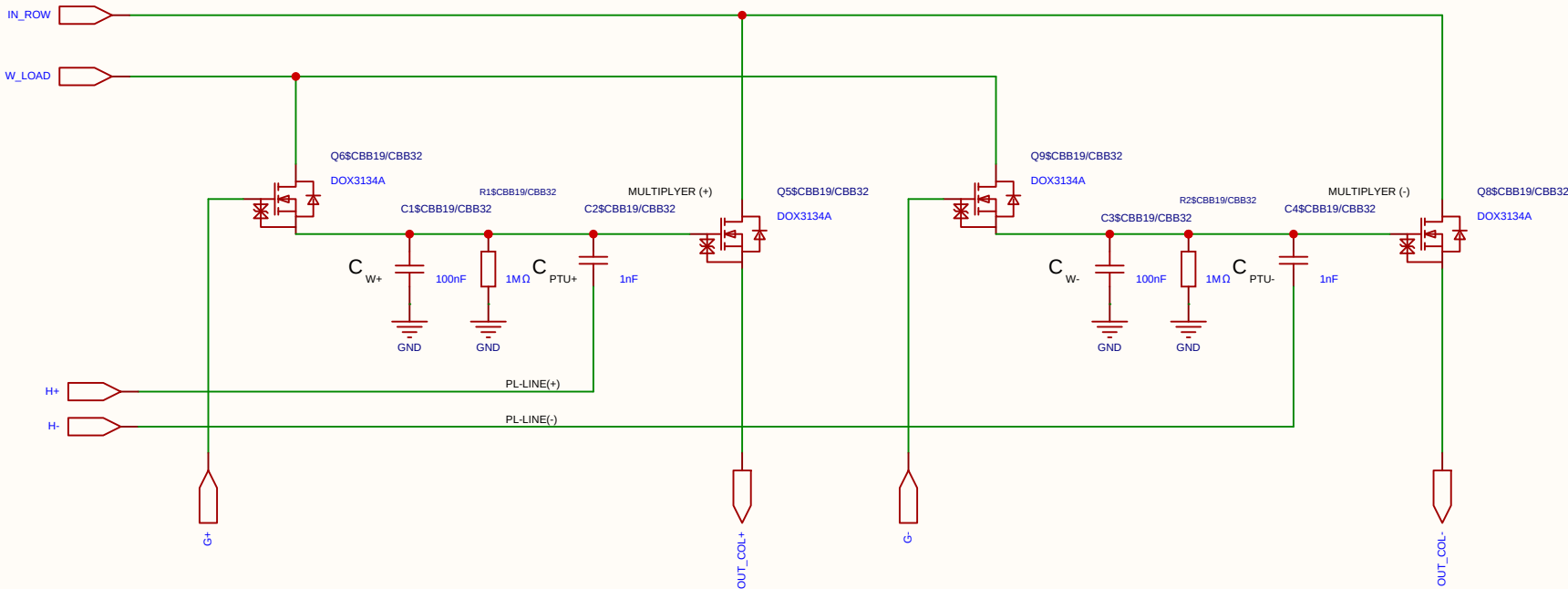
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

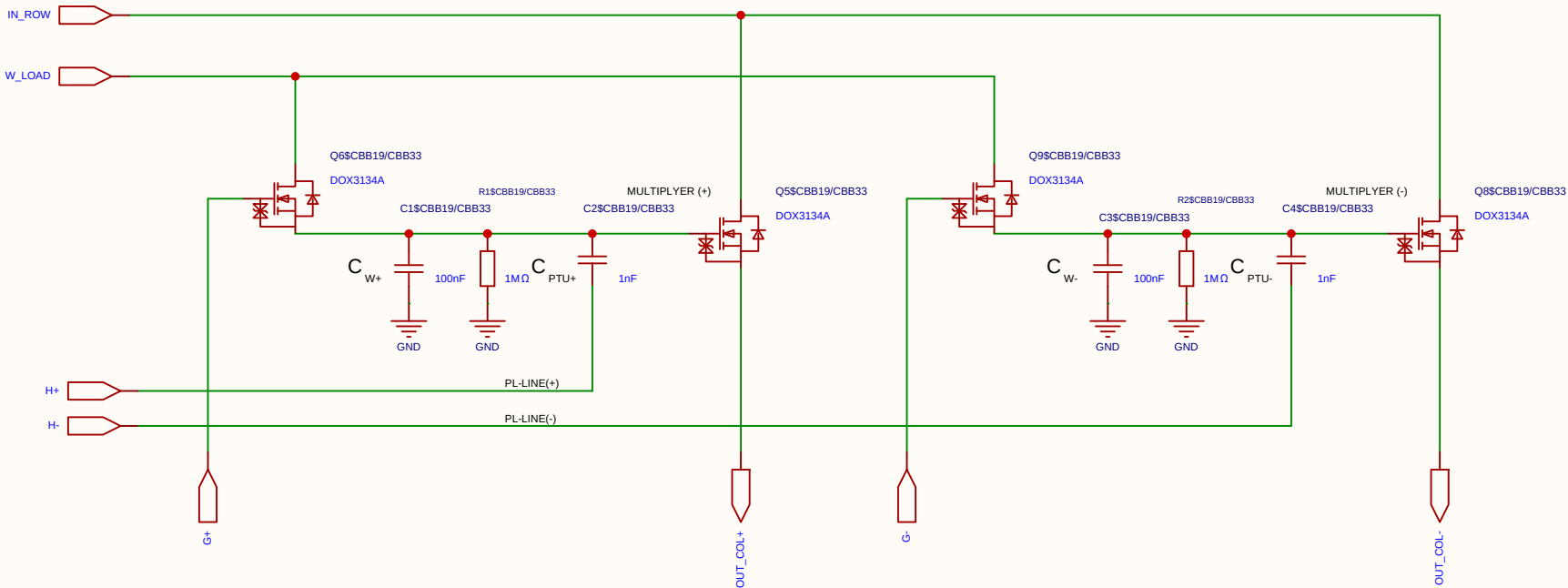
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
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EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

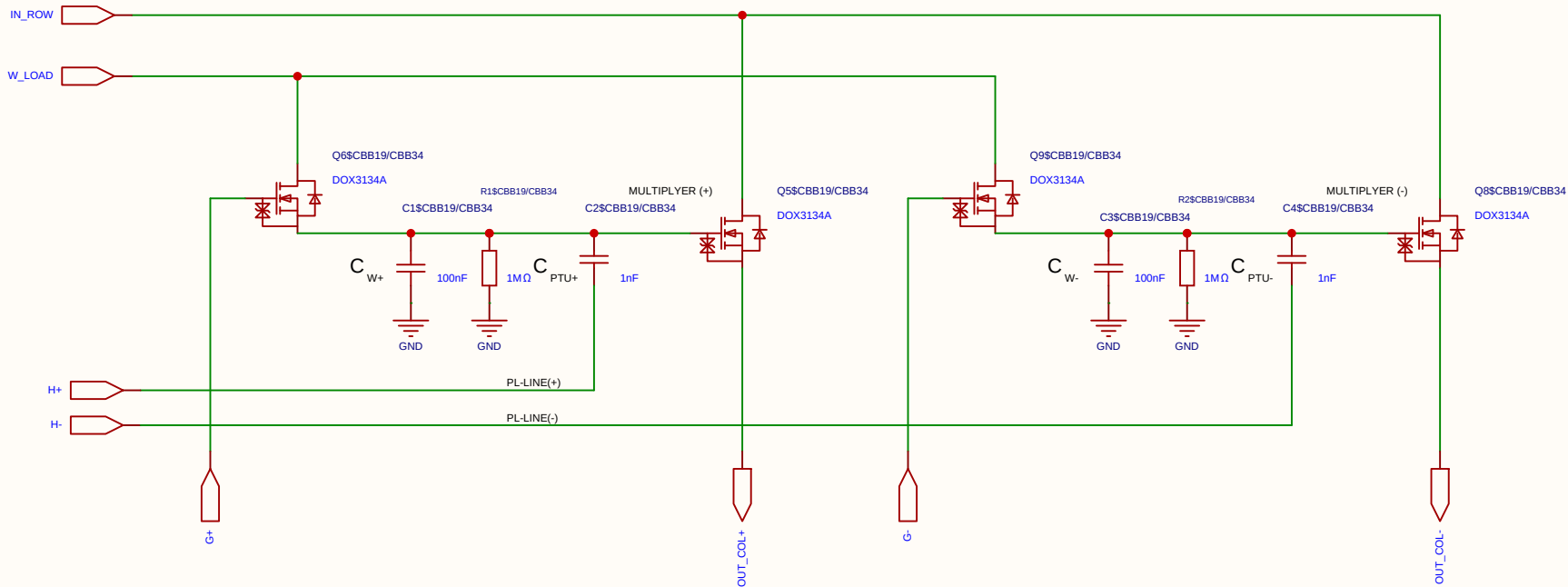
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
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Drawn	Olle Welin	Inverted_MAML_6x6			
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		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

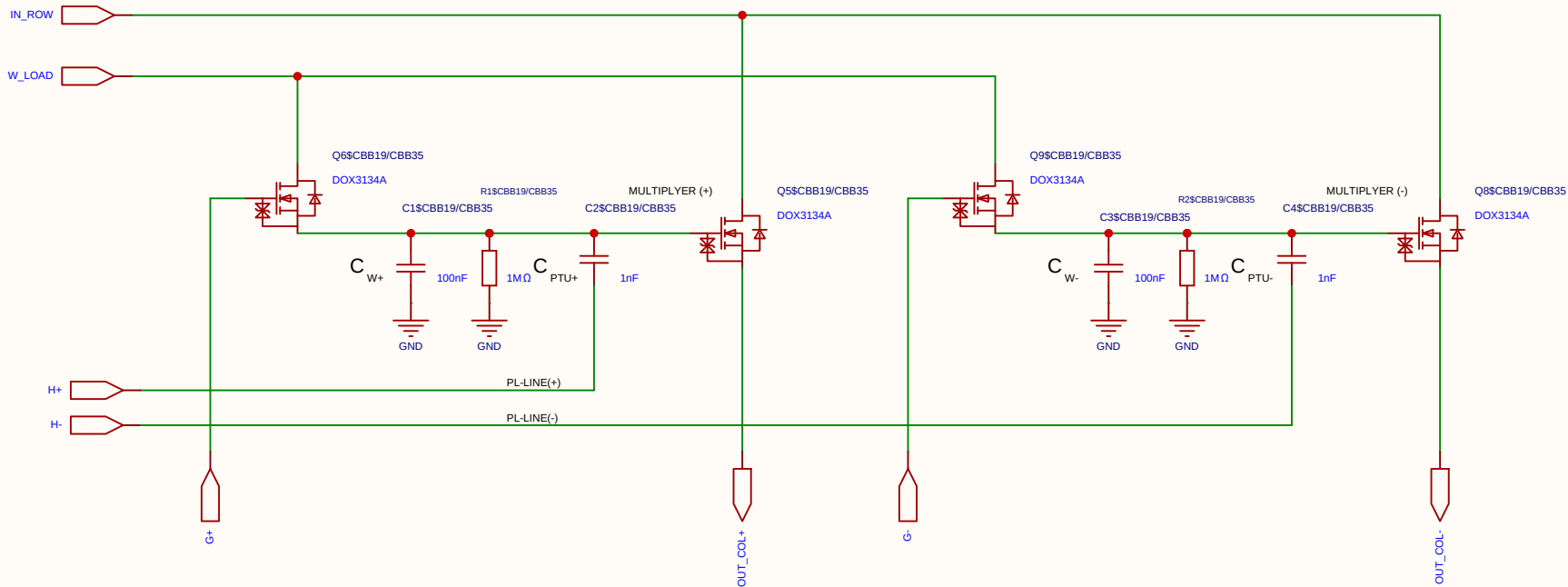
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
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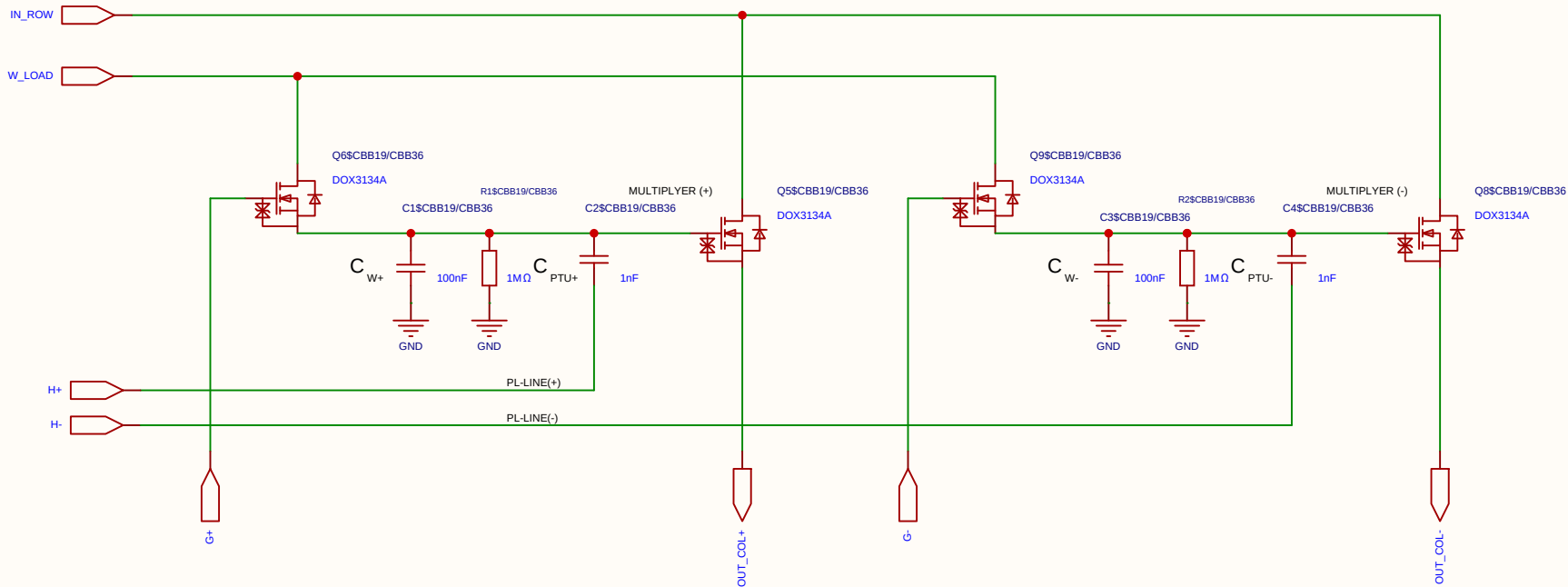
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
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Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
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Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

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2. Resistor Network (Per PL Row)

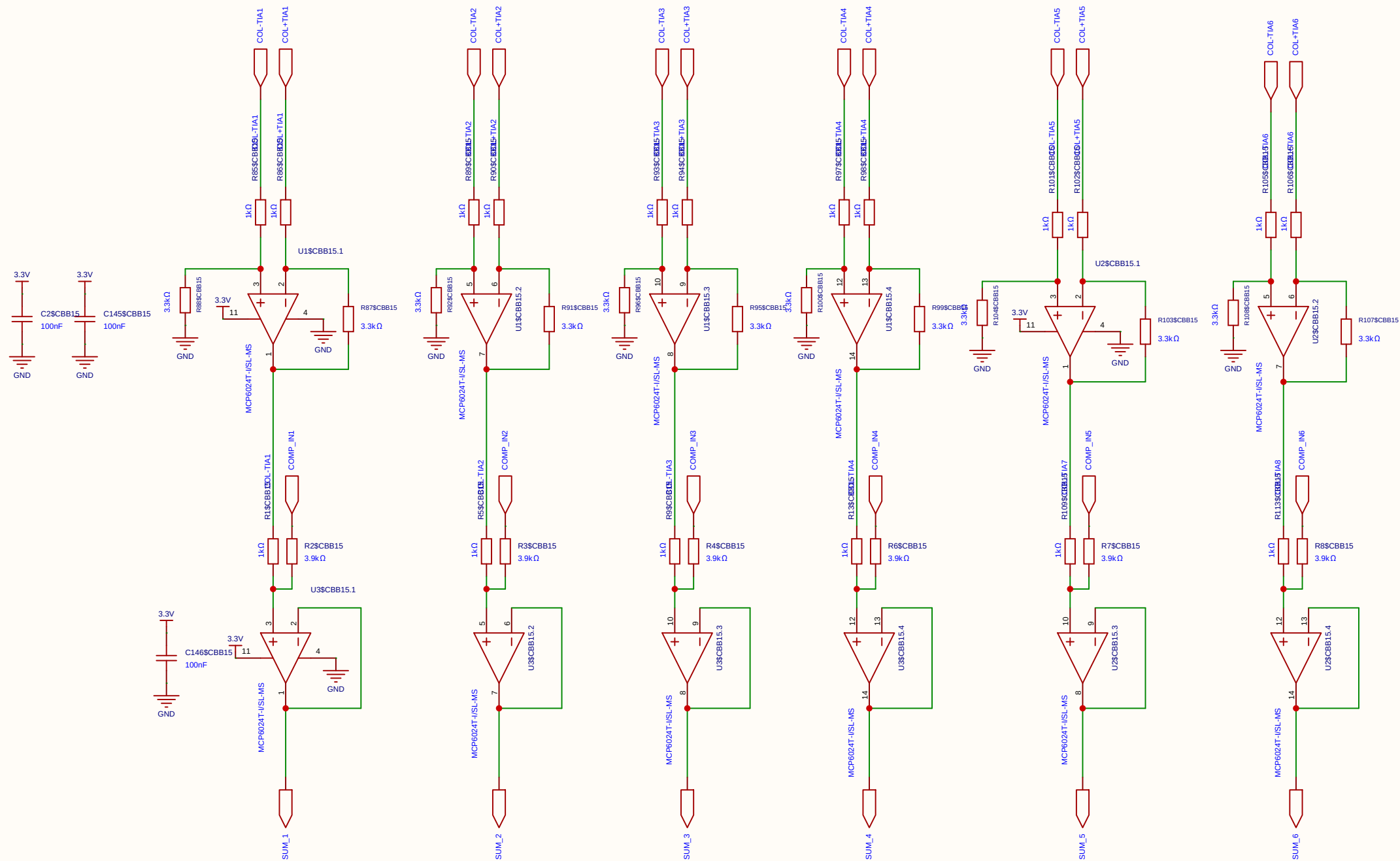
This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

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Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

• **Calibration Example:** To match a 3.3 V ADC, the resistors are configured for a Gain of 6.6×.

$V_{ADC} = 6.6 \times (1.65 \text{ V} - V_{out_TIA})$



Schematic	SUM35			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	sum35
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	